

Newborough Warren Methods Supplement

Hollingham (2026) — Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales.

This document accompanies `report.pdf` and `Supplementary_Material.pdf`. It is the per-script methodological record of the analytical pipeline.

Document version: 1.7.6 (June 2026).

Pipeline: `run_analysis.py` runs 43 steps across 17 phases — 41 analytical steps across 16 analytical phases (Phases 1–16), plus two display/utility steps excluded from the analytical count: Script 26c (step 32, the MSL5 report-format figure-rendering companion to Scripts 26 and 26b, documented in §S.18c) and Script 27 (step 43, the greyscale figure-conversion utility, documented in Appendix A). Phase 12 covers the residual diagnostics (Scripts 22–24); Phase 13 covers the van Willegen MSL analyses (Script 26 observational aggregation, Script 26b UKCP18 climate projections, and Script 26c report-format figures); Phase 14 covers the cluster framework diagnostics: Script 28 (C3 detrend check) and Script 29 (within-C3 variance attribution) from the 2026-05-29 post-review pass, together with Script 30 (C4 constrained- β_3 triangulation sensitivity, added 2026-06-23), all documented in §S.19. Phase 15 covers the observed differential-change and climate-envelope analyses: Script 32 (secular differential water-table movement, report Figure 59), Script 33 (climate-swing amplification and dry-year spring floor, report Figure 60), and Script 35 (per-well climate-sensitivity coefficient), all documented in §S.20. Phase 16 covers four supplementary standalone diagnostics: Script 24b (cluster-stratified residual climatology), Script 31 (independent $k=5$ partition validation), Script 31b (cluster separation versus recoverability), and Script 34 (MSL5 two-window sensitivity demonstration), all documented in §S.21. Two further post-review scripts were inserted into earlier phases: Script 11c (P_{flood} achievability map, Phase 3 step 13, documented in §S.9.3) and Script 14b (bootstrap year-of-crossing, Phase 4 step 16, documented in §S.8.5).

F.1 Scope and audience

The main report's §3 *Methods* is calibrated to a journal-paper length: approximately 25 lines per method, with a single software-reference sentence pointing to the public repository at https://github.com/newbroman/Newborough_Hydrology. That length is appropriate for a paper. It is not enough for the audiences who will actually use this work.

This supplement is for those audiences. Four are anticipated. **Hydrogeological and ecohydrological researchers** rebuilding parts of the analysis on their own sites need the per-script methodological detail that the main report compresses. **Conservation managers at Newborough Warren NNR**, and at Natural Resources Wales more broadly, need to know what each result is and is not telling them — particularly which findings are robust, which are conditional on a methodological choice that could reasonably have gone the other way, and which are honestly tentative. **Future site investigators** — including, almost certainly, future PhD or MSc students at the warren — need a starting point that does not require them to reconstruct the methodological reasoning from the code. And **the author**, three years on, needs to be able to defend any particular methodological choice without re-deriving why it was made.

The supplement is therefore structured as a script-by-script walk through the 41-step analytical pipeline (Phases 1–16 in `run_analysis.py`; Script 26c, the MSL5 report-format figure-rendering companion, is the thirty-second step and is documented in §S.18c, and Script 27, the greyscale figure-conversion utility, is the forty-third step and is documented in Appendix A — both are display/utility steps excluded from the analytical count), with each chapter covering the seven concerns described in the original task brief: motivation, inputs, methodology, site-specific choices and rationale, outputs, limitations and known caveats, and where the result appears in the report. Some chapters are short (figure-only or post-processing scripts); some are longer (the structurally important Scripts 01, 02, 03, 10 suite, 17, 21, 25, 26). Total document length is approximately 110 pages.

This document is **not** a tutorial on Python or scientific computing. It is not a literature review — references appear where they are required for methods provenance, not as a research review. It is not a redo of the main report's results; results appear only as concrete examples of what the methods produce. And it is not exhaustive code documentation — the scripts in `src/` are the code documentation. This supplement is the methodological narrative.

The relationship between this document, the main report, and the existing supplementary material is:

- `docs/report/report.pdf` — the principal scientific deliverable. Canonical statement of what was found.
- `docs/report/Supplementary_Material.pdf` — quantitative supporting material referenced from the report (extended tables, additional figures).
- *This document* — methodological supplement. Canonical statement of how the work was done.

Cross-references work in both directions. The supplement points to the report wherever a result is discussed (§4.2, Table 3c, Figure 11b); the report’s §3 *Methods* should be read in conjunction with the relevant chapters here when methodological detail is needed.

Where this supplement and the main report disagree, the main report is canonical for results and the supplement is canonical for methods. Where the supplement and the source code disagree, the source code is canonical — the supplement describes how each script works, but the script itself is the implementation.

F.2 Field protocol, bucketing convention, and date semantics

The 88-well dipwell network is read monthly by the author. Readings are taken at the **end of each month** — typically the last day of that month, or the first day or two of the following month. Each reading is the water level *for the month just ended*: a measurement taken on 1 May 2026 represents the **April 2026** water level.

Climate data from RAF Valley Meteorological Station (53°14′32″N, ≈16 km from the site) is a monthly total for the same calendar month. Rainfall (mm) and minimum and maximum temperatures (°C) are tabulated month-by-month from 1930 to present. Potential evapotranspiration is computed inside the pipeline (Script 00) using the Thornthwaite method at the RAF Valley latitude, returning a monthly PET total in millimetres.

Bucketing

Readings on physical dates $\leq$ day 15 of month M are bucketed into month $M-1$ because they belong to the previous month’s water table. Readings on physical

dates $>$ day 15 of month M are bucketed into month M . The cutoff is at day 15 because field readings are nearly always taken within the first week of the month or on the last day of the previous month; day 15 is comfortably in the middle of the gap and matches the field convention without ambiguity.

This bucketing is implemented in Script 01 (`01_data_prep.py`). Once bucketed, every monthly timestamp in the pipeline is the first of the month (YYYY-MM-01).

The -01 day component is a pandas formatting artefact and does not refer to the 1st of the month. A row labelled 2007-07-01 is “July 2007” — it contains the end-of-July water level and July’s climate data.

History: why HEADLINE_LAG was 1 and is now 0

Before April 2026, the bucketing in Script 01 used nearest-month assignment: a reading on 1 September was bucketed to September even though it represented August’s water level. The pipeline compensated by applying a one-month lag to rainfall in the SSM, so that September’s labelled water-table change was modelled against August’s labelled rainfall — the correct physical pairing despite the mislabelled month. This was the `HEADLINE_LAG = 1` regime.

The April 2026 bucketing fix corrected the convention to match the field protocol ($\text{day} \leq 15 \rightarrow$ previous month). Once water-table readings are correctly labelled, the SSM should regress the August water-table change against the August rainfall, with no lag. `HEADLINE_LAG` was therefore changed to 0. All regression coefficients are numerically identical to the pre-fix lag-1 results, because the physical pairing is unchanged — only the labelling differs. Hydrograph x-axis labels are now one month earlier than in pre-fix figures; all other numerical outputs are unaffected.

Note on maOD invariance to ground-surface change

The maOD (above-Ordnance-Datum) water-table elevation is a physical quantity independent of the ground surface above it. Removing material from the surface — for example by intentional scraping for slack restoration, as at CEH18 and CEH21 in October 2023 — does not move the water table; the maOD reading is the same before and after. Era-specific data handling is therefore only required where depth-below-ground is the quantity of interest, not where maOD water-table elevations are being averaged or compared. Script 11b (S.9) uses this principle: the DEM elevation at scraped wells is corrected by the scrape depth, but the full pre-and-post observed maOD record contributes to the water-table averaging.

Reporting dates in the report and supplement

Pipeline output CSVs retain the YYYY-MM-01 index for pandas compatibility. The reader never sees these directly. In text — both in the report and in this supplement — monthly data is referred to by month and year only: “July 2007”, “the winter 2017–18 peak”, “the summer 2022 minima”. On figure axes, `matplotlib.dates.DateFormatter('%Y')` (for plots spanning several years) or `'%b %Y'` (for shorter ranges) is used in preference to `'%Y-%m-%d'`, so axis tick labels read as years or “Jul 2007” rather than as specific days. The `CHANGELOG_date_formatting_sweep.md` entry records the audit that brought all figure axes into compliance with this convention.

Hydrology year conventions

Most of the supplement reports time in calendar months and calendar years, as above. Two specific analytical contexts use *hydrology years* instead. The summer transfer functions in Script 11 Sections 2 and 4 (S.9) use an October-start hydrology year (1 October $y-1$ to 30 September y) — the conventional choice for catchment-scale annual water balances. The van Willegen MSL aggregation in Script 26 (S.18), the Tool A spring MSL transfer function in Script 11 Section 5 (S.18b), and the Tool B UKCP18 MSL5 projection in Script 26b (S.18b) all use van Willegen 2025’s **hydrology year B** (1 June $y-1$ to 31 May y). The June-start convention places the entire October-to-May antecedent forcing window — winter peak, winter rainfall, spring rainfall, spring PET — inside a single hydrology year alongside the spring response window it predicts. A reading dated 2010-04 therefore belongs to hydrology year 2010 under both conventions; a reading dated 2009-07 belongs to hydrology year 2010 under hydrology-year-B (because 1 June 2009 falls inside the year-2010 window) but to hydrology year 2009 under the October-start convention. The two conventions co-exist in the pipeline; chapters that use one or the other are explicit about which.

Concrete example

The row labelled 2007-07-01 for well CEH9 contains the following fields in the SSM design matrix (constructed by `model_utils.build_ssm_frame()`):

Field	Value	Meaning
$h(t)$	−0.610 m	End-of-July reading (taken late July / early August)
$h(t-1)$	−0.440 m	End-of-June reading

Field	Value	Meaning
Delta_h	−0.170 m	Water-table change <i>during</i> July
P	0.1019 m	July rainfall total
PET	0.0985 m	July PET total
h_disp_prev	3.260 m	3.7 + (−0.440) — displacement above drainage datum at end of June

The SSM uses this row to explain July’s 170 mm drop using July’s rainfall, July’s PET, and the water-table position at the *start* of July (= end of June).

F.3 The state-space model — displacement formulation

The state-space model (SSM) is the methodological core of the analysis. Every cluster characterization, every BACI step, every forecasting threshold, every scenario response ultimately rests on it. This section gives the canonical form.

Equation

The fitted equation is

$$\Delta h(t) = \beta_1 \cdot P(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot (D + h(t-1))$$

where:

- **$\Delta h(t)$** is the change in water table during month t (m, signed; negative when the water table falls).
- **$P(t)$** is the rainfall during month t (m). Note: under HEADLINE_LAG = 0, no lag is applied. In the pre-bucketing-fix regime this term was $P(t-1)$.
- **$PET(t)$** is the Thornthwaite PET during month t (m).
- **$h(t-1)$** is the water table at the *end* of month $t-1$ (m, signed; negative below ground surface).
- **D** is the drainage datum, DRAINAGE_DATUM = 3.7 m. See *Drainage datum* below.
- **$\beta_1, \beta_2, \beta_3$** are positive coefficients fitted by ordinary least squares (no intercept) using `utils.model_utils.fit_ssm()`.

The quantity $D + h(t-1)$ is the *displacement* of the water table above the drainage datum at the start of month t . With $D = 3.7$ and a typical end-of-previous-month head of -0.4 m, displacement is 3.3 m; with a deeper end-of-month head of -2.0 m, displacement is 1.7 m. The β_3 term says: the deeper the water table sits below ground at the start of a month, the smaller the drainage during that month — Darcy-consistent.

Sign conventions

All three β values are reported as **positive** in the output CSV (`03_master_data.csv`, columns `beta_1_recharge`, `beta_2_atmospheric_draw`, `beta_3_drainage`). Sign conventions are baked into the OLS design matrix, not into the coefficient signs.

In the design matrix constructed by `build_ssm_frame()`:

- The β_1 column is $+P$. A positive β_1 means rainfall raises the water table.
- The β_2 column is $-PET$. A positive β_2 means PET lowers the water table (the negation in the column produces a negative contribution to Δh).
- The β_3 column is $-(D + h_{prev})$. A positive β_3 means displacement above the datum drives drainage downward (Δh decreases as displacement increases).

Two of these signs are hard-asserted by `model_utils.assert_physical_signs()`: a fitted $\beta_1 \leq 0$ or $\beta_2 \leq 0$ halts the pipeline because either is physical nonsense. $\beta_3 > 0$ is soft-asserted — a negative β_3 is anomalous and worth investigating but does not halt the pipeline, because under unusual partitions or short data windows it could reflect a genuine statistical artefact rather than a physical impossibility.

Why $h(t-1)$, not $h(t)$

The drainage term uses the water-table position at the *end* of the previous month, not the contemporaneous level. Two reasons.

First, $h(t)$ is the dependent variable (via $\Delta h = h(t) - h(t-1)$). Using it simultaneously as a predictor creates simultaneity bias — the model would predict the change using the result of that change. The OLS would still fit, but the β_3 estimate would be biased and the interpretation broken.

Second — and physically — drainage *during* a month is driven by the head at the *start* of that month, not the head at the end. The end-of-month head is the result of drainage, not its cause. Using start-of-month head in the predictor row captures the physical dependence correctly.

Why not the within-month mean? The within-month mean $\frac{1}{2}[h(t-1) + h(t)]$ is, in principle, a more faithful approximation to the time-integrated drainage flux — total monthly drainage is the integral of a head-dependent rate, so the trapezoidal form is nearer the exact integral than the forward (start-of-month) form. It is nonetheless not usable, for a fundamental reason: with a single end-of-month reading the within-month mean is unobserved and must be reconstructed as $\frac{1}{2}[h(t-1) + h(t)] = (D + h(t-1)) + \Delta h/2$. That reconstruction carries $h(t)$ — the dependent variable, via $\Delta h = h(t) - h(t-1)$ — back into the drainage predictor, reinstating exactly the simultaneity bias the $h(t-1)$ form is chosen to avoid, and biasing β_3 (attenuation toward zero in general; spurious negative β_3 in the noise-dominated limit). The reconstruction adds no new information beyond the two endpoints; it merely relocates $h(t)$ from response to predictor. The forward–trapezoidal discrepancy that is given up by retaining the start-of-month form is second-order: it scales with $\beta_3 \cdot \Delta h$ (a few mm per month at the fitted coefficients) and, crucially, is an **exact reparameterisation** of β_3 rather than an estimation bias. On monthly data the two discretisations differ only by a global rescaling $\beta_{3_trap} = \beta_{3_fwd} / (1 - \beta_{3_fwd}/2)$, which at the fitted β_3 values is a 1–5 % numerical relabelling of β_3 that leaves the model fit, residuals, and cluster contrasts entirely unchanged. The start-of-month form is therefore not a convenience approximation — it is the only implementable choice given the observational protocol, and it costs nothing in model performance.

Darcy interpretation guard. β_3 is the monthly head-space recession constant — the monthly drainage fraction per metre of displacement above the datum — aggregating hydraulic conductivity K , saturated thickness b , specific yield S_y and flow-path geometry L (Boussinesq-linearised drainage). Its reciprocal $t_R = 1/\beta_3$ is the head-space e-folding timescale (S.12). The datum plays the role of the Darcy base head: β_3 and the datum are the slope and intercept of one flux–head line, so they are partially confounded. β_3 therefore **cannot** be decomposed into K and L independently, nor can the absolute base be located, from the SSM alone. The model accordingly must **not** be presented as a calibrated Darcian gradient model; it is a lumped linear-reservoir model with empirically determined recession constants. The surface-following convention for the datum (discussed below) is consistent with local drainage toward a base that tracks the dune topography rather than a regional sea-level gradient — the test for this is in §S.3 (datum sensitivity) and the supporting evidence is that the per-well optimal datum depth varies modestly with ground elevation (slope +0.15 per metre of elevation, strongly surface-following) rather than tracking an absolute mAOD base.

The implementation in `model_utils.build_ssm_frame()` is

`h_disp_prev = DRAINAGE_DATUM + h.shift(1)`

which produces displacement above the datum *at the end of the previous month*. Every script that fits an SSM uses this function — there is no script that reimplements the displacement calculation locally.

Drainage datum (3.7 m)

`DRAINAGE_DATUM = 3.7` m is the depth below ground surface used as the displacement reference. The value was selected to give comfortable β_3 identification at the forest clusters (C4 and C5), where β_3 is hardest to pin down. At the live empirical minimum of 1.7 m — the shallowest depth at which all five clusters simultaneously satisfy $\beta_3 > 0$ with $p < 0.05$ — C4's β_3 p-value sits at the significance edge (0.040). At 3.7 m it drops to 0.0027, with C5 also gaining substantially (β_3 p-value from 4×10^{-11} to 4×10^{-16} and R^2 from 0.648 to 0.680). The trade-off is small R^2 penalties at C1 Lake Edge (−0.052) and C2 Dune (−0.029), where β_3 is over-determined and remains significant at $p < 10^{-25}$ at either depth. 3.7 m also aligns with the Script 16 water-balance sensitivity analysis, which matters for keeping the displacement reference consistent across the pipeline. The full sensitivity sweep is in `03_state_space_model.py` (output `03_08_datum_sensitivity.csv` and `.png`) and the trade-off is discussed at the per-cluster level in chapter S.3.

The role of the datum is to shift the reference for the drainage term. Without it (i.e. with $h(t-1)$ instead of $(D + h(t-1))$ in the design column), the C3, C4, and C5 clusters produced negative β_3 under the head-below-ground convention. This was a sign-convention artefact, not a physical anomaly: it reflected that the OLS was correlating drainage with a quantity that crossed zero rather than staying on one side of a fixed reference. Setting the reference 3.7 m below ground places every observation comfortably on the positive side of the datum, restoring physical interpretability without changing the predictive content of the model.

Δh is invariant under the choice of datum (the datum cancels in first differences). The coefficients β_1 and β_2 are also invariant. Only β_3 shifts numerically, in a way that preserves its physical interpretation as a Darcy drainage coefficient.

Per-well optimal datums (where each well is fitted with whatever datum maximises its R^2) show a coherent spatial gradient: C1 Lake Edge wells optimize at a median of ~0.8 m, C2 Dune at ~0.9 m, C3 Western Residual at ~1.7 m, C5 Coastal Forest at ~1.9 m, and C4 Main Forest at ~2.7 m. The pattern reflects deepening effective drainage base from the lake-adjacent eastern margin toward the forested western clusters. The uniform 3.7 m datum produces a network

mean R^2 penalty of +0.028 across the network, heaviest at C1 Lake Edge (+0.069) and lightest at C4 Main Forest (+0.001) — making the uniform datum a clean choice for cluster-level work, while a reader with a single-well question can read the per-well optimum off `03_09_well_optimal_datums.csv`. Per-well datum results are also discussed in §3.4 of the main report and at length in chapter S.3.

Implementation: `model_utils.fit_ssm()`

The authoritative implementation is `src/Utils/model_utils.py`. Every per-well, per-cluster, or per-network SSM fit in the pipeline goes through `fit_ssm()` (no-intercept, “Model A”) or `fit_ssm_intercept()` (with-intercept, “Model B”). Scripts must not reimplement OLS locally.

The function signatures are

```
fit_ssm(h_series, climate, lag=None, window=None,
        drainage_datum=DRAINAGE_DATUM, min_obs=MIN_OBS,
        provenance=None, exclude_interpolated=False)
fit_ssm_intercept(h_series, climate, lag=None, window=None,
                  drainage_datum=DRAINAGE_DATUM, min_obs=MIN_OBS,
                  provenance=None, exclude_interpolated=False)
```

with defaults that produce the canonical SSM specification. The returned dict carries long-form keys (`beta_1_recharge`, `beta_2_atmospheric_draw`, `beta_3_drainage`), their p-values, R^2 , n , and a residual series. The keys are intentionally verbose so that downstream code is self-documenting and cross-script searches (`grep beta_1_recharge`) return only genuine matches.

The `provenance=` and `exclude_interpolated=` kwargs control interpolated-row handling. Default behaviour: with `exclude_interpolated=False` (the default), interpolated rows from `01_wells_provenance.csv` remain in the SSM fit and the canonical $\beta_1/\beta_2/\beta_3$ coefficient table is preserved. Setting `exclude_interpolated=True` with a provenance series supplied restricts the fit to measured rows only, as a documented sensitivity path; the chapter that consumes this — §S.3 — describes when it would be invoked.

Two regimes: no-intercept (A) and with-intercept (B)

The headline SSM is Model A — no intercept. The justification: the SSM is a physically motivated decomposition of Δh into rainfall, PET, and drainage contributions, and there is no remaining physical mechanism for a constant bias. A non-zero intercept under Model A would indicate that one or more β coefficients is absorbing a constant lateral inflow/outflow that the model does not explicitly represent.

Model B (with intercept α) is fitted in Scripts 07, 08, 22, and 24 as a diagnostic. A statistically significant α at a particular well flags that well as poorly described by the headline SSM — typically because lateral subsidies (ridge-derived recharge, lake exchange) or local hydraulic effects matter at that location. The intercept audit in Script 08 (LCSC vs Traditional Linear Model benchmarking) uses this distinction to identify wells where the SSM's physical decomposition outperforms a more flexible but unphysical linear model.

Sanity-check assertions

`assert_physical_signs()` is called at every cluster fit in Script 03. It returns a (`hard_violations`, `soft_warnings`) tuple. Hard violations halt the pipeline after the relevant diagnostic outputs (LOO, bootstrap, datum sensitivity) have been written, so the investigator has the supporting tables to diagnose the failure rather than just a stack trace. Soft warnings ($\beta_3 \leq 0$) are printed and recorded; they do not halt.

The principle is that the pipeline should fail loudly when its physical assumptions are violated. A negative β_1 is not a “degraded fit” — it is physical nonsense that should never silently propagate into a report number.

F.4 Configuration, constants, and the cluster partition

All values that are constant across the pipeline live in `src/utils/config.py`. Where a scientific decision determines a value — the canonical 24% canopy interception fraction, the 3.7 m drainage datum, the Curreli ecohydrological thresholds, the UKCP18 climate scaling factors — that decision is recorded against the constant. Scripts must not redefine these values locally; the principle is that there is one place to change a value, and the change propagates everywhere.

What is in `config.py`

The full inventory:

Constant	Value	Origin
CLUSTER_LABELS	{1: "C1 (Lake Edge)", 2: "C2 (Dune)", 3: "C3 (Western Residual)", 4: "C4 (Main Forest)", 5: "C5 (Coastal Forest)"}	k=5 partition, this study

Constant	Value	Origin
CLUSTER_COLOURS	{1: "#1a6faf", 2: "#2ca02c", 3: "#d62728", 4: "#7f77dd", 5: "#8B4513", 6: "#0072B2"}	Canonical palette; ID 6 reserved
CLUSTER_COLOURS_BW	greyscale equivalents at ≈ 40 -unit luminance spacing	Journal print mode (BW_MODE)
CLUSTER_MARKERS	{1: 'o', 2: 's', 3: '^', 4: 'D', 5: 'P'}	Cluster-shape symbols on spatial maps
DRAINAGE_DATUM	3.7 m	Sensitivity analysis, Script 03 (03_08)
HEADLINE_LAG	0	Post-bucketing-fix value (was 1 historically)
FOREST_INTERCEPTION	0.24	Freeman (2008), Newborough Corsican pine
FOREST_CIDS	(4, 5)	k=5 partition mapping
BROADLEAF_INTERCEPTION	0.15	Komatsu et al. (2011) annual mean
BROADLEAF_B2_SUMMER	1.0750	Script 21 monthly β_2 profile, May–Oct mean (canopy-on window)
BROADLEAF_B2_WINTER	0.8817	Script 21 monthly β_2 profile, Nov–Apr mean (canopy-off window)
REFERENCE_CUTOFF_DATE	"2026-02-01"	Reference-network selection in Script 01
RAF_VALLEY_LAT_DEG	53.25	53°14'32"N, used for Thornthwaite PET
SD15b, SD15b_REC, SD16, SD16_REC	0.61, 0.75, 0.98, 1.20 m	Curreli et al. (2013) summer slack thresholds
SD15b_WINTER, SD16_WINTER	0.10, 0.25 m	Curreli winter flooding limits
UKCP18_DRY_*,	seasonal P and PET	UKCP18 RCP8.5 Wales,

Constant	Value	Origin
UKCP18_WET_*	scaling factors	50th percentile, 2050s
DEM_VMIN, DEM_VCENTER, DEM_VMAX	0.0, 12.0, 35.0 m AOD	TwoSlopeNorm anchors for DEM rendering

The B&W rendering mode (BW_MODE) is toggled by the NRG_BW_MODE environment variable, which `run_analysis.py` sets when invoked with `--greyscale-full`. When active, scripts switch to the CLUSTER_COLOURS_BW palette, apply BW_HATCHES to bar charts, use BW_LINESTYLES for multi-series line plots, and call the hillshade DEM loader for map backgrounds. The convenience functions `get_cluster_colour()`, `get_bar_hatch()`, `get_line_style()`, `get_line_colour()`, and `get_cmap()` provide colour-mode and B&W-mode behaviour in a single call, so a script does not need to branch on BW_MODE itself.

The k=5 partition

The reference network is partitioned into five behavioural clusters using Ward's linkage on correlation distance between cluster-mean hydrographs (Script 02). The cluster IDs and labels are:

	ID	Label	Anchor wells	Block label	Substrate / vegetation
	1	C1 Lake Edge	ceh5, ceh11	Eastern Block	Lake- adjacent, finer sediments
	2	C2 Dune	d10	Eastern Block	Mature open dune, shallow till
	3	C3 Western Residual	nw1	Western Block	Deep aeolian sand
	4	C4 Main Forest	ceh2	Forest	Corsican pine on deep sand
	5	C5 Coastal Forest	ceh16, nw9	Coastal Forest	Corsican pine, coastal margin

02_clustering.py carries a CLUSTER_ID_ANCHORS dict that pins Ward's raw (arbitrary) output to these canonical IDs. A guard at module load asserts that CLUSTER_ID_ANCHORS and CLUSTER_LABELS agree. Membership counts (live, May 2026 partition under 01_data_prep.py v1.3.0 which blacklists the tidal pdfs well) are C1 = 7, C2 = 24, C3 = 21, C4 = 9, C5 = 5; total 66 wells in the reference network.

Partition history

The k=5 partition supersedes an earlier k=6 partition that included a separate “Lake” cluster for Llyn Rhos-Ddu. Under k=5, Llyn Rhos-Ddu is treated as a fixed-head boundary feature rather than a cluster — it is the dominant local drainage sink for C1 Lake Edge wells but is not itself a behavioural cluster. The old C5 (Coastal, n=1, single tidally-influenced well) and old C6 (Lake, n=1) were dropped at the partition step as physically unreliable singletons. Their names re-appear under k=5 with different membership — old C5 ≠ new C5.

References to “C6 Lake” or “tidal exclusion” framing of C5 elsewhere in the project file store are from the superseded k=6 partition and should be ignored.

The identity-vs-integer keying principle

This principle warrants its own section here, because future partition changes are not impossible (a sixth cluster could be identified, or the C5 Coastal Forest subset could be reabsorbed into C3 if the coastal-retreat signal proves to be exogenous), and the principle determines how downstream code survives.

Things keyed on **cluster identity** — labels, colours, markers, anchor wells, well-to-cluster membership tables — survive a partition change cleanly. When clusters are re-numbered, these values stay attached to the same physical cluster and are reassigned to the new ID automatically.

Things keyed on **cluster integer** — Python dicts mapping 1, 2, ... to physical quantities like specific yield, peak month, trend value, flood frequency, residual SE — are physical inputs to downstream arithmetic. When the partition changes, the dict key is just an integer; it does not follow a cluster around. **Each entry must be checked individually**: does the value still apply to whatever physical cluster has that integer ID under the new partition?

The convention going forward: if a dict is keyed by integer cluster ID and holds anything other than labels, colours, or markers, treat it as physical data tied to a specific cluster. When the partition changes, walk through every such dict and verify each entry; do not assume.

Several dicts in the codebase are physical-data-keyed-by-integer and need verification under any future partition change: SUMMER_TRENDS and FLOOD_FREQ in Script 16, RESIDUAL_PCT_SE in the water-balance work, and CLUSTER_PEAK_MONTH in Script 11. These dicts are now read dynamically from upstream pipeline CSVs where possible (03_cluster_peak_months.csv for the peak months, 16_water_bal_table.csv for residuals), but any remaining hardcoded entries should be revisited if the partition changes.

Specific yield: two values per cluster

Two independent methods produce per-cluster Sy estimates, and both are reported in the report. The supplement chapters for Scripts 16 (water balance), 17 (WTF cluster Sy), and 18 (WTF spatial Sy) describe each in detail. For reference here:

Cluster	Fetter mass-balance Sy	WTF cluster median Sy (corrected, Script 17)	WTF well-level cluster mean (Script 18)
C1 Lake Edge	0.08	0.223	0.217
C2 Dune	0.12	0.234	0.227
C3 Western Residual	0.12	0.259	0.257
C4 Main Forest	0.12	0.227 (corrected) / 0.215 (uncorrected)	0.202
C5 Coastal Forest	0.12	(corrected variant produced by Script 17 under FOREST_CIDS = (4, 5); see Phase 6 chapter)	(see Phase 8 chapter)

The Fetter values are the operational Sy used in the water-balance arithmetic of Script 16 (Table 3b in the main report). The WTF values are the empirical estimates from rising-limb event analysis (Script 17, Table 3c). They are not in conflict; they are independent estimates from different methods, and the report cites both. The Phase 6 chapter explains why the corrected and uncorrected C4 medians both appear in Table 3c, and the Phase 8 chapter explains why the well-level cluster mean (0.202 for C4) is lower than the cluster-aggregate value (0.227).

Scripts in the post-Phase-2 region — naming and step assignment

The scripts in the post-Phase-2 region of the pipeline carry filename prefixes, orchestrator step numbers, and chapter assignments aligned as follows:

- **Script 11c** (11c_pfflood_achievability.py) — Per-well P_flood achievability categorical priority map. Operationalises Conclusion 4's $\lambda < 1.5$ priority criterion. Step 13/43, Phase 3. Routed from the 2026-05-29 post-review cascade. Documented in S.9.3.
- **Script 14b** (14b_year_of_crossing.py) — Bootstrap year-of-crossing for Curreli (2013) ecological thresholds. Step 16/43, Phase 4. Routed from the 2026-05-29 post-review cascade. Documented in S.8.5.
- **Script 25** (25_coastal_gradient.py) — Coastal-retreat gradient analysis. Step 26/43, Phase 11. Documented in S.15.
- **Script 26** (26_van_willegen_msl.py) — Van Willegen 2025 observational 5-year MSL aggregation. Step 30/43, Phase 13. Documented in S.18.
- **Script 26b** (26b_van_willegen_msl_projections.py) — UKCP18 RCP8.5 MSL5 climate projections (Tool B). Step 31/43, Phase 13. Documented in S.18b.
- **Script 26c** (26c_msl5_report_figures.py) — MSL5 report-format figures for §4.8.4 and §4.10.1 of the main report. Display-only companion to Scripts 26 and 26b; reads only canonical outputs from Scripts 26, 26b, and 19; no recomputation. Step 32/43, Phase 13. Excluded from the supplement's 41-step analytical count. Documented in S.18c.
- **Script 28** (28_c3_detrend_check.py) — C3 detrend check; quantitative validation of the aquifer-architecture framing. Step 33/43, Phase 14. Routed from the 2026-05-29 post-review cascade. Documented in S.19.1.
- **Script 29** (29_c3_within_variance_check.py) — Within-C3 variance attribution. Step 34/43, Phase 14. Routed from the 2026-05-29 post-review cascade. Documented in S.19.2.
- **Script 30** (30_c4_constrained_fit.py) — C4 constrained- β_3 triangulation sensitivity. Step 35/43, Phase 14. Added 2026-06-23. Documented in S.19.3.
- **Script 32** (32_differential_movement.py) — Secular differential water-table movement; report Figure 59. Step 36/43, Phase 15. Documented in S.20.1.
- **Script 33** (33_envelope_amplification.py) — Climate-swing amplification and dry-year spring floor; report Figure 60. Step 37/43, Phase 15. Documented in S.20.2.

- **Script 35** (`35_per_well_amplification.py`) — Per-well climate-sensitivity coefficient (discrete companion to the Figure 60a surface). Step 38/43, Phase 15. Documented in S.20.3.
- **Script 24b** (`24b_residual_climatology.py`) — Cluster-stratified residual climatology; supplementary diagnostic. Step 39/43, Phase 16. Documented in S.21.1.
- **Script 31** (`31_cluster_validation.py`) — Independent $k=5$ partition validation; supplementary diagnostic. Step 40/43, Phase 16. Documented in S.21.2.
- **Script 31b** (`31b_separation_vs_recoverability.py`) — Cluster separation versus recoverability; supplementary diagnostic. Step 41/43, Phase 16. Documented in S.21.3.
- **Script 34** (`34_window_sensitivity.py`) — MSL5 two-window sensitivity demonstration. Step 42/43, Phase 16. Documented in S.21.4.
- **Script 27** (`27_greyscale_figures.py`) — Post-pipeline greyscale figure-rendering utility. Step 43/43, Phase 17. Excluded from the supplement’s 41-step analytical count. Documented in Appendix A.

MSL aggregation — constants and conventions

A new family of constants in `utils/config.py` parameterises the van Willegen 5-year mean spring water level (MSL5) aggregation introduced at Script 26 and consumed by Tools A and B in S.18b. These constants are stable across the pipeline (changing them would re-define the metric) and are documented here as the canonical source.

- `MSL_SPRING_MONTHS = (3, 4, 5)` — March, April, May; the spring window from van Willegen 2025 Table 2.
- `MSL_HYDRO_YEAR_START_MONTH = 6` — June; the start of van Willegen’s hydrology year B (1 Jun $y-1$ to 31 May y). See F.2 for the two-hydrology-year-convention discussion.
- `MSL_DEFAULT_WINDOW_YEARS = 5` — the 5-year averaging window length; van Willegen et al. (2025) sensitivity-tested this choice and found cross-correlation stabilises at 5 years.
- `MSL_MIN_MONTHS_PER_SPRING = 3` — strict 3 of 3 completeness rule for a single annual MSL to be valid.
- `MSL_MIN_YEARS_IN_WINDOW = 5` — strict 5 of 5 completeness rule for a 5-year MSL window to be valid. Stricter than van Willegen 2025; consistent

with the BACI summer-minima `min_measured=2` pattern in Scripts 09c and 10d.

- `MSL_TRAJECTORY_START_YEAR = 2014` — earliest window-end included in the cluster trajectory and per-quadrat-well figures, set to exclude windows that draw on the pre-2010 network of fewer than 35 wells. Per-well CSVs retain the full record from window-end 2009 onwards.
- `VW_QUADRAT_WELLS` — tuple of 17 lowercase piezometer IDs with co-located permanent vegetation quadrats from van Willegen 2025 Table 1: `ceh1`, `ceh4`, `ceh5`, `ceh8`, `ceh9`, `ceh22`, `ceh23`, `ceh24`, `ceh26`, `nw2`, `nw3`, `nw4`, `nw5`, `nw6`, `nw7`, `t41`, `wmc2`. Sixteen of these pass the strict completeness rules; T41 fails because of insufficient recent record.

A pair of intervention-line-colour constants for the trajectory figures also lives in `utils/config.py`:

- `INTERVENTION_COLOUR_SCRAPE = "#7b3294"` — purple; used for the April 2015 CEH36 scrape and the October 2023 CEH18/CEH21 re-scrape.
- `INTERVENTION_COLOUR_CLEARFELL = "#e66101"` — orange; used for the December 2017 pine clearfell.

UKCP18 multipliers used by Tool B are central-estimate Wales 50th-percentile values for RCP8.5 at the 2050s and 2080s. The 2050s multipliers exist in `utils/config.py` as the `UKCP18_DRY_*` / `UKCP18_WET_*` constants (already used by Script 19's scenario viewer); the 2080s multipliers are currently hardcoded in Script 26b's `UKCP18_SCENARIOS` dictionary. Centralising the 2080s constants in `config.py` is documented as a follow-up in the Tool B changelog and in §S.18b.6.

pipeline_params.py — the consolidated scenario parameter file

A subset of `config.py` constants is supplemented at run time by values that are computed by the pipeline itself and consumed by downstream scenario scripts. These live in a per-run CSV at `outputs/01_data_prep/pipeline_scenario_params.csv`, managed by `utils/pipeline_params.py`.

The architecture is producer-consumer with explicit provenance:

- **Script 01** writes the initial file at the end of data preparation, populating summer climate means, head displacements, and forest flags directly; defaulting β_1 , β_2 , β_3 , S_y , and the clearfell/thinning β_2 multipliers if upstream outputs don't exist yet.
- **Script 03** updates β_1 , β_2 , β_3 once it has fitted them.

- **Script 10e** updates the clearfell and thinning β_2 multipliers from the BACI-corrected Edge-tier ratio.
- **Script 17** updates S_y from the WTF cluster medians.
- **Downstream consumers** (Scripts 09b, 09d, 19, 21) call `load_params()` to read the consolidated file in one go.

Each value carries a `source_*` column flagging whether it is from "pipeline" (a script has updated it) or "defaults" (it is still a placeholder from the initial write). Where any consumer reads a placeholder, a warning is printed recommending a second pipeline run. The two-pass workflow described in `PIPELINE_README.md` exists for this reason: the scenario figures use S_y and β_2 multipliers that are computed later in the pipeline, so the first pass uses fallbacks and the second pass uses canonical values.

F.5 Shared utility modules

Eight modules in `src/utils/` carry the shared functionality that the analytical scripts depend on. The module inventory:

`config.py`

Constants and palette. F.4 covers its contents in full.

`paths.py`

Single source of truth for every input and output file path. The module is organized into four blocks:

1. **Root directories** — `ROOT_DIR`, `DATA_DIR`, `OUT_DIR`, and per-script subdirectories `DIR_00` through `DIR_27`, plus `DIR_26B` for Script 26b's outputs. A helper function `make_all_dirs()` creates any missing directories. Scripts call this at the top of their main function rather than checking individual directories.
2. **Data inputs** — paths to raw CSV and GIS files in `data/`: `DATA_WELLS_RAW` (the cleaned monthly dipwell CSV), `DATA_LOCATIONS_RAW`, `DATA_CLIMATE_RAW`, the KML files (`Features.kml`, `streams.kml`, `clearfell.kml`, `broadleaf_restock.kml`), the DEM (`newborough_dem.tif`), and the precomputed coastal-distance CSV (`DATA_DIST_COAST`).
3. **Intermediate files** — outputs written to `OUT_DIR` root by upstream scripts and read by downstream scripts. These are named with the `INT_` prefix: `INT_WELLS_CLEAN`, `INT_CLIMATE`, `INT_LOCATIONS`, `INT_MASTER_DATA`,

INT_REGIONAL_AVG, INT_CLUSTER_AVG_MAOB, INT_CLUSTER_PEAK_MONTHS, and so on.

4. **Final outputs** — figures, tables, and report CSVs written to per-script subdirectories. These are named with the OUT_NN_ prefix to indicate which script writes them: OUT_03_MECHANISTIC_TABLE, OUT_10E_COEFF_SHIFTS, OUT_25_FIT_PARAMETERS, and so on.

The naming convention is rigid: every path constant has a single owner (the script that writes it) and many readers. A grep for OUT_03_MECHANISTIC_TABLE finds the writer (Script 03) and every reader (Scripts 07, 11, 16, 19, 21). No script reads or writes a hardcoded path; every path goes through `utils.paths`.

`model_utils.py`

The SSM specification. F.3 covers `build_ssm_frame()` and `fit_ssm()`; the full function inventory:

- `build_ssm_frame(h_series, climate, lag, window, drainage_datum)` — align well and climate data, compute the SSM predictor columns (`h`, `h_prev`, `Delta_h`, `P`, `PET`, `h_disp_prev`), drop NaN, apply optional windowing. The data-alignment helper every script was previously duplicating.
- `fit_ssm(...)` — no-intercept OLS (Model A / headline SSM). Returns a dict with `beta_1_recharge`, `beta_2_atmospheric_draw`, `beta_3_drainage`, their p-values, R^2 , `n`, and the residual series.
- `fit_ssm_intercept(...)` — with-intercept OLS (Model B), returning the same dict plus α and its p-value. Used by Scripts 07, 08, 22, 24.
- `simulate_ssm(h0, P, PET, b1, b2, b3, drainage_datum)` — iterative forward simulation using the displacement recurrence $h(t) = (1 - \beta_3) \cdot h(t-1) + \beta_1 \cdot P(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot D$. Used by Scripts 07, 08, 09, 15.
- `pflood_lambda(h_target, h_0, b1, b2, b3, months, P_clim, PET_clim, drainage_datum)` — iterated closed-form `P_flood` threshold with the datum drain correction. Solves for the rainfall multiplier λ that brings `h` from `h_0` to `h_target` over a specified horizon. Returns λ , `P_flood` in mm, the collapsed linear-form slope and intercept (so `P_flood` = `A`·`d` + `B`), and the weighted sums. Used by Scripts 11 and 11b.
- `monthly_perturbation(b1, b2_base, b2_scen_arr, P_eff_base, P_eff_scen, monthly_PET)` — single-step forcing-change response. Computes $\Delta h(m) = \beta_1 \cdot (P_{scen}(m) - P_{base}(m)) - (\beta_2_{scen}(m) -$

β_2 _base)·PET(m). Used by Script 21 for forestry scenarios; replaced an earlier equilibrium formulation ($\Delta h / \beta_3$) that produced physically implausible magnitudes.

- `assert_physical_signs(fit, context)` — sign-rule enforcement. Returns hard violations ($\beta_1 \leq 0, \beta_2 \leq 0$) and soft warnings ($\beta_3 \leq 0$).
- `get_metrics(obs, sim), get_r2(obs, sim)` — NSE/RMSE/bias and Pearson R^2 helpers.
- `compute_intercept_audit(target_well_name, df_clean, df_climate)` — Model A vs Model B comparison for a single well, used by Scripts 07 and 08 in the LCSC benchmarking.

Two named thresholds: `MIN_OBS = 30` (minimum aligned rows for a per-well fit, applied inside `fit_ssm` and `fit_ssm_intercept`) and `LCSC_DATA_LIMIT = 100` (most-recent-window length for per-well fits in the intercept audit).

`data_utils.py`

Three short helpers shared across scripts:

- `normalize_well_name(value)` — lowercase, strip, remove spaces. Used wherever well names are joined between data sources.
- `parse_met_date(date_str)` — parse Met Office “Mon YY” strings (e.g. “Jan 95”, “Dec 26”) with the 2000-cutover handled internally.
- `clean_well_series(series, max_depth=4.0)` — drop unphysical depth values (> 4.0 m below ground) to NaN, then linearly interpolate single missed-visit gaps only (`limit=1`). The 4 m maximum is the `MAX_PHYSICAL_DEPTH` constant — set at the deepest plausible water table for the warren given site geometry and historical records.
- `calculate_cusum(series, baseline_mean)` — cumulative sum of departures from a baseline mean. The CUSUM diagnostic used by Scripts 09 and 10.

`clearfell_common.py`

Shared module for the Script 10 clearfell BACI suite (10a–10l). Provides:

- **Well tier definitions.** `IMPACT_WELLS = ['wmc3']` (the sole impact well spanning all three eras: pre-scraping, post-scraping, post-felling). `EDGE_WELLS = ['ceh31', 'ceh20', 'ceh30', 'ceh16']` (compartment edge). `FOREST_CONTROL_WELLS = ['ceh32', 'ceh34', 'ceh33', 'nw10', 'ceh2']` (C4 interior, unaffected by felling). `COASTAL_CONTROL_WELLS = ['ceh19', 'ceh17']` (C5 interior, distinct β_2 regime).

CLIMATE_CONTROL_WELLS = ['ceh9', 'nw7', 'nw6', 'nw5', 'wmc2'] (C3, climate-only counterfactual). C3_WARREN_WELLS = ['ceh1', 'nw1', 'nw2', 'nw11'] (the balanced four-well shielded western-dune set used as the second control zone by the four-zone sub-scripts 10k and 10l). The TIERS dict groups the five clearfell tiers into the BACI network used by 10a.

- **Intervention dates.** INTERVENTION_DATE = 2017-12-01 (clearfell); SCRAPING_DATE = 2015-04-01; SCRAPING_DATE_2 = 2023-10-01.
- **Spatial constants and distance functions** for scraping-propagation modelling (the $\lambda = 300$ m exponential weight).
- **Data loaders.** load_clearfell_data() reads wells, climate, and master coefficients in one call.
- **BACI helpers.** compute_baci_displacement(), compute_cwb() (climate water balance), distance_weighted_scraping(), annual_summer_minimum().
- **load_clearfell_b2_multiplier()** — the BACI-corrected Edge-tier β_2 ratio, computed dynamically from 10e_01_coefficient_shifts.csv rather than hardcoded. The thinning multiplier is derived as half-perturbation. Used by Scripts 19 and 21 to ensure their forestry scenarios use the canonical BACI-derived values rather than stale hardcoded constants.

The suite-shared module makes it possible to change the impact-well list, the BACI dates, or the distance-weight function in one place, and have all twelve sub-scripts pick up the change consistently.

[scraping_common.py](#)

Analogous shared module for the Script 09 scraping suite (09a–09e). Provides:

- **Intervention dates** (shared with clearfell where overlapping).
- **Well groups for BACI** (IMPACT_WELLS, PAIRED_CONTROLS_MAP, CLIMATE_CONTROLS, DONOR_CANDIDATES).
- **Tier assignments** (TIER1 controls vs TIER2 impacts).
- **Era definitions** — WELL_ERAS dict keyed by well, with (start, end) tuples per era. Different wells have different era boundaries: CEH36's eras are 1_Baseline / 2_Pure_Scraping / 3_Felling_Pulse around the 2015 and 2017 events; CEH21's eras are 1_Baseline / 2_Coastal_Drawdown / 3_After_Scraping around the 2017 felling and the 2023 second-scraping event. This structure allows each sub-script to apply consistent era windows without duplicating definitions.

- **Plot style constants** — ERA_COLORS, ERA_MARKERS, ERA_LINESTYLES, and MPL_DEFAULTS.
- **Data loaders.** load_scraping_data() returns wells (reference + extended) and climate.
- **Cluster-parameter loader.** load_cluster_params() consolidates β coefficients (from Script 03), Sy (from Script 17), and head displacement (from Script 01) into a single per-cluster dict.
- **Scenario computation.** compute_scenario_bars(cluster_params, summer_P, summer_PET, ...) produces the per-cluster volumetric scenario bars (mm/month) for clearfell, thinning, broadleaf, climate-dry, and climate-wet scenarios using the Option 3 monthly perturbation formulation. This is the single source of truth for scenario values; Scripts 09d, 19, and 21 all call it.

pipeline_params.py

Consolidated scenario parameter file management. F.4 describes the architecture. The function inventory:

- write_initial_params(wells_clean, climate) — called by Script 01. Writes summer climate means, head displacements, forest flags directly; reads from existing upstream outputs (Scripts 03, 10e, 17) opportunistically, defaulting where they don't exist.
- update_betas(beta_by_cluster) — called by Script 03 after fitting.
- update_b2_multipliers(clearfell_mult, thinning_mult) — called by Script 10e.
- update_specific_yield(sy_by_cluster) — called by Script 17.
- update_h_disp(h_disp_by_cluster),
update_peak_months(peak_by_cluster) — auxiliary updates.
- load_params() — called by Scripts 09b, 09d, 19, 21. Returns a dict containing per-cluster β , Sy, h_disp, peak month, forest flag, and the summer climate means.

The _DEFAULTS dict at the top of the module records the placeholder values used on a first pipeline run ($\beta_1 = 3.5$, $\beta_2 = 1.5$, $\beta_3 = 0.025$, Sy = 0.25, clearfell_b2_mult = 1.10, thinning_b2_mult = 1.05, peak_month = 2). These are chosen as central plausible values rather than as scientifically meaningful — they exist to keep the pipeline runnable on a fresh install before upstream scripts have produced canonical outputs.

map_utils.py

GIS and map-plotting helpers shared by every spatial-output script (04, 07, 08, 11b, 12, 13, 18, 19, 20). Key functions:

- `load_dem_layer(ax, data_dir)` — coloured terrain colormap onto an existing axes. Used by point-symbol metric maps.
- `load_dem_hillshade(ax, data_dir, alpha, vert_exag, zorder)` — greyscale hillshade (LightSource, azdeg=315, altdeg=35). Used where a metric surface is overlaid semi-transparently.
- `add_kml_features(ax, data_dir)` — overlays `Features.kml`, `streams.kml`, `clearfell.kml`. Returns legend handles.
- `add_osm_basemap(ax, gdf)` — OSM fallback when the DEM is unavailable.
- `add_idw_surface(ax, df, value_col, xi, yi, method, ridge_mask_threshold, dem_e_arr, dem_n_arr, dem_data, cmap, norm, alpha, zorder)` — inverse-distance-weighted interpolation of a per-well metric onto a regular grid, with an optional ridge mask (cells where the DEM sits more than `ridge_mask_threshold` metres above the IDW-interpolated well-DEM surface are masked, preventing extrapolation onto the bedrock ridge where there are no wells). Returns the `pcolormesh` object, grid coordinates, and the masked surface.
- `plot_metric_map(map_df, value_col, title, output_path, cmap, data_dir, vmin, vmax)` — full publication-quality spatial map: DEM background, KML overlays, cluster-shape markers, dual colorbars, legend. The high-level wrapper that the spatial scripts call.

The module routes through `BW_MODE` for B&W mode rendering, switching to hillshade DEM, greyscale colormaps, and hatched legend markers when active.

How the modules compose

A typical analytical script imports from several utility modules:

```
from utils.config import DRAINAGE_DATUM, CLUSTER_LABELS, FOREST_CIDS
from utils.paths import (
    INT_WELLS_CLEAN, INT_CLIMATE, INT_MASTER_DATA,
    OUT_NN_FIGURE, OUT_NN_TABLE,
)
from utils.model_utils import fit_ssm, build_ssm_frame,
assert_physical_signs
from utils.data_utils import clean_well_series, normalize_well_name
```

The pattern is intentional: constants from `config`, paths from `paths`, OLS from `model_utils`, and small per-row transformations from `data_utils`. Where the

script implements one of the BACI or scraping analyses, it additionally imports from `clearfell_common` or `scraping_common` for the suite-shared logic. Where it is a spatial output, it imports `map_utils` for the rendering helpers. Scripts do not import from each other — every shared piece of state passes through an intermediate CSV or through one of these utility modules.

Phase 1 — Core LCSC Chain

S.1 Script 01 — Data preparation

Step 1 / 27. Phase 1.

Motivation

Script 01 is the gateway to every downstream step in the pipeline. Each of the 27 scripts that follow reads from one or more of its eight output files, and no other script touches the raw data CSVs (with two documented exceptions that re-read raw inputs for specialist purposes: Script 09a for BACI windowing and Script 24 for sunshine-hour series). The chapter therefore covers two distinct jobs that Script 01 performs together: a *technical* job — converting three raw data sources into a self-consistent set of monthly time series with a single date convention, a single well-name convention, and well elevations in metres above Ordnance Datum — and a *scientific* job, which is to declare the membership of the reference network and the extended network that subsequent analyses operate on. Three methodological choices here have particular weight in the rest of the document: the day-15 bucketing convention introduced in F.2, the hardcoded reference-network whitelist that excludes the Forest Enterprise clearfell wells, Llyn Rhos-Ddu, and two singleton-outlier CEH wells, and the single-month interpolation `limit=1` policy with its per-cell provenance partner file `01_wells_provenance.csv`. All three are described in summary in the front matter; this chapter is the full description.

Inputs

File	Origin
<code>data/Newborough_Cleaned_For_Model.csv</code>	Field-collected monthly dipwell readings, manually pre-cleaned by the author. Wells are columns, reading dates are rows in DD/MM/YYYY form.

File	Origin
data/Well_locations_height.csv	DGPS-surveyed easting, northing, ground elevation, and pipe-top upstand per well. The Pipe_Top_Elev column is the independently measured reference datum for field readings.
data/RAF_Valley_Climate.csv	Met Office monthly returns from RAF Valley station, ≈16 km north-east of the site, 1930–present. Provides monthly maximum and minimum temperature, air-frost days, rainfall, and sunshine hours.

Methodology

The script runs four blocks in sequence, preceded by a metadata sanity check.

The sanity check passes both the well-name set from `Well_locations_height.csv` and the well-name set from the dipwell CSV through `normalize_well_name` (lower-case, whitespace-stripped) and prints any mismatch. The function is also used by every downstream script that joins on well name, so the same normalization rule applies throughout the pipeline.

Climate processing. The Met Office CSV has Mon YY header dates (e.g. “Jan 95”, “Dec 26”). These are parsed by `parse_met_date` with a two-digit-year cutover handled internally: years ≤ 26 are interpreted as 21st century, otherwise 20th century. The cutover is rolled forward each year as part of the annual data update. Rainfall is converted from millimetres to metres, and the Met Office “- - -” missing-data marker is replaced by zero. Treating missing rainfall as no rainfall rather than as unknown rainfall is a conservative choice; long gaps in the RAF Valley record are rare and the 1930–present series is otherwise complete, so the alternative — propagating NaNs through every downstream water-balance calculation — would lose more information than it preserves. Mean monthly temperature is taken as the arithmetic mean of the recorded monthly maximum and minimum. The maximum-temperature column header occasionally migrates between “Max Temp ©” and “Max Temp (C)” between Met Office releases; both forms are handled.

Potential evapotranspiration is then computed by the local `thornthwaite_pet_m()` function (specified below). The cleaned monthly P (m) and PET (m) are written to `01_climate.csv` along with the source temperatures.

Bucketing. The raw well CSV is transposed so that rows become dates and columns become wells. The day-15 bucketing convention is then applied: a reading taken on day ≤ 15 of month M is assigned to month $M-1$ (the previous month's water table); a reading on day > 15 of month M is assigned to month M . The cutoff is empirical — field practice is that monthly readings are taken either in the last few days of a month or in the first week of the following month, and the day-15 boundary sits cleanly in the gap between these clusters. Once bucketed, multiple readings in the same month are averaged. The convention and its implications for downstream alignment with the climate record are covered in F.2; Script 01 is where it is implemented and where the change from the legacy nearest-month bucketing was made.

The canonical implementation in `01_data_prep.py` is

```
d = pd.to_datetime(wells.index, dayfirst=True, errors="coerce")
prev = (d.to_period("M") - 1).to_timestamp()
this = d.to_period("M").to_timestamp()
wells.index = np.where(d.day <= 15, prev, this)
```

`(d.to_period("M") - 1).to_timestamp()` correctly subtracts one calendar month for any day-of-month. The seemingly equivalent `dt - pd.offsets.MonthBegin(1)` should **not** be used: `pd.offsets.MonthBegin` rolls to the start of the *current* month when the input is not already day-1, so for any reading taken between days 2 and 15 it returns the same month instead of the previous month — silently corrupting every such reading. Any parallel-pipeline check that re-implements bucketing should copy the code form above verbatim.

A single-well merger handles the historical NW8 $\rightarrow$ NW8B replacement: NW8B values take precedence where available, with NW8 as the fallback, and the result is stored on a single NW8 column.

Well cleaning. Each well column is then passed through `clean_well_series` (in `utils/data_utils.py`), which masks readings more negative than `MIN_PHYSICAL_DEPTH = -4.0` m and then fills any single-month gap (one consecutive NaN bridged between two measurements) by time-based linear interpolation. The interpolation `limit` parameter is set to 1. Multi-month gaps are left as NaN downstream. The -4 m floor is a safety margin: the deepest plausible water table at Newborough is around 3 m below ground, so a reading more

negative than -4 m is treated as unphysical and dropped before downstream analysis sees it.

The `limit=1` rule permits interpolation across single missed-visit gaps only. A looser `limit=3` setting — bridging gaps of up to three consecutive months — was tested against a site-wide audit of the 17-well clearfell-BACI panel from 2007 onwards and rejected. Under `limit=3`, 124 monthly cells (3.2 % of the BACI panel) were filled by interpolation; of these, 52 were two- or three-month runs and a disproportionate share spanned the Jun–Sep drawdown season — exactly the season in which linear interpolation between May and October endpoints systematically flattens the summer minimum that the analysis is built around. Three well-years (WMC3 2019, NW6 2019, NW7 2019) acquired a phantom summer minimum despite zero measured Jun–Sep readings: a 154-day field-access gap at all three Climate-control wells would be bridged by a single straight line from each well’s May reading to its October reading. Under `limit=1`, those 52 multi-month cells become NaN and are excluded from downstream analyses. The 72 cells still bridged by `limit=1` (1.85 % of the BACI panel) are all single missed-visit gaps, which match the field interpretation of one delayed monthly visit and do not span a drawdown season. The annual Forest $\times$ Impact ANCOVA-BACI step from Script 10a is robust in direction to the choice between the two settings; the summer-only contrast (Jun–Sep refit) is sensitive to it, because the multi-month summer interpolations the looser rule admits fall directly on the metric. The empirical consequences are taken up in S.6 (scraping summer minima) and S.7 (clearfell BACI).

Positive readings are not masked. The dune slacks at Newborough regularly flood above the dipwell pipe top, and on those visits the surveyor records the standing-water level above the pipe rim. These are real water-level observations, not measurement artefacts, and the SSM, the clustering, and the flood-threshold work all depend on them. The current dataset contains around 1,400 such readings, with a median of +13 cm above pipe top, a 99th percentile of +53 cm, and a maximum of +73 cm. They pass through the cleaning step unchanged.

Provenance tracking. Alongside the cleaned wells file, Script 01 emits `01_wells_provenance.csv` — a sibling table of identical shape (same row index, same column set as `01_wells_clean.csv`) recording per-cell origin as one of three string values: "measured" for a value present in the raw bucketed series, "interpolated" for a value created by `clean_well_series` linear-fill, "missing" for a cell still NaN after cleaning. The provenance file is the public record of what was inferred versus what was observed. Across the full 80-well reference +

extended panel from 2005 onwards (250 monthly rows $\times$ 80 columns = 20,000 cells), the current state is 14,948 measured, 274 interpolated, 4,778 missing. Restricted to the 17-well clearfell-BACI panel from 2007-01 onwards (229 months $\times$ 17 wells = 3,893 cells), the state is 3,399 measured, 72 interpolated, 422 missing.

The provenance file is consumed by analysis scripts that have an interpretive reason to discount cells with low measured-coverage. Two consumers are wired in the current pipeline: Script 10d (clearfell summer minima) and Script 09c (scraping summer minima) load the provenance file via `clearfell_common / scraping_common` and apply `min_measured=2` to `annual_summer_minimum`, dropping any (well, year) row whose Jun–Sep series contains fewer than two measured months. The new `n_interpolated` column in `10d_01_summer_minima.csv` and `09c_01_summer_minima.csv` flags the surviving rows for reviewer inspection. Script 03's `build_ssm_frame / fit_ssm` accept an optional provenance argument and an `exclude_interpolated=False` flag that defaults to retaining all rows in the canonical SSM fit. Retaining all rows is a separate matter from the interpolation limit setting, which alters `01_wells_clean.csv` itself: the canonical $\beta_1/\beta_2/\beta_3$ cluster-coefficient table is computed under `limit=1` (see §S.3). The `exclude_interpolated` flag is documented as an available sensitivity path, run on demand.

Network partitioning. The cleaned table is then partitioned into two networks. The full cleaned table (`01_wells_clean.csv`) retains every well with at least `MIN_MONTHS_THRESH = 100` non-NaN months. From this set:

- The **reference network** (`01_wells_reference.csv`) is the subset that meets all three of the following criteria: at least 100 monthly observations, a final record date on or after `REGENCY_DATE = REFERENCE_CUTOFF_DATE = 2026-02-01` (from `utils.config`), *and* appears on the hardcoded `REFERENCE_NETWORK_WHITELIST` of 66 well names.
- The **extended network** (`01_wells_extended.csv`) is every well with at least `MIN_EXTENDED_MONTHS = 24` observations, *except* those on `EXTENDED_NETWORK_BLACKLIST = frozenset({"llynrhos", "pdfs"})`. Wells that meet the automatic reference-eligibility criteria but are not on the whitelist are *demoted* — they are excluded from the reference network and routed to the extended network.

Setting `REFERENCE_NETWORK_WHITELIST = None` restores fully automatic selection (any well meeting the auto-criteria enters the reference network). The whitelist is

the live setting and exists to pin the reference network deterministically to the published partition, regardless of subsequent data additions.

The four categories of exclusion from the reference network are:

1. **FE1–FE4 and LIS1.** These wells lie inside the Forest Enterprise clearfell footprint and record a non-stationary management regime — the SSM’s stationarity assumption is not appropriate for them. They are routed to the extended network, where they form the treatment arm of the Script 10 BACI suite. The decision is methodological, not a judgement on data quality.
2. **Llyn Rhos-Ddu and pdfs.** Llyn Rhos-Ddu records lake stage, not water-table response to recharge. It is excluded from both networks via `EXTENDED_NETWORK_BLACKLIST` because the SSM forcing terms (P and PET) do not describe a lake’s hydrology in the same way they describe a dune-slack water table. The pdfs well was added to the blacklist on 2026-05-24: its hydrograph carries a tidal-influence signature that makes it unrepresentative of water-table behaviour, the same exclusion principle applied to CEH3 and CEH22 below. With pdfs blacklisted and one further extended well (p1) brought back in once its elevation row was completed, the classified network settles at 88 wells (66 reference + 22 extended).
3. **CEH3 and CEH22.** Ward’s hierarchical clustering identifies these wells as singleton outliers at low partition orders, consistent with tidal-signal contamination on top of the climate-forcing response that the SSM is designed to capture. CEH22 sits at 3.3 m ground elevation, near the lower end of the elevation distribution; CEH3 has the most pronounced tidal signature on inspection (working assessment by the author). Both remain in the extended network for per-well analyses where their behaviour is of interest in its own right.
4. **Other automatic candidates.** Wells that meet the record-length and recency criteria but were not part of the originally published 2026 reference network are routed to the extended network rather than added to clustering. This is the protective half of the whitelist mechanism: adding a well to the reference network requires a code change, which guards against silent partition shifts when the underlying data is updated.

Elevation conversion. The cleaned well table is then re-expressed in metres above Ordnance Datum and written to `01_wells_clean_maod.csv`. Conversion uses the `Pipe_Top_Elev` column of `Well_locations_height.csv` directly: because

raw depths are negative by convention, the conversion is the addition $\text{ma0D} = \text{Pipe_Top_Elev} + \text{depth}$, which subtracts the depth-below-pipe-top to yield upward-positive elevation. `Pipe_Top_Elev` is used in preference to the reconstructed value `DGPS_Ground_Elev + Upstand_m`; the two disagree for the majority of wells, and `Pipe_Top_Elev` is the independently surveyed value. The conversion sign is sanity-checked on a small sample of wells against the original raw depth, with a printed pass/fail summary in the script log.

A condensed elevation lookup (`01_well_elevations.csv`: ground elevation, upstand, pipe-top elevation per well) is written for use by Scripts 03 (upstand correction) and 10–21.

Thornthwaite PET (`thornthwaite_pet_m`). PET is computed locally inside Script 01 rather than imported from a utility module, on the principle that the climate-processing step is self-contained. The implementation follows Thornthwaite (1948) with the Thornthwaite & Mather (1955) day-length and month-length correction. In outline:

The heat index I is the annual sum of monthly heat indices $i = (T/5)^{1.514}$ for months with monthly mean temperature $T \geq 0$ °C. The exponent a is the cubic polynomial in I of Thornthwaite (1948). Unadjusted monthly PET in millimetres is then

$$\text{PET}_u = 16 \left(\frac{10T}{I} \right)^a$$

with the boundary condition $\text{PET} = 0$ for $T \leq 0$ °C. For $T \geq 26.5$ °C the formula is replaced by the Camargo et al. high-temperature linearisation, which prevents the polynomial form from blowing up at temperatures that are rare but not impossible at the site. The adjusted monthly value applies the K correction

$$K = \frac{N}{12} \cdot \frac{d}{30}$$

where N is the mean monthly daylight hours computed from the site latitude (`RAF_VALLEY_LAT_DEG = 53.25°` from `utils.config`) and the solar declination for the mid-month day, and d is the number of days in the month. The output is monthly PET in **metres** per month. The pipeline-wide convention is that both P and PET are in metres; converting at the producer rather than the consumer side is more reliable than scattering unit conversions through 27 downstream scripts.

Scenario-parameter seed. Finally, Script 01 calls `pipeline_params.write_initial_params()` to write the initial state of

01_data_prep/pipeline_scenario_params.csv. This file is the consolidated scenario-parameter store described in F.4: a per-cluster table of SSM coefficients, specific yields, displacement values, peak month, and summer climate seeds, written here with placeholders and then updated in later passes by Scripts 03 (cluster SSM coefficients), 10e (clearfell coefficient decomposition), and 17 (peak-month seasonal stats). The Script 01 → downstream-scenario handshake is a producer-consumer pattern; the file is the contract, the script is one of four writers, and the scenario-running scripts (09b, 09d, 19, 21) are the consumers.

Outputs

File	Contents	Downstream consumers
01_locations.csv	E, N, ground elevation, pipe-top elevation, and upstand per well	Scripts 02–25 (any spatial analysis)
01_climate.csv	Monthly P (m), PET (m), source temperatures	All SSM-fitting and water-balance scripts
01_wells_clean.csv	Bucketed, cleaned, interpolated depth time series	Scripts 02, 03, 09–21
01_wells_provenance.csv	Per-cell provenance partner to 01_wells_clean.csv: values {measured, interpolated, missing}	Scripts 03 (optional exclude_interpolated sensitivity), 09c, 10d
01_wells_clean_maod.csv	The same series in m AOD	Scripts 03 (centroid m AOD), 20
01_wells_reference.csv	Reference-network subset (66 wells)	Scripts 02 (clustering), 06 (extension audit)
01_wells_extended.csv	Extended-network subset (22 wells)	Scripts 06, 09b, 10h, 18, 25
01_well_elevations.csv	Ground elevation, upstand, pipe-top per well	Scripts 03 (upstand correction), 10–21
01_data_prep/pipeline_scenario_params.csv	Per-cluster β , S_y , h_{disp} , peak month, summer climate seeds	Scripts 09b, 09d, 19, 21

All paths resolve through `utils/paths.py` (`INT_LOCATIONS`, `INT_CLIMATE`, `INT_WELLS_CLEAN`, `INT_WELLS_PROVENANCE`, `INT_WELLS_CLEAN_MAOD`, `INT_WELLS_REFERENCE`, `INT_WELLS_EXTENDED`, `INT_WELL_ELEVATIONS`, `DIR_01`) and are never hardcoded in the script body.

Limitations and known caveats

- **The reference-network whitelist is hardcoded.** Adding or removing a well requires a code change and a CHANGELOG entry. The trade-off is deterministic reproducibility against operational ease, and this script deliberately prioritises the former; the alternative — an algorithmic selection that drifts silently when the data is updated — has worse failure modes for a paper-of-record analysis.
- **The bucketing convention assumes monthly readings.** Multiple readings in the same bucketed month are averaged, which would mask any genuine intra-month dynamics if higher-frequency data became routinely available. The current monthly cadence is set by the field protocol and the script's design follows that cadence; sub-monthly logging would require a different implementation.
- **Thornthwaite PET is a simple empirical estimator.** It uses temperature and day length, not net radiation or vapour-pressure deficit, and is known to underestimate summer atmospheric demand for forested surfaces. The site-wide implications are taken up in S.10 (depth-dependent PET) and S.16 (residual seasonality), and §5.3 of the report discusses what this means for the cluster-mean β_2 values. For a long-term monthly model on a temperate site with the data the Met Office routinely publishes, Thornthwaite is the defensible choice; the limitation is acknowledged rather than corrected here.
- **The 4 m depth floor catches nothing in the current dataset.** No reading in `Newborough_Cleaned_For_Model.csv` is more negative than -2.87 m. The floor is a forward-looking safeguard against stray entries from future data drops or damaged-sensor records, not a constraint on the current data. It is implemented as a signed floor (`series >= -4.0`) against the negative-valued depth series.

Where the result appears in the report

- §3.1 *Data preparation* — bucketing convention, network partition.
- §3.1.2 — the FE/LIS exclusion rationale and the Llyn Rhos / CEH3 / CEH22 routing.

- §3.2 — Thornthwaite PET specification.
- §3.3 and onward — implicit, in every analysis that reads from the Script 01 outputs.

Cross-references

- **F.2** — bucketing convention, date semantics, the `HEADLINE_LAG = 0` decision and its history. Refer to F.2 rather than re-deriving.
- **F.3** — SSM equation form and sign conventions; relevant background to the network-partition decisions but not re-stated here.
- **F.4** — reference-network membership counts (66 wells; $C1 = 7$, $C2 = 24$, $C3 = 21$, $C4 = 9$, $C5 = 5$) and the `pipeline_scenario_params.csv` producer-consumer architecture.
- **F.5** — `paths.py`, `data_utils.py`, and `pipeline_params.py` module roles.
- **S.6** (sub-script 09c) — consumes `01_wells_provenance.csv` for the `min_measured=2` summer-minima rule.
- **S.7** (sub-scripts 10a, 10d) — consumes `01_wells_provenance.csv` for the same rule.
- **S.7** (sub-script 10e) — updates `pipeline_scenario_params.csv` with the clearfell coefficient decomposition.
- **S.10** — depth-dependent PET correction that addresses the Thornthwaite under-estimation noted above.

S.2 Script 02 — Behavioural clustering

Step 2 / 27. Phase 1.

Motivation

Script 02 produces the cluster partition that almost every downstream analysis inherits. It applies Ward’s hierarchical clustering on Pearson correlation distance to the 66-well reference network defined by Script 01, cuts the resulting tree at $k = 5$, and writes a per-well cluster assignment to `02_cluster_stats.csv`. Three methodological decisions made here propagate through the rest of the supplement: the choice of Ward’s linkage on correlation distance (rather than Euclidean or some other behavioural metric); the choice of $k = 5$ (rather than 4, 6, or some other partition order); and the anchor-well remap that pins Ward’s arbitrary integer outputs to canonical, stable cluster IDs.

Alongside the clustering itself the script computes a richer validation suite — silhouette and Calinski-Harabasz across $k = 2 \dots 10$, bootstrap stability across $k =$

4...7, co-assignment heatmaps — and a second analysis bundle that does not strictly belong to clustering: per-well and per-cluster seasonal-amplitude descriptors with a climate-normalized variant that excludes three identified drought summers. The amplitude descriptors feed §4.2 of the report directly.

Inputs

File	Origin
01_wells_reference.csv	Reference-network depth time series, 66 wells (Script 01)
01_wells_clean.csv	Full cleaned table — used inside <code>make_cluster_hydrograph_wb_figure()</code> for the water-balance panel of figure 02_03 (Script 01)
01_climate.csv	Monthly P and PET (m), used in the same figure (Script 01)

Methodology

The script proceeds in seven blocks executed in order in `__main__`:

1. Load and prepare reference data. `01_wells_reference.csv` is read with `parse_dates=True`, coerced to numeric, and any columns that are entirely NaN are dropped. The result is a DataFrame with monthly rows and one column per reference well.

2. Build the correlation-distance matrix. Distance between two wells i and j is $1 - \rho(i, j)$ where ρ is the Pearson correlation of the two monthly depth series over their common non-NA months. The full reference time series is passed through; there is no detrending or windowing step before the correlation is computed. This is intentional: the goal of the partition is to capture *behavioural* similarity (do two wells rise and fall together over the full record), and the same long-term wetting or drying signal that a linear detrend would remove is itself a behavioural property that distinguishes coastal-erosion-affected wells from the rest of the network. Removing it would reduce the dimensionality of “behaviour” that the algorithm sees.

The matrix is computed by `_correlation_distance(): corr = wells.corr()` (which uses pairwise non-NA observations per cell), then `1 - corr`, then clipped to non-negative, then symmetrised against small numerical asymmetries, then converted to the condensed form scipy’s linkage expects. The square form is retained for use as the precomputed distance input to `silhouette_score`.

3. Ward’s linkage and the k-sweep validation. Ward’s variance-minimisation linkage is applied to the condensed distance vector. The cluster validation runs over $k = 2 \dots 10$, recording silhouette score (on the precomputed correlation-distance matrix), Calinski-Harabasz score (on the raw wells-as-rows feature matrix with within-well mean-imputation of gaps for the CH computation only), and the Ward merge distance at the k -th-to-last linkage step. Outputs land in `02_02_validation_plots.png` (silhouette + merge distance at the chosen k) and `02_02b_validation_k_sweep.png` (all three metrics across the full k range).

4. Bootstrap stability diagnostics (`bootstrap_cluster_stability`). For each k in `K_RANGE_BOOTSTRAP = (4, 5, 6, 7)`, the reference network is resampled with replacement `N_BOOTSTRAP = 1000` times. Each bootstrap re-fits Ward’s at k on the resampled distance matrix. Pairwise co-assignment counts are accumulated: numerator counts bootstraps in which both wells appear and land in the same cluster, denominator counts bootstraps in which both appear. The per-well stability is the median co-assignment probability between the well and its cluster-mates in the reference (full-sample) fit, which is the “does this well stick with its neighbours” interpretation. Reproducibility is fixed by `BOOTSTRAP_SEED = 20260424`. Outputs include per-well stability scores, per-cluster summary statistics, and a co-assignment heatmap for each k .

5. Canonical-ID remap (`_remap_cluster_ids_by_anchor`). `fcluster` returns arbitrary integer IDs that depend on the order Ward’s encounters merges, so the same partition can come back with cluster integers permuted between runs. The remap function locates each canonical anchor well in the raw labelling, identifies the raw cluster it belongs to, and reassigns that raw ID to its canonical ID. Two integrity checks are enforced: anchor wells listed for the same canonical ID must land in the same raw cluster (otherwise the partition assumptions have been violated and `ValueError` is raised), and no raw cluster can be claimed by two different canonical IDs (also a `ValueError`). A guard at module load asserts that `CLUSTER_ID_ANCHORS` and `utils.config.CLUSTER_LABELS` describe the same set of cluster IDs; if they disagree, `RuntimeError` is raised before any analysis runs.

The canonical anchors are:

Cluster ID	Label	Anchor wells
1	C1 (Lake Edge)	ceh5, ceh11
2	C2 (Dune)	d10
3	C3 (Western Residual)	nw1

Cluster ID	Label	Anchor wells
4	C4 (Main Forest)	ceh2
5	C5 (Coastal Forest)	ceh16, nw9

These anchors are *interior* wells — wells that on visual inspection of the hydrograph cluster most clearly to each behavioural type — not the wells with the highest within-cluster correlation. Anchoring to interior wells guards against centroid drift between runs silently relabelling the partition when the data is updated.

6. Cluster assignment write-out. The remapped labels are saved to `02_cluster_stats.csv` with columns `Match_ID` (normalized well name), `Name_Original`, `Cluster` (canonical integer ID), and `Cluster_Label` (human-readable label from `CLUSTER_LABELS`). This file is the contract for downstream consumers — Script 03 reads it to define the per-cluster SSM fits, and most analysis scripts from 04 onwards depend on it.

7. Dendrogram and cluster-hydrograph figures. The dendrogram (`02_01_dendrogram.png`) is produced with `color_threshold` set to the $k = 5$ merge height; tick labels are coloured by canonical cluster ID. The cluster-hydrograph water-balance figure (`02_03_cluster_hydrographs_wb.png`) shows cluster-mean depth-below-pipe time series with an annotated water-balance panel above. The water-balance panel uses `DETREND_START = 2004-12-01` and `DETREND_END = 2025-12-01` to compute a long-term mean of the climate water balance ($P - PET$) for visual reference on the figure; these constants are local to the figure and play no role in the clustering distance computation.

Amplitude descriptors (`compute_cluster_amplitude_descriptors`). A secondary descriptor of cluster behaviour, intentionally separated from the partition itself. For each well the script computes the seasonal amplitude $p_{90} - p_{10}$ over the full record, over the pre-2018 sub-window ($< \text{AMP_SPLIT_DATE} = 2018-01-01$), and over the post-2018 sub-window. The split date is anchored to the 2018 UK-wide summer drought — one of the three identified drought summers and the most consequential meteorologically — which falls a few weeks after the December 2017 experimental clearfell. The pre-window therefore captures the pre-2018 baseline (before both the meteorological inflection and the management intervention); the post-window captures the drought-dominated, post-felling period. A window stat requires at least `AMP_MIN_OBS_PER_WIN = 24` non-NaN months or it returns NaN.

Two parallel climate-normalized variants drop the Jun–Sep monthly observations for three identified drought summers — `DROUGHT_SUMMERS = (2005, 2018, 2022)` — from the relevant window before recomputing `p90 - p10`. The summers were identified empirically against the 1931–2017 RAF Valley Jun–Sep rainfall mean of 260 mm ($\sigma = 70$ mm) using a one-sigma-below-mean threshold of approximately 190 mm. The list is hardcoded so the normalization is auditable: the values should only be revised if the underlying RAF Valley series is updated. Note that the normalization drops drought summers — not all dry summers — and the unnormalised variant is also reported so the reader can see both.

Per-well descriptors are written to `02_08_cluster_amplitude_per_well.csv`. Per-cluster aggregation is **median-of-medians across wells** within each cluster (not aggregation on a cluster-mean hydrograph), with the per-cluster summary written to `02_09_cluster_amplitude_summary.csv` and a boxplot of post-2018 amplitude distributions to `02_10_cluster_amplitude_boxplot.png`. A defensive check at the top of the function rejects any `cluster_df` that contains non-canonical IDs (i.e. that has not been passed through the `remap`), so the amplitude file cannot drift away from the partition.

Site-specific choices and rationale

- **Ward’s linkage.** Single-linkage produces chained clusters in this data — the algorithm strings together intermediate wells and the lake-edge / dune distinction is lost. Complete-linkage produces overly compact clusters that split otherwise coherent behavioural groups across the partition boundary. Ward’s variance-minimisation is the standard choice for small-network behavioural clustering and produces clusters that align with the site’s hydrogeology on inspection.
- **Correlation distance, not Euclidean.** The clustering question is about behavioural similarity, not absolute level. Two wells with identical seasonal cycles but mean depths 1 m apart should cluster together — and do under correlation distance. Euclidean distance on raw depths would separate them.
- **No detrending before correlation.** Long-term trends in individual wells are part of the behavioural signal the partition is meant to capture. The coastal-edge wells in C5 share a coastal-erosion-influenced trend signature that a linear-trend removal would erase; the eastern lake-buffer wells share a different trend signature. Pearson correlation on raw monthly series captures both seasonal and inter-annual co-variation, which is the intended target. The `DETREND_START/DETREND_END` constants in the script

are scoped to the water-balance panel of figure 02_03 and do not touch the wells.

- **k = 5, not k = 6.** Earlier work used a k = 6 partition with a separate single-well “Lake” cluster (Llyn Rhos-Ddu, n = 1) and a separate single-well “Coastal” cluster. Under the published partition, Llyn Rhos-Ddu is excluded from the reference network on physical grounds (S.1) and treated as a fixed-head boundary feature; the current C1 (Lake Edge) captures the wells *adjacent* to the lake. The mathematical case for k = 5 rests on the Ward’s merge-distance curve, which shows a clear elbow at k = 4–5: the drop from k = 4 → k = 5 ($\approx 0.19 \rightarrow 0.17$) is the last substantial descent before the curve flattens through k = 6–10. Silhouette is monotonically declining from k = 2 (≈ 0.46) through k = 5 (≈ 0.40) and is not on its own an argument for any specific k beyond 2 — this is typical of behavioural-clustering datasets where the data has hierarchical structure rather than well-separated point clouds. k = 5 is the partition order that sits at the merge-distance elbow, resolves the hydrogeologically meaningful C4 / C5 split (which k = 4 collapses), keeps every cluster at ≥ 5 members, and has an interpretable site reading. The full history is consolidated in F.4.
- **Hardcoded anchor wells, not centroid- or correlation-based.** Anchors are interior wells chosen by visual inspection — not the wells with the highest within-cluster mean correlation. Centroid-based anchoring drifts as cluster membership shifts; correlation-based anchoring can be dominated by a tightly correlated sub-group within a heterogeneous cluster. Interior-well anchoring is the most defensive choice and produces deterministic cluster IDs across runs.
- **Drought-summer normalization in the amplitude descriptor.** The post-2018 window contains three drought summers (2005, 2018, 2022) that disproportionately inflate raw p90 – p10 in any well that responds to those summers. The climate-normalized variant drops the Jun–Sep months of those years, identified empirically against the 1931–2017 RAF Valley mean rather than against a model-derived definition; this preserves auditability and means the list does not change if a future analysis revises the PET model. The unnormalised variant is reported alongside so the reader can see what the normalization does.
- **N = 1000 bootstraps.** At N = 200, the per-well stability CIs are noticeably wider; at N = 5000 they tighten to within ~ 0.001 of the N = 1000 estimates at a runtime cost that is not justified for a diagnostic descriptor. 1000 is the

standard published choice for bootstrap-stability diagnostics on networks of this size.

Outputs

File	Contents
02_cluster_stats.csv (INT_)	Per-well: Match_ID, Name_Original, Cluster, Cluster_Label
02_clustering/ 02_01_dendrogram.png	Ward's dendrogram, tick labels coloured by canonical cluster
02_clustering/ 02_02_validation_plots.png	Merge-distance elbow and silhouette score at the chosen k
02_clustering/ 02_02b_validation_k_sweep.png	Silhouette + Calinski-Harabasz + merge distance across $k = 2...10$
02_clustering/ 02_03_cluster_hydrographs_wb.png	Cluster-mean depth hydrographs with water-balance annotation panel
02_clustering/ 02_04_bootstrap_stability_summary.csv	Per-cluster median pairwise co-assignment at each k in K_RANGE_BOOTSTRAP
02_clustering/ 02_05_bootstrap_stability_per_well.csv	Per-well stability scores across k
02_clustering/ 02_06_coassignment_heatmap_k{k}.png	Co-assignment heatmap, one per k in K_RANGE_BOOTSTRAP
02_clustering/ 02_07_cluster_membership_k{k}.csv	Per-well majority assignment at each k
02_clustering/ 02_08_cluster_amplitude_per_well.csv	Per-well amplitude descriptors
02_clustering/ 02_09_cluster_amplitude_summary.csv	Per-cluster median-of-medians amplitude summary
02_clustering/ 02_10_cluster_amplitude_boxplot.png	Post-2018 amplitude distribution boxplot, one box per cluster

Limitations and known caveats

- **C3 (Western Residual) is the lowest-stability cluster, not C1.** Bootstrap diagnostics give median within-cluster co-assignment of approximately 0.50 for C3 at $k = 5$, against ≥ 0.93 for C1, C2, C4, and C5. C3 contains two geographically distinguishable sub-populations — a low-ground southern

coastal fringe (ceh4, ceh18, ceh21, ceh36, ceh42) and the west-side open dune wells (nw1, nw2, nw5–7, nw11, nw13) — that Ward’s does not separate at any k from 5 to 9 because they share a common behavioural signature at the resolution the algorithm achieves. This is treated as a landscape/behaviour distinction in §3.2 rather than as a clustering failure.

- **CEH11 is the weakest member of C1 (Lake Edge).** Individual-well bootstrap co-assignment is approximately 0.68 for ceh11 at $k = 5$, against ≥ 0.95 for the other six C1 members. CEH11 is included on physical grounds (its lake-adjacent location matches the cluster’s character) but is flagged as a borderline membership in both the script comment block above CLUSTER_ID_ANCHORS and in the report.
- **The partition is not stable under all parameter perturbations.** Removing the drought-summer mask, or shifting MIN_RECORD_MONTHS from 100 to 80, produces small membership changes at the boundaries of C1 and C5 (the two smallest clusters). The published partition is the canonical one and is the basis for every downstream analysis; alternative parameter choices produce qualitatively similar five-cluster structures but are not committed.
- **The anchor remap fails loudly under sufficient perturbation.** If a future data change causes Ward’s output to place two canonical anchors in the same raw cluster, or to split one anchor pair across two raw clusters, _remap_cluster_ids_by_anchor raises ValueError rather than silently producing an ambiguous partition. The error is the intended behaviour; resolution is to inspect the dendrogram and either revise the anchors (if the new partition is a refinement of the old) or accept that the partition has genuinely changed and recompute everything that depends on it.
- **MIN_RECORD_MONTHS = 100 is duplicated from Script 01.** Script 01’s MIN_MONTHS_THRESH = 100 defines the reference-network membership; Script 02’s MIN_RECORD_MONTHS = 100 is used only inside the cluster-hydrograph water-balance figure to filter the *full* (not reference) wells table for the short-profile panel. The two values happen to coincide and match by convention; if Script 01’s threshold were ever changed, Script 02’s local constant would need to be revisited.

Where the result appears in the report

- §3.2 *Cluster definition* — Ward’s, correlation distance, $k = 5$, anchor remap, validation suite.

- §4.2 *Cluster characterization and description* — built from the cluster-hydrograph figure and the amplitude descriptors.
- Table 2 — cluster mechanistic coefficients (β_1 , β_2 , β_3) per cluster, sourced from Script 03 but indexed by the partition defined here.
- Figures 6–8 — dendrogram, validation, and cluster-hydrograph figures from 02_clustering/.

Cross-references

- **F.4** — $k = 5$ partition table with anchor wells and member counts; identity-vs-integer-keying principle, which `_remap_cluster_ids_by_anchor` operationalises.
- **F.5** — `paths.py`, `config.py`, and `data_utils.py` module roles.
- **S.1** — reference-network selection: the 66-well subset that this script consumes.
- **S.3** — SSM fitting on the cluster partition. Script 03 reads `02_cluster_stats.csv` and is the primary downstream consumer.
- **S.4** — Pearson-affinity audit and extended-network analysis (grouped chapter covering Scripts 04, 05, 06), which quantify how confidently each well's cluster assignment can be defended.
- **F.4** — partition history ($k = 6 \rightarrow k = 5$ transition, dropped single-well clusters, identity-vs-integer-keying principle) is consolidated in F.4 of this supplement.

S.3 Script 03 — State-space regression and LCSC

Step 3 / 27. Phase 1.

Motivation

Script 03 is where the state-space model gets fitted to the data. Per-well coefficients are written to `03_master_data.csv` (consumed by Scripts 07, 08, 11, 16, 19, 21, and the rest); cluster-centroid coefficients are written to `03_03_cluster_mechanistic_coefficients.csv` (the source of Table 3 in the main report). Five quantitative validation outputs sit alongside the headline fits: a lag diagnostic that tests whether the centroid responds best to the current month's rainfall or to an earlier one, a 1,000-replicate bootstrap that asks how sensitive each cluster's β estimate is to the particular set of member wells, a leave-one-out sweep that checks for single-well domination, a pre/post-2018 split-window for C1 Lake Edge, and a drainage-datum sensitivity sweep that runs both at the cluster centroid and per-well. The chapter walks each in turn. The SSM

equation, sign conventions, displacement formulation, and the role of `DRAINAGE_DATUM = 3.7 m` are covered in full in F.3; this chapter does not re-derive them. The Lumped Catchment Storage Coefficient (LCSC) — the rescaled β_1 used as the report’s headline mechanistic descriptor — is introduced here for the first time.

Inputs

File	Origin
<code>01_climate.csv</code>	Monthly P (m) and PET (m) (Script 01)
<code>01_wells_clean.csv</code>	Cleaned depth time series, depth-below-pipe-top convention (Script 01)
<code>01_wells_clean_maod.csv</code>	Same series in m AOD (Script 01)
<code>01_well_elevations.csv</code>	Pipe-top elevation and upstand per well (Script 01)
<code>01_locations.csv</code>	Easting, northing, ground elevation, pipe-top elevation per well (Script 01)
<code>02_cluster_stats.csv</code>	Per-well cluster assignment (Script 02)
<code>02_08_cluster_amplitude_per_well.csv</code>	Per-well amplitude descriptors (Script 02) — optional input to the heterogeneity flag in the summary table; falls back to hard-coded values if absent
<code>01_wells_provenance.csv</code>	Per-cell provenance flags from Script 01 — optional input, loaded only when a caller invokes <code>fit_ssm(..., exclude_interpolated=True)</code> for the measured-only sensitivity. The canonical centroid fits do not consume this file.

Methodology

The script runs eleven blocks in `main()`. The order matters: per-well fits → centroid construction → centroid headline fits → diagnostic suite (lag, datum, bootstrap, leave-one-out, C1 split) → summary table → figures and regional-average exports. The hard-halt on a centroid sign violation is deferred to the very end so that the diagnostic tables and figures get written *before* the pipeline stops — these are exactly the tables an investigator needs to diagnose the failure.

Centroid construction — `build_cluster_centroids` and

`build_upstand_lookup`. Each cluster centroid is the arithmetic mean of its member wells' depth series, with one twist: each member well is upstand-corrected before averaging. A well's raw depth reading is depth below the pipe top, which sits some `Upstand_m` above the ground surface; subtracting the upstand converts the series to depth below ground, the common datum shared by every well. The upstand lookup (`build_upstand_lookup`) is built from `01_well_elevations.csv`; wells without an upstand entry are averaged uncorrected, with a console count so the investigator can audit the warning. An `upstand_audit()` diagnostic prints the wells with `Upstand_m > 0.30` — under the live network, CEH2 (~0.71 m), NW2 (~0.59 m), and t41a (~0.40 m) are the three flagged; CEH2 is deliberately tall to remain visible in the forest understorey, NW2 and t41a are the other two ordinary cases. The flag is informational only.

Per-well fits — `per_well_fits`. Each reference-network well is fitted individually via `model_utils.fit_ssm()` (F.3 covers the function's interface), with the well's depth series upstand-corrected first so the `DRAINAGE_DATUM = 3.7 m` displacement reference is anchored consistently to the ground surface. The per-well fit uses `window=LCSC_DATA_LIMIT = 100` — the most recent 100 valid monthly rows — so that all per-well coefficients describe behaviour over the same recent window. This is the LCSC window subsequently used by Script 07 and 08; the rationale for choosing it here is downstream consistency. Per-well sign violations ($\beta_1 < 0$ or $\beta_2 < 0$) are reported but do not halt the pipeline — only centroid violations halt, because centroid fits are the values the report cites. A second LCSC estimate, the *empirical* LCSC, is computed alongside the regression LCSC for cross-check: rainfall divided by Δh on each unambiguous recharge event ($\Delta h > 0.02$ m), with the screen for spurious ratios (LCSC_raw outside [0.05, 1.0]) and a 10-event minimum. The regression LCSC is the one carried forward.

The per-well master table (`03_master_data.csv`, written by this block) is the single most consumed Script 03 output. Scripts 07, 08, 11, 16, 17, 19, and 21 all read it.

Centroid headline fits — `centroid_headline_fits`. This is the canonical Table 3 result. Each cluster centroid is fitted with `lag = HEADLINE_LAG = 0` and `window = None` — the full record, not the 100-month window used by per-well fits. The reasoning for using the full record at cluster level is that the centroid already averages over within-cluster noise, and the longer window simply tightens the standard errors. LCSC is computed as $100 / \beta_1$ (in mm per 10 cm of rise; F.4 explains the unit). `assert_physical_signs()` is called on each centroid fit and

accumulates hard violations ($\beta_1 \leq 0$, $\beta_2 \leq 0$) and soft warnings ($\beta_3 \leq 0$). Under the live data, no violations or warnings fire: all five clusters return positive β_1 , β_2 , β_3 with vanishingly small p-values (β_1 p $\ll 10^{-45}$ everywhere; β_3 p ranges from 6×10^{-35} for C1 down to 1.7×10^{-3} for C4, the weakest case).

The cluster-centroid coefficients under the live partition are:

Cluster	n	β_1	β_2	β_3	R^2	LCSC (mm per 10 cm)
C1 (Lake Edge)	236	4.58	0.96	0.090	0.732	21.8
C2 (Dune)	247	3.98	1.77	0.066	0.747	25.1
C3 (Western Residual)	248	3.57	1.85	0.058	0.813	28.0
C4 (Main Forest)	236	2.52	2.55	0.020	0.685	39.7
C5 (Coastal Forest)	238	2.41	1.32	0.045	0.680	41.5

These are read directly from `03_03_cluster_mechanistic_coefficients.csv`. The triplet plus LCSC is the “mechanistic signature” of each cluster.

Interpolated-row handling. `build_ssm_frame()` and `fit_ssm()` accept optional `provenance=` and `exclude_interpolated=` kwargs (see S.1 for the underlying provenance file). The Script 03 canonical fits are run with the default `exclude_interpolated=False`: every monthly row from `01_wells_clean.csv` enters the design matrix, regardless of whether the underlying cell is measured or interpolated. The canonical fit retains all rows rather than restricting to a measured-only characterization; the `exclude_interpolated=True` path is the documented sensitivity (next paragraph). The cluster-centroid coefficient table above is computed from `01_wells_clean.csv` under the `clean_well_series`

interpolation limit=1 policy (S.1). The interpolated-cell footprint is small: aggregated over each cluster's full member panel, the interpolated cell fraction is 1.3–1.6 % per cluster (22 cells in C1's 7-well × 250-month panel, 88 in C2's 25-well × 250-month panel, 73 in C3, 31 in C4, 16 in C5); the SSM aggregates over hundreds of monthly rows per cluster and is correspondingly robust to this footprint. The interpolation limit setting affects the centroid coefficients only at the second decimal, and no cluster changes rank, sign, or mechanistic signature: the LCSC ordering $C1 < C2 < C3 < C4 < C5$ and the forest-versus-open-dune β_2 contrast are insensitive to it.

The `exclude_interpolated=True` path is the documented sensitivity. A caller that wishes to refit the cluster centroid SSM on measured-only rows can pass the provenance series (loaded from `01_wells_provenance.csv`) alongside the cleaned centroid: `fit_ssm(h_centroid, climate, provenance=cluster_provenance, exclude_interpolated=True)` will drop any month in which the centroid cell was produced by interpolation. The chapter does not run this sensitivity. The provenance-aware path is consumed elsewhere in the pipeline — Script 10d (clearfell summer minima) and Script 09c (scraping summer minima) apply a related `min_measured=2` rule per (well, year); see S.6 and S.7 for those consequences. Within Script 03, the sensitivity remains an on-demand check that future investigators can run against the published coefficient table.

The LCSC. $LCSC = 100 / \beta_1$. Interpreted as the number of millimetres of rainfall required to raise the water table by 10 cm at the cluster centroid. A cluster with $LCSC = 22$ (C1) takes 22 mm of rainfall to raise its water table 10 cm; a cluster with $LCSC = 42$ (C5) takes nearly twice that. The framing reverses β_1 's direction so that *larger numbers mean less responsive water tables* — easier to communicate to a non-modeller audience than the bare β_1 . The name “Lumped Catchment Storage Coefficient” was chosen because $LCSC = 100 / \beta_1$ is dimensionally equivalent to an effective lumped storativity multiplied by 100 (loosely: how many tens of millimetres of rain end up in storage per millimetre of head rise, expressed as a percentage). The construct is introduced for the first time in this script; every subsequent chapter that cites a cluster's “recharge responsiveness” is referring to LCSC.

Lag diagnostic — lag_diagnostic. Each cluster centroid is refitted at rainfall lags 0, 1, 2, 3 months. The result is a 5×4 grid of fits (`03_04_lag_diagnostic.csv`). The diagnostic is not a model-selection step — `HEADLINE_LAG` is fixed in `config.py` — but it is a check that the bucketing-fix-era data (F.2) actually places the rainfall–response pairing where it belongs. Under the live data, **all five clusters**

maximize R^2 at lag 0: C1's R^2 drops from 0.732 at lag 0 to 0.037 at lag 1, C2 from 0.747 to 0.136, C3 from 0.813 to 0.197, C4 from 0.685 to 0.384, C5 from 0.680 to 0.221. At lag 3 the β_1 sign goes wrong for three of five clusters. This confirms the F.2 reasoning: once Script 01 buckets readings by field convention, current-month rainfall is the right predictor.

Bootstrap — bootstrap_centroid_fits. For each cluster, draw `n_members` wells with replacement, build the upstand-corrected centroid over that resample, refit `fit_ssm()`. Repeat `N_BOOTSTRAP = 1000` times with `BOOTSTRAP_SEED = 20260424`. The replicates yield a distribution per cluster from which 2.5 % and 97.5 % percentile CIs are computed for β_1 , β_2 , β_3 , R^2 , and LCSC, plus the fraction of replicates with $\beta_1 > 0$. The well-level (not month-level) resample is deliberate: the question being asked is *how sensitive is the cluster β estimate to which wells happen to be in the cluster?* — not *how sensitive is a single well's β to which months are observed?* The latter is the per-well CI question that the OLS p-values already answer. Under the live data, all five clusters' β_1 CIs are comfortably positive (β_1 frac positive = 1.000 in every cluster), the CIs are narrow (C1: 4.25–4.93; C5: 2.20–2.65), and the bootstrap β median is within 1.3 % of the centroid headline β_1 everywhere (the largest gap is C4 at 1.2 %; the other four clusters are within 0.4 %). The bootstrap output is also the source of the error bars on `03_01_mechanistic_signatures.png`.

Leave-one-out — leave_one_out_fits. For each cluster with ≥ 4 members, remove each well in turn, refit the centroid SSM on the remaining members. Useful for spotting clusters dominated by a single well. The full table is `03_06_leave_one_out.csv`. Under the live data, no cluster shows catastrophic single-well dependence: β_1 ranges from leaving out the most-influential well are within ~ 7 % of the headline. The single most-influential well by R^2 recovery is **CEH2 in C4 Main Forest** — dropping it shifts β_1 from 2.52 to 2.41 and increases R^2 from 0.685 to 0.729. That recovery (+0.043) is the largest in the LOO sweep; CEH2 is also co-largest on β_1 shift (tied with NW10, both ≈ 0.11). The substantive interpretation, taken up in S.7, is that CEH2 sits at a ridge-recharge interface where the SSM's single-equation form does not fully capture the local subsidy; removing it from the centroid produces a fit closer to the “pure” forest signal. C5's β_1 LOO range (2.34–2.54) is similar in width to C4's, despite C5 having only five members, but no single C5 well is as influential as CEH2 in C4.

C1 split-window — c1_split_window_diagnostic. C1 Lake Edge underwent a regime change in 2018 — the Llyn Rhos-Ddu drawdown event reset the lake stage that buffers most of the cluster. The split-window diagnostic refits the C1 centroid

separately on the pre- and post-2018 sub-windows, and bootstraps each side by well-resampling. Under the live data the two windows do not produce overlapping β CIs: pre-2018 β_1 is 4.36 (CI 4.07–4.64), post-2018 β_1 is 4.93 (CI 4.47–5.40) — the centroid responds more strongly to rainfall in the post-drawdown regime, consistent with a lake that no longer absorbs the rainfall pulse before it reaches the dipwell network. β_3 also shifts (0.080 pre, 0.104 post), suggesting drainage now operates against a less-buffered head. The two β_2 envelopes are also distinguishable (1.23 pre, 0.55 post) but with wider CIs. The headline cluster β values average over both regimes; the split-window result is a flag in the limitations section, not a recommendation to abandon the single-fit cluster characterization.

Datum sensitivity (cluster-centroid) — datum_sensitivity_analysis. The methodologically important piece. The drainage-datum sweep runs `drainage_datum` from 0.5 m to 8.0 m in 0.1 m steps, refitting all five centroids at each step. Two thresholds frame the result. The *first* is the minimum depth at which all five clusters simultaneously produce $\beta_3 > 0$ with $p < 0.05$ — under the live data this is **1.7 m**, limited by C4 Main Forest, whose β_3 p-value crosses into significance there ($p = 0.040$ at 1.7 m, 0.057 at 1.6 m). The *second* is whether the datum that delivers that empirical minimum is the right one to use. It is not. At 1.7 m, C4's β_3 identification is right at the significance boundary; going deeper to 3.7 m drops C4's β_3 p-value from 0.040 to 0.0027 — a fifteenfold improvement in significance — and lifts its R^2 by +0.007. C5 Coastal Forest gains too (β_3 p from 4×10^{-11} to 4×10^{-16} , R^2 +0.032). The cost is borne by the well-determined eastern clusters: C1 Lake Edge loses 0.052 in R^2 and C2 Dune loses 0.029. But C1 and C2 are nowhere near the significance boundary at either depth — their β_3 p-values stay around 10^{-35} and 10^{-25} respectively. C3 is essentially indifferent ($\Delta R^2 = -0.002$).

3.7 m therefore reflects a deliberate trade: a small amount of fit at the eastern clusters, where β_3 is over-determined, exchanged for substantially better β_3 identification at the two forest clusters, where it is hardest to pin down. The value also aligns with the Script 16 water-balance sensitivity analysis, which matters because consistent datum across the pipeline keeps the displacement reference comparable between the SSM coefficient fits and the water-balance arithmetic. The β_1 and β_2 coefficients are invariant in this sweep — only β_3 shifts numerically while preserving its sign and physical interpretation. The three-panel sensitivity figure (03_08_datum_sensitivity.png) shows β_3 vs reference depth (top), R^2 vs reference depth (middle), and aggregate $R^2 + \text{AIC}$ vs reference depth (bottom). The “all $\beta_3 > 0$ & sig” band is shaded green across each axis; the canonical 3.7 m sits inside the band rather than at its edge. This is the diagnostic that lets a reader

satisfy themselves that the datum choice has been made with the trade-off explicit rather than tuned to a single optimum.

Datum sensitivity (per-well) — `well_datum_sensitivity`. The same sweep, applied to each well individually ($66 \times 76 = 5,016$ fits). For each well three optima are recorded: the *primary* optimal datum (minimum depth at which $\beta_3 > 0$ and $p < 0.05$), the *secondary* optimal datum (minimum depth at which $\beta_3 > 0$ with any p), and the *R²-maximizing* datum (highest R^2 across the sweep, regardless of β_3 sign). The full sweep is `03_09_well_datum_sensitivity.csv`; the per-well optima with their β coefficients are `03_09_well_optimal_datums.csv`. The motivation: under a uniform 3.7 m datum, the headline β_3 values are all positive and significant, but the optimum at any individual well may differ. The per-well sweep makes the local optimum visible.

Under the live data, the per-well primary optima are mostly pinned at the lower bound of the sweep (0.5 m) for C1, C2, and C3 wells — meaning the data alone do not require a deep datum at these wells, but they tolerate one without breaking sign. The more informative diagnostic is the *R²-maximizing* datum, whose per-cluster medians are: C1 ≈ 0.8 m, C2 ≈ 0.9 m, C5 ≈ 1.9 m, C3 ≈ 1.6 m, C4 ≈ 2.7 m. The eastern lake-adjacent and dune wells optimize at shallow datums; the forest and coastal-forest clusters at deeper ones. The pattern is loosely consistent with a deepening effective drainage base from lake margin to dune interior, and is shown spatially by `03_10_well_datum_r2max_map.png`. The R^2 -gain map (`03_10_well_r2_gain_map.png`) makes visible the cost of using a uniform datum: the network mean R^2 gain from per-well optimisation over the uniform 3.7 m is +0.028, ranging from +0.001 at C4 (almost no penalty) to +0.069 at C1 (the cluster most penalised by the uniform datum). A reader who wants the deepest per-well fit can read it off `03_09_well_optimal_datums.csv`; the uniform-datum work is the report's headline.

Six of 66 wells never achieve $\beta_3 > 0$ with $p < 0.05$ at any datum in the sweep — these are the wells at which the SSM's drainage term cannot be cleanly identified. They are routed into the spatial analysis (Scripts 07, 08) where their uncertainty is shown rather than hidden.

Spatial datum maps — `make_well_datum_maps`. `plot_metric_map` from `map_utils` (F.5) produces two publication-quality spatial maps: per-well R^2 -maximizing datum and per-well R^2 -gain over the uniform datum. DEM background, KML overlays (`Features.kml`, `streams.kml`, `clearfell.kml`), cluster-shape markers from `config.CLUSTER_MARKERS`, dual colorbars, legend — all the

standard rendering described in F.5. These are the only spatial maps Script 03 produces; per-well β maps are Script 07's job.

Summary table and signatures figure — `build_summary_table`, `make_signatures_figure`. The summary table merges the centroid mechanistic coefficients with the bootstrap CIs and an amplitude-heterogeneity flag (which loads from Script 02's `02_08_cluster_amplitude_per_well.csv` if present, with a hard-coded fallback). The amplitude flag is a heuristic: a cluster is flagged “heterogeneous” if the post-2018 ratio of its highest to lowest per-well $p_{90} - p_{10}$ amplitude exceeds 1.5. The flag drives no analytical choice — it is a reader-facing diagnostic. The three-panel signatures figure (`03_01_mechanistic_signatures.png`) is the bar chart of β_1 , β_2 , β_3 per cluster with bootstrap error bars, used as Figure 8 in the main report.

Regional-average exports — `export_regional_averages`, `export_regional_averages_maod`, `export_cluster_peak_months`. Three files for downstream consumption: the cluster-centroid hydrograph in depth-below-ground (`03_regional_averages.csv`), the same in m AOD (`03_regional_averages_maod.csv`), and a one-row-per-cluster table of the calendar month in which the centroid's long-term mean depth is shallowest (`03_cluster_peak_months.csv`). The peak-month table is the forecasting-horizon endpoint used by Script 11 and 11b — the canonical “when in the year is the cluster's water table at its annual peak?”.

Site-specific choices and rationale

- **Per-well upstand correction inside `per_well_fits`, not just at the centroid.** Before April 2026 the script only upstand-corrected the centroid. Wells with tall upstands — CEH2 at ~0.71 m, NW2 at ~0.59 m, t41a at ~0.40 m — had their depth series fitted against a DRAINAGE_DATUM anchored to the ground surface even though the depth values themselves were relative to the pipe top. This produced systematic, well-by-well displacement errors at the few wells where the offset was large enough to matter. The per-well correction (`h_corrected = wells_clean[col] - u` before each `fit_ssm()` call) closes that gap. The change is small for most wells (median upstand is well under 30 cm) but unambiguous for the three flagged wells.
- **Cluster centroid as arithmetic mean of upstand-corrected member wells.** The simplest defensible aggregation. Median, record-length-weighted mean, and PCA-derived centroids were all considered during the rebuild and rejected as adding complexity without measurable benefit on

this dataset — the within-cluster amplitude distributions are well-behaved enough that the mean is not pulled around by outliers.

- **window = LCSC_DATA_LIMIT = 100 for per-well fits; window = None (full record) for centroid fits.** The two-window design reflects the two different purposes. Per-well fits feed the spatial coefficient maps (Script 07) and the LCSC benchmarking (Script 08), both of which use the most-recent 100-month window so that what they show is contemporary behaviour rather than a long-term average. Cluster centroid fits feed the report's mechanistic table and the forecasting work, where the longest possible record tightens the standard errors and captures behaviour across the full instrumental period. The 100-month window choice itself was made to align with the LCSC analysis in the report (Section 3.3.1).
- **MIN_OBS_PER_WELL = 30.** Standard minimum for a no-intercept OLS fit on first differences. Below 30 aligned rows the per-well coefficient CIs widen sharply and the β estimates are dominated by noise. The constant is also defined as `MIN_OBS = 30` inside `model_utils`. Script 03 passes `MIN_OBS_PER_WELL` explicitly through to `fit_ssm()` at every per-well call site (`per_well_fits`, `well_datum_sensitivity`, `leave_one_out_fits`), so changing the Script 03 value is the right way to retune the per-well minimum if needed.
- **Bootstrap resamples wells, not months.** `N_BOOTSTRAP = 1000` replicates draw the cluster's member wells with replacement, build the upstand-corrected centroid over the resample, and refit. This matches the question being asked: how sensitive is the cluster β estimate to which wells happen to make up the cluster? — not the per-well-monthly-noise question that the OLS standard error already answers. `BOOTSTRAP_SEED = 20260424` fixes reproducibility.
- **The lag diagnostic refits at lags 0–3 regardless of HEADLINE_LAG.** Diagnostic, not model-selection. Under the live bucketing-fix-era data, all five clusters maximize R^2 at lag 0 and the headline value is correct. The diagnostic is preserved because (i) the bucketing fix is recent enough that the reader may legitimately want to satisfy themselves the headline lag is right, and (ii) a future data update that drifted the bucketing could in principle reintroduce a lag signal; the diagnostic catches that.
- **Datum sweep 0.5–8.0 m in 0.1 m steps; canonical value 3.7 m.** 76 depths $\times$ 5 clusters = 380 fits at the centroid level (fast); $76 \times 66 = 5,016$ fits at the

per-well level (a few minutes on a current laptop). 0.5 m is the shallowest depth where the dune-slack record contains any meaningful headroom above the datum; 8.0 m is comfortably below any plausible drainage base. The canonical 3.7 m value sits above the live empirical minimum (1.7 m, at which all five clusters first simultaneously satisfy $\beta_3 > 0$ with $p < 0.05$). Going from 1.7 m to 3.7 m takes the two forest clusters from β_3 identification near the significance edge (C4 $p = 0.040$ at 1.7 m) to comfortably-determined fits (C4 $p = 0.0027$, C5 $p = 10^{-16}$), at the cost of a small R^2 penalty at the two clusters where β_3 is over-determined (C1 $\Delta R^2 = -0.052$, C2 $\Delta R^2 = -0.029$); C1 and C2 stay astronomically significant on β_3 at either depth. 3.7 m also aligns with the Script 16 water-balance sensitivity analysis, where downstream consistency matters. The β_1 and β_2 coefficients are invariant in this sweep — only β_3 shifts numerically while preserving its sign and physical interpretation.

- **C1 split at 2018-01-01.** Chosen because the post-2018 amplitude analysis (Script 02) showed that four of seven C1 wells had post-2018 $p_{90} - p_{10}$ ranges meaningfully wider than their pre-2018 values, *even after climate normalization* — i.e. a structural change in cluster behaviour rather than a climate forcing effect. The Llyn Rhos-Ddu drawdown event provides the physical mechanism. The split-window diagnostic tests whether the change shows up in the β coefficients (it does), and the analysis is preserved as a known caveat rather than as a recommendation to fit two separate C1 models.
- **Sign violations halt the pipeline only at the *centroid* level, and only *after* the diagnostic outputs are saved.** A failing centroid fit is the only kind that propagates into the report's headline values. The decision to defer the halt until after the diagnostic tables and the signatures figure are written is deliberate: the LOO, bootstrap, and datum-sensitivity outputs are exactly the diagnostics an investigator needs to understand the failure. Halting before they are saved would give the user a stack trace and no diagnostic context. Per-well sign violations are reported as console diagnostics; they may flag wells worth visiting in the spatial chapters but do not invalidate the broader analysis.

Outputs

File	Contents	Principal downstream consumers
03_master_data.csv	Per-well β_1 , β_2 , β_3 , p -	Scripts 07, 08, 11, 16, 17,

File	Contents	Principal downstream consumers
(INT_)	values, R^2 , n, LCSC, cluster ID, easting/northing	19, 21
03_regional_averages.csv (INT_)	Cluster-centroid monthly hydrographs (depth below ground) + P, PET in mm	Scripts 11, 16, 21
03_regional_averages_maod.csv (INT_)	Same in absolute m AOD	Script 21 (forestry scenarios)
03_cluster_peak_months.csv (INT_)	Long-term mean peak month per cluster	Scripts 11, 11b
03_state_space_model/03_01_mechanistic_signatures.png	3-panel β bar chart with bootstrap CIs	Diagnostic / summary-document figure; not a numbered figure in the main report
03_state_space_model/03_02_cluster_summary_table.csv	Headline per-cluster summary with bootstrap CIs and amplitude flag	Report Table (supplementary)
03_state_space_model/03_03_cluster_mechanistic_coefficients.csv	Centroid β coefficients with p-values, R^2 , n, LCSC, datum	Scripts 11, 11b, 16, 19, 21; Report Table 3
03_state_space_model/03_04_lag_diagnostic.csv	Centroid fits at lags 0–3 per cluster	Report §3.3
03_state_space_model/03_05_bootstrap_ci.csv	Bootstrap median and 95 % CIs per cluster	Signatures figure error bars
03_state_space_model/03_06_leave_one_out.csv	LOO centroid fits per cluster member	Report §3.3 / supplementary
03_state_space_model/03_07_c1_split_window.csv	C1 pre/post-2018 fits with bootstrap CIs	Report §4.2 (C1 caveat)
03_state_space_model/03_08_datum_sensitivity.{csv,png}	Cluster-level datum sweep	Report §3.4
03_state_space_model/03_09_well_datum_sens	Per-well datum sweep	Report §3.4

File	Contents	Principal downstream consumers
itivity.csv	(5,016 fits)	
03_state_space_model/ 03_09_well_optimal_data- tums.{csv,png}	Per-well primary, secondary, R^2 -max datum + β at each	Report §3.4
03_state_space_model/ 03_10_well_datum_r2max- map.png	Spatial map of per-well R^2 -maximizing datum	Report Figure (§3.4)
03_state_space_model/ 03_10_well_r2_gain_map- p.png	Spatial map of R^2 gain vs uniform datum	Report Figure (§3.4)

All paths resolve through `utils/paths.py` (INT_MASTER_DATA, INT_REGIONAL_AVG, INT_CLUSTER_AVG_MAOB, INT_CLUSTER_PEAK_MONTHS, OUT_03_*, DIR_03).

Limitations and known caveats

- The fitted β values are reliable for ranking clusters, not for quantifying absolute flux magnitudes.** The OLS estimator applied to a lumped monthly model carries two structural biases that inflate the absolute coefficient values: (i) collinearity between monthly P and PET inflates both β_1 and β_2 simultaneously, producing an upward bias whose magnitude cannot be recovered from the regression alone; (ii) the monthly climate regressors are area-averages over a spatially heterogeneous forcing field — a form of errors-in-variables that further biases β away from the true site-scale coefficients. The net effect is that β_1 , β_2 , and β_3 as reported in Table 3 are biased somewhat high in absolute terms. The cluster contrasts — C4 $\beta_2 >$ C5 $\beta_2 >$ C3 β_2 ; the open-dune β_1 gradient C1 $>$ C2 $>$ C3 — are robust ordinal findings. Statements such as “the C4 forest cluster intercepts 2.55 m of water-table head per metre of PET” are not warranted; statements such as “C4 has a higher atmospheric draw coefficient than any open-dune cluster” are. Any downstream analysis that converts β values to volumetric fluxes (Scripts 16, 19, 21) should be read with this caveat in mind: the absolute magnitudes carry the OLS bias, so the scenario comparisons are relative rather than calibrated.
- The SSM is a single-equation model.** It is a lumped decomposition of Δh into rainfall, PET, and drainage contributions. It does not represent boundary-layer hydraulics, lateral subsidies (ridge recharge into C4, lake exchange at C1), tidal couplings, or non-stationary effects (coastal retreat at C5). Where any of these matter for the science, they appear as systematic

residuals diagnosed downstream — Scripts 22 (residual lag structure), 23 (ridge recharge), 24 (residual seasonality), and 25 (coastal gradient).

- **The centroid averages over within-cluster heterogeneity.** Where a cluster contains wells with very different mean amplitudes — notably C4, which includes CEH14 at the ridge crest — the centroid coefficients describe the cluster mean rather than any single well. The leave-one-out output makes the well-by-well sensitivity visible; the heterogeneity flag in the summary table flags clusters where the within-cluster range exceeds a 1.5× ratio.
- **C1 Lake Edge changed regime in 2018.** The pre/post-split-window diagnostic shows non-overlapping β CIs across both windows. The single-fit C1 cluster characterization in Table 3 is the average across both regimes. Where C1 results are quoted in the main report, the split-window outcome should be cited alongside as a caveat.
- **Per-well fits at the network edges have small n.** Wells with ~30–50 valid rows produce β with wide CIs. Six wells in the live network never achieve $\beta_3 > 0$ with $p < 0.05$ at any datum in the sweep. Per-well coefficient maps in Script 07 annotate uncertainty so the reader can distinguish robust spatial structure from sample-size artefacts.
- **The uniform 3.7 m datum is a deliberate trade, not an empirical optimum.** It sits above the live empirical minimum (1.7 m) and is the value at which the two forest clusters' β_3 identification is comfortably significant rather than near the threshold. The cost of going to 3.7 m rather than staying at the empirical minimum is an R^2 reduction of -0.052 at C1 Lake Edge and -0.029 at C2 Dune, both of which remain astronomically significant on β_3 regardless of depth. The benefit is a twentyfold improvement in C4 β_3 significance and substantially better fit at C5. For cluster-level work the uniform 3.7 m is the right choice; for any single-well question the per-well datum outputs (03_09_well_optimal_datums.csv) should be consulted — the mean network R^2 penalty from using the uniform datum at every well, rather than the per-well R^2 -maximizing value, is $+0.028$ across the network, heaviest at C1 Lake Edge ($+0.069$) and lightest at C4 Main Forest ($+0.001$).
- **C4 Main Forest has the weakest β_3 identification of the five clusters.** Even at the canonical 3.7 m datum, C4's β_3 p-value (0.0027) is several orders of magnitude larger than the other four clusters' (which sit at 10^{-16} to

10^{-35}). The datum choice was deliberately set deeper than the empirical minimum to move C4 away from the significance boundary, and the move worked — but C4 remains the cluster where the SSM most underdescribes the local hydrology. CEH14's ridge-recharge subsidy is the canonical missing term, taken up in S.16.

Where the result appears in the report

- §3.3 *State-space regression* — methodology.
- §3.4 *Drainage datum sensitivity* — 03_08_datum_sensitivity.png and the per-well sensitivity discussion.
- Table 3 — cluster mechanistic coefficients (from 03_03_cluster_mechanistic_coefficients.csv).
- 03_01_mechanistic_signatures.png (cluster β bar chart) — a diagnostic figure used in the summary documents; not a numbered figure in the main report (the report's Figure 8 is the cluster hydrographs).
- §4.2 — cluster characterization, drawing on the LCSC values and the cluster-mean signatures.
- §4.4 / Figure 45b — per-well β_1 map (Script 07 produces; values come from 03_master_data.csv).
- §4.2 *C1 caveat* — 03_07_c1_split_window.csv.
- Subsequent §4 and §5 sections — implicit, in every analysis that reads SSM outputs.

Cross-references

- **F.3** — the full SSM specification (equation, displacement formulation, sign conventions, drainage datum rationale, `model_utils.fit_ssm()` interface). Refer to F.3 throughout rather than re-deriving.
- **F.4** — the $k=5$ partition that the centroids and per-well fits use; per-cluster member counts (C1=7, C2=26, C3=19, C4=9, C5=5; $n=66$ total).
- **F.5** — `model_utils.py`, `paths.py`, and `map_utils.py` module roles.
- **S.1** — input data preparation; upstand source; reference-network membership; 01_wells_provenance.csv (the optional input consumed when `exclude_interpolated=True` is invoked).
- **S.2** — cluster partition that defines the centroids; amplitude descriptors that feed the heterogeneity flag in 03_02_cluster_summary_table.csv.
- **S.7** — spatial per-well coefficient maps that consume 03_master_data.csv.
- **S.8** — LCSC vs Traditional Linear Model benchmarking.

- **S.9** — forecasting thresholds that consume `03_03_cluster_mechanistic_coefficients.csv` and `03_cluster_peak_months.csv`.
- **S.16** — supplementary diagnostics that pick up the systematic SSM residuals (ridge recharge at C4/CEH14, residual seasonality).

Phase 2 — Pearson Membership Audit

S.4 Scripts 04, 05, 06 — Cluster visualization and Pearson affinity audits

Steps 4–6 / 27. Phase 2.

Motivation (chapter-level)

Phase 2 answers two related questions that Phase 1 leaves open. First, where in the warren does each of the five behavioural clusters actually sit on the ground — what does the partition look like as a map rather than a dendrogram? Script 04 produces the orientation figure. Second, how confidently is each well in the cluster the partition has placed it in, and how should the ~22 extended-network wells (FE1–4, LIS1, the demoted CEH wells, and the perimeter p* wells) be slotted into the same five-cluster scheme when they were not part of Ward’s input? Scripts 05 and 06 are the confidence-quantification layer: per-well Pearson correlations against the five cluster centroids, gap statistics between best-match and runner-up, and a status flag that separates wells comfortably inside their cluster from wells on the boundary. The three scripts are grouped here because together they form a coherent visualization-and-audit package consumed in §4.2 and §4.3 of the main report. Script 04 visualises the partition; Scripts 05 and 06 quantify how strong the partition’s claim on each well actually is.

The methodology shared by Scripts 05 and 06 — z-score each well’s depth series before correlating, build cluster centroids as the mean of the z-scored member series, compute Pearson correlation against each centroid, gap = best – second-best — is described once at the chapter level and not repeated in the two sub-sections. The cluster-identity conventions (IDs 1–5, labels, colours, markers) are inherited from F.4 and not re-derived.

Sub-script 04 — Cluster orientation map (04_cluster_visualisations.py)

Motivation

A single publication-quality map showing the five-cluster partition of the reference network on the Newborough Warren DEM, with each well as a cluster-coloured, cluster-shaped marker against the site's hydrological features (lake, streams, clearfell footprint). It is an orientation figure, not an analytical output: it carries no numbers, only the spatial arrangement of cluster membership that all subsequent spatial work assumes the reader has internalised.

Inputs

File	Origin
02_cluster_stats.csv	Per-well cluster assignment (Script 02)
01_locations.csv	Easting, northing, ground elevation per well (Script 01)
data/Features.kml, streams.kml, clearfell.kml	Site GIS overlays
data/newborough_dem.tif	Coloured terrain background

Methodology

The script is 77 lines and does one thing: it joins 02_cluster_stats.csv to 01_locations.csv on the normalized well name (data_utils.normalize_well_name()), extracts the integer cluster ID from the Cluster column (via cluster_id_from_value() to tolerate either "C3" or 3 formats from upstream), and plots each well as a matplotlib.pyplot.scatter point. The DEM is loaded by map_utils.load_dem_layer(); the three KML overlays are added by map_utils.add_kml_features(). If the DEM file is not available, map_utils.add_osm_basemap() provides an OpenStreetMap fallback so the figure remains plottable in environments without the GIS data files. Each cluster's marker shape comes from config.CLUSTER_MARKERS (circle for C1, square for C2, triangle for C3, diamond for C4, plus for C5; see F.4) and its colour from config.CLUSTER_COLOURS. Well labels are placed by adjustText to minimize overlap, with thin black callout lines where the algorithm has had to move a label far from its point. The map extent is fixed at easting 240,100–243,900 and northing 362,200–365,800 (British National Grid, EPSG:27700) and the axes are forced to equal aspect so distances on the map are isotropic. The figure is rendered at 300 dpi and written to 04_01_core_architecture_map.png.

The reference network only (66 wells) is plotted — the extended-network wells appear on Script 06’s integration map instead. Two legends are placed in opposite corners: cluster assignments in the lower left and KML site features (lake outline, stream, clearfell polygon) in the upper right. When the script runs in greyscale mode (`BW_MODE = True`), the DEM colour bar is suppressed because the underlying greyscale DEM raster does not carry a meaningful colour mapping; the cluster colours fall back to `CLUSTER_COLOURS_BW` via the same identity-keyed lookup, so the figure remains legible without colour.

Outputs

File	Contents	Downstream consumer
04_cluster_visualisations/ 04_01_core_architecture_map.png	Spatial cluster map, reference network	Report Figure (§4.2 cluster orientation)

Sub-script 05 — Pearson affinity (reference network, 05_pearson_affinity.py)

Motivation

For each of the 66 reference-network wells, Ward’s clustering has produced a cluster assignment; the question Script 05 answers is *how confidently does that assignment sit*. Two wells in the same cluster can have very different relationships to the cluster’s centroid — one might be a tight match, another might correlate almost as strongly with a neighbouring cluster. The Pearson affinity audit computes the gap between best-match and assigned-cluster correlation for every well and produces a four-way status classification: Core (best-match = assigned, comfortable margin), Fuzzy (best-match = assigned but small margin), Spy (best-match $\neq$ assigned — Ward’s and Pearson disagree), and Unclassified (insufficient data to compute a correlation). The output is read by Scripts 11b (spatial threshold maps) and indirectly underpins the per-well confidence rendering on subsequent spatial figures.

Inputs

File	Origin
01_wells_clean.csv	Cleaned depth time series (Script 01)
02_cluster_stats.csv	Per-well cluster assignment (Script 02)
01_locations.csv	Easting, northing per well (Script 01)

Methodology

The script is 259 lines. After loading the wells matrix and reducing it to the 66 reference-network wells named in `02_cluster_stats.csv`, it z-scores each well's depth series row-wise (`zscore_rows`) and builds five cluster centroids as the row-mean of the z-scored member series per cluster (`build_centroids`). Z-scoring per well, before averaging, is the operative choice: it removes each well's mean depth and amplitude so the resulting correlations are sensitive to *shape* (when the well rises and falls, relative to the rest of the cluster) rather than *level* (how deep below ground its mean sits). The standard deviation uses `ddof=0` (population formula), consistent with the bootstrap convention used in Script 02 and Script 03.

For every well, the script then computes a Pearson correlation against each of the five z-scored centroids via `safe_pearson()`, which requires at least 24 paired non-missing observations and returns NaN if either series is constant over the overlap window. The 24-month minimum matches the `MIN_EXTENDED_MONTHS` threshold used by Script 01 to admit an extended-network well — a well that did not produce enough data to be admitted to the network has no business producing a Pearson value here either. Reference-network wells comfortably exceed this minimum, so the threshold is in practice operative only for Script 06's extended-network rows.

The classification step is in `classify()`. For each well, the assigned-cluster correlation is read from the corresponding `Cluster_n` column; the best-match cluster is whichever has the highest valid correlation; the gap is best-match minus the highest *other* (non-assigned) correlation. A well where best-match $\neq$ assigned is labelled Spy; a well where best-match = assigned and the gap exceeds 0.05 is Core; a well where best-match = assigned but the gap is at or below 0.05 is Fuzzy. The 0.05 threshold is hardcoded in `classify()` and reappears as `DELTA_THRESH = 0.05` at module scope in Script 06; in the current build the two are not linked, but the value is identical.

A secondary diagnostic — the *Mean Cluster Affinity* (MCA) flag — is computed separately. A well is MCA-flagged if three or more of its five cluster correlations exceed 0.90 (`MCA_Count_r_gt_0_90 >= 3`). The label `MCA_Cluster_Label` records which three (e.g. C2/C3/C5). The flag identifies wells whose shape signature is so generic that several cluster centroids match it strongly — typically wells in the central dune field where C1, C2, and C3 hydrographs all rise and fall in unison and the cluster boundary is a matter of mean depth and amplitude rather than shape. Under the live data, 46 of 66 reference wells are MCA-flagged, illustrating

that the partition rests on dimensions other than month-to-month co-variation alone (notably mean depth and amplitude — see F.4 on the no-detrending choice in Script 02).

Under the live partition, the audit table classifies 17 wells as Core, 46 as Fuzzy, and 3 as Spy: CEH4 (assigned C3, best-match C5, gap -0.0118), CEH21 (assigned C3, best-match C5, gap -0.0054), and NW13 (assigned C3, best-match C2, gap -0.0015). All three spies have margins of 0.012 or smaller, i.e. they sit on the C3 boundary with barely any preference between C3 and the neighbour Pearson picks. The spy flag is descriptive, not corrective — Ward's centroid is built from *all* members and accounts for amplitude as well as shape, where Pearson on z-scored series sees only shape, so the two methods can disagree by tiny margins on boundary wells without either being wrong.

The script writes one CSV — `05_pear_membership_audit.csv` (the INT_PEAR_AUDIT intermediate) — containing the audit columns plus the five per-cluster Pearson values, the MCA flag, and the MCA cluster label. Two figures follow. The first is a bar chart of per-cluster Pearson values for an illustrative subset of reference wells — one or two from each cluster — that shows the five-bar affinity profile against all five centroids and is read as the per-well analogue of the cluster-centroid signature plots in Script 03 (`05_pear_02_affinity_chart_reference.png`). The second is a spatial confidence map (`05_pear_01_spatial_confidence_map.png`) with each well drawn at its location, coloured by its best-match cluster, with marker shape encoding status (Core=circle, Fuzzy=diamond, Spy=star); Fuzzy and Spy markers carry the secondary-cluster integer as a small overlay label so the reader can see which alternative the well is wavering toward. MCA-flagged wells receive an additional open black surround marker, with the surround shape encoding the specific MCA combination (e.g. C2/C3/C5), so the reader can see at a glance where in space each MCA pattern occurs. The DEM and KML overlays are loaded by the same `map_utils` helpers as Script 04.

Site-specific choices and rationale

- **Pearson, not Spearman or Kendall.** Ward's clustering in Script 02 uses correlation distance, $1 - \rho$ (Pearson). Auditing the partition with a different correlation measure would mean Pearson set the partition and a non-Pearson statistic judged it — an inconsistency that would muddy interpretation of any disagreement. Using the same coefficient closes that loop.

- **Z-scored centroids; mean, not median.** The Script 03 SSM also fits to the mean centroid (F.3), so the audit's centroid construction matches the fitting convention. Shape-only comparison via z-scoring is consistent with the no-detrending choice in Script 02 (S.2): both the partition and the audit operate on the behavioural-pattern axis of “wells that rise and fall together”, not on the absolute-depth axis.
- **24-month minimum on the Pearson overlap.** Matches the extended-network admission threshold. A series shorter than two years cannot resolve an annual seasonal cycle reliably and the Pearson value against an annually-cycling centroid becomes a function of which months happen to be present.
- **Gap threshold of 0.05 for Core vs Fuzzy.** Empirical, not principled — chosen to separate the comfortable interior of each cluster (~26% of reference wells) from the broad boundary band (~70%). Sensitivity to this value is low: at 0.03 the Core count rises to the high thirties, at 0.10 it falls to a handful. The threshold's role is descriptive labelling, not gatekeeping.
- **Three-cluster MCA threshold at $r > 0.90$.** Identifies wells whose hydrograph correlates strongly with several cluster centroids — a sign that month-to-month shape alone does not separate them from the alternatives, and the cluster boundary in their case rests on amplitude or mean depth. The threshold of 0.90 and the count of 3 are both empirical; together they pick out the central-dune wells whose generic seasonal cycle correlates well across most clusters.

Outputs

File	Contents	Downstream consumer
05_pear_membership_audit.csv (INT)	Per-well audit table (66 rows, 16 cols)	Scripts 11b, 18 (read indirectly via the sitewide audit)
05_pearson_affinity/05_pear_02_affinity_chart_reference.png	Bar chart of per-cluster Pearson values for an illustrative reference-well subset	Report §4.3
05_pearson_affinity/05_pear_01_spatial_confidence_map.png	Spatial map of best-match cluster + status	Report §4.3

Limitations

- **Pearson similarity captures shape, not magnitude.** A well whose mean depth has drifted systematically from the cluster mean is not caught by the audit. The drift would show up as an amplitude or mean-shift difference between the well and the centroid, neither of which contributes to a Pearson value once both have been z-scored.
- **A small gap is descriptive, not corrective.** A Fuzzy well, or even a Spy well with a 0.01 gap, is not misclassified by Ward's. It is a well that sits in transitional behavioural territory — the cluster boundary is genuinely indistinct for it, and the audit reports that indistinctness rather than resolving it.
- **MCA is descriptive of the *common* behavioural signal.** A high MCA count means the well's hydrograph is generic enough to track several centroids; it is not a partition criticism, because Ward's resolves these wells using the parts of the behaviour that *do* differ between clusters (amplitude, deep-summer minima, response timing at longer lags), which the z-scored Pearson does not see.

Sub-script 06 — Pearson affinity (sitewide / extended, 06_pearson_extended.py)

Motivation

Script 05 audits the 66 reference-network wells. Script 06 extends the same logic to the full 88-well sitewide network — the 66 reference wells plus 22 extended wells (FE1–4, LIS1, the demoted CEH wells with shorter records or singleton-outlier behaviour, and the perimeter p1/p2/pe/pw wells; Llyn Rhos-Ddu and pdfs are blacklisted upstream in Script 01). Extended-network wells were excluded from Ward's input by design, because either their records are too short to anchor a stable distance to the reference wells or, in the CEH3/CEH22 case, they carry a non-stationary signal (tidal contamination, S.1) that Ward's clustering would either ignore or distort. The Pearson audit gives each of them a cluster label retroactively: the cluster whose centroid their shape best matches, with the gap statistic carried alongside as a calibration of confidence. Downstream consumers — Script 11b for spatial thresholds, Script 18 for WTF Sy interpolation — read the sitewide audit and decide for themselves how much weight to give a Pearson-assigned well; Script 06 does not gate-keep.

Inputs

File	Origin
01_wells_reference.csv	Reference-network subset, 66 wells (Script 01)
01_wells_extended.csv	Extended-network subset, 22 wells (Script 01)
02_cluster_stats.csv	Reference-network cluster assignments (Script 02)
01_locations.csv	Easting, northing per well (Script 01)

Methodology

The script is 363 lines and follows the same shape as Script 05, with three differences that are worth pinning down.

First, the z-scoring is applied to the *combined* reference + extended series (`z_all = zscore_rows(pd.concat([ref_wells, ext_wells]))`), and the reference centroids are then extracted from this combined matrix rather than z-scoring the reference set alone. In practice the two approaches give virtually identical centroids because each well is z-scored row-wise — adding the extended wells to the matrix does not alter any reference well’s own z-score. The construction is convenient rather than substantive: the extended wells need to be z-scored anyway, and doing it in a single call keeps the indexing simple.

Second, the classification has five status labels rather than four, because the network split is binary and not every status applies to both sides. Reference-network wells with a Ward’s assignment can be `Ref_Core` (best-match = assigned, $\text{gap} \geq 0.05$), `Ref_Fuzzy` (best-match = assigned, $\text{gap} < 0.05$), or `Ref_Spy` (best-match $\neq$ assigned). Extended-network wells have no Ward’s assignment to be a spy of and receive either `Ext_Core` ($\text{gap} \geq 0.05$ against the best-match cluster) or `Ext_Fuzzy` ($\text{gap} < 0.05$). The `DELTA_THRESH = 0.05` and `MCA_THRESH = 0.90` constants are exposed at module scope here, in contrast to Script 05 where the gap threshold is buried in the `classify()` function. The values are identical between the two scripts.

Third, the spatial output is an *integration map*, not a confidence map. Reference and extended wells share the same plot, distinguished by marker fill — reference wells are filled in their best-match cluster colour (or dark grey in BW mode), extended wells are filled in light grey, and any well classified `Ref_Spy` is drawn with a white fill and a thickened black edge so the reader can see at a glance

which Ward's assignments the Pearson audit disagrees with. Cluster identity is encoded by marker *shape* (the canonical CLUSTER_MARKERS from F.4), so colour is freed up to carry network-membership information without ambiguity. Well labels are placed by `adjustText` with the same offset-and-repel parameters tuned for the dense reference-network labelling. The audit-bar chart for the extended wells is the only figure that Script 06 produces with the reference wells absent — bars for FE1–4, LIS1, the demoted CEHs, the p-wells, NW8/8b/12 — and uses the same five-cluster colour palette so the reader can read off both the assignment and the runner-up at a glance.

Under the live data, all 22 extended-network wells receive a valid Pearson best-match — there are no rows lost to the 24-month threshold. The status breakdown is 19 `Ext_Fuzzy` and 3 `Ext_Core`. The three `Ext_Core` wells are NW12 (best-match C4, gap 0.123), CEH3 (best-match C5, gap 0.175), and P2 (best-match C1, gap 0.112); each is well-separated from its neighbours and has a confident assignment. The 19 `Ext_Fuzzy` wells include all four FE clearfell wells (each assigned to either C4 or C5 with gaps under 0.04, consistent with their forest-margin location), LIS1 (best-match C4), and the perimeter wells (P1, PE, PW), which sit at best-match C2 or C3 with small gaps, indicating they lie on the eastern open-dune / western-residual boundary. CEH3 and CEH22 — the two wells routed to the extended network on tidal-contamination grounds (S.1) — receive different best-match clusters under the audit: CEH3 best-matches C5, the coastal-forest cluster whose reference members (CEH16, NW9) themselves carry a partial tidal signature, while CEH22 best-matches C2 by the narrowest of margins. CEH3 takes `Ext_Core` status (gap 0.175) because the C5 fit is unambiguous; CEH22 takes `Ext_Fuzzy` with a gap of 0.0011 — among the smallest in the entire audit — and a low best-match correlation (0.857), its tidal signature strong enough that it does not align cleanly with any one centroid shape. CEH22's near-tie sits between C2 and C5 (correlations within 0.001 of each other, with C3 a close third), so although the audit nominally assigns it to C2 the assignment carries essentially no behavioural preference; this near-indeterminacy, rather than the specific cluster label, is the informative result. Under the previous (`limit=3`) interpolation policy CEH22 best-matched C5; the `limit=1` change moved the cluster centroids slightly and flipped this coin-flip assignment — expected behaviour for a well this marginal, not an instability in the partition. The pdfs well, which carried the lowest best-match correlation sitewide in earlier audits, no longer appears: it was added to `EXTENDED_NETWORK_BLACKLIST` in Script 01 on 2026-05-24 on tidal-influence grounds (S.1) and is removed before the extended-network CSV is written.

Site-specific choices and rationale

- **Reference centroids drive the audit; extended wells are not included in centroid construction.** A centroid built from short, possibly non-stationary records would carry exogenous variance into the cluster mean. The reference network is the authoritative behavioural definition, and extended wells are admitted only as evaluated objects, never as evaluators.
- **No confidence threshold for assignment.** Every extended-network well receives a cluster label and a gap value; the downstream consumer decides what to do with the gap. Script 11b downweights Pearson-assigned wells in the spatial-threshold rendering; Script 18 treats them as supplementary in WTF Sy interpolation. Centralising the decision would force a one-size-fits-all gate where the downstream contexts require different ones.
- **Llyn Rhos-Ddu and pdfs are filtered upstream in Script 01.** Reflected here only as an absence — neither appears in any row of 06_pear_membership_audit_sitewide.csv because Script 01's `EXTENDED_NETWORK_BLACKLIST = frozenset({"llynrhos", "pdfs"})` removes them before the extended-network CSV is written. The rationale is in S.1; the audit inherits the decision rather than re-evaluating it.
- **Cluster identity by shape, not colour, on the integration map.** Network identity (reference vs extended vs spy) needs a separate channel from cluster identity, and humans read marker fill more reliably than marker shape. Putting cluster on shape and network on fill keeps the two readable simultaneously.

Outputs

File	Contents	Downstream consumer
06_pear_membership_audit_sitewide.csv (INT)	Sitewide per-well audit (89 rows, 14 cols)	Scripts 11b, 18
06_pearson_extended/06_pear_01_affinity_chart_extended.png	Bar chart of extended-network affinities	Report §4.3
06_pearson_extended/06_pear_02_integration_map.png	Sitewide integration map	Report §4.3 (Figure 12)

Limitations

- **Short overlap windows can produce misleading correlations.** A well with only 30 monthly observations against a centroid spanning the full reference window has limited statistical power; a chance alignment between an annual cycle and a short overlap can produce a high Pearson

value that does not reflect long-term cluster fidelity. The 24-month minimum protects against the worst cases but does not eliminate the effect at the lower end. Consumers of the audit table can use the per-cluster r_{C^*} columns and the gap to judge robustness on a case-by-case basis.

- **Pearson best-match is not a substitute for Ward's clustering.** A well classified as C5 by Pearson with a 0.85 correlation and a 0.02 gap should not be treated, in mechanistic inference (SSM fits, water balance, scenario analysis), as a full member of C5 in the way that an Ward's-assigned reference well is. The audit table carries the values; the scientific call is downstream.
- **Tidal-signal wells (CEH3 and CEH22) sit on the coastal/tidal behavioural margin.** CEH16 and NW9, both reference wells in C5, are the coastal-margin pair whose hydrographs include the most coastal-process influence in the reference partition. CEH3 shares a tidal-influence signature with those wells, so the C5 centroid is its closest available match in correlation space — a genuine rather than coincidental match, with the audit identifying the same coastal/tidal axis that motivates the C5 cluster. CEH22, by contrast, sits in a near-tie: its best-match cluster is C2 with a gap of only 0.0011 over C5, so its assignment carries no real behavioural preference. Both wells' audit results are an assessment of *shape similarity to a centroid*, not of fitness for inclusion in a reference cluster for downstream mechanistic work. CEH3 and CEH22 carry more pronounced tidal contamination than the reference C5 wells (the basis on which S.1 routes them to the extended network), so their audit assignments are informative for spatial interpretation but do not promote them to reference-cluster membership.

Site-specific choices and rationale (chapter-level)

- **Three scripts, one analytical idea.** Script 04 visualises Ward's partition. Scripts 05 and 06 quantify confidence in that partition. Together they answer “where is each well, and how confidently is it where the partition says it is?” The grouping reflects the analytical unit, not a code-organisation accident.
- **Pearson coefficient and z-scored centroids in both audit scripts.** Pearson because the partition itself uses Pearson distance (S.2). Z-scored centroids because the comparison should isolate shape — when a well rises and falls — from amplitude and mean depth, which are themselves cluster-distinguishing features that the partition has already handled. Both choices propagate the methodological vocabulary of S.2 into Phase 2.

- **No CEH3/CEH22 in Script 05.** Script 01's REFERENCE_NETWORK_WHITELIST excludes them on tidal-contamination grounds (S.1); they appear only in Script 06's extended-network analysis, where their low best-match correlations (0.86 for both) are themselves a signature of the routing decision.
- **The same cluster colours and shapes across all three figures.** F.4's CLUSTER_COLOURS and CLUSTER_MARKERS are the single source of truth, so the orientation map (Script 04), the spatial confidence map (Script 05), and the sitewide integration map (Script 06) all use the same visual vocabulary. A reader who has internalised the colour-shape scheme from Figure 4.2 reads §4.3's figures without effort.

Limitations and known caveats (chapter-level)

- **The Pearson audit is the project's only per-well cluster-confidence quantification.** There is no formal bootstrap confidence interval on per-well affinity, only the bootstrap CI on the cluster-level partition (S.2). Per-well CIs would be a natural extension — e.g. resampling each well's monthly observations and recomputing all five Pearson values — but are not in scope here. The audit table's gap statistic is the closest the pipeline gets to a per-well confidence measure.
- **The sitewide integration map does not visualize the Pearson value itself.** Marker shape encodes best-match cluster, marker fill encodes network membership (reference / extended / spy). A reader who wants the specific correlation value or gap for an individual well must read 06_pear_membership_audit_sitewide.csv. The trade is a map that stays legible at print resolution against carrying every diagnostic on the figure.
- **The audit assumes the five-cluster partition is the right one.** If the partition were re-cut at $k = 4$ or $k = 6$, the audit logic would still run but the conclusions would be different. The partition is itself defended in S.2; the audit inherits that defence and does not re-litigate it.

Where the result appears in the report

- §4.2 cluster orientation figure — 04_01_core_architecture_map.png.
- §4.3 Pearson affinity audit — discussion of borderline wells and the Spy/Fuzzy/Core distribution, drawing on 05_pear_membership_audit.csv and the spatial confidence map.
- §4.3 / Figure 12 — 06_pear_02_integration_map.png (sitewide integration map).

Cross-references

- **F.4** — `k=5` partition, cluster IDs, labels, colours, markers.
- **F.5** — `map_utils.py` (DEM and KML rendering helpers used by all three scripts).
- **S.1** — reference-network whitelist, extended-network construction, `EXTENDED_NETWORK_BLACKLIST`, CEH3/CEH22 tidal routing.
- **S.2** — Ward’s clustering that produces `02_cluster_stats.csv`; the no-detrending choice that motivates z-scored-shape-only auditing.
- **S.9** — Script 11b (the threshold-map chapter) consumes `06_pear_membership_audit_sitewide.csv` for the extended-network rendering.
- **S.12** — Script 18 (WTF spatial) consumes the sitewide audit for extended-network Sy interpolation.

S.5 Scripts 07, 08 — Spatial coefficient maps and LCSC model benchmarking

Steps 7 and 8 / 27. Phase 2 — Pearson Membership Audit.

Scripts 07 and 08 sit immediately after the Pearson affinity audits and together form the spatial-diagnostic layer on the per-well SSM fits. Both consume the per-well coefficients from `03_master_data.csv` produced in S.3. Script 07 renders the four β_1 , β_2 , β_3 , and R^2 surfaces across the site so that spatial structure in each coefficient can be inspected directly. Script 08 quantifies how much explanatory power the SSM gains over a deliberately weak counterfactual (a “Traditional Linear Model” with intercept but no drainage term), producing the benchmarking summary that appears as Table 3 of the main report along with a CEH6 hydrograph showdown and two spatial improvement maps.

The grouping is tight. Both scripts read per-well coefficients from the same canonical store; both produce site-scale spatial outputs over the same DEM-and-KML basemap; both rely on the same `map_utils` helpers (F.5). The methodological divergence is what they ask of the coefficients: Script 07 takes the values as given and visualises them as interpolated surfaces; Script 08 refits SSM and TLM on a constrained 100-month window and compares performance per well at the well locations themselves.

Inputs (shared)

Path resolution for both scripts goes through `utils.paths`. The canonical input constants and their roles:

Constant	Resolved file	Used by
INT_MASTER_DATA	03_master_data.csv (per-well β , R^2 , n, p-values, coordinates, cluster ID)	Scripts 07 and 08
INT_WELL_ELEVATIONS	01_well_elevations.csv (DEM ground elevations for ridge masking)	Script 07
INT_WELLS_CLEAN, INT_WELLS_REFERENCE	01_wells_clean.csv, 01_wells_reference.csv	Script 08
INT_CLIMATE	01_climate.csv (monthly P and PET)	Script 08
INT_CLUSTER_STATS	02_cluster_stats.csv (cluster IDs for the metric map merge)	Script 08
DATA_DIR (KMLs + DEM raster)	Features.kml, streams.kml, clearfell.kml, newborough_dem.tif	Both

Naming and pathing conventions are project-wide. Every input goes through an INT_ constant defined in `paths.py`; no script reads a hardcoded filename. Cluster colour, marker shape, and label dictionaries (CLUSTER_COLOURS, CLUSTER_MARKERS, CLUSTER_LABELS) come from `config.py`. Scenario parameters and per-cluster mechanistic constants used downstream are consolidated through `pipeline_params.py` (F.4) — neither Script 07 nor Script 08 modifies that store, but they share its variable naming.

Sub-script 07 — Spatial coefficient mapping

Motivation. The SSM produces a triple (β_1 , β_2 , β_3) per well. Whether those coefficients vary as smooth functions of position, as cluster-level steps, or as noise around cluster means is a question that only a spatial render can answer. Script 07 produces the four maps that ground the main report’s §4.4 spatial discussion: β_1 recharge sensitivity, β_2 atmospheric draw, β_3 drainage rate (log-scaled, as percentage per month), and the R^2 fit-quality map.

The script replaces a former `07_boundary_intercept.py` that ran a Model A vs Model B intercept audit. Under the displacement formulation (F.3), the headline

SSM fits well across all clusters (Script 08 median iterative NSE = 0.75), so the intercept audit added little beyond what direct coefficient mapping reveals. The current Script 07 is a pure visualization: no refit, no statistical test, no Model B fit. Per-well coefficients are inherited unchanged from `03_master_data.csv`.

Methodology. Per metric (β_1 , β_2 , β_3 as percentage, R^2), the script:

1. Loads `INT_MASTER_DATA` and merges DEM ground elevations from `INT_WELL_ELEVATIONS` on the normalized well name.
2. Calls `map_utils.add_idw_surface()` to interpolate the per-well metric onto the project's standard 50 m grid (`np.arange(240200, 243800, 50) × np.arange(362200, 365800, 50)`). The implementation routes through `scipy.interpolate.griddata` with `method='linear'` — a Delaunay-triangulation-based piecewise-linear interpolation. The grid matches the one used by Scripts 11b, 19, and 20.
3. Renders the interpolated surface as a semi-transparent `pcolormesh` layer (`alpha=0.65`) on top of a greyscale DEM hillshade (`map_utils.load_dem_hillshade`, vertical exaggeration $\times 3$).
4. Applies a ridge mask. Grid cells where the DEM raster elevation exceeds the interpolated well-elevation surface by more than 1.0 m are set to NaN, preventing the surface from extending across inter-dune ridges where no well measurements exist. In B&W output mode the mask is disabled in favour of an annotation that warns the reader to interpret ridge-top values with caution.
5. Overlays Features and clearfell KML features (streams are omitted for these maps), then plots the well markers themselves at each well's (E, N) — each marker uses the cluster's colour (`config.CLUSTER_COLOURS`) and cluster shape (`config.CLUSTER_MARKERS`) at a uniform size. The markers indicate cluster membership; the metric value at each well is read off the underlying interpolated surface, not the marker.
6. Adds metric-value contours for β_1 (every 0.5 from 2.0 to 6.5), β_2 (every 0.5 from 0.5 to 3.5), β_3 (at 0.5, 1.0, 2.0, 5.0, 10.0 %), and R^2 (every 0.10 from 0.50 to 0.90).

All four panels (β_1 , β_2 , β_3 , R^2) follow the same render path. The R^2 panel is an interpolated surface in the same sense as the β panels — a reader looking at the β_3 map can take the corresponding R^2 map and read off where the underlying fits are stronger or weaker.

β_3 uses a log scale because the network range spans nearly two orders of magnitude — C4 Main Forest median around 2 %/month (and one well, CEH14, returning negative β_3 ; see *Limitations*), C1 Lake Edge median around 10 %/month and reaching 13.5 %/month at its strongest wells. A linear scale would collapse the lower- β_3 cluster bulk into a narrow band at the bottom of the colour ramp. The map also rescales β_3 from its decimal form to percentage ($\times 100$) for intuitive reading.

A cluster summary table is also printed to console and written to `07_coefficient_summary.csv`. The table reports per-cluster mean, std, min, and max for each β and for R^2 — a per-well-derived summary that complements (rather than replaces) the cluster-centroid Table 3 that Script 03 fits directly. A companion file, `07_coeff_05_cluster_ranges.csv`, records the per-cluster minimum and maximum of β_1 , β_2 and β_3 (with n) as a compact per-cluster range summary, so the tabulated coefficient ranges have a dedicated, reviewer-findable source CSV.

Site-specific choices and rationale.

- **Linear (Delaunay) interpolation rather than IDW.** The helper is called `add_idw_surface` for historical reasons, but the live default is `method='linear'`. Linear interpolation produces a continuous surface that passes exactly through each well's coefficient value and varies smoothly between neighbours. True IDW would smooth across the network even within Delaunay triangles, which is poorly motivated here because the per-well coefficients are point measurements, not noisy realisations of a continuous field. The misnomer is on the to-fix list (end-of-chapter notes).
- **Ridge mask threshold of 1.0 m.** The threshold removes grid cells where the DEM raster sits more than 1 m above the elevation surface the wells interpolate to — i.e. inter-dune ridges and the bedrock ridge along the northern boundary. A 1 m threshold keeps everything within the warren's working amplitude masked-in (the highest dipwell sites sit on slack-floor terraces) while excluding the dune-ridge tops and the bedrock-ridge area where no wells exist.
- **Cluster colour and shape on markers, with no per-well R^2 encoding.** Markers are filled with the cluster colour and use the cluster shape at uniform size. The metric value at each well is encoded in the interpolated surface beneath the marker, not on the marker itself. A reader of the β_1 map can identify which cluster each well belongs to without needing a

separate cluster-orientation figure (S.4); per-well uncertainty is read off the dedicated R^2 panel.

- **R^2 rendered as the fourth interpolated panel rather than as a marker overlay on the β panels.** Each of the four panels carries its own information independently. The β panels show coefficient values; the R^2 panel shows where the underlying fits are robust. A reader interrogating a spatial pattern in β_3 should cross-reference the R^2 panel at the same locations.

Outputs.

File	Contents
07_coefficient_summary.csv	Per-cluster summary (mean, std, min, max) for each β and R^2
07_spatial_coefficients/ 07_coeff_01_beta1_recharge.png	β_1 map
07_spatial_coefficients/ 07_coeff_02_beta2_atm_draw.png	β_2 map
07_spatial_coefficients/ 07_coeff_03_beta3_drainage.png	β_3 map (log scale, percentage)
07_spatial_coefficients/ 07_coeff_04_r2_quality.png	R^2 map
07_spatial_coefficients/ 07_coeff_maps_data.csv	Per-well data underlying the maps

Limitations and known caveats.

- **Per-well coefficient values for wells with small n carry wide confidence intervals.** The R^2 map serves as a soft proxy for fit quality across the network — wells with $R^2 < 0.6$ should be read as carrying real spatial information only where neighbouring wells in the same cluster show concordant values.
- **Interpolation between point measurements is a visual aid, not a mechanistic claim.** A continuous surface drawn across the site should not be read as evidence of a continuous physical gradient. Where cluster-level steps appear (for example the β_2 jump from C1 Lake Edge at 0.63 to C4 Main Forest at 2.64), the discrete cluster structure is the underlying signal; the smoothness of the interpolated surface between them is graphical, not physical.
- **CEH14 produces a negative β_3 at the canonical 3.7 m datum** ($\beta_3 = -0.016$, $R^2 = 0.58$; see 07_coeff_maps_data.csv). It is the only well in the

network where the soft physical-sign warning fires under the canonical datum. The well sits at the bedrock ridge crest and is dominated by lateral subsidy that the SSM does not represent; the negative β_3 is the spatial map's signal that something physical is missing here rather than that the well is malfunctioning. The phenomenon is taken up in S.16.

Where the result appears in the report.

- §4.4 *Spatial coefficient distribution* — the four maps as Figures 15a–d.
- §3.4 *Drainage datum sensitivity* — the per-well R^2 map provides cross-check context for the per-well datum sensitivity discussion.

Sub-script 08 — LCSC model benchmarking

Motivation. The headline SSM is a three-parameter physically motivated model. A reasonable counter-question is: what does the third parameter (β_3 , the drainage feedback) actually buy over a simpler model that lacks it? Script 08 quantifies this by fitting both the SSM and a “Traditional Linear Model” (TLM, also called Model A and Model B in the historical intercept-audit naming) on a 100-month most-recent window per well, then comparing performance on three metrics: one-step R^2 , iterative R^2 , and iterative NSE.

The TLM is the simplest unphysical alternative:

$$\Delta h(t) = \alpha + \beta_1 \cdot P(t) - \beta_2 \cdot PET(t)$$

Crucially the TLM has an *intercept*. Without $\beta_3 \cdot (D + h(t-1))$ in the right-hand side, an intercept absorbs whatever constant lateral subsidy would otherwise be unrepresented. The TLM is therefore not the worst possible alternative — it has one degree of freedom more than the SSM (an intercept) but lacks the displacement feedback. This makes the comparison conservative: if the SSM beats the TLM, it does so despite the TLM's extra fitted parameter.

The TLM is not put forward as the published model and is not claimed to be optimal in any sense. It is the counterfactual: a deliberately weak alternative that quantifies how much explanatory power the SSM contributes beyond a structurally simpler unphysical baseline. More elaborate alternatives (PEARL, MODFLOW with calibrated boundary conditions, Bayesian state-space formulations) would in principle fit better than either model at specific wells. The benchmarking does not engage with those — its remit is to justify the SSM's additional complexity against the structurally simplest possible competitor.

Methodology. For each well in the reference network (excluding three wells in EXCLUDED_WELLS_NORM = {'ceh7', 'ceh8', 'ceh37'}, which carry inconsistent records that fail the benchmarking's data-quality bar — see Site-specific choices below):

1. The script aligns the well series with monthly climate via `build_ssm_frame()` (F.3), with `window=100` so only the most recent 100 aligned months are kept.
2. The SSM design matrix is built directly (`[+P, -PET, -h_disp_prev]` with no intercept) and fit by OLS. The TLM design matrix is `sm.add_constant([P, -PET])` and fit by OLS. Both fits use `HEADLINE_LAG = 0` from config — no rainfall lag.
3. **One-step simulation** uses the observed previous-month water level as the recurrence anchor: at each month t , the model predicts $h(t)$ from $h_{\text{obs}}(t-1)$ plus its own Δh prediction. This metric measures diagnostic fit (how well the model recovers month-to-month changes given the truth at the previous step).
4. **Iterative simulation** is the autonomous forecast: starting from h_{obs} at the window's first month, the model predicts h forward step-by-step, with each month's prediction feeding into the next as h_{prev} . The SSM iterative simulation uses `model_utils.simulate_ssm()`. The TLM iterative is simpler — without a feedback term, it just accumulates Δh from climate forcing alone. This metric measures forecasting stability, which is the harder test.
5. For each well and metric, the script computes one-step R^2 , iterative R^2 , iterative NSE, and iterative RMSE for both models, plus per-well $\Delta = (\text{SSM} - \text{TLM})$ for each metric.

Three figures are produced:

- A **dual-panel showdown** for CEH6 — top panel is one-step diagnostic fit, bottom is iterative forecasting. CEH6 was chosen as the showcase because it produces a clear visual contrast between the two models without being the most extreme case in the network. The TLM drifts to $\text{NSE} = -1.12$; the SSM holds at $\text{NSE} = +0.66$.
- A **spatial ΔR^2 map** showing the SSM's per-well improvement in iterative R^2 over the TLM. The map is a point-scatter rendering with no interpolation: each well is plotted at its (Easting, Northing) as a cluster-shape marker filled by ΔR^2 value, on top of the standard DEM-and-KML basemap. The per-well improvement values are themselves the diagnostic —

interpolating between wells would smooth across the localised structure that the map exists to make visible.

- A **spatial Δ NSE map** in the same style, with the diverging signed-NSE-improvement colour scale clipped to ± 1.5 to keep the bulk of the network on the visible scale (CEH14's -3.64 outlier would otherwise compress everything else into a few colour bins; raw values are preserved in the underlying CSV).

A summary table is written for the main report (`08_lcsc_04_table3_benchmark_summary.csv`), and a per-well stats table is written for downstream reference (`08_lcsc_model_stats.csv`).

Site-specific choices and rationale.

- **100-month window** rather than the full record. The choice keeps both fits on a recent, well-sampled segment of the record and avoids letting older data (with sparser sampling, pre-clearfell baselines, and possible measurement-protocol drift) dominate. The number matches `LCSC_DATA_LIMIT = 100` in `model_utils.py`, which is the same window the per-well intercept-audit historically used.
- **Excluded wells: CEH7, CEH8, CEH37.** These three wells have inconsistent records — gaps, regime changes, or sampling discontinuities that disqualify them from a 100-month-window benchmarking exercise. The exclusion is recorded as a fixed rule in `EXCLUDED_WELLS_NORM` at the top of the script. The exclusion is empirical and field-driven rather than algorithmic: documenting the per-well reason in an in-code comment at the exclusion site is on the to-fix list.
- **CEH6 as the showcase well.** CEH6 was chosen because it produces a clear visual contrast between the TLM and SSM iterative trajectories without being the most extreme well in the network. A more extreme case would compress the visual difference into the same plot range; CEH6's $\text{NSE}_{\text{TLM}} \approx -1.1$ / $\text{NSE}_{\text{SSM}} \approx +0.7$ produces two trajectories that diverge visibly but both stay on a readable scale. No other methodological consideration entered the choice.
- **Lake sentinel excluded from the max-improvement reporting.** The Table 3 “Max NSE improvement” row excludes the lake-side LlynRhos sentinel well. The lake sentinel has a tightly bounded surface-water signal that makes its TLM-vs-SSM comparison atypical of warren hydrology.
- **The TLM iterative simulation has no feedback term.** That is the deliberate point. The TLM fits a constant Δh -per-month from climate alone,

with no recovery mechanism if water level drifts off the observed trajectory. Without a drainage term the TLM cannot represent the physical fact that drier sites drain less slowly than wet sites; without the displacement formulation the iterative simulation accumulates whatever climate-driven Δh the OLS estimated, and drifts. This is the model the SSM is being compared against — the simplest specification under which “memory of the water-table position” is absent.

Outputs.

File	Contents
08_lcsc_model_stats.csv	Per-well one-step R^2 , iterative R^2 , iterative NSE for both models, with per-well improvements
08_model_benchmarking/ 08_lcsc_01_ceh6_showdown.png	CEH6 dual-panel showdown (one-step top, iterative bottom)
08_model_benchmarking/ 08_lcsc_02_r2_improvement_map.png	Spatial ΔR^2 (iterative) point-scatter map
08_model_benchmarking/ 08_lcsc_03_nse_improvement_map.png	Spatial Δ NSE point-scatter map
08_model_benchmarking/ 08_lcsc_04_table3_benchmark_summary.csv	Six-row summary table for the main report (Table 3)

The live Table 3 summary records: median one-step R^2 of 0.91 (TLM) vs 0.92 (SSM); median iterative R^2 of 0.66 vs 0.80; median iterative NSE of 0.16 vs 0.75. The one-step gap is small — both models recover diagnostic fit. The iterative-NSE gap is large — the SSM contributes most of its value in forecasting stability, not in one-step diagnostic fit. Forty-three of 66 reference wells have iterative NSE > 0 under the TLM; 65 of 66 do under the SSM. The single well where SSM iterative NSE is negative is CEH14 — the same ridge-crest well that produces the negative β_3 in Script 07’s map. The two diagnostics agree on which well the canonical SSM does not describe.

Limitations and known caveats.

- **The TLM is the simplest unphysical alternative, not a comprehensive comparator.** Wells where the SSM iteratively underperforms the TLM (CEH13 by Δ _NSE = -0.11 , CEH14 by Δ _NSE = -3.64) sit at the C4 Main Forest’s bedrock-ridge interface. The shortfall there is not that the TLM is

better; both models perform poorly. The TLM happens to drift less catastrophically because its intercept absorbs some of the systematic ridge-recharge subsidy that the SSM treats as residual. Neither model represents the physical mechanism (lateral inflow from the bedrock ridge); diagnosing it requires the supplementary analyses in S.16.

- **One-step R^2 gap is small by construction.** Both models share the same predictors P and PET; the one-step metric is dominated by the climate-driven variance, which both decompositions capture. The benchmarking's headline number is the iterative NSE, where the SSM's structural advantage is unmasked.
- **The 100-month window encodes a temporal-stationarity assumption.** Where the underlying hydrology is non-stationary (C5 Coastal Forest's coastal retreat, C1 Lake Edge's 2018 regime change), restricting the fit to recent months partially mitigates the issue, but the comparison still asks "which model fits the last 100 months better" rather than "which model is correct over the full record". The non-stationary regimes are taken up in S.7 (clearfell BACI) and S.15 (coastal gradient).

Where the result appears in the report.

- §4.4 *Model benchmarking* — Table 3 of the main report draws directly from 08_lcsc_04_table3_benchmark_summary.csv.
- §4.4 / Figure — CEH6 showdown panel.
- §4.4 / Figure — ΔR^2 and ΔNSE spatial maps.

Site-specific choices and rationale (chapter-level)

- **Both scripts consume 03_master_data.csv as the per-well coefficient table.** No script in this chapter refits the SSM on the headline (full-record) configuration; Script 07 inherits values from S.3, and Script 08's refits are on a deliberately restricted 100-month window for the model-comparison question. The headline cluster-level coefficients used elsewhere in the pipeline (S.9 thresholds, S.11 water balance, S.14 forestry scenarios, S.7 clearfell BACI) are S.3's full-record values, not the 100-month values from S.8.
- **Both scripts produce site-scale spatial outputs over the same basemap.** The DEM-and-KML basemap, the cluster-shape and cluster-colour marker convention, the variable-name conventions inherited from pipeline_params.py and config.py: all of these are project-wide. A reader can compare Script 07's β_3 map with Script 08's ΔNSE map directly because

both are rendered on the same extent with the same cluster-shape marker convention.

- **The two scripts diverge on the question of interpolation.** Script 07's metric maps are continuous interpolated surfaces (Delaunay-linear) over the standard 50 m grid; the surface is the answer. Script 08's improvement maps are point-scatter plots: each well is a single coloured marker with no interpolation between wells. This is deliberate. Script 07's β values are correlated as members of the same cluster, so an interpolated surface between nearby wells in the same cluster is informative; Script 08's per-well improvement values encode where the canonical SSM specifically struggles, which is a per-well property, and interpolating would smooth across the localised structure that the maps exist to make visible.

Limitations and known caveats (chapter-level)

- **Both spatial maps inherit the cluster partition's spatial structure.** Where the maps show step-like changes between adjacent regions, the underlying signal is that the wells on each side belong to different clusters with different mechanistic signatures (S.2, S.3). The partition is itself defended in S.2; this chapter inherits that defence rather than re-litigating it. If the cluster partition were re-cut at a different k , both Script 07's coefficient maps and Script 08's improvement maps would change.
- **CEH14 is the consistent outlier across both scripts.** Script 07's β_3 map flags it as the single negative- β_3 well at the canonical datum; Script 08's iterative NSE flags it as the worst-fitting well in the network ($\Delta_{\text{NSE}} = -3.64$). Both signals point to the same physical mechanism — bedrock-ridge subsidy that the displacement-formulation SSM does not represent. The investigation of that subsidy is in S.16 (ridge recharge lag hypothesis test).
- **The benchmarking is a counterfactual, not a comprehensive model comparison.** The chapter does not claim the SSM is optimal — only that it improves substantially over the simplest unphysical alternative. More elaborate alternatives exist; their evaluation is out of scope.

Cross-references

- **F.3** — SSM equation form, sign conventions, drainage datum, `model_utils.fit_ssm()` interface. Both scripts use this implicitly through `03_master_data.csv` (Script 07) or directly through `build_ssm_frame()` and OLS (Script 08).

- **F.5** — `map_utils.py` (DEM hillshade, KML overlay, grid interpolation helpers) and `model_utils.simulate_ssm()` (Script 08's SSM iterative simulation).
- **S.3** — produces `03_master_data.csv`, the input to both scripts. The per-well datum sensitivity discussion in S.3 also contextualises CEH14's β_3 anomaly.
- **S.4** — Pearson affinity audit (the immediately preceding chapter); the two chapters together carry the §4.2–§4.4 spatial discussion in the main report.
- **S.16** — supplementary diagnostic chapter on ridge recharge at CEH14, picking up the residual signal that Script 07 and Script 08 both flag.
- **S.9** — Script 11b uses the same 50 m grid for spatial threshold mapping; the grid resolution is a project-wide convention.

Phase 3 — Model Diagnostics and Intervention Analysis

S.6 Script 09 suite (a–e) — Scraping intervention

Step 9 / 27 (sub-scripts 9a–9e). Phase 3 — Model Diagnostics and Intervention Analysis.

In April 2015, approximately 0.2 m of surface soil was mechanically excavated over an area of approximately 0.3 ha at CEH36 in the felled forest compartment. The intervention tested whether mechanical lowering of the ground surface could restore dune-slack hydrology — bringing the water table closer to the surface and reversing the drying that the forest canopy had imposed on the underlying slacks. This is one of two management experiments at the warren; the other is the December 2017 clearfell, covered in S.7. A second, smaller scraping event in October 2023 at CEH18 and CEH21 introduces an additional era boundary for those two wells but is not the chapter's headline. The Script 09 suite is the entire methodological evaluation of the scraping intervention.

The suite is five sub-scripts answering five distinct questions. *09a — Hierarchical paired BACI* (the core analysis) asks whether scraping worked at the scraped well itself. *09b — Scraping propagation* asks whether it propagated uphill into the surrounding forest as a detectable shift in SSM coefficients. *09c — Summer minima* asks whether the intervention improved ecologically critical summer-minimum depths, not just annual means. *09d — Scenario comparison* asks whether scraping was a good management choice compared to alternative interventions at the same site. *09e — Robustness* gives three independent estimates of the headline step change to verify it is not an artefact of any single

method's assumptions. The orchestrator `run_09_scraping.py` invokes the sub-scripts in this order; 09b must precede 09d because 09d consumes 09b's centroid CSV indirectly via the shared scenario engine. All five share infrastructure through `utils/scraping_common.py`.

The suite's principal results — the headline +0.129 m benefit at CEH36 (paired-BACI Pure-Scraping era, vs CEH4), the propagation signal across wells 247–776 m uphill, and the alternative-intervention comparisons — populate §4.5 *Scraping intervention* in the main report, including Table 6 (β_3 era coefficients per well, mapped from `09_scrape_04b_beta3_era_summary.csv`) and Figures 15–18, 20–22, 24, and 25.

Sub-script 09a — Hierarchical paired BACI

Motivation. The core scraping analysis. A naïve paired BACI of CEH36 against CEH4 — the obvious local control well, ≈ 100 m to the south — would overstate the scraping benefit, because CEH4 is itself drying due to progressive coastal retreat affecting the western coastline. A two-tier hierarchical design separates the *scraping signal* at CEH36 from the *coastal drainage signal* contaminating the local controls.

Methodology. Tier 1 evaluates the local controls (CEH4, CEH22) against the regional mean — a five-well climate-only composite (`CLIMATE_CONTROLS = ceh9, nw7, nw6, nw5, wmc2`). If the local controls dry faster than the regional baseline, Tier 1 has demonstrated a coastal drainage signal at the controls themselves. The result in the live data is unambiguous: CEH4 drifts to a CUSUM terminal value of -10.7 m relative to the regional mean, and CEH22 to -19.1 m. The local controls are themselves carrying a coastal signal. Tier 2 evaluates the impact wells (CEH36, CEH18, CEH21) against the paired local controls. Once Tier 1 has established the coastal signal, the Tier 2 BACI shift at CEH36 vs CEH4 is the *pure scraping effect* net of that signal.

Within each tier, per-well analysis runs in two layers. First, the well-minus-control time series is averaged within each era window (from `WELL_ERAS`, described in *Suite-level methodology* below), and the change in mean between successive eras is the BACI step. Second, the SSM (F.3) is refit per-well per-era via direct OLS on the era's monthly observations, with the same $[+P, -PET, -(D + h_{prev})]$ design matrix as Script 03. Per-era β_3 estimates with 95 % confidence intervals come from an “isolated” refit: the script first fits the full SSM to extract β_1 and β_2 , subtracts the climate-driven Δh component, then regresses the remaining drainage component on $-h_{disp_prev}$ with an intercept to recover an

unbiased β_3 and its CI. The isolated β_3 estimates populate the report's per-well β_3 era table (Table 6 in the main report).

The live numbers tell the story. CEH36's β_3 rises from 0.096 (Baseline, $p = 0.011$) to 0.142 (Pure Scraping, $p = 0.001$) to 0.124 (Felling Pulse, $p < 0.001$) — a structural drainage signal consistent with the surface having been lowered by 0.2 m. The paired BACI shift at CEH36 vs CEH4 is +0.129 m for the Pure Scraping era and +0.024 m for the Felling Pulse era. The net benefit against CEH21 — the coastal benchmark, secondary impact, but the well whose own trajectory most closely represents what CEH36 would have done without scraping — is +0.144 m.

Site-specific choices.

- **Two-tier rather than direct impact-vs-climate.** Tier 1 confirms the coastal signal at the local controls; Tier 2 nets it out at the impact wells. Skipping Tier 1 would conflate scraping with coastal retreat.
- **CEH21 as the coastal benchmark for the net-benefit calculation.** CEH21 is itself a secondary scraping recipient in 2023, but in the Pure Scraping era (2015–2017) it is an undisturbed coastal-retreat-affected well. Its drying baseline is the most relevant comparator for what CEH36 would have done without scraping; the net-benefit calculation accordingly uses CEH21 rather than CEH4.
- **The paired CEH36 / CEH4 shift is the principal headline.** The +0.129 m Pure-Scraping shift is the figure quoted throughout §4.5. The slightly larger net benefit against CEH21 (+0.144 m) is reported alongside for context but is not the primary number.

Outputs.

File	Contents
09_scrape_01_full_parameters.csv	Per-well, per-era SSM coefficients (β_1 , β_2 , β_3)
09_scrape_02_beta3_significance.csv	Isolated β_3 estimates with 95 % CIs and p-values
09_scrape_03_baci_shifts.csv	Paired BACI step changes between eras
09_scrape_04_net_benefits.csv	Net benefits at impact wells vs CEH21 benchmark
09_scrape_04b_beta3_era_summary.csv	Formatted β_3 era summary (the Table 6 source in the main report)

File	Contents
09_tier1_final_cusum.csv	Tier 1 CUSUM terminal values
09_scrape_05_tier1_background_dri ft.png	Tier 1 BACI hydrographs + CUSUM panel
09_scrape_06_tier2_scraping_signa l.png	Tier 2 BACI hydrographs + CUSUM panel
09_scrape_07_beta3_confidence.png	β_3 era estimates with 95 % CIs across wells
09_scrape_report_numbers.csv	All citable values for \$4.5

Sub-script 09b — Scraping propagation

Motivation. Did the ground-scraping at CEH36 propagate uphill into the surrounding forest as a detectable change in SSM coefficients? The physical hypothesis: increased drainage at the scraped site creates a hydraulic gradient that draws water from upgradient wells, manifesting most clearly as elevated β_3 at neighbouring wells north and northwest of CEH36 in the local groundwater-flow direction.

Methodology. Split-window SSM fitting on a fixed pre-scraping window (start of record to April 2015) versus a fixed post-scraping window (April 2015 to December 2017, before the felling pulse). Per-well, the SSM is fit on each window via `model_utils.fit_ssm()` (F.3) with a minimum-observations threshold of 12 months per window. The raw shift $\Delta\beta = \text{post} - \text{pre}$ is then BACI-corrected against the centroid raw shift of seven distant control wells (NW1, NW2, NW11, NW13, WMC4, D25, WMC2), which sit at 852–1070 m from CEH36 — far enough that they are outside the scraping’s hydraulic-gradient reach, close enough that they share the same regional climate forcing. The BACI-corrected shift at each well is the propagation signal at that well.

Ten uphill wells north and northwest of CEH36 enter the analysis (CEH31, WMC3, NW6, NW7, CEH30, CEH20, CEH33, CEH9, CEH34, CEH1), at distances 247–776 m. Wells south or coastward of CEH36 (CEH4, CEH18, CEH21, CEH22, NW5) are excluded as downhill of the scraping in the flow direction — a change at a downhill well is not attributable to scraping in the same way, because scraping draws water *uphill* via the local hydraulic gradient, not downhill. The outer C5 coastal wells (NW9, CEH16, CEH19, CEH17) are also excluded as confounded by coastal boundary effects. FE1–4 and LIS1 are excluded because they have no pre-scraping record. CEH39 was included in an earlier 11-well pass of the analysis but

has been dropped from the live set on baseline-length grounds ($n_{\text{pre}} = 24$, too short for a reliable pre-scraping SSM fit at this well).

After per-well fitting, wells are aggregated into three centroid groups: scraped (CEH36 alone), non-forest uphill (CEH31 with the C3 uphill wells), and forest uphill (the C4 wells). The centroid time series is averaged across wells and refit on the same split windows, producing a centroid-level pre β_3 , post β_3 , and BACI-corrected percentage shift.

The live centroid summary gives BACI-corrected β_3 shifts of +7.6 % (non-forest uphill, 6 wells), +9.4 % (forest uphill, 4 wells), and +11.0 % (all uphill, 10 wells). At the individual-well level the shifts span -24 % (CEH34) to +29 % (WMC3 at 262 m) with no monotonic distance decay: CEH9 at 571 m gives +21 % and WMC3 at 262 m gives +29 %, while NW7 at 383 m gives +5 % and NW6 at 284 m gives +0.5 %. The centroid-level signal is the robust summary; per-well shifts at this n are noisy. All three centroids are positive, individual wells are noisy, and there is no clean distance gradient.

Site-specific choices.

- **The 852–1070 m control distance band.** Far enough that the controls are outside the scraping's hydraulic-gradient reach; close enough that they share regional climate forcing.
- **Excluding downhill wells.** Flow-direction physics: scraping draws water uphill via the local hydraulic gradient. A change at a downhill well after the intervention is not attributable to scraping in the same way.
- **Centroid averaging.** The 31-month post-scraping window is short relative to typical SSM fit windows. Per-well coefficients have wide CIs at this n ; centroid averaging within distance bands compresses the noise and gives the propagation signal at a scale where it can be defended.
- **The cluster-weighted scraping bars in the scenario chart.** The propagation analysis feeds a scraping scenario bar into the cross-cluster scenario comparison figure. C3 wells use the non-forest uphill centroid shifts; C4 wells use the forest uphill centroid shifts; C5 uses the C3+CEH31 centroid shifts (same western coastal zone). Bars are then weighted by the fraction of each cluster within 800 m uphill of CEH36 (C3: 29 %, C4: 78 %, C5: 100 %, C1 and C2: 0 %).

Scenario figures are volumetric, not summer-minimum. The cross-cluster scenario figure (09b_05) and the CEH36 scenario figures (09d_01, 09d_02) report each scenario as an equilibrium **volumetric** change (mm water-equivalent per

month), the quantity produced directly by `scraping_common.compute_scenario_bars()` (§F.5, the Option-3 engine used throughout the scenario work). Earlier versions converted this flux to a summer-**minimum** depth by dividing by the cluster specific yield and multiplying by a per-cluster amplification factor (the OLS slope of annual summer-minimum head on annual-mean head). That conversion was withdrawn on 2026-07-02: the SSM equilibrium framework carries no transient and therefore cannot resolve a true summer minimum, and the mean→minimum amplification slope is least reliable at precisely the forested clusters, where the winter-rise/summer-stagnation asymmetry means a mean-level change does not propagate cleanly to the summer trough (the +120 mm mean-level clearfell recovery yields a non-significant summer-minimum step; §S.7). All scenario figures are consequently on one volumetric scale, directly comparable across 09b and 09d.

Converting a volumetric bar to a water-table **head** change (mm), as used for the Curreli et al. (2013) thresholds, requires dividing by an appropriate specific yield; this is not straightforward, because the WTF specific yields are reliable for ranking clusters but high in absolute terms (§S.12, Appendix B), so any head equivalent is approximate. The figures therefore carry a caption note to that effect rather than presenting a head axis.

The scraping bar remains an *empirical* BACI shift measured directly at the scraped site (CEH36 versus CEH18 for 09b_05; CEH36 versus CEH4 for the 09d observed bar), rendered on the volumetric axis by multiplying the observed head shift by the relevant specific yield. It is placed on the same axis as the SSM-equilibrium scenarios for comparison but is derived independently and labelled as such.

Implementation. The scenario values flow from a single source of truth, `scraping_common.compute_scenario_bars()` (v1.5.0), consumed directly by 09b (09b_05, all scenarios) and Script 21 (21_forestry_06, the forest-management scenarios only), so the per-cluster forestry volumetric values are byte-identical between the two outputs by construction. The earlier flux→summer-minimum helpers (`summer_amplification_factors()`, `flux_to_summer_min_mm()`, `scenario_summer_min_bars()`) are retained in `scraping_common.py` but are no longer called by any script; they are marked for removal in a future output-tidy pass.

Outputs.

File	Contents
09b_01_individual_well_baci.csv	Per-well pre/post β coefficients and BACI-corrected shifts
09b_02_centroid_summaries.csv	Group centroid BACI shifts with percentage β_3 change
09b_03_ceh36_equilibration.jpg	CEH36 climate-corrected post-scraping trajectory
09b_04_scenario_comparison. {csv,jpg}	Cross-cluster scenario bar chart with scraping bars
09b_05_summer_scenario_comparison. {csv,png}	Volumetric cross-cluster scenario comparison (equilibrium mm water-equiv/month)
09b_report_numbers.csv	Citable values for §4.5

Sub-script 09c — Summer minima

Motivation. Annual mean shifts can hide ecologically critical summer minima. Dune-slack flora at Newborough are constrained by summer drying; an intervention that improves the annual mean but leaves summer minima unchanged is ecologically less valuable than one that lifts the minimum. The script measures whether scraping affected the critical Jun–Sep minimum depth, not just the annual mean.

Methodology. Per-well annual summer-minimum depth (the minimum Jun–Sep value each year) is computed pre- and post-scraping at CEH36, CEH18, CEH21, with CEH4 and CEH22 carried as controls. Each well’s “gap” each year is the well’s summer minimum minus a control centroid’s summer minimum. Two control centroids run in parallel — a *climate-only* centroid (the five climate-control wells averaged) and a *paired* centroid (CEH36’s paired control CEH4; CEH21’s paired control CEH22) — giving the dual-control name. The gap series is compared via Welch t-test pre- versus post-intervention, with the 2015 cut-off year as the boundary. The methodology mirrors Script 10d (S.7), applied to the scraping timeline rather than the clearfell timeline.

The dual-control design now produces two clearly significant results pointing the same direction. CEH36’s summer-minimum gap against the climate-control centroid widens by +159 mm post-scraping ($p = 0.007$, **); the gap against the paired control CEH4 widens by +195 mm ($p = 0.004$, **). The two shift magnitudes

are within 36 mm of each other. Both controls tell the same story at similar magnitude: at CEH36, scraping lifted the summer minimum.

The interpolation policy matters to this result. The five climate-control wells include NW6 and NW7, which have multi-month Jun–Sep gaps in several years; under a loose interpolation limit those gaps are bridged by straight lines from May to October endpoints, imputing phantom summer values into the control centroid (S.1). The `limit=1` policy excludes those multi-month bridges, so the climate-control centroid here rests on measured Jun–Sep data, and the `min_measured=2` rule (below) drops any control-well year that cannot honestly supply a summer minimum.

Site-specific choices.

- **Dual-control design.** The paired control is the most direct comparator (same hydrogeological setting, same coastal exposure) but is itself affected by coastal retreat. The climate-only control is unaffected by coastal retreat but is hydrogeologically more distant. Showing both lets the reader see how much of the apparent benefit at the paired-control level is amplified by the control’s own deepening.
- **Year boundary at 2015 rather than monthly boundary.** Summer minima are annual quantities. The April 2015 scraping date falls within the 2015 hydrological year; including 2015 in the post-scraping group rather than the pre-scraping group is the protocol (`POST_YEAR = SCRAPING_DATE.year`).

Outputs.

File	Contents
<code>09c_01_summer_minima.csv</code>	Per-well, per-year summer minima and gap series; the <code>n_interpolated</code> column records how many of each (well, year)’s surviving Jun–Sep months were filled by <code>limit=1</code> interpolation (see S.1 for the underlying <code>01_wells_provenance.csv</code>). Rows with fewer than two measured Jun–Sep months are dropped by the <code>min_measured=2</code> rule.
<code>09c_02_summer_minima_shifts.csv</code>	Per-well Welch t-test results, both controls

File	Contents
09c_03_summer_minima_climate_ctrl.png	Three-panel: raw minima, impact gaps, control gaps (climate control)
09c_04_summer_minima_paired.png	Paired-control comparison: CEH36 vs CEH4
09c_report_numbers.csv	Citable values for §4.5

Sub-script 09d — Scenario comparison

Motivation. Was scraping a good management choice for this site? Using CEH36’s own SSM coefficients, its WTF-derived S_y , and its mean head displacement, the script computes the equilibrium volumetric response to alternative interventions — clearfell, thinning, broadleaf conversion, UKCP18 dry and wet climate scenarios — under both annual-mean and summer forcing, and compares each to the observed scraping benefit. The comparison is a like-for-like, single-site equivalent.

Methodology. Per-well parameters for CEH36 come from 03_master_data.csv (β coefficients, cluster ID), 17_wtf_well_sy.csv (S_y median), and 01_wells_clean.csv (mean depth for h_{disp}). Scenario perturbations use the Option 3 monthly formulation: each scenario modifies the SSM monthly flux $\Delta h = \beta_1 \cdot P_{eff} - \beta_2 \cdot PET - \beta_3 \cdot h_{disp}$ by changing the rainfall reaching the water table (interception), the β_2 multiplier (canopy transpiration), or the PET and P scaling (climate). The flux difference is multiplied by S_y and converted to mm water equivalent per month. The same Option 3 engine is used in `scraping_common.compute_scenario_bars()` (the suite-shared scenario function) and by Script 21 (the cross-pipeline forestry scenarios chapter, S.14).

The observed scraping value is computed differently — directly from the paired BACI step at CEH36, converted to volumetric via S_y . This is the like-for-like volumetric equivalent of the observed empirical step rather than an SSM-derived counterfactual.

The live monthly scenario bars at CEH36 give (annual-mean forcing): scraping (observed) +45.2 mm/month, clearfell (hypothetical) +15.4, thinning 50 % (hypothetical) +7.7, broadleaf (hypothetical) +4.5, climate-dry –13.8, climate-wet +7.7. Under summer (July–September) forcing: clearfell +13.5, thinning +6.7, broadleaf +2.9, climate-dry –14.4, climate-wet +7.8. The forestry scenarios are *hypothetical* in the sense that CEH36 is in C3 (not forested under current land use); the question they answer is “what would each intervention have produced

at CEH36’s hydrogeological setting if CEH36 were forested?” — a like-for-like comparison framework, not a prediction.

Two forcings, and the off-site scraping drawdown. The scenario dict is computed twice — once under annual-mean P and PET (09d_01) and once under summer (July–September) P and PET (09d_02) — so the two figures are genuinely distinct rather than a single forcing re-expressed. Both are volumetric and directly comparable. Each figure also carries a modelled *off-site* scraping bar: the neighbour drawdown the scrape drain imposes on the surrounding water table, from the same steady-state drain cone that feeds the Script 20 spatial maps (edge magnitude H_0 anchored to the measured CEH36 response, decay length λ read live from `20_report_numbers.csv`, $\lambda = 225$ m). The bar is drawn at 100 m; a dark reference line across it marks the milder drawdown at 250 m (near the nearest real uphill well, 247 m). The 100 m point lies inside the near field the 88-well network cannot resolve, so this bar is explicitly modelled and captioned as such. Because the summer figure is an equilibrium response to summer forcing and *not* a summer minimum, the observed paired-BACI summer-minimum shift at CEH36 (+165 mm) is reported in the caption for context rather than plotted as a bar.

Site-specific choices.

- **CEH36’s own well-level Sy rather than the cluster median.** CEH36’s WTF Sy is closer to its actual hydrogeology than the C3 cluster median. Cluster medians enter the per-cluster scenario bars in 09b’s chart; the per-well scenario chart here uses the per-well Sy for sharper representation of CEH36’s specific response.
- **Forestry scenarios as hypothetical at CEH36.** CEH36 is in C3 (unforested at the time of scraping; the surrounding compartment was clearfelled in December 2017). Reporting “clearfell at CEH36” as a hypothetical lets the reader compare the monthly flux change a clearfell *would* have produced at this hydrogeological setting against what the scraping actually achieved.

Outputs.

File	Contents
09d_01_scenario_comparison. {csv,jpg}	Equilibrium volumetric scenarios under annual-mean forcing at CEH36
09d_02_summer_scenario_comparison . {csv,png}	Equilibrium volumetric scenarios under summer (Jul–Sep) forcing at CEH36

Sub-script 09e — Robustness

Motivation. Three independent estimates of the CEH36 Pure-Scraping step change confirm that the headline +0.129 m benefit is not an artefact of any single method’s assumptions — specifically not an artefact of CEH4’s own progressive deepening, which is the most plausible single-source contamination of the paired BACI estimate.

Methodology. Three estimators on the same CEH36 series:

1. **Raw paired BACI.** CEH36 minus CEH4, averaged over the baseline era versus the Pure Scraping era. The simplest possible estimator.
2. **Synthetic control.** CEH36 minus a weighted composite of donor wells from DONOR_CANDIDATES (eleven candidate wells outside the scraping influence — CEH1, CEH2, CEH5, CEH6, CEH9, CEH11, CEH16, CEH17, CEH19, CEH22, CEH24). Donor weights are fitted by OLS on the pre-scraping baseline, then the same weighted composite is computed across the full record. The step is the post-scraping minus pre-scraping mean of the (CEH36 – synthetic) gap series.
3. **SSM forward residual.** The SSM is calibrated on CEH36’s pre-2015 baseline via OLS on $\Delta h = \beta_1 \cdot P - \beta_2 \cdot PET - \beta_3 \cdot (D + h_{\text{prev}})$ (with HEADLINE_LAG = 0), then iterated forward from the last baseline observation as an autonomous forecast: at each month, the predicted Δh is added to the previous predicted level. The residual (observed minus predicted) over the Pure Scraping era is the step change.

The live estimates: raw BACI +0.129 m, synthetic control +0.137 m (with 11 donors), SSM forward residual +0.081 m. The first two are tightly convergent (within 7 mm); the SSM forward residual gives a noticeably smaller estimate, partly because the model’s drainage feedback term iteratively brings the predicted level back toward equilibrium as the post-2015 elevation rises, absorbing some of the observed step into the forecast trajectory. The methodological point is that all three are positive and substantial: the +0.129 m benefit is not an artefact of CEH4’s deepening, which only affects the raw BACI estimate.

Outputs.

File	Contents
09e_report_numbers.csv	The three step estimates with units and notes

File	Contents
09_scrape_08_ceh36_robustness.png	Three-panel: gap series, SSM forward residual, summary bar

Suite-level methodology

The era system. Two intervention dates — April 2015 (scraping at CEH36) and December 2017 (clearfell of the surrounding compartment) — are the principal era boundaries for the scraping site itself. A third event — the October 2023 second scraping at CEH18 and CEH21 — defines an additional boundary for those two wells only. Era structures are therefore well-specific: CEH36 and CEH4 (its paired control) have three eras (Baseline, Pure Scraping, Felling Pulse); CEH18, CEH21, and CEH22 have three eras with different structure (Baseline, Felling Pulse or Coastal Drawdown, After Scraping). The per-well era windows are encoded in `scraping_common.WELL_ERAS` as start-inclusive, end-exclusive tuples and propagate identically across 09a, 09c, 09d, and 09e.

scraping_common as the shared infrastructure. The five sub-scripts import their well groups, era boundaries, intervention dates, climate-control list, donor candidates, data loaders, and the scenario-bar engine from a single module. Specifically: `SCRAPING_DATE`, `INTERVENTION_DATE`, `SCRAPING_DATE_2`, `WELL_ERAS`, `IMPACT_WELLS`, `PAIRED_CONTROLS_MAP`, `CLIMATE_CONTROLS`, `DONOR_CANDIDATES`, `TIER1_WELLS`, `TIER2_WELLS`, `SUMMER_MONTHS`, and the helpers `load_scraping_data()`, `load_cluster_params()`, `load_summer_climate()`, `compute_scenario_bars()`, and `compute_scenario_bars_from_params()`, plus `scenario_cluster_sy()` (used to render the empirical scraping bar on the volumetric axis) and `load_annual_climate()` (added v1.5.0, the annual-mean companion to `load_summer_climate()` used by 09d's annual-forcing figure). The former flux→summer-minimum helpers (`summer_amplification_factors()`, `flux_to_summer_min_mm()`, `scenario_summer_min_bars()`) remain in the module but are unused after the 2026-07-02 volumetric re-basis. The BACI structure, era boundaries, well-group memberships, and scenario engine are defined once and propagate consistently. `scraping_common.py` is the scraping-suite analogue of `clearfell_common.py` (S.7).

Site-specific choices and rationale (suite-level)

- **CEH36 is the single scraped well.** The 2015 ground-lowering was a localised 0.3 ha excavation at one well. CEH18 and CEH21 are secondary impacts only by virtue of the 2023 second scraping; in the headline 2015

analysis they are tier-2 impact wells in the sense of being adjacent to forest management but not themselves scraped.

- **scraping_common.WELL_ERAS defines well-specific era windows.** Different wells experience different eras because the 2023 second scraping affects CEH18 and CEH21 but not CEH36 or CEH4. The era system encodes this directly rather than imposing a uniform per-well era structure that would conflate the second-scraping signal with the felling-pulse era at the directly-scraped well.
- **Climate-only control list (CLIMATE_CONTROLS = ceh9, nw7, nw6, nw5, wmc2).** Five wells across the climate-control set in clearfell_common (the same wells used as Climate Ctrl in S.7's clearfell BACI). The choice keeps the regional baseline consistent across the two intervention analyses.
- **The propagation-analysis distance bands (uphill 247–776 m, control 852–1070 m).** Set empirically from the well-location data. The uphill band starts at CEH31 (247 m, the nearest uphill well) and extends to CEH1 (776 m, the most distant well that still sits within the local hydraulic-gradient catchment of CEH36). The control band is everything beyond ≈ 850 m — far enough that the controls share regional climate forcing but are outside the scraping's gradient reach.
- **Provenance-aware summer minima.** 09c consumes 01_wells_provenance.csv (S.1) and applies a min_measured=2 rule per (well, year): a Jun–Sep row whose summer series contains fewer than two measured months is dropped from the per-well summer-minima table, on the principle that a year with effectively no measured Jun–Sep data cannot honestly supply a summer minimum however well the wider record bridges. The min_measured=2 choice is the same value used by 10d in the clearfell suite; the consistency is deliberate, since the two scripts share clearfell_common / scraping_common infrastructure and use the same five climate-control wells. The new n_interpolated column on 09c_01_summer_minima.csv flags any surviving row whose summer minimum was selected from a series containing at least one limit=1-interpolated month, so reviewers can see which year's value rests on a single-missed-visit gap-fill versus an entirely-measured Jun–Sep series.

Limitations and known caveats (suite-level)

- **Single scraped well in the headline.** CEH36 is the only well where ground was actually lowered in 2015. WMC3 in the clearfell suite is similarly constrained, but here the constraint is even tighter — at the scraped well the $n = 1$ limit is structural to the intervention, not just to the network.

Inference about scraping's effects is necessarily about *this* site rather than scraping as a general intervention.

- **The pure-scraping window is short.** Roughly 32 months between April 2015 and December 2017. Per-well SSM fits on this window have wide CIs at the per-well level. The reported step changes are robust at the cluster-mean level (09b) and at the headline CEH36 BACI (09a, 09e); per-well shifts in the propagation analysis are noisy.
- **The propagation analysis attributes coefficient changes to scraping rather than to concurrent change.** The split-window design assumes that the only systematic change between the pre- and post-scraping windows is the intervention itself. Climate-driven coefficient drift between the windows is netted out by the BACI correction against distant controls, but other concurrent processes — gradual coastal retreat, undocumented forestry activity, instrumentation changes — are partially absorbed into the correction and partially attributed to the intervention.
- **The October 2023 second scraping affects CEH18 and CEH21.** Era 3 effects at these wells in 09a, 09c and the propagation analysis are taken up as supplementary; they do not alter the headline at CEH36.
- **The SSM forward-residual estimator in 09e gives a smaller step than raw and synthetic estimators.** All three are positive and substantial, but the SSM forward residual at +0.081 m is materially below the raw BACI (+0.129) and synthetic control (+0.137) estimates. The most plausible source of the divergence is the drainage-feedback term iteratively absorbing the elevated post-2015 water table back toward equilibrium across the forecast horizon; the divergence is a property of the estimator rather than evidence the step is smaller than reported.

Outputs (consolidated)

File	Sub-script	Contents
09_scrape_01_full_parameters.csv	09a	Per-well, per-era SSM coefficients
09_scrape_02_beta3_significance.csv	09a	Isolated β_3 estimates with CIs
09_scrape_03_baci_shifts.csv	09a	Paired BACI step changes
09_scrape_04_net_benefits.csv	09a	Net benefits vs CEH21
09_scrape_04b_beta3_era_summary.csv	09a	Formatted β_3 era summary (Table 6)

File	Sub-script	Contents
		source)
09_tier1_final_cusum.csv	09a	Tier 1 CUSUM terminal values
09_scrape_05_tier1_background_drift.png	09a	Tier 1 BACI + CUSUM figure
09_scrape_06_tier2_scraping_signal.png	09a	Tier 2 BACI + CUSUM figure
09_scrape_07_beta3_confidence.png	09a	β_3 era estimates with CIs
09_scrape_report_numbers.csv	09a	All citable §4.5 values
09b_01_individual_well_baci.csv	09b	Per-well BACI-corrected shifts
09b_02_centroid_summaries.csv	09b	Group centroid shifts
09b_03_ceh36_equilibration.jpg	09b	CEH36 equilibration trajectory
09b_04_scenario_comparison.{csv,jpg}	09b	Cross-cluster scenario chart
09b_05_summer_scenario_comparison.{csv,png}	09b	Volumetric cross-cluster scenario comparison (equilibrium mm water-equiv/month)
09b_report_numbers.csv	09b	Citable §4.5 values
09c_01_summer_minima.csv	09c	Per-well, per-year summer minima
09c_02_summer_minima_shifts.csv	09c	Welch t-test results (both controls)
09c_03_summer_minima_climate_ctrl.png	09c	Three-panel summer minima vs climate control
09c_04_summer_minima_paired.png	09c	Paired-control comparison
09c_report_numbers.csv	09c	Citable §4.5 values

File	Sub-script	Contents
09d_01_scenario_comparison.{csv,jpg}	09d	Equilibrium volumetric scenarios under annual-mean forcing at CEH36
09d_02_summer_scenario_comparison.{csv,png}	09d	Equilibrium volumetric scenarios under summer (Jul–Sep) forcing at CEH36
09e_report_numbers.csv	09e	Three-method robustness step estimates
09_scrape_08_ceh36_robustness.png	09e	Three-panel robustness figure

Where the result appears in the report

- §4.5 *Scraping intervention* — the entire section draws on this suite.
- Table 6 — per-well β_3 era coefficients, sourced from 09_scrape_04b_beta3_era_summary.csv.
- Figures 15–18, 20–22, 24, 25 — multiple figures from the suite.

Cross-references

- **F.3** — SSM equation, displacement formulation, sign conventions, `model_utils.fit_ssm()`. All five sub-scripts fit SSMs through this interface (09a directly via OLS using the same design matrix; 09b, 09e via `fit_ssm()`).
- **F.4** — `pipeline_params.py` consolidation. 09b and 09d both prefer the consolidated params file (`pipeline_params.load_params()`) and fall back to individual loaders if it is absent.
- **F.5** — `scraping_common.py` is documented in detail in this chapter’s *Suite-level methodology* section; the front-matter listing is a brief role summary.
- **S.1** — input data preparation, well-cleaning policy, `01_wells_provenance.csv` (consumed by 09c via `scraping_common.load_scraping_data()` for the `min_measured=2` summer-minima rule).
- **S.3** — produces `03_master_data.csv` and `03_03_cluster_mechanistic_coefficients.csv`, consumed by 09a (per-well coefficients) and the `load_cluster_params()` loader (cluster-level coefficients).

- **S.7** — Script 10 clearfell BACI suite, the parallel intervention analysis. `clearfell_common.py` supplies the β_2 multipliers used by 09d's scenario calculations via `load_clearfell_b2_multiplier()`. The interpolation policy moves this chapter's 09c summer-minima step and S.7's summer-only Forest $\times$ Impact step in opposite directions, because the two analyses use the same NW6 / NW7 climate-control wells in different roles; the §S.7 chapter expands on the clearfell-side picture.
- **S.12** — Script 17 WTF Sy, consumed by 09d (CEH36-specific Sy) and by `load_cluster_params()` (cluster-median Sy).
- **S.14** — Script 21 forestry scenarios; uses the same `scraping_common.compute_scenario_bars()` engine for cross-cluster forestry projections.

S.7 Script 10 suite (a-l) — Clearfell BACI

Step 10 / 30 (sub-scripts 10a–10l). Phase 3 — Model Diagnostics and Intervention Analysis.

The Script 10 suite is the second of the two management-intervention analyses at the warren. Where Script 09 (S.6) analyses the April 2015 scraping at CEH36, Script 10 analyses the December 2017 clearfell of approximately 0.6 ha of Corsican pine (*Pinus nigra* var. *maritima*) inside the western forest block. The intervention removed canopy interception and transpiration over a single compartment with the explicit aim of restoring dune-slack hydrology in the post-felled compartment and the adjacent slacks. The suite is the entire methodological evaluation of that intervention: twelve sub-scripts orchestrated by `run_10_clearfell.py`, with one sub-script (10c) producing supplementary spatial diagnostics that sit outside the main results chain.

The published BACI design uses a five-tier comparison network of 17 wells: one impact well inside the felled compartment (WMC3), four edge wells immediately adjacent (CEH31, CEH20, CEH30, CEH16), five C4 Main Forest interior controls (CEH32, CEH34, CEH33, NW10, CEH2), two C5 Coastal Forest controls (CEH19, CEH17), and five C3 Western Residual climate-only controls (CEH9, NW7, NW6, NW5, WMC2). The four FE wells inside the felled compartment (FE1–FE4) are not in the canonical network because they lack pre-intervention baseline records; FE1 and FE2 are reintroduced by 10h via donor-regression synthetic extension. NW8 and NW8B are excluded from all analyses because of data-quality issues in their pre-intervention records. CEH42 was considered for the climate tier but its 3.4-year pre-felling baseline is too short, so it is excluded as well.

The suite asks eleven questions across its eleven primary sub-scripts. How can the pre-fell window be extended back for impact-tier wells whose own record starts after the canonical pre-fell start date — specifically CEH34, whose Aug-2010 first observation truncates the Forest-control centroid (10i)? Did the clearfell raise the mean monthly water table inside and adjacent to the compartment, after controlling for climate forcing and the residual influence of the earlier scraping (10a)? Where in the network is the step-change spatially concentrated, and where is it absorbed by background drift (10b)? Did the clearfell improve the ecologically critical summer-minimum water table at impact and edge wells (10d)? Which SSM pathway — recharge sensitivity β_1 , atmospheric draw β_2 , or drainage feedback β_3 — shifted at the felled and edge wells after the intervention, as a mechanistic-direction diagnostic alongside the statistical step estimate (10e)? Does the headline ANCOVA result survive being recomputed by independent estimators — per-well SSM forward residual, and zone-level synthetic control (10f)? What do the diagnostic checks — NW10 broadleaf succession, a radial transect from the compartment, rolling-window SSM coefficients — say about the trajectory of the recovery (10g)? Does the result strengthen, weaken, or hold when the synthesised FE-well records are added back into the impact centroid (10h)? Does the result survive a direct Impact-vs-Edge contrast that uses only the closest spatial control and identifies the felling step without the easting $\times$ time covariate that the headline ANCOVA relies on (10j)? And does a single four-zone panel model — fitting the Impact, Edge and a shielded second-control western-dune zone against the Forest control in one internally-consistent regression — agree on the felling step at monthly resolution (10k) and at the summer minimum (10l)? The twelfth sub-script, 10c, sits outside this question chain as a supplementary spatial diagnostic of the C4–C5 forest partition.

The headline ANCOVA-BACI result is reported in the main report as Table 7 and Figures 26–28 (§4.6 *Clearfell intervention*); spatial maps (Figures 23, 31), summer minima (Figure 29), coefficient shifts (Figure 32), the radial transect (Figure 33), and the supplementary forest-zone analysis (Table 16) draw on the other sub-scripts. The compositional mapping is recorded in the live PIPELINE_README.md table.

Inputs (suite-shared)

File	Origin	Used by
01_wells_clean.csv, 01_wells_extended.csv	Script 01	All sub-scripts
01_climate.csv	Script 01	10a, 10d, 10e, 10f, 10g, 10h, 10j, 10k

File	Origin	Used by
03_master_data.csv	Script 03	All (per-well coefficients, locations, cluster IDs)
Well_info.csv	raw data	10h (FE well locations not in master)
07_coeff_maps_data.csv	Script 07	10c
06_pear_membership_audit_sitewide.csv	Script 06	10c
Features.kml, clearfell.kml, newborough_dem.tif	raw data	10b, 10c (spatial maps)
10a_* outputs	Script 10a	10f, 10h (ANCOVA comparison)
10d_01_summer_minima.csv	Script 10d	10j (annual-summer-minimum contrast), 10l (four-zone summer minima)
10e_01_coefficient_shifts.csv	Script 10e	downstream Scripts 19, 21 via load_clearfell_b2_multiplier()
10i_01_ceh34_hindcast.csv	Script 10i	10a, 10b, 10e, 10h via clearfell_common.apply_ceh34_hindcast(); 10d, 10f, 10j, 10k, 10l do not consume

All loading goes through `clearfell_common.load_clearfell_data()`, which merges the reference and extended well frames, normalises column names to lowercase, and validates the five tiers against the live data. Per-well locations are sourced first from the master data, with `Well_locations_height.csv` as a fallback for wells not in the reference network.

Sub-script 10a — Three-counterfactual ANCOVA-BACI

Motivation. This is the headline analysis. The question is whether the December 2017 clearfell produced a step-change in the water-table displacement of the impact and edge wells relative to nearby unaffected controls, after climate forcing and the residual scraping effect are controlled out.

Methodology. 10a runs the same ANCOVA model six times — three control definitions (Forest, Climate, Combined) crossed with two target zones (Impact, Edge) — yielding six ANCOVA results per intervention. The three controls are:

- **Forest:** the five C4 Main Forest interior wells (CEH32, CEH34, CEH33, NW10, CEH2). The cleanest counterfactual mechanistically: matched canopy, matched substrate, matched climate band.
- **Climate:** the five C3 Western Residual climate wells (CEH9, NW7, NW6, NW5, WMC2). Open-dune wells under shared climate forcing; the unconfounded climate baseline.
- **Combined:** all 12 Forest, Coastal, and Climate control wells pooled. Larger sample, less mechanistically targeted.

(Note: a standalone “Coastal Forest” counterfactual is *not* run in 10a — the C5 Coastal wells enter only through the Combined pool. The two Coastal wells are mechanistically distinct from the C4 interior, with lower β_2 and a confounding coastal-retreat signal that 10b’s climate correction is designed to absorb. Treating C5 as a third standalone counterfactual would give it more weight in the inference than its sample size and confound structure can carry.)

For each (control, zone) pairing the script:

1. Computes the BACI displacement timeseries: target-zone monthly mean minus control-zone monthly mean.
2. Adds a centred climate-water-balance (CWB) covariate, computed by `compute_cwb()` as the demeaned cumulative ($P - PET$) anomaly in mm.
3. Adds a distance-weighted scraping covariate as the *difference* of target and control centroid weights, with weight $\exp(-d / \lambda)$ and $\lambda = 300$ m by default. The differential form is essential because the BACI itself is target-minus-control.
4. Adds the December 2017 felling dummy, the CWB $\times$ felling interaction, and where geometrically valid, an easting $\times$ time interaction that absorbs the coastal-retreat gradient.
5. Adds a sensitivity to the October 2023 re-scraping (Model 3 includes the second scraping pulse; the difference in AIC from the baseline Model 2 records whether the second pulse measurably improves the fit).

The easting $\times$ time term is included only when the easting range across the union of target and control wells exceeds 200 m. For the Forest control runs, the C4 wells cluster too tightly in easting for the term to be informative, and it is dropped. For the Climate and Combined runs the range is large enough to fit the

gradient cleanly. This is automatic — `build_ancova_frame()` writes a boolean `has_easting` flag and `run_ancova()` reads it.

Sensitivity is checked by re-running every (control, zone) pairing at $\lambda = 200$ m and $\lambda = 500$ m. The headline values are at $\lambda = 300$ m, which sits roughly at the centroid of the within-network distances.

The headline output table records the felling step (in metres), its 95 % confidence interval, the scraping step, the easting coefficient, R^2 , the model N, the Oct 2023 step and its AIC differential, and the climate-background subtraction in mm. The headline result is the Forest-control step at each zone:

- **Impact (WMC3):** +120 mm, 95 % CI [+50, +189], $p < 0.001$, $R^2 = 0.274$, $N = 162$.
- **Edge (four wells):** +33 mm, 95 % CI [−16, +82], $p = 0.193$, $R^2 = 0.485$, $N = 158$.

The Climate-control runs give a small non-significant Impact step (−13 mm, $p = 0.599$) and an Edge step of −109 mm, 95 % CI [−172, −47], $p < 0.001$. The easting $\times$ time correction is significant at $p < 0.001$ in both Climate runs (and in the Combined runs) but is dropped in the Forest runs by the 200 m easting-range rule (see *Site-specific choices* below). The Combined-control runs give an Impact step of +73 mm ($p < 0.001$) and an Edge step of −21 mm ($p = 0.136$), pooling Forest, Coastal, and Climate controls; the Combined Impact estimate is tighter than either Forest or Climate alone because of the larger control sample. Each (control, zone) pair is exported as a row in `10a_01_ancova_comparison_table.csv`. Full coefficients, easting $\times$ time effects, and the per-pair model diagnostics are in `10a_02_ancova_full_coefficients.csv`. The corrected BACI displacement timeseries — observed minus the fitted climate and easting effects — is exported in `10a_03_baci_timeseries.csv` and underlies the §4.6 figures.

The Climate-control Edge step is small in magnitude but its sign is opposite to the Forest-control Edge step. The two controls answer different counterfactual questions: the Forest control isolates the clearfell signal against canopy-matched substrate-matched wells, while the Climate control compares the forest-edge wells against open-dune climate-baseline wells that sit further from the coast. The negative Climate-Edge step reflects residual coastal-zone drift between the C3 climate wells, which sit further west and away from the coast, and the four C4-edge wells, which sit closer to the receding coastline. The easting $\times$ time correction in the ANCOVA absorbs the linear spatial gradient across the network and is significant at $p < 0.001$ in the Climate runs, but it does not remove the

residual zone-level drift between two well clusters at different distances from a coastline retreating episodically through the post-felling era (Script 25 documents the gradient in detail).

Directly-fitted summer ANCOVA step. 10a v1.3.0 also fits the Forest-control Impact specification on a Jun–Sep subset (N = 52 months), under the same model terms as the annual headline. The fitted summer step is +46 mm, 95 % CI [–69, +162], $p = 0.44$, $R^2 = 0.337$ — not significant at conventional thresholds. A CWB-dropped sensitivity variant (+123 mm, $p = 0.058$, $R^2 = 0.098$) is emitted to `10a_report_numbers.csv` as `ANCOVA_Forest_Impact_clearfell_step_summer_noCWB`; the full specification is preferred on Δ AIC grounds.

The annual Forest $\times$ Impact step is the durable headline of the clearfell analysis. Its sample size (162 months) carries the inference. The Jun–Sep subset is only 52 months and the same specification on that subset does not reach significance. The summer signal corroborates the direction of the annual step but is not retained as a separate statistical claim. The arithmetic-construct fallback (`BACI_ANNUAL  $\times$  1.5034`) that was retained transiently as a Script 21 fallback and removed in Script 21 v1.0.3 / 10a v1.3.0 is therefore not replaced by a summer-specific scaling.

Curvature (CWB² $\times$ felling) sensitivity variant. The headline ANCOVA fits a *linear* CWB $\times$ felling interaction, which tests whether felling changed the linear sensitivity of the water table to climate forcing — the canopy-buffering hypothesis in its simplest form. That interaction is non-significant at both Forest-control zones (Impact $p = 0.40$, Edge $p = 0.16$): on a whole-range linear slope, the pre- and post-felling climate sensitivities are not distinguishable. A linear slope is, however, a blunt instrument for a hypothesis about climate *extremes* — a buffering effect concentrated at the wet or dry tails can average to near-zero across the full CWB range. 10a v1.4.0 therefore also fits a non-linear extension on the full-data Forest-control Impact and Edge frames: the headline design matrix plus a centred CWB² main effect and a CWB² $\times$ felling interaction. The CWB² $\times$ felling term is significant at both zones (Impact -2.75×10^{-6} m mm⁻², $p = 0.016$; Edge -2.22×10^{-6} m mm⁻², $p = 0.008$), the curvature model improves on the linear model by Δ AIC -4.2 (Impact) and -10.7 (Edge), and the joint F-test of the two added curvature terms is significant at both zones. The CWB² *main* effect is non-significant at both zones — the pre-felling relationship is linear; only the post-felling relationship bends. The negative interaction sign means the post-felling response is concave in CWB, i.e. the felling uplift is larger in dry (low-CWB) conditions. This is a reported sensitivity variant only: the headline clearfell step

and the linear model are unchanged, and the curvature coefficients, the re-referenced clearfell step (which differs from the headline step because the step is evaluated at mean CWB under a curved fit — +146 mm Impact, +50 mm Edge), the curvature-model R^2 , the ΔAIC , and the joint-F statistic are emitted to `10a_report_numbers.csv` under the `ANCOVA_Forest_{zone}_coeff_cwb2_x_fell / _curv_*` keys. The result indicates the felling response is climate-state-dependent in a way the linear interaction averages away; it is consistent with — but not on its own proof of — a dry-period canopy-buffering mechanism, since climate-state-dependence has alternative explanations (post-felling non-stationary drift, the coastal-retreat gradient) that the variant does not exclude. The main report (§4.6) reports it neutrally as a flagged preliminary finding, not as a confirmed buffering effect.

Sub-script 10b — Spatial step-change maps

Methodology. For each well in the warren-wide network with at least six observations in each of the three eras (pre-scraping, post-scraping, post-felling), the script computes era-mean depths and two step-changes: a scraping step (post-scrape minus pre-scrape) and a felling step (post-felling minus post-scrape). The full network of approximately 90 wells is rendered as an IDW-interpolated surface on a 40 m grid with the standard DEM hillshade and KML feature overlay (`map_utils.add_idw_surface`, `load_dem_hillshade`, `add_kml_features`). Four publication figures result: raw and climate-corrected versions of both step-changes.

The climate correction uses the median step at a deliberately chosen subset of four C3 western wells (NW5, NW6, NW7, CEH1) — intentionally different from 10a’s full C3 climate-control set. These four wells share both the western climate signal and the coastal-retreat boundary position of the intervention zone, so subtracting their median step from every well removes climate plus coastal-retreat drift in a single operation. The script’s docstring records this choice and the per-well counts at each era are exported.

Output	Contents
<code>10b_spatial_step_data.csv</code>	Per-well era means, step values, climate-corrected steps, well counts
<code>10b_spatial_scrape_raw.png / 10b_spatial_scrape_corrected.png</code>	Scraping era maps
<code>10b_spatial_fell_raw.png / 10b_spatial_fell_corrected.png</code>	Clearfell era maps

The clearfell-era corrected map is the canonical Figure 31 in the main report; the scraping-era equivalent is Figure 23. The per-well step CSV is also consumed by Script 09b (scraping propagation) for the network-wide propagation modelling.

Sub-script 10d — Summer minima (dual-control)

Motivation. The annual June–September minimum water table is the ecologically critical metric for dune-slack vegetation. A clearfell that raises the mean monthly water table but leaves the summer minimum unchanged delivers a different kind of result than one that does both. 10d evaluates the summer-minimum response separately and through two independent control structures.

Methodology. For each well in the 17-well network, `annual_summer_minimum()` extracts the deepest (most negative) June–September water level per year, requiring at least two observations in the window. The script then compares the well’s summer-minimum series against two control centroids — the Forest control centroid (mean of the five C4 wells’ annual minima, requiring at least two wells per year) and the Climate control centroid (mean of the five C3 wells) — yielding a “gap” series per well per control. Pre-felling years (2007–2017) are compared with post-felling years (2018–2025) by Welch t-test on the gaps.

`annual_summer_minimum()` and `forest_control_centroid_summer_min()` consume `01_wells_provenance.csv` (§S.1) and apply a `min_measured=2` rule: a (well, year) summer minimum is admitted to the panel only when at least two of the four Jun–Sep months are actual field measurements rather than interpolations. Single-month interpolations (the residuals under the `limit=1` cleaning policy) are allowed to contribute, but a `n_interpolated` column in `10d_01_summer_minima.csv` flags every (well, year) row that carries at least one such cell. The rule excludes (well, year) combinations whose summer minimum would otherwise be supported only by interpolated values — most notably at WMC3, NW6, and NW7 where Jun–Sep 2019 was entirely unmeasured.

A pooled mixed-effects model with a random intercept per well (`statsmodels.regression.mixed_linear_model.MixedLM`) gives a tier-level clearfell-attributable step with proper uncertainty. The Impact tier (one well, WMC3) falls back to OLS.

The result is reported with care because it contains both forest-positive and forest-negative components: the mean monthly recovery in 10a is statistically significant, but the summer-minimum signal at impact wells is null and at edge wells is a negative shift that does not reach significance at $\alpha = 0.05$. Specifically:

- **Forest-control mixed model: Impact step = -1 mm, $p = 0.99$ ($n = 1$, OLS).** No improvement in summer minima at the impact well.
- **Forest-control mixed model: Edge step = -64 mm, $p = 0.12$.** The edge-tier summer minima are deeper post-felling than pre-felling relative to the Forest control, but the shift is not significant at conventional thresholds.

The Climate-control mixed-model Edge step is $+48$ mm ($p = 0.28$) — the negative shift seen under the Forest control does not reproduce against the open-dune climate baseline, which is the disconfirmation. The Coastal Control tier shows the largest negative shift across both controls (Forest control: -176 mm, $p = 0.059$; Climate control: -42 mm, $p = 0.31$), which is the coastal-retreat signal that Scripts 25 and 10c document at length.

Sub-script 10e — SSM coefficient shift diagnostic

Motivation. The ANCOVA in 10a is the statistical estimate of the clearfell step magnitude. 10e is a complementary *mechanistic-direction* diagnostic: for each well it asks which SSM pathway moved after felling — recharge sensitivity (β_1), atmospheric draw (β_2), or drainage feedback (β_3) — and in which direction. It is a qualitative pattern diagnostic, not a second estimator of the step magnitude. The clearfell magnitude result is, and remains, the 10a ANCOVA BACI step.

Methodology. For each of the 17 network wells, the SSM frame is built via `build_ssm_frame()` (F.3) and split at the December 2017 felling date into Before and After eras. Both eras are fitted with the canonical no-intercept SSM through `model_utils.fit_ssm()` — the same Model A form used by Script 03 and described in S.3. The Before fit additionally carries a scraping dummy (1 from April 2015 onward), supplied via the `extra_regressors` keyword of `fit_ssm()`, to absorb the residual scraping influence within the pre-felling baseline; the After fit is the standard three-column SSM. For each well the script computes the per-coefficient deltas $\Delta\beta = \text{After} - \text{Before}$ and reports them per well and per tier.

The script does **not** attempt to reconstruct or predict the 10a BACI step from the coefficient shifts. A $\Delta\beta$ -projected predicted step ($\Delta h_{\text{predicted}} = \Delta\beta_1 \cdot \text{mean}_P - \Delta\beta_2 \cdot \text{mean}_{\text{PET}} - \Delta\beta_3 \cdot \text{mean}_h_{\text{disp}}$) is not commensurable with the observed 10a step: the 10a step is the coefficient on a felling dummy in an ANCOVA fitted to the *control-differenced* centroid (target centroid minus control centroid), with regional climate forcing removed by construction, whereas a $\Delta\beta$ projection would be built from single-well, undifferenced, per-era fits with no control subtraction and on a different climate projection basis. Reconciling the two would require absorbing a residual into an intercept shift $\Delta\alpha$ of unidentified physical content.

Script 10e therefore reports the $\Delta\beta$ pattern as a direction-and-sign diagnostic and leaves the step magnitude to 10a; the era fits use the canonical no-intercept form, consistent with the rest of the pipeline.

Under the v1.4.0 fits, the per-tier mean coefficient shifts are: at the Impact tier (WMC3, one well) $\Delta\beta_1 = -0.10$, $\Delta\beta_2 = -0.28$, $\Delta\beta_3 \approx 0.00$; at the Edge tier $\Delta\beta_1 = -0.27$, $\Delta\beta_2 = -0.03$, $\Delta\beta_3 = -0.01$. The interpretation is qualitative — the felled and edge wells show a post-felling reduction in recharge sensitivity, with the largest single shift at the Impact well being in β_2 — and is read as supporting mechanistic context for the 10a result, not as an independent quantification of it.

The β_2 multiplier exported via `clearfell_common.load_clearfell_b2_multiplier()` is a *differenced* tier quantity:

$$\text{clearfell_}\beta_2\text{_multiplier} = \text{Edge_ratio} - \text{Climate_Ctrl_ratio} + 1.0$$

where each “ratio” is the tier mean of $b2_after / b2_before$ across that tier’s wells. The construction takes the Edge-tier ratio and subtracts the Climate-control-tier ratio to remove background climate drift, recentring the result near 1.0; the thinning multiplier is the half-perturbation. Both values are read dynamically from `10e_01_coefficient_shifts.csv` by Scripts 19 and 21; the previous static-constant approach is retired. *Note: the individual per-tier $b2_after / b2_before$ ratios are differently scaled under no-intercept fits than under intercept-bearing fits — the raw ratios should not be read as direct canopy-effect magnitudes; only the Edge-minus-Climate differenced quantity is interpreted.*

Sub-script 10f — Robustness

Two independent estimators broadly support the ANCOVA result at the Impact tier.

SSM forward residual. For each network well, the SSM (no scraping dummy, three columns) is calibrated by OLS on the pre-scraping era (where available, requiring at least 36 calibration months) and iterated forward through the scraping and felling eras to generate a counterfactual trajectory. The residual (observed minus predicted) is normalized by subtracting the mean of the control-tier residuals. The era-mean residuals are tested via Welch t-test. The Impact mean step is +56 mm (positive, but smaller in magnitude than 10a’s +120 mm Forest-control step at WMC3; the SSM forward-residual estimator is a per-well calibration against a counterfactual built from the pre-scraping era and is more sensitive than the ANCOVA to month-by-month variation); Edge mean step is

essentially zero (+4 mm), reflecting that the Edge wells are well-explained by their own pre-scraping SSM fit and the felling produces a residual within climate noise — consistent with 10a's small non-significant Forest-control Edge step.

Synthetic control. A donor pool of six wells outside the BACI network (CEH1, CEH5, CEH6, CEH10, CEH11, CEH24) is used to build a synthetic counterfactual at each zone. The donor weights are fitted by OLS on the pre-scraping baseline (no intercept). The gap series (observed zone mean minus synthetic) is segmented at the felling date; the step is the post-felling minus post-scraping gap. The Impact step is +99 mm, $p = 0.001$; the Edge step is +41 mm, $p = 0.22$. The three Impact-tier estimators span +56 to +120 mm (SSM forward residual, synthetic control, 10a ANCOVA Forest control). All three are positive; the synthetic-control and ANCOVA results are significant at conventional thresholds, while the SSM forward-residual estimate is positive but lies at the low end of the range without independent significance. The methodological convergence the script exists to test holds in direction; the magnitude spread is wider than within-rounding, with the SSM residual estimator sitting roughly 74 mm below the ANCOVA headline.

Two independent estimators agreeing with 10a's headline at the Impact tier in direction and broad magnitude is the main message. The Edge tier diverges more across methods, reflecting the larger heterogeneity of edge wells and the differential climate-and-coastal-retreat correction that 10a's full ANCOVA design is built to handle.

Sub-script 10g — Diagnostics

Three short diagnostics that justify network choices made elsewhere and characterize the trajectory of the recovery.

NW10 broadleaf trend. NW10 sits at a broadleaf-pine margin and was included in the Forest Control tier despite being borderline. The script tests its 2019–2025 summer-minimum anomaly trend relative to a pine-interior composite (CEH2, CEH32, CEH33, CEH34). The trend is -46 mm/year, $p = 0.024$, indicating progressive broadleaf-margin drawdown. The mean 2010–2021 anomaly (+288 mm above the pine composite) confirms NW10's distinctly shallower historical baseline. The trend is documented but NW10's inclusion in the Forest Control tier is retained — the 2019–2025 drawdown is a real broadleaf-succession signal that the chapter discusses, not a data-quality issue.

Radial transect. A three-panel figure plotting WMC3 (45 m from compartment centroid), the four edge wells (152–229 m), and two reference wells (CEH34 at 306 m, CEH2 at 428 m) over time. Panel A shows depth anomaly relative to scrape-era

mean as 6-month rolling means; panel B shows zone anomalies relative to the transect mean; panel C is a step-vs-distance scatter. The fitted gradient is -0.4 mm per 100 m ($p = 0.186$) — no significant spatial gradient in the felling step across the transect, consistent with a recovery effect spatially concentrated at the compartment edge rather than gradiently dissipating with distance.

Rolling SSM coefficients. 48-month rolling-window SSM fits at the Impact centroid (WMC3), C3 (open dune) centroid, and C4 (forest interior) centroid. The diagnostic question is whether the Impact centroid β_1 shifts toward C3 post-felling — the expected behaviour if the canopy removal restores open-dune recharge sensitivity. The pre-felling Impact β_1 rolling mean is 2.48; post-felling 2.57; the C3 post-felling rolling mean is 3.49. The Impact β_1 rises slightly toward C3 but remains well below the C3 baseline, consistent with partial canopy recovery on a one-compartment, six-year time horizon and with the limited statistical power of rolling 48-month windows applied to a single well.

Sub-script 10h — Synthetic FE-well extension BACI

Motivation. WMC3 is the only impact-zone well that spans all three eras. FE1–FE4 sit inside or at the immediate edge of the felled compartment but have no pre-clearfell baseline (FE1 and FE2 from July 2015, FE3 and FE4 from 2017). The published 10a headline therefore rests on a single impact well. 10h extends FE1 and FE2 backwards using donor regression on Forest Control wells unaffected by the clearfell, then reruns the ANCOVA with these synthetic baselines spliced in. The extension brings the Impact-tier N from one well to three and provides an independent test of the 10a Impact result.

(FE3 and FE4 are not synthesised. Both start in 2017 — too late for a calibration window with adequate length.)

Methodology. Three donor candidates (CEH34, CEH2, CEH33) are jointly regressed against FE1 and FE2 on the pre-clearfell overlap window (July 2015 to November 2017, 29 months). The donor regression is multi-donor OLS with intercept, fitted independently per FE well. The calibrated relationship is then used to hindcast each FE well backward to August 2010, gaining 49 pre-scraping months of synthetic record per well. The calibration R^2 is 0.998 for FE1 and 0.994 for FE2, with RMSEs of 18 and 25 mm respectively. The synthetic record is spliced with the actual FE observations from July 2015 onward.

Three impact-centroid variants are then run through the identical 10a ANCOVA framework:

- **Variant A:** WMC3 + FE1 + FE2 (three-well centroid).
- **Variant B:** WMC3 + FE2 (two-well centroid, excluding FE1).
- **Variant C:** WMC3 alone (reproduces 10a Impact).

The reason for Variant B is that FE1 sits approximately 20 m outside the clearfell boundary in standing forest, while FE2 is inside the felled compartment. FE1's post-felling divergence is +9 mm ($p = 0.15$); FE2's is +28 mm ($p < 0.001$). Averaging FE1 into the impact centroid dilutes the clearfell signal because FE1 is not actually clearfell-impacted.

Variant	Control	Step (mm)	95 % CI	p
A (WMC3 + FE1 + FE2)	Forest	+85	[+39, +131]	< 0.001
A	Climate	-59	[-120, +2]	0.062
B (WMC3 + FE2)	Forest	+103	[+45, +162]	< 0.001
B	Climate	-36	[-90, +19]	0.201
C (WMC3)	Forest	+120	[+50, +189]	< 0.001

Variant C (WMC3-only) is the headline result for the impact tier; it reproduces 10a's Forest-control Impact step (+120 mm, $p < 0.001$) and is the form quoted as the chapter's primary impact-zone estimate. Variants A and B are robustness checks built on synthetic pre-clearfell baselines for FE1 and FE2 via donor regression on Forest Control wells. Variant B is a two-well centroid (WMC3 at the compartment edge; FE2 inside the compartment) against forested controls, returning a Forest-control step of +103 mm, $p < 0.001$. Variant A (which adds FE1) gives +85 mm, $p < 0.001$ against the Forest control; the script reports it but does not recommend it as a robustness anchor because FE1 sits in standing forest outside the clearfell boundary.

The three Impact-tier variants span +85 to +120 mm against the Forest control, all positive and individually significant. Both synthetic-extension variants are smaller in magnitude than the WMC3-only headline, so the synthetic extension is best read as a robustness check on direction and sign rather than as an independent magnitude estimator. The methodological convergence available from 10h is: the clearfell signal is positive and significant at the Impact tier under every plausible centroid composition, and the WMC3-only step is not a one-well artefact in its sign or its direction. The magnitude attenuation under synthetic-baseline inclusion is noted but not given a physical interpretation in this chapter.

Sub-script 10i — CEH34 donor-regression hindcast

Motivation. CEH34 is a Forest Control well used in the 10a Forest-control centroid and in the 10b spatial step maps. Its monthly record starts on 1 August 2010, later than the August 2010 first observation of several other Forest-control wells. Under the pre-fell window start in force when 10i was written (1 July 2010) this truncation affected the Forest-control centroid directly: including CEH34 with its truncated record either reduced the pre-fell N (and so the precision of the climate-corrected pre-fell mean) or introduced a composition shift around August 2010 as CEH34 joined the centroid. Following the PRE_FELL_START migration to 1 January 2011 (`clearfell_common` v1.7.0), CEH34's record start falls before the window and the in-window centroid is no longer truncated; the hindcast splice is retained for cross-version reproducibility and for analyses that opt into the earlier start. 10i removes the original asymmetry by hindcasting CEH34 backward to the pre-CEH34-record window using donor regression on a single donor well from the climate-control set.

Methodology. The donor well is CEH9. CEH2 has a stronger empirical correlation with CEH34 ($r^2 = 0.97$ against CEH9's $r^2 = 0.89$), but CEH2 is itself a Forest Control well that contributes to the BACI Forest Control centroid; hindcasting CEH34 against CEH2 and then adding both to the Forest Control set would amount to upweighting CEH2's contribution to the pre-fell baseline — a partial double-count. CEH9 sits in the Climate Control tier and is therefore independent of the Forest Control centroid; its slightly weaker correlation reflects the strong site-wide groundwater synchrony at Newborough rather than any methodological weakness. The fit is OLS on the pre-clearfell overlap window (August 2010 to November 2017):

$$\text{CEH34}(t) = \alpha + \beta \cdot \text{CEH9}(t) + \varepsilon$$

The pre-clearfell window is used deliberately. Fitting in the post-fell era would risk inheriting any clearfell-related divergence between CEH34 and CEH9 into the calibrated relationship; the pre-fell-only fit gives the unconditional donor regression that pre-dates the intervention. The fitted relationship is then applied to CEH9's pre-CEH34 record (May 2006 to July 2010, 51 monthly cells) to produce a synthetic CEH34 trajectory. The output is a spliced series — synthetic for dates before 1 August 2010, observed afterwards — with a source flag distinguishing the two.

Downstream consumption. The spliced series is exposed via `clearfell_common.load_ceh34_hindcast_series()` (added in `clearfell_common`

v1.2.0). Scripts that opt in call the loader explicitly: 10a, 10b, 10e, and 10h consume the hindcast and gain a four-year extension to the Forest-control centroid's pre-fell window. Scripts 10d, 10f, and 10j do not consume the hindcast — 10d works at annual summer-minimum resolution where the 2010 part-year would contribute no annual statistic anyway; 10f's per-well SSM forward residual fits each well independently and is not affected; 10j uses the Edge tier as control rather than the Forest tier so CEH34 is not in its design at all.

Script 10i (v1.0.0): CEH34 donor-regression hindcast against CEH9; pre-clearfell-only OLS calibration; spliced output consumed by 10a, 10b, 10e, 10h.

Sub-script 10j — Direct Impact-vs-Edge contrast

Motivation. The three-counterfactual ANCOVA in 10a (Forest, Climate, Combined control tiers) accounts for climate forcing via the cumulative water balance (CWB) covariate and for coastal-retreat drift via an easting $\times$ time interaction whenever the well-set spans more than 200 m easting. Both covariates carry independent physical justification — CWB tracks atmospheric forcing month by month, and the easting $\times$ time term reflects the documented coastal-retreat gradient at the western margin (Script 25; CEH4 retreat rate of order 29 mm per year over 2010–2024, with documented six-week losses during Storm Brendan, January 2020). However, the headline ANCOVA's identification of the felling step is materially dependent on the easting $\times$ time specification: a sensitivity run that omits the term (10j internal diagnostic, reported below) reduces the Forest-control Impact step from its full-specification value to a small positive figure that is not statistically distinguishable from zero. The dependence is mechanistically defensible — coastal retreat is real, and well-documented — but a sceptical reading of the headline benefits from a corroborating estimator whose identification does not pass through any easting $\times$ time covariate at all.

The Edge tier provides that estimator by design. The four Edge wells (CEH16, CEH20, CEH30, CEH31) sit within or immediately adjacent to the felled compartment but experienced markedly less of the felling treatment than WMC3 itself, while sharing nearly every confounder with the Impact tier: the same coastal-retreat gradient (the Edge eastings span the WMC3 easting), the same monthly climate forcing, the same regional groundwater drift, the same Cluster 4 forest-canopy interception during the pre-fell era. The 2015 scraping is the one major exception — WMC3 sits within the scraping-influence footprint while the Edge wells, located further east of the scraped compartment, do not — and the contrast must therefore retain a tier-asymmetric scraping term.

Conceptually the contrast is a two-zone pooled BACI in which the Edge tier is the spatial buffer. If the Impact and Edge wells would have moved together in the absence of the felling, the difference in their step changes at December 2017 is the felling response stripped of every shared confounder. This is a weaker and more defensible identifying assumption than the per-zone modelling required for 10a's three ANCOVA designs.

Methodology. 10j runs two parallel analyses at monthly-mean and annual-summer-minimum resolution.

The monthly-mean contrast pools the Impact and Edge wells into a long-format panel of water-table depth observations from PRE_FELL_START (1 January 2011) onward, and fits

$$h(i,t) = \alpha + \beta_{\text{cwb}} \cdot \text{CWB}(t) + \beta_{\text{S}} \cdot \text{Scraped1}(t) + \beta_{\text{P}} \cdot \text{Post}(t) + \gamma_{\text{S}} \cdot Z(i) \cdot \text{Scraped1}(t) + \gamma_{\text{P}} \cdot Z(i) \cdot \text{Post}(t) + \delta_{\text{cwb}} \cdot Z(i) \cdot \text{CWB}(t) + \mu(i) + \varepsilon(i,t)$$

where $h(i,t)$ is the monthly water-table depth at well i , time t ; $\text{CWB}(t)$ is the centred cumulative water balance for month t (the same series used by 10a's ANCOVA); $\text{Scraped1}(t)$ is an indicator equal to 1 for $t \geq \text{SCRAPING_DATE}$ (1 April 2015); $\text{Post}(t)$ is an indicator equal to 1 for $t \geq \text{INTERVENTION_DATE}$ (1 December 2017); $Z(i)$ is an indicator equal to 1 for the Impact tier (zero for Edge); and $\mu(i)$ is a well-level fixed effect. Estimation is by OLS with cluster-robust standard errors clustered on i (well), which absorbs the within-well temporal autocorrelation.

The headline coefficient is γ_{P} — the differential felling step (Impact minus Edge). The auxiliary coefficient γ_{S} is the differential scraping step (Impact minus Edge), retained because the scraping treatment is asymmetric across the two tiers; omitting it forces the scraping signal at WMC3 to load onto β_{P} and biases the felling-step estimate downward. The $Z(i) \cdot \text{CWB}(t)$ interaction permits the climate sensitivity to differ between the two tiers and is included for parity with 10a's cwb_x_fell term. The Impact main effect is collinear with the well-fixed-effects block (the Impact tier comprises a single well, WMC3) and is dropped from the design.

The annual summer-minimum contrast operates on the same panel of wells but on the (well, year) summer-minimum frame produced by 10d (10d_01_summer_minima.csv), which already encodes the shared cleaning rules — Jun–Sep minimum, requiring at least two measured months per summer, with the $n_{\text{interpolated}}$ flag identifying summers whose minimum is supported by interpolated cells. 10j restricts to the rows with $n_{\text{interpolated}} = 0$ (measured-only summers) and fits

$$m(i,y) = \alpha + \beta_P \cdot \text{Post}(y) + \gamma_P \cdot Z(i) \cdot \text{Post}(y) + \mu(i) + \varepsilon(i,y)$$

where $m(i,y)$ is the Jun–Sep minimum depth at well i , year y ; $\text{Post}(y) = 1$ for $y \geq 2018$. The scraping term is omitted at this resolution: under the measured-only rule the Impact tier contributes only two pre-2015 summers (2011 and 2013), which is too few to identify a scraping step with stable inference. The model collapses to OLS with cluster-robust standard errors on well (the Impact tier’s single well makes a formal random-intercept specification redundant).

The two coefficient estimates (γ_P at monthly and annual resolution) and their standard errors are written to `pipeline_site_observations.csv` via four new registry entries — `impact_vs_edge_clearfell_monthly_step`, `impact_vs_edge_clearfell_monthly_step_se`, `impact_vs_edge_clearfell_summer_step`, `impact_vs_edge_clearfell_summer_step_se` — to make the values available to downstream consumers as live pipeline numbers rather than cached constants.

Easting-term sensitivity (internal diagnostic). 10j also runs a sensitivity check on 10a in which the easting $\times$ time covariate is omitted from each of the six ANCOVA configurations. The result is reported in the main-report editorial queue as a robustness paragraph for §4.6; 10j does not write the sensitivity to a pipeline CSV because it is a diagnostic on a different script’s specification rather than an output of 10j itself. The qualitative finding — that the Forest-control Impact step collapses to a small non-significant figure when the easting term is removed, while two other estimators (synthetic control and SSM forward residual) continue to detect a positive felling step — supports the interpretation that the easting $\times$ time term is correctly absorbing coastal-retreat physics rather than over-fitting a felling-era residual. The direct Impact-vs-Edge contrast adds a third estimator to that set whose identification does not pass through easting $\times$ time at all.

Relationship to the other 10-suite estimators. The five monthly-mean estimators of the Impact felling step (10a Forest-control headline, 10f synthetic control, 10f SSM forward residual, 10a joint-fit Forest-control sensitivity, and 10j direct contrast) sample the same underlying response through different counterfactual designs and therefore answer subtly different questions. 10a’s separately-fitted Forest-control ANCOVA estimates the Impact response against a mechanistically-matched but spatially-distant counterfactual; 10f’s synthetic control estimates against a data-driven optimal combination of donor wells; 10f’s SSM forward residual estimates against the Impact tier’s own pre-scraping SSM fit; 10j estimates against the closest spatial control available. The Edge contrast is the smallest of the five in magnitude — by construction, because both Impact and

Edge experienced any unmodelled wet-period drift that the climate-corrected estimators net out, but the Edge wells did not experience the felling treatment to anything like the same degree as WMC3. The contrast is what survives once every shared confounder cancels, and it is the most conservative single statement of the felling response at WMC3.

The monthly contrast is larger in magnitude and statistically significant; the summer-minimum contrast is smaller and not statistically distinguishable from zero. This is consistent with the wider Script 10 finding (10a annual recovery significant, 10d summer minima at the Impact tier null) and reflects the seasonal phenology of the clearfell response: the felling raised the mean of the year by reducing canopy interception and direct atmospheric draw on the saturated zone, but did not detectably lift the summer minimum, which is set by the depth of the dry-period drawdown rather than by the mean. The two coefficient estimates together describe a felling response that is real in the seasonal mean but ecologically attenuated at the summer extreme — the resolution that matters for Curreli's SD15b / SD16 wet-slack thresholds.

Inputs. Wells, climate, master, locations, and valid tier dictionaries are loaded via `clearfell_common.load_clearfell_data()`. The summer-minima frame is read from `OUT_10D_DATA(10d_01_summer_minima.csv)`; 10j requires that 10d has already run in the pipeline pass. All dates and well lists are sourced from `clearfell_common` (`INTERVENTION_DATE`, `SCRAPING_DATE`, `PRE_FELL_START`, `IMPACT_WELLS`, `EDGE_WELLS`); the CWB series is built via `clearfell_common.compute_cwb()`. No site-specific values are hardcoded.

Outputs. Per-resolution results in `10j_01_monthly_contrast_results.csv` and `10j_02_summer_contrast_results.csv` (one row each with coefficients, standard errors, 95 % confidence intervals, p-values, R^2 , and sample sizes); standard ReportNumbers CSV in `10j_report_numbers.csv`; visualization in `10j_03_contrast_timeseries.jpg` (two-panel: zone centroids and the raw Impact-minus-Edge differential) and `10j_04_summer_minima_contrast.jpg` (per-well and tier-mean annual summer-minimum trajectories). Four site-observation entries written to `pipeline_site_observations.csv` via `update_site_observation()`. No figures are saved at PNG resolution; the JPEG quality follows the suite-wide convention.

Script 10j (v1.0.0): direct Impact-vs-Edge BACI contrast at monthly and annual-summer-minimum resolution.

Sub-scripts 10k and 10l — Four-zone pooled-panel BACI

Motivation. The three-counterfactual ANCOVA in 10a fits the same model six times, once per (control, zone) pair, and each fit has its own dependent variable — the BACI displacement series is the target centroid minus a control centroid, and the control centroid differs between the Forest, Climate, and Combined configurations. The six step estimates therefore do not share a coefficient vector or a covariance matrix, and they are not subtractable: a reader cannot legitimately recover “Impact relative to Edge” by differencing the Forest $\times$ Impact and Forest $\times$ Edge steps, because the two fits do not share a baseline. Sub-script 10j addressed this for the two-zone Impact-vs-Edge case by stacking the Impact and Edge wells into a single long-form panel and reading the differential felling step off one interaction term. Sub-scripts 10k and 10l generalise the 10j method from two zones to four, so that every zone-vs-zone contrast comes from one internally-consistent fit. They are a methodological cross-check on the principal 10a analysis: 10a remains the principal clearfell BACI analysis of the report, and 10k/10l test whether its felling signal survives re-estimation in a single jointly-fitted panel.

10k is the monthly-mean four-zone model (Phase 1); 10l is the annual summer-minimum four-zone model (Phase 2). Both are built and live. They supplement, and do not replace, 10a (the principal three-counterfactual ANCOVA) and 10j (whose two-zone Impact-vs-Edge contrast cross-validates 10k’s derived Impact-minus-Edge contrast).

The four zones. All wells from four zones are stacked into one panel, and zone becomes a four-level factor with the Forest control as the reference level:

Zone	Wells	Role
Forest control (reference)	CEH32, CEH34, CEH33, NW10, CEH2	Mechanistically-matched counterfactual — same Corsican pine canopy and forest hydrogeology as the felled compartment
C3/Warren	CEH1, NW1, NW2, NW11	Second control — open western dune, expected to behave like the Forest control
Edge	CEH31, CEH20, CEH30,	Differential felling

Zone	Wells	Role
	CEH16	contrast — compartment edge
Impact	WMC3	Headline felling contrast — in-situ at the felled compartment

The Forest-control reference is the C4-interior unfelled set, so the headline Impact coefficient reads as a like-with-like Impact-minus-Forest contrast with the forest-versus-open-dune hydrological difference kept out of it. The C3/Warren zone is a newly-defined four-well set: it is the balanced subset of the C3 Western Residual cluster that reaches back to the panel baseline and lies more than 500 m from the felled compartment. NW13 and WMC4, two further C3 candidates, are dropped because their records begin in February 2012, some nineteen months after the panel baseline, which would leave the zone internally short-changed against the other three. The Coastal-control wells (CEH19, CEH17) are excluded from the four-zone panel entirely: they carry the coastal-retreat confound and are not a clean counterfactual.

C3/Warren is designated a **second control** zone, not a second impact-like zone. Forest-management perturbations at the warren propagate south-westward off the bedrock ridge; there are no C3 wells in that propagation sector, so the western-dune zone is hydraulically shielded from the felled compartment. The expectation is therefore that the C3/Warren differential felling step is near zero — a confirmatory result that the panel design is sound, not a finding. A clearly non-zero C3/Warren step would be treated as a flag (possible unshielded propagation or another confound), not as a felling effect.

Methodology. 10k fits a long-form monthly panel — one row per (well, month), every well in the four zones, every month from PRE_FELL_START (1 January 2011) onward. The dependent variable is the well’s own monthly water-table depth, not a differenced centroid; the zone factor and per-well fixed effects do the differencing inside one regression. The design carries a mean-centred cumulative-water-balance covariate and its three zone interactions, a binary April-2015 scraping indicator and its three zone interactions, a December-2017 felling indicator and its three zone interactions, a single network-wide easting × time covariate, and the per-well fixed-effects block. Estimation is by OLS with cluster-robust standard errors on well. The zone main effects are perfectly collinear with the well fixed effects — every well belongs to exactly one zone — and are absorbed rather than entered as separate columns. The three felling-

interaction coefficients (C3/Warren, Edge, Impact, each relative to the Forest reference) are the headline outputs. The October 2023 re-scraping is omitted from the primary model; the easting $\times$ time term is retained as a network-wide covariate and is a robustness control, not an erosion decomposition — the coastal-retreat magnitude is estimated independently by Script 25.

10l fits the same four zones on the annual Jun–Sep summer-minimum frame. The summer panel is an order of magnitude smaller — annual rather than monthly — and the scraping term is dropped at this resolution, as in 10j’s summer model, because the Impact tier carries too few pre-felling summers to support it. 10l is the committed Phase 2 of the four-zone design and reads its summer-minimum frame on the same provenance-aware basis as 10d.

Primary and derived contrasts. The four-zone model estimates three **primary** contrasts — the C3/Warren, Edge, and Impact felling steps, each relative to the Forest reference. These are direct model coefficients; their standard errors and p-values are interpreted normally. Every zone-vs-zone contrast — Impact-minus-Edge, Impact-minus-C3/Warren, Edge-minus-C3/Warren — is an exact linear combination of the three primary coefficients (for example, Impact-minus-Edge equals the Impact step minus the Edge step). These **derived** contrasts have exact point estimates and the arithmetic identity holds, but their standard errors depend on the covariance between the two coefficients being differenced. When two quiet, control-like zones are differenced their coefficients are strongly positively correlated, the variances partly cancel, and the derived contrast acquires an artificially narrow confidence interval and a small p-value that is not independent evidence of a felling effect. The live pairwise-contrast CSVs (10k_02_pairwise_contrasts.csv, 10l_02_summer_pairwise_contrasts.csv) deliberately segregate the primary p column from the derived p_derived column for this reason. The chapter — and the main-report §4.6 text — report the three primary zone-vs-Forest contrasts as the results; cite the Impact-minus-Edge derived contrast only with its 10j cross-validation as the warrant for it, not its p_derived; and never cite the C3/Warren-bearing derived contrasts’ p_derived as significance.

Monthly result (10k). The four-zone joint fit ($R^2 = 0.856$, $N = 2475$ well-months) gives a primary Impact-minus-Forest monthly felling step of **+0.032 m (+32 mm)**, **95 % CI [+0.010, +0.053], p = 0.004** — a significant post-felling rise in the felled compartment’s monthly-mean water table. The primary Edge-minus-Forest step is -0.030 m, $p = 0.128$ (not significant) and the primary C3/Warren-minus-Forest step is $+0.005$ m, $p = 0.647$ (not significant — the near-zero second-control result

the shielding design predicts). The Impact-minus-Edge derived contrast is +0.061 m; it is not cited via its `p_derived` but is corroborated independently by 10j's two-zone monthly Impact-vs-Edge step of +0.063 m, $p = 0.0002$ — the agreement between the two estimators, which sample the same response through different designs, is a validation of the four-zone fit. The easting-sensitivity refit confirms the headline is robust to the covariate: dropping the easting $\times$ time term moves the Impact step only from +32 mm to +33 mm with the significance unchanged.

The four-zone joint Impact-minus-Forest step (+32 mm) is smaller in magnitude than the +120 mm Forest $\times$ Impact step from the principal 10a ANCOVA. The two estimators are not in conflict: they differ in how climate variance is partitioned. The joint panel estimates climate sensitivity from the full cross-zone monthly record, with each zone carrying its own cumulative-water-balance interaction, so a portion of the monthly variance that the separately-fitted differenced-centroid ANCOVA attributes to the felling step is instead attributed to climate in the joint fit. The cross-check is consistent with 10a in the respects that matter for the report's conclusion: the felling response is positive and statistically significant under both estimators, and only the magnitude is sensitive to the partitioning choice.

Summer result (10l). The four-zone summer-minimum joint fit ($R^2 = 0.724$, $N = 181$ well-years) gives a primary Impact-minus-Forest summer-minimum step of **−0.007 m (−7 mm)**, 95 % CI [−0.023, +0.008], $p = 0.36$ — **a null**. The primary Edge-minus-Forest summer step is −0.029 m, $p = 0.162$ and the primary C3/Warren-minus-Forest step is +0.012 m, $p = 0.257$ — neither significant. No zone contrast resolves a significant differential felling response at the summer minimum. The summer-minimum result is the more conservation-relevant of the two: the Jun–Sep minimum is the driest, most ecologically critical point of the year and the level against which the Curreli SD15b / SD16 slack thresholds are defined. The two-part four-zone finding is therefore that the felling produced a measurable rise in the felled compartment's monthly-mean water table but no detectable change at the summer minimum; both parts are reported with equal weight.

Inputs. Wells, climate, master, locations, and the valid tier dictionaries are loaded via `clearfell_common.load_clearfell_data()`; 10l additionally reads the summer-minimum frame from `10d_01_summer_minima.csv`. The four zones, the `C3_WARREN_WELLS` constant, and all dates are sourced from `clearfell_common`; the cumulative-water-balance series is built via `clearfell_common.compute_cwb()`. No site-specific values are hardcoded.

Outputs. 10k writes the three-row primary results to `10k_01_four_zone_results.csv`, the six ordered pairwise contrasts to `10k_02_pairwise_contrasts.csv` (with the primary/derived `contrast_type` column), the easting-sensitivity refit to `10k_03_easting_sensitivity.csv`, the standard ReportNumbers CSV to `10k_report_numbers.csv` (new `FourZone_*` keys, no collision with 10a's `ANCOVA_*` or 10j's `ImpactVsEdge_*` keys), and three figures — zone-centroid hydrographs, the differential series stacked against the Forest centroid, and a coefficient forest-plot of the six pairwise contrasts. 10l writes the parallel summer-minimum set (`10l_01_four_zone_summer_results.csv`, `10l_02_summer_pairwise_contrasts.csv`, `10l_report_numbers.csv`, and the summer trajectory and forest-plot figures). All figures follow the suite-wide JPEG convention.

Script 10k (v1.0.0): four-zone monthly pooled-panel BACI. Script 10l (v1.0.0): four-zone summer-minimum pooled-panel BACI.

Supplementary — Sub-script 10c — Forest zone analysis

Marked supplementary in `run_10_clearfell.py` and written to a separate output directory (`outputs/10c_forest_zone_analysis/`) rather than the suite's main directory. The analysis asks whether the C4/C5 partition reflects a substrate or topographic transition or is arbitrary within a continuous gradient. Per-well SSM coefficients from Script 07 are regressed against three spatial predictors (DEM elevation, distance from ridge crest at $E = 241750$, $N = 364500$, distance from coast); the C4–C5 boundary is mapped with elevation context; and the clearfell treatment wells (FE1–FE4, WMC3, LIS1) are located in β_1 – β_2 space against the surrounding forest cluster. The outputs feed Table 16 in the main report and provide context for the suite's treatment of C4 and C5 as mechanistically distinct tiers in 10a. The analysis is not part of the §4.6 clearfell result and is not given further weight in this chapter.

Methodology (suite-level) — `clearfell_common.py`

The suite-shared module sits at `src/utis/clearfell_common.py` and provides everything that needs to be consistent across the twelve sub-scripts. Changing an impact-well list, an intervention date, or a distance-weight function in one place propagates to all of them.

Well tier definitions. Five module-level lists define the canonical network:

```
IMPACT_WELLS = ['wmc3']; EDGE_WELLS = ['ceh31', 'ceh20', 'ceh30',  
'ceh16']; FOREST_CONTROL_WELLS = ['ceh32', 'ceh34', 'ceh33', 'nw10',  
'ceh2']; COASTAL_CONTROL_WELLS = ['ceh19', 'ceh17'];
```


CLIMATE_CONTROL_WELLS = ['ceh9', 'nw7', 'nw6', 'nw5', 'wmc2']. Concatenated as ALL_NETWORK_WELLS (17 wells). A TIERS dict groups them for iteration. The lowercase names match the column-name normalization that load_clearfell_data() applies on import. A sixth list, C3_WARREN_WELLS = ['ceh1', 'nw1', 'nw2', 'nw11'], defines the shielded western-dune second-control zone used by the four-zone sub-scripts 10k and 10l; it is not part of the 17-well clearfell network and is not in ALL_NETWORK_WELLS.

Intervention dates. INTERVENTION_DATE = 2017-12-01; SCRAPING_DATE = 2015-04-01; SCRAPING_DATE_2 = 2023-10-01. FELLING_YEAR = 2017. These three dates appear throughout the suite as era boundary masks (wells.index >= INTERVENTION_DATE, etc.) — the suite has no WELL_ERAS dict; era boundaries are inline masks against the date constants.

Data loading. load_clearfell_data() returns (wells, climate, master, well_locations, valid_tiers). It merges the reference and extended well frames, lowercases the column names, applies clean_well_series() to each column, reads climate with parse_dates=True, parses 03_master_data.csv, and builds well_locations from the master data (with a Well_locations_height.csv fallback for wells not in the master). It also validates the five tiers, emitting a warning for any well in a tier that is missing from the data.

BACI helpers. compute_baci_displacement(wells, target_list, control_list) returns the target-centroid minus control-centroid timeseries. compute_cwb(climate, baseline_start, baseline_end) returns the centred cumulative (P – PET) anomaly in mm. compute_control_centroid() returns the monthly mean of an arbitrary control list.

Distance-weighted scraping covariate. distance_from_ceh36() and distance_from_fell_centroid() are the two Euclidean distance functions, using CEH36_EASTING = 241161, CEH36_NORTHING = 363306 and FELL_CENTROID_EASTING = 241210, FELL_CENTROID_NORTHING = 363607. scraping_weight(d, λ) returns $\exp(-d / \lambda)$ with default SCRAPING_DECAY_LAMBDA = 300 m. distance_weighted_scraping() builds the per-well covariate: zero before the scraping date, exponential weight after. build_scraping_covariate_centroid() builds the centroid covariate as the mean of per-well weights across a tier — used by 10a and 10h to construct the differential scraping covariate.

Summer-minimum extraction. annual_summer_minimum(series, start_year, end_year) returns {year: float} of June–September minima, requiring at least

two observations per year (SUMMER_MONTHS = [6, 7, 8, 9]).

`forest_control_centroid_summer_min()` builds the multi-well summer-minimum centroid, requiring at least two wells per year by default.

BACI-corrected β_2 multiplier. `load_clearfell_b2_multiplier(verbose=True)` reads `10e_01_coefficient_shifts.csv`, computes per-tier mean `b2_after / b2_before` ratios, and returns the BACI-corrected clearfell multiplier `Edge_ratio - Climate_Ctrl_ratio + 1.0` and the thinning multiplier `1.0 + (clearfell - 1.0) / 2.0`. Fallback values of 1.10 (clearfell) and 1.05 (thinning) are returned if the 10e file is unreadable. The function returns three values — multiplier, thinning multiplier, tier-ratios dict — so all callers can inspect provenance.

Reporting utility. `ReportNumbers` is a simple accumulator class with `add(parameter, value, unit, well, era, note)` and `save(path)`. Every sub-script uses it to write a per-script `*_report_numbers.csv` exported into the main directory; `run_10_clearfell.py` then merges all per-sub-script files into `10_consolidated_report_numbers.csv`.

Site-specific choices and rationale (suite-level)

- **The 17-well BACI network excludes FE1–FE4, NW8, and NW8B.** FE1–FE4 lack pre-clearfell baselines (FE1 and FE2 start July 2015; FE3 and FE4 in 2017). NW8 and NW8B have data-quality issues in their pre-intervention records. CEH42 is excluded by length-of-baseline rule (3.4 years pre-felling, below the threshold). The exclusions are codified in the network constants — there is no separate exclusion list.
- **Single impact well in the published headline (WMC3).** Like the scraping suite (S.6) where the impact-zone is CEH36 alone at the published level, the clearfell suite has structural $n = 1$ at the published Impact tier. 10h's synthetic extension via FE1/FE2 donor regression is the principal robustness check; both synthetic-extension variants (Variant A and Variant B) return positive, individually significant or marginally so Forest-control steps, confirming the direction and sign of the WMC3-only headline but at smaller magnitude.
- **Three control structures (Forest, Climate, Combined), not a single counterfactual.** Each isolates a different aspect of the clearfell signal. Forest controls for canopy effects; Climate is the unconfounded climate baseline; Combined pools the three control tiers for tighter intervals at the cost of mechanistic specificity. The three are reported together rather than reduced to one because they answer different questions and the agreement (or disagreement) across them is itself the headline. There is no standalone

“Coastal Forest” counterfactual — the C5 wells enter only inside Combined, because the C5 tier carries a confounding coastal-retreat signal whose explicit handling is more clearly done by the differential easting $\times$ time correction inside the ANCOVA than by treating C5 as a third counterfactual.

- **Easting $\times$ time correction in the ANCOVA.** Coastal retreat at Newborough is episodic and acute (approximately 50 m of retreat between 2014 and 2020, with the acute loss during Storm Brendan in January 2020). The easting $\times$ time correction in the BACI ANCOVA is a physically real spatial-gradient term, included automatically where the easting range across the wells in a single (target, control) pairing exceeds 200 m. For the Forest-control runs this fails (the C4 wells all sit close together in easting) and the term is dropped; for the Climate and Combined runs it is included and is significant at $p < 10^{-10}$ in every case. The fuller treatment of the coastal-erosion gradient is in chapter S.15 (Script 25); the easting $\times$ time term in 10a is the suite’s local handling of the same signal.
- **Climate-correction wells for the spatial maps (10b) are intentionally different from the ANCOVA climate controls (10a).** 10b uses the four-well western subset NW5, NW6, NW7, CEH1 because these share the western climate signal and coastal-retreat position with the impact zones, absorbing both climate and coastal drift in a single median subtraction. 10a’s full five-well C3 set is used in the ANCOVA where the explicit covariate structure handles the climate signal separately. The two purposes — spatial correction for a map, full covariate handling for a step estimate — call for different reference sets.
- **Fixed-membership control centroid.** As of `clearfell_common v1.7.0`, each control centroid is computed only over months in which every roster well has a value; a month with any roster well missing is excluded from the BACI series. The earlier implementation took a NaN-skipping monthly mean, so the centroid silently re-weighted whenever a control well went offline. The dominant case was the joint outage of NW10 and CEH2 from September 2011 to September 2012, during which the Forest-control centroid was the mean of only CEH32, CEH33, and CEH34. The fixed-membership rule removes that artefact. It lowers the Forest $\times$ Impact annual clearfell step from +0.135 m to +0.120 m (both $p < 0.001$) and the model R^2 from 0.37 to 0.27; the result remains positive and highly significant. A coordinated change in the same `clearfell_common v1.7.0` commit migrated the shared `PRE_FELL_START` cutoff from 1 July 2010 to 1 January 2011, so the legacy 10-series sub-scripts share the four-zone scripts’ pre-felling start; the audit established this as a consistency change rather

than the artefact fix (the 2010 install-ramp months were already outside the ANCOVA window after the cumulative-water-balance inner-join). The audit basis is `AUDIT_10series_PRE_FELL_START.md`.

- **Provenance-aware summer minima.** 10d consumes `01_wells_provenance.csv` (§S.1) and applies a `min_measured=2` rule to `annual_summer_minimum()`: a (well, year) summer minimum is admitted to the panel only when at least two of the four Jun–Sep months are actual field measurements. Single-month interpolations under `limit=1` are allowed to contribute and are flagged in the `n_interpolated` column of `10d_01_summer_minima.csv`. The rule excludes (well, year) combinations whose minimum would rest only on interpolated values. The same provenance file and rule is consumed by §S.6 sub-script 09c.

Limitations and known caveats (suite-level)

- **Single impact well in the published headline.** 10h’s synthetic extension partially mitigates this but the synthetic baselines themselves carry calibration uncertainty ($R^2 \approx 0.99$ across 29 months) and the assumption that the donor regression is stationary backward through pre-scraping years. The synthetic-extension variants confirm the direction and sign of the WMC3-only headline but return smaller magnitudes; the synthetic extension is read as a direction-and-sign robustness check rather than an independent magnitude estimator.
- **Mean monthly recovery is significant; summer minima are not.** This is the central scientific neutrality point for the suite. 10a’s Forest-control Impact step is +120 mm, $p < 0.001$; 10d’s Forest-control mixed-model Impact step is –1 mm, $p = 0.99$ — no improvement (and no penalty) in summer minima at the impact well. The Forest-control Edge summer-minimum mixed-model step is –64 mm, $p = 0.12$ — a negative shift that does not reach significance at $\alpha = 0.05$ against the Forest control, and that does not reproduce against the Climate-control baseline (+48 mm, $p = 0.28$). The mean-recovery result is real, and the summer-minima result is real; the chapter presents both with equal weight rather than emphasising the recovery and softening the null summer signal.
- **Six-year post-felling window with a coastal-erosion event mid-way.** Just over six years from December 2017 to the analysis cut-off, with Storm Brendan in January 2020 inside the post-felling window. The easting $\times$ time correction handles the spatial-gradient component but the temporal-step component of Storm Brendan is partially absorbed into the CWB $\times$ felling interaction term rather than being separately identified.

- **The β_2 multiplier exports a clearfell perturbation but uses no winter/summer seasonal split.** The current `load_clearfell_b2_multiplier()` returns a single annual-mean β_2 multiplier from the BACI-corrected Edge ratio. A seasonal split — winter ≈ 0.88 (leaves off), summer ≈ 1.11 (full leaf) for broadleaf, with a corresponding asymmetric profile for evergreen pine — would match the monthly profile that Script 21's Option 3 perturbation requires. The implications for §4.9.6 are flagged in the project's `BETA2_DECOMPOSITION_UPDATED.md`.
- **10c is supplementary, not part of the headline chain.** Its outputs feed Table 17 but the §4.6 clearfell result rests on 10a (with 10h's extension), supported by 10b, 10d, 10e, 10f, 10g. Readers should not treat 10c's per-well coefficient regression as evidence about the clearfell — it is evidence about the C4–C5 cluster boundary.
- **Forest β_2 uses a single annual mean in the suite-shared multiplier.** As above; flagged for future seasonal-split work.

Outputs (consolidated)

Sub-script	File	Contents
10a	10a_01_ancova_comparison_table.csv	6-row summary: 3 controls $\times$ 2 zones; clearfell step, CI, p, R^2 , N
10a	10a_02_ancova_full_coefficients.csv	Full model coefficients per (control, zone)
10a	10a_03_baci_timeseries.csv	Raw and climate-corrected BACI displacement timeseries
10a	10a_04..08*.png, 10a_S1..S3*.png	Forest-control 3-panel BACI figures, CUSUM, climate-sensitivity scatter; supplementary 3-control panels
10b	10b_spatial_step_data.csv	Per-well era means, scraping and felling step values, climate-corrected steps
10b	10b_spatial_*_raw.png, 10b_spatial_*_correct	Four spatial step-change maps

Sub-script	File	Contents
10c	ed.png 10c_forest_zone_correlations.csv	β coefficient regression results vs elevation, distance from ridge, distance from coast
10c	10c_forest_zone_cluster_summary.csv	C4 vs C5 t-test summary statistics
10c	10c_forest_zone_analyses/ 10c_01..04*.png/.txt	β_1 – β_2 scatter, β_2 elevation regression, boundary map, summary
10d	10d_01_summer_minima.csv	Per-well, per-year summer minimum with control centroids; n_interpolated column flags rows containing single-month interpolated Jun–Sep cells
10d	10d_02_summer_minima_shifts.csv	Per-well pre/post shifts, t-test summary
10d	10d_03_mixed_model_results.csv	Mixed-effects pooled step estimates per tier
10d	10d_04_summer_minima_forest_ctrl.png, 10d_05_*climate*.png	4-panel summer-minimum figures
10e	10e_01_coefficient_shifts.csv	Per-well Before / After β coefficients and $\Delta\beta$ across the five BACI tiers. Source for load_clearfell_b2_multiplier()
10e	10e_03_coefficient_shifts.png	Before/after coefficient figure
10f	10f_01_ssm_residual_results.csv	Per-well SSM forward-residual step estimates

Sub-script	File	Contents
10f	10f_02_synthetic_control_results.csv	Zone-level synthetic-control step estimates
10g	10g_01_nw10_broadleaf_trend.csv	NW10 summer-min anomaly trend data
10g	10g_02_clearfell_transsect.png, 10g_03_clearfell_transsect_steps.csv	Radial transect figure and step-vs-distance data
10g	10g_04_rolling_coefficients.csv	48-month rolling SSM coefficients (Impact, C3, C4)
10h	10h_01_synthetic_calibration.csv	Donor regression diagnostics (R^2 , RMSE, hindcast)
10h	10h_02_ancova_comparison_table.csv	Three variants $\times$ three controls ANCOVA results
10h	10h_03_ancova_full_coefficients.csv	Full coefficients per (variant, control)
10h	10h_04_baci_timeseries.csv	BACI timeseries per variant
10h	10h_05..10*.png	Donor regression validation, three-panel BACI per variant, CUSUM, sensitivity
10i	10i_01_ceh34_hindcast.csv	Spliced CEH34 series with source flag ('hindcast' or 'observed'); consumed by 10a/10b/10e/10h via <code>apply_ceh34_hindcast()</code>
10i	10i_02_donor_regression.csv	OLS fit parameters (α , β) and residual diagnostics for the $CEH34 \leftarrow CEH9$ regression
10i	10i_03_hindcast_diagnostic.png	Three-panel donor-

Sub-script	File	Contents
		regression diagnostic figure
10j	10j_01_monthly_contrast_results.csv	One-row summary of the monthly-mean Impact-vs-Edge contrast: coefficients, SE, CI, p, R^2 , N
10j	10j_02_summer_contrast_results.csv	One-row summary of the annual-summer-minimum Impact-vs-Edge contrast (measured-only summers): coefficients, SE, CI, p, R^2 , N
10j	10j_03_contrast_time_series.jpg	Two-panel figure: zone centroids and raw Impact-minus-Edge differential
10j	10j_04_summer_minimum_contrast.jpg	Per-well and tier-mean annual summer-minimum trajectories
10k	10k_01_four_zone_results.csv	Three primary zone-vs-Forest monthly felling contrasts: step, SE, CI, p, scraping step, CWB interaction, R^2 , N
10k	10k_02_pairwise_contrasts.csv	Six ordered pairwise felling contrasts with a contrast_type column segregating primary from derived
10k	10k_03_easting_sensitivity.csv	Full model re-fit with and without the easting $\times$ time covariate — robustness diagnostic
10k	10k_04_..06*.jpg	Zone-centroid

Sub-script	File	Contents
		hydrographs, differential series vs Forest centroid, pairwise-contrast forest-plot
10l	10l_01_four_zone_summer_results.csv	Three primary zone-vs-Forest summer-minimum felling contrasts: step, SE, CI, p, well-year counts, R^2 , N
10l	10l_02_summer_pairwise_contrasts.csv	Six ordered pairwise summer-minimum contrasts with primary/derived contrast_type column
10l	10l_03_c3warren_summer_minima.csv	C3/Warren-zone per-well summer-minimum series
10l	10l_04_..05*.jpg	Zone summer-minimum trajectories, summer-minimum pairwise-contrast forest-plot
All	10[a,d,e,f,g,h,i,j,k,l]_report_numbers.csv → 10_consolidated_report_numbers.csv	All citable values from each sub-script, consolidated by run_10_clearfell.py

Where the result appears in the report

- §4.6 *Clearfell intervention* — entire section.
- **Table 7** (per the live PIPELINE_README.md table-mapping) — Clearfell ANCOVA-BACI results, source: 10a_report_numbers.csv.
- **Table 8** — Independent corroboration of the BACI ANCOVA easting × time covariate against the Script 25 coastal-retreat gradient model, source: 25_04_baci_corroboration.csv.
- **Table 9** — Per-well summer minimum shifts, source: 10d_02_summer_minima_shifts.csv.
- **Table 10** — Mixed-effects clearfell step by tier, source: 10d / mixed-effects output.

- **Table 11** — Before/after clearfell SSM coefficients for all 17 BACI-network wells, source: `10e_01_coefficient_shifts.csv`.
- **§4.6.3 / §4.6.4** — Four-zone pooled-panel BACI: the primary monthly clearfell result (10k) and the summer-minimum response (10l), sources: `10k_report_numbers.csv`, `10l_report_numbers.csv`.
- **Table 17** — Forest zone spatial predictors, source: `10c_forest_zone_analysis.py` (`10c_04_forest_zone_summary.txt`; underlying correlations in `10c_forest_zone_correlations.csv`).
- **Figure 23** — Scraping-era spatial step map (10b).
- **Figures 26–28** — CWB vs BACI displacement, Forest-control BACI Impact and Edge tiers (10a).
- **Figure 29** — Summer minima vs Forest control (10d).
- **Figure 31** — Clearfell-era spatial step map (10b).
- **Figure 32** — Before/after SSM coefficients across the 17-well network (10e).
- **Figure 33** — Clearfell radial transect, step vs distance (10g).
- **§4.6 robustness paragraph** — easting × time sensitivity and the direct Impact-vs-Edge contrast (10j); values sourced from `pipeline_site_observations.csv` (entries `impact_vs_edge_clearfell_monthly_step` and `impact_vs_edge_clearfell_summer_step`) rather than from a cached constant.
- **§4.6 / §5 conservation-management paragraph** — seasonal asymmetry between the monthly-mean and summer-minimum Impact-vs-Edge contrasts (10j), corroborating the 10d summer-minima null result against the 10a annual-mean positive result.

Cross-references

- **F.3** — SSM equation, displacement formulation, sign conventions. 10e and 10f fit SSMs through this; 10e diverges from `fit_ssm()` by adding a scraping dummy for the Before fit.
- **F.4** — `pipeline_params.py` consolidation, including the BACI-corrected β_2 multiplier consumed by Scripts 19 and 21.
- **F.5** — `clearfell_common.py` brief role summary (full detail in this chapter's *Methodology (suite-level)* section). The `site_observations.py` registry is the route by which 10j's four contrast values, and the four-zone primary zone-vs-Forest steps from 10k and 10l, reach downstream consumers without caching.

- **S.1** — produces `01_wells_provenance.csv` consumed by 10d (and by 09c in §5.6) under the `min_measured=2` rule; 10j inherits the rule by restricting to `n_interpolated = 0` rows of 10d's summer-minima output.
- **S.3** — produces `03_master_data.csv` consumed by all sub-scripts.
- **S.6** — Script 09 scraping suite, the parallel intervention analysis. `scraping_common.py` is the analogous shared module; the structural parallel (single impact well in the published headline, climate-water-balance covariate, era-based design) is deliberate. Script 09b also reads `10b_spatial_step_data.csv` for the network-wide propagation modelling. Sub-script 09c shares the `min_measured=2` provenance-aware summer-minima rule with this chapter's 10d. The scraping suite does not have a direct analogue of 10j because the scraping intervention does not have an Edge tier as cleanly separable from its Impact tier.
- **S.12** — Script 17 WTF Sy, consumed in the volumetric translation used downstream (not directly in 10d; the volumetric step appears in §4.6 narrative via cluster-level Sy values).
- **S.14** — Script 21 forestry scenarios, which consume 10e's BACI-corrected β_2 multiplier via `load_clearfell_b2_multiplier()`.
- **S.15** — Script 25 coastal-retreat gradient, which informs the easting $\times$ time correction in 10a's ANCOVA. 10j's identification design is the corroborator that bypasses that correction; the two estimators are designed to be cross-checked rather than to agree by construction.

Phase 4 — Climate and Spatial Context

S.8 Scripts 00, 12, 13, 14 — Climate baseline and site-overview figures

Steps 13–16 / 27 (Phase 4 in `run_analysis.py`). Phase 4 — Climate and Spatial Context.

Motivation

The four scripts in this chapter produce the report's foundational orientation material: the climate baseline that §2 leans on for site setting, the DEM map that opens §2 as Figure 1, the experimental-design diagram that opens §3 as the BACI network figure, and the observed-trajectory figures in §5 that frame the climate-change discussion. None of these scripts produces a CSV that feeds another analytical chain — they are visualization and baseline-context outputs. The

chapter accordingly compresses the methodological detail and concentrates on what each script chooses to render and why.

Three of the four scripts (00, 12, 13) sit at Phase 4's start as orientation tools that run alongside the main analysis. Script 14 sits later in the phase, reading from `03_regional_averages.csv` to project cluster-centroid summer minima forward to 2040 and to evaluate observed winter maxima against the Curreli et al. (2013) flooding thresholds.

Inputs

File	Used by	Origin
<code>01_climate.csv</code>	00	Script 01
<code>01_wells_clean.csv</code>	00	Script 01
<code>data/RAF_Valley_Climate.csv</code>	00 (Figures 4, 5)	Raw climate input
<code>data/Well_locations_height.csv</code>	12, 13	Raw locations input
<code>data/newborough_dem.tif</code>	12, 13	Raw DEM input
<code>data/*.kml</code> (clearfell, broadleaf restock, etc.)	12, 13	Raw site features
<code>03_regional_averages.csv</code>	14	Script 03

Scripts 12 and 13 are the two pipeline scripts that legitimately read from `Well_locations_height.csv` directly rather than from a Script 01 intermediate. The exception is recorded in F.5: site-mapping figures need raw coordinates and the full locations register, including wells excluded from the analytical network, which the Script 01 intermediate does not retain.

Sub-script 00 — Climate summary

Script 00 generates two complete output sets in a single run via `_run_all()`. The **full-record** set covers the entire RAF Valley climate series (December 1930 to present, ~95 years) and produces the climate timeseries figure, well-network summary figure, summer warming-trend figure, and three matching CSV tables. The **monitoring-period** set repeats the climate and network outputs over the well-record overlap window (roughly 2004 onwards), restricting climate and wells to the date span where the reference network has observations. The two

filename groups are distinguished by a `_short` suffix on the monitoring-period outputs.

The well filter retains wells with at least `MIN_RECORD_MONTHS = 100` valid records up to `REFERENCE_CUTOFF_DATE`. Llyn Rhos-Ddu is hard-excluded by name because it is a lake-stage measurement rather than a water-table observation; this mirrors the `EXTENDED_NETWORK_BLACKLIST` in Script 01. Annual climate aggregation (`make_table1_annual_climate`) computes annual P, annual PET, and the P/PET ratio per year, with a `Months_complete` count and a long-term-mean summary row. The well-network table (`make_table2_well_network`) writes per-well record start/end, valid month count, mean, standard deviation, seasonal amplitude (August mean minus February mean), and the per-hydrological-year summer minimum and winter maximum.

The monitoring-period timeseries figure uses a four-panel layout (P, PET, cumulative levelled P–PET balance, network-mean well level) with an intervention line at January 2018 for the clearfell. The cumulative balance is detrended against the mean (P–PET) over a December 2004–December 2025 reference window (`DETREND_START`–`DETREND_END`); this removes the long-run climatological bias so the cumulative curve sits on zero rather than drifting linearly. The detrending mean is annotated on the figure for transparency.

The summer warming-trend figure (`00_03_summer_warming_trend.png`) is the only Script 00 figure that reads from the raw `RAF_Valley_Climate.csv` rather than from `01_climate.csv` — the pipeline-filtered climate file contains P and PET only, not maximum temperatures. The figure plots the JJA (June–July–August) mean maximum temperature for each year of the full 95-year record as red/blue bars relative to the pre-2013 mean, with a linear OLS trend overlay and the post-2013 mean as a horizontal reference. Only years with all three summer months recorded are used. The figure is produced on the full-record profile only — running it over the monitoring-period subset would remove the long baseline that makes the recent warming interpretable. Output CSVs accompany each figure; for the warming-trend figure the table includes the regression slope, intercept, R^2 , p-value, pre/post-2013 means, and the post-2013 anomaly. This figure appears in the main report as Figure 4 in §4.1.1 *Climate record*.

Sub-script 12 — Site overview map

Script 12 produces Figure 1 of the main report: a DEM hillshade over the full warren extent with every well in `Well_locations_height.csv` (currently 97 wells, including extended-network wells and lake-stage points) plotted as a red

marker with a name label. The script reads the raw locations CSV, projects to EPSG:27700 (British National Grid), and overlays the local 1 m DEM (newborough_dem.tif) using rasterio with a custom topographic colormap. The colormap's `set_under('dodgerblue')` paints any sub-zero elevation as water, and `TwoSlopeNorm` anchors the colour scale at 0 m sea level, 12 m at the slack-dune transition, and the true ridge maximum at the top. KML site features (forestry boundaries, broadleaf restock blocks, scrape sites) are added through `map_utils.add_kml_features` (F.5).

If the local DEM is absent or rasterio is not installed, the script falls back to a contextily OpenTopoMap basemap framed around the well bounding box plus a 300 m buffer. The fallback exists so the pipeline runs end-to-end on a reviewer's machine without the 1 m DEM; the published Figure 1 uses the local DEM render.

Sub-script 13 — Experimental design map

Script 13 produces the BACI / scraping design figure used in §3 of the main report. The well coordinates come from the same raw CSV as Script 12, restricted by a tight bounding box (240,500–243,500 E × 362,700–364,800 N) that frames the clearfell zone and the scraping sites without the open dune to the south-east. CEH12 is dropped on coordinate grounds (out-of-area).

Each well's experimental role is assigned by `assign_category` using a strict priority order: clearfell tiers first (Impact → Edge → Forest Control → Coastal Control → Climate Control), then BACI exclusions (FE1–4, LIS1, NW8B, CEH42), then scraping roles (impact, local paired control, regional control), then background. The clearfell tier definitions are imported directly from `clearfell_common.IMPACT_WELLS`, `EDGE_WELLS`, `FOREST_CONTROL_WELLS`, `COASTAL_CONTROL_WELLS`, and `CLIMATE_CONTROL_WELLS` (S.7); under the live network this gives $1 + 4 + 5 + 2 + 5 = 17$ BACI wells. Tier colours come from `clearfell_common.TIER_COLOURS` so the BACI map and the BACI analytical figures share a palette. Each tier gets a distinct marker shape (triangle, diamond, square, plus, circle) so the figure reads in greyscale as well as colour. The clearfell centroid, the felling boundary KML, and the CEH2→CEH34→WMC3 clearfell transect are overlaid; for the scraping analysis, dashed red lines connect each impact site to its local paired control.

The DEM hillshade underlay uses the same rasterio + custom topographic colormap as Script 12. The legend is grouped under three headings (clear-fell BACI, topographical scraping, analytical linkages) with empty `Line2D` separators acting as section breaks. Excluded BACI wells are drawn as grey crosses so the

reader can see where the rejected wells sit relative to the felling boundary — this is the figure’s most useful diagnostic feature for anyone wanting to understand why the BACI network excludes the FE wells.

Sub-script 14 — Climate projections

Script 14 reads `03_regional_averages.csv` (cluster-centroid hydrographs in m below ground) and computes per-cluster annual summer minima and winter maxima on a hydrological-year basis (year starts 1 October). Summer months are April–September; winter months are October–March; a minimum of three valid monthly readings is required for an annual minimum or maximum to be recorded. For each cluster, summer minima are fitted with `scipy.stats.linregress` and the trend is projected linearly from the first observation year to 2040, with a 95% confidence interval for the mean response (`fit_trend`). Winter maxima are computed and plotted as observation only — no projection is fitted on the winter side because the report’s framing is that summer minima are the climate-trajectory variable of interest, and winter exceedance is reported as an observed-frequency count against the Curreli flooding thresholds rather than as a forward projection. The observed-period winter trend (OLS slope, R^2 , p-value, n-years) is nonetheless computed for each cluster and persisted to `14_winter_trend_stats.csv`; it is descriptive only and carries no projection, but it is written to a pipeline CSV so that any winter-trend figures cited in the report are sourced from a pipeline output rather than recomputed by hand from the raw annual extremes.

The three figure outputs are `14_climate_trajectory_summer.png` (summer panel, all five clusters, with the SD15b wet-slack and SD16 dry-slack viability lines at -0.61 m and -0.98 m), `14_climate_trajectory_winter_flooding.png` (winter panel against the SD15b_WINTER and SD16_WINTER flooding lines at -0.10 m and -0.25 m, with an in-figure exceedance-frequency box), and `14_climate_trajectory_stacked.png` (the two panels combined). A fourth output is an interactive Plotly HTML scatter of mean annual summer minimum versus mean annual winter maximum per well, coloured by cluster, with the Curreli thresholds overlaid. This is published as an Online Supplementary Tool on the project’s GitHub Pages site (https://newbroman.github.io/Newborough_Hydrology/) rather than as a static report figure. Four CSV companions are written: `14_summer_trend_stats.csv` (per-cluster summer-minimum slope, R^2 , p-value, n-years), `14_winter_trend_stats.csv` (the equivalent per-cluster winter-maximum statistics, descriptive only), `14_annual_extremes.csv` (annual minima and

maxima per cluster), and `14_winter_exceedance.csv` (winter exceedance counts and percentages per cluster).

The framing in the report is deliberate: the OLS linear extrapolation is **not** a climate forecast — it is an observed-trajectory baseline against which the UKCP18-based forecast in Script 21 (S.14 of this supplement) can be compared. Script 14 answers “where is the cluster heading at its observed rate of change”; Script 21 answers “where could the cluster end up under a UKCP18 scenario.” The pairing is explicit in §5 of the main report.

Site-specific choices (suite-level)

- **Two profiles in Script 00 (full and short) rather than one.** The well-network analysis runs over 2004–present; the climate context the report’s §2 sets out runs over the full 95-year record. Producing both lets the report quote climate normals against either window without re-running the script. Both profiles are always generated; the function names retain the legacy “profile” terminology for clarity but there is no CLI flag to select one — the orchestrator gets both.
- **The summer warming-trend figure reads the raw climate CSV, not `01_climate.csv`.** This is the only Script 00 output that needs temperatures (P and PET alone do not give a warming signal); Script 01’s intermediate carries P and PET only. The departure from the “everything reads from intermediates” rule is intentional and limited to this one figure.
- **DEM optional, OpenTopoMap fallback.** Scripts 12 and 13 both produce a usable figure without the local DEM raster, so the pipeline runs end-to-end on a reviewer’s machine without the 1 m DEM. The published Figure 1 uses the local DEM.
- **Script 13 uses imported tier definitions, never literals.** The script’s BACI tier lists come from `clearfell_common.py`; changing a tier membership in one place automatically updates the design figure and the analytical scripts together. The excluded-well list (FE1–4, LIS1, NW8B, CEH42) is one of the few short literal lists in the script — these are the wells the clearfell analysis dropped on baseline-stationarity and small-n grounds (S.7).
- **Script 14 uses simple OLS rather than a more elaborate trend model.** The figure’s purpose is the observed-trajectory baseline; UKCP18 scenarios (consolidated in `pipeline_params.py` and used by Script 21 in S.14) carry the forecast.
- **No projection on the winter side.** The winter panel reports observed maxima against the Curreli flooding thresholds and an exceedance-

frequency count; it does not extrapolate. The asymmetric treatment matches the report's framing.

Outputs

File	Contents	Downstream consumer
00_climate_summary/ 00_01_climate_timeseries[_short].png	Monthly P, PET, balance, well level	Report \$2 (Figure)
00_climate_summary/ 00_02_well_network_summary[_short].png	Per-well record bar / summary	Report \$2 (Figure)
00_climate_summary/ 00_03_summer_warming_trend.png	RAF Valley JJA max-temp anomaly, 1931–2025	Report \$4.1.1, Figure 4
00_climate_summary/ 00_01_annual_climate_summary[_short].csv	Annual P, PET, P/PET ratio	Report \$2 (Table)
00_climate_summary/ 00_02_well_network_summary[_short].csv	Per-well record stats, summer min, winter max	Report \$2 (Table); Script 14 references OUT_00_WELL_NETWORK_TABLE
00_climate_summary/ 00_03_summer_warming_stats.csv	Annual JJA means + OLS trend parameters	Report \$4.1.1
12_figure_site_overview/ 12_01_dem_site_overview.png	DEM + 97-well site map	Report Figure 1
13_figure_experimental_design/ 13_01_experimental_setup_map.png	Five-tier BACI + scraping design map	Report \$3 (BACI design figure)
14_climate_projections/ 14_climate_trajectory_summer.png	Summer-minimum trajectories with 95% CI to 2040	Report \$5
14_climate_projections/ 14_climate_trajectory_winter_flooding.png	Winter-maximum observations vs Curreli winter thresholds	Report \$5
14_climate_projections/ 14_climate_trajectory_stacked.png	Combined summer + winter panel	Report \$5
14_climate_projection	Per-cluster summer-	Report \$5

File	Contents	Downstream consumer
s/ 14_summer_trend_stats .csv	minimum OLS parameters	
14_climate_projection s/ 14_winter_trend_stats .csv	Per-cluster winter- maximum OLS parameters (descriptive only)	Report §5
14_climate_projection s/ 14_annual_extremes.csv	Annual summer min / winter max per cluster	Report §5
14_climate_projection s/ 14_winter_exceedance. csv	Winter exceedance counts per cluster	Report §5
14_climate_projection s/ 14_seasonal_extremes_ scatter.html	Interactive per-well scatter, summer min vs winter max	Online Supplementary Tool (GitHub Pages)

All paths resolve through `utils/paths.py` (`OUT_00_*`, `OUT_12_DEM_OVERVIEW`, `OUT_13_EXPERIMENTAL_MAP`, `OUT_14_*`).

Where the result appears in the report

- §2 *Site setting* — Figure 1 from Script 12 (DEM render); climate-baseline figures and tables from Script 00.
- §3 *Methodology* — BACI design map from Script 13.
- §4.1.1 *Climate record* — Figure 4 (RAF Valley summer warming trend) from Script 00.
- §5 *Discussion / climate trajectory* — summer- and winter-trajectory figures and exceedance summary from Script 14; the Script 00 warming-trend figure (§4.1.1 / Figure 4) provides the long-baseline temperature context that §5 draws on.
- Online Supplementary Tool — seasonal-extremes scatter from Script 14, hosted at the project’s GitHub Pages site.

Cross-references

- **F.3** — SSM formulation that drives the cluster centroids that Script 14 reads.
- **F.4** — Curreli thresholds (`SD15b`, `SD16`, `SD15b_WINTER`, `SD16_WINTER`); cluster colours and labels; `REFERENCE_CUTOFF_DATE`.

- **F.5** — `map_utils.add_kml_features` rendering helper used by Scripts 12 and 13.
- **S.1** — `01_climate.csv` and `01_wells_clean.csv` consumed by Script 00.
- **S.3** — `03_regional_averages.csv` consumed by Script 14.
- **S.7** — `clearfell_common.py` tier definitions consumed by Script 13.
- **S.14** — Script 21 forestry scenarios, which use UKCP18 climate constants from `pipeline_params.py` for the forecast model that Script 14's linear extrapolation does not provide.

S.8.5 Script 14b — Bootstrap year-of-crossing (post-review addition, Phase 4 step 16)

Script 14b was added on 2026-05-29 following the post-review pass on the main report. Conclusion 11 of the report previously read “*summer minimum trends indicate C1 summer minima approaching the SD16 dry slack viability threshold around 2030–2032 under current trajectories*”, with the qualitative date band taken from a reading of Script 14's per-cluster summer-trend slopes but without a stated confidence interval. Script 14b replaces the qualitative band with a bootstrap CI on the crossing year per cluster × threshold, anchored in the same annual summer-minimum series that Script 14 produces.

Procedure. For each of the five clusters, the per-cluster annual summer-minimum series (`outputs/14_climate_projections/14_annual_extremes.csv` from Script 14) is fitted as a linear trend over the observed years. A non-parametric bootstrap ($n = 1000$ replicates) resamples years with replacement; for each resample the trend is refitted and the year at which the linear extrapolation crosses each Curreli threshold (SD15b = 0.61 m below ground, SD16 = 0.98 m below ground) is computed. The script tabulates 5th, 50th and 95th percentile crossing years per cluster × threshold and renders a five-panel figure showing observed minima, the OLS trend with 90% bootstrap CI cone, threshold lines and crossing-year CI bands.

Headline result (2026-05-29). The C1 SD16 crossing year, central to Conclusion 11, has a median of 2022 with a 90% confidence interval of 2017–2042 (5th–95th percentile of the bootstrap distribution). Four of the eight most recent observed C1 summer minima have already exceeded the SD16 depth (2018, 2019, 2022, 2025); 2023 sits just above the threshold at -0.92 m. The cluster is no longer approaching the threshold but oscillating across it. The wide confidence interval reflects substantial year-to-year variability in summer minima against a shallow trend slope of -9.7 mm yr^{-1} , not measurement uncertainty in the underlying

slope estimate. C2 has crossed SD16 in the trend sense around 2011, with the CI extending into the pre-monitoring period (the cluster's observed minima have been below SD16 throughout the record). C3 lies above SD16 and its non-significant trend reaches the threshold beyond the 2080 horizon. C4 and C5 sat below SD16 throughout the observed record — the forest-zone clusters do not host slack vegetation, so the threshold projection is reported for completeness rather than as an ecological signal.

Inputs. `outputs/14_climate_projections/14_annual_extremes.csv` (Script 14 output); Curreli thresholds from `utils.config` (SD15b, SD16).

Outputs. Sharing `paths.DIR_14` with Script 14: - `14b_year_of_crossing.csv` — per-cluster × threshold table with median + 5th/95th percentile crossing years, OLS slope, intercept, and current-year (2025) predicted depth. - `14b_year_of_crossing.png` — five-panel figure (one per cluster) showing observed minima, OLS trend, 90% bootstrap CI cone (5th–95th percentile), threshold lines and crossing-year CI bands. - `14b_year_of_crossing_results.md` — memo with headline table, decision rule, and caveats.

No new path constants were added in `paths.py`; Script 14b shares `paths.DIR_14` with Script 14 since the input lives there.

Limitations. Linear extrapolation. The bootstrap captures sampling uncertainty in slope and intercept but not model-form uncertainty (the assumption that the linear trend extrapolates cleanly into a regime where summer-min approaches a drainage-controlled basement or where climate trajectory diverges from observed). The per-cluster centroid summer-minimum averages over wells with different ground elevations within each cluster, so the threshold is an effective threshold against the centroid, not against any specific well. Year-resampling bootstrap preserves the trend signal but does not preserve year-to-year autocorrelation; for trends with strong autocorrelation a block bootstrap would be marginally wider. Inspection of the summer-minimum residuals against the per-cluster linear trends does not show strong autocorrelation in the C1 series, so a block bootstrap is unlikely to materially widen the CI.

Cross-references.

- §7 Conclusion 11 of the main report — the “around 2030–2032” qualitative date band is replaced by the bootstrap CI from this script (median 2022, 90% CI 2017–2042 for C1 SD16).

- §4.10.1 of the main report — the figure 14b_year_of_crossing.png lands here as supporting evidence for the seasonal-prediction discussion in §5.2.4.

S.9 Scripts 11, 11b — Forecasting and spatial thresholds

Steps 11 and 12 / 27. Phase 4 — Climate and spatial context.

The pair implements the §3.6 *Forecasting* methodology of the main report at two scales. Script 11 produces cluster-level temporal forecasts — peak winter flood and summer drought transfer functions, plus the iterated closed-form P_{flood} that is the chapter’s headline management tool. Script 11b applies the same threshold framework per-well, producing spatial maps of where the network sits relative to the Curreli et al. (2013) thresholds, the BACI-derived recovery limits, and the per-well P_{flood} surface (the §3.6.3 figure). The pair share the Curreli thresholds (F.4), the cluster β from Script 03’s mechanistic table, hydrological-year indexing, and the iterated P_{flood} derivation in `model_utils.pflood_lambda()`; they differ only in aggregation level. Refer to F.3 for the SSM, F.4 for the constants and partition, and F.5 for the shared utilities.

Inputs (suite-level)

File	Origin
03_03_cluster_mechanistic_coefficients.csv	Script 03 — cluster centroid $\beta_1, \beta_2, \beta_3$ and p-values
03_master_data.csv	Script 03 — per-well $\beta_1, \beta_2, \beta_3$ (used in 11b for reference wells)
03_regional_averages.csv	Script 03 — cluster centroid hydrographs plus monthly P_{mm} , PET_{mm}
03_cluster_peak_months.csv	Script 03 — per-cluster mean peak-of-record month
01_wells_clean_maod.csv	Script 01 — reference network maOD series (used by 11b)
01_wells_clean.csv	Script 01 — reference network depth-below-pipe-top series (used by 11b for per-well climatology in the forecaster bundle)
01_wells_extended.csv	Script 01 — extended network raw depths (used by 11b)

File	Origin
01_well_elevations.csv	Script 01 — DEM_Ground_Elev, Pipe_Top_Elev (used by 11b)
01_locations.csv	Script 01 — easting/northing (used by 11b)
06_pear_membership_audit_sitewide .csv	Script 06 — extended well cluster assignments
data/Features.kml, data/site_boundary.kml	Site overlay and interpolation domain

Sub-script 11 — Cluster-level temporal forecasting

Motivation

Script 11 takes Script 03's cluster-centroid SSM coefficients and answers five management-relevant forecasting questions. What rainfall is required to recover the slack from a known summer minimum (Section 2, empirical peak-flood transfer function)? What rainfall threshold triggers winter flooding to the slack floor (Section 3, the iterated P_{flood} , the chapter's headline)? Given a winter peak, how dry will the following summer get (Section 4, summer-drought transfer function)? Given the same winter peak, what will the *spring* mean be — the metric the van Willegen 2025 monitoring framework uses (Section 5, added at Script 11 v1.1.1; documented in full in S.18b as Tool A)? And — restated for the transcript — what are the cluster-centroid coefficients that everything downstream rests on (Section 1)? The sections are independent and write to separate output CSVs.

Methodology

Shared infrastructure. All four sections operate on 03_regional_averages.csv, which carries monthly cluster-centroid hydrographs (C1–C5 and block aliases Lake_Edge, Eastern_Block, Western_Block, Forest, Coastal_Forest — one alias per cluster under $k=5$) alongside monthly P_{mm} and PET_{mm} . Hydrological year indexing — $\text{Hydro_Year} = \text{year} + (\text{month} \geq 10)$ — keeps each recharge cycle (winter rainfall → spring peak → summer drawdown → autumn minimum) in one year. $\text{MIN_VALID_MONTHS} = 3$ requires three observed months in both Apr–Sep and Oct–Mar for a hydrological year to enter a regression. A `_Tee` context manager mirrors `print()` output to 11_forecast_01_results.txt so regression summaries are recoverable without re-running.

Section 1 — Mechanistic state-space equations. Reads

03_03_cluster_mechanistic_coefficients.csv, extracts cluster centroid β_1 , β_2 , β_3 , R^2 and p-values, and converts β_1 and β_2 from m/m to m/mm (divide by 1000) for dimensional consistency with the climatological P and PET in millimetres used by Section 3. β_3 is dimensionless and used as-is, with an `abs()` guard. The section is the bridge from S.3's analytical output to the §3.6 forecast equations; under the live partition all five clusters report.

Section 2 — Peak flood transfer functions. For each block column in the regional-averages frame, the script computes a per-hydrological-year triple — summer minimum ($h_{\min}$, Apr–Sep minimum of the centroid hydrograph), cumulative winter rainfall (P_{winter} , Oct–Mar sum of P_{mm}), and winter maximum (h_{peak} , Oct–Mar maximum) — and fits an OLS regression with intercept

$$h_{\text{peak}} = \beta_1 \cdot P_{\text{winter}} + \beta_2 \cdot h_{\min} + c$$

β_1 here is the regression slope on winter rainfall, not the SSM β_1 of Section 1 (the symbol-reuse is unfortunate but harmless because the regression is separately reported). Output `11_forecast_winter_transfer_functions.csv` (Table 6). Forecast use: given an observed summer minimum and a forecast winter rainfall (e.g. a UKCP18 percentile), predict the winter peak.

Section 3 — Critical rainfall thresholds (the iterated P_flood). The headline forecasting tool of the chapter. The April 2026 revision replaced an earlier single-step inversion of the SSM with an iterated closed-form solution, because the single-step form neglected winter PET ($\approx 30\%$ of winter atmospheric demand at RAF Valley) and implicitly treated the six-month recharge season as one SSM timestep — an approximation that produced physically implausible thresholds at the cluster extremes. The iterated form starts from the monthly recurrence with displacement-formulation drainage, written in collapsed form as

$$h(t) = \alpha \cdot h(t-1) + \beta_1 \cdot \lambda \cdot \bar{P}(t) - \beta_2 \cdot \bar{PET}(t) - \beta_3 \cdot D, \alpha = 1 - \beta_3$$

where $\bar{P}(t)$ and $\bar{PET}(t)$ are the long-term monthly climatologies indexed by calendar month, λ is the rainfall multiplier being solved for, and $D = \text{DRAINAGE_DATUM} = 3.7$ m. Iterating from a summer-minimum head h_0 over n months gives

$$h_n = h_0 \cdot \alpha^n + \beta_1 \cdot \lambda \cdot S_P - \beta_2 \cdot S_E - D \cdot (1 - \alpha^n)$$

with the drainage-weighted climatology sums

$$S_P = \sum_i \alpha^{(n-1-i)} \cdot \bar{P}(m_i), S_E = \sum_i \alpha^{(n-1-i)} \cdot \bar{PET}(m_i)$$

Setting $h_n = h_target = 0$ (the slack-floor / ground-surface target) and solving for λ :

$$\lambda = (h_target - h_0 \cdot \alpha^n + \beta_2 \cdot S_E + D \cdot (1 - \alpha^n)) / (\beta_1 \cdot S_P), P_flood = \lambda \cdot \sum \bar{P}_i$$

The $+D \cdot (1-\alpha^n)$ term in the numerator is the accumulated effect of the displacement-formulation drainage constant; without it the threshold is systematically too low. The forecast horizon for each cluster runs from October to that cluster's historical peak-of-record month, loaded from `03_cluster_peak_months.csv`. Under the current partition the horizons are short — three to five months — because Newborough's slacks peak in late winter rather than late spring. The closed-form is collapsed for spreadsheet use into

$$P_flood = A \cdot d + B$$

where d is the well's depth below ground at summer minimum entered as a positive metre value (so $h_0 = -d$); A and B are written out per cluster in `11_forecast_pflood_threshold_equations.csv` with the spreadsheet formula `string=(A·A2+B)`. A reviewer-friendly summary CSV `11_forecast_pflood_summary.csv` carries A , B , λ , horizon, peak month, and P_clim_total in fewer columns. The legacy single-step value is retained in column `P_flood_old_single_step_mm` for continuity but is not used downstream.

The implementation calls `model_utils.pflood_lambda()` (F.5), which is the single source of truth for the closed form and is also the function 11b calls for the per-well map. A negative or non-finite λ flags the target as unreachable from h_0 under positive rainfall, and the script writes NaN to `P_flood` for that cluster.

Section 4 — Summer drought transfer functions. The mirror-image regression: for each block, fit

$$h_min_summer = \beta_1 \cdot P_summer + \beta_2 \cdot h_max_winter + c$$

over the same hydrological-year indexing, predicting the summer minimum from antecedent winter peak and summer rainfall. Output `11_forecast_summer_transfer_functions.csv` (Table 7). The forecast use is to bound the drought risk implied by a known winter peak under a UKCP18 summer climate, complementing Section 2's flood-direction prediction.

Section 5 — Spring MSL transfer functions (Tool A). Tool A sits alongside the Phase 13 van Willegen 5-year MSL aggregation (Script 26, S.18). For each cluster, fit $MSL_y = \alpha \cdot h_max_winter + \beta \cdot P_win_to_spr + \gamma \cdot PET_win_to_spr +$

intercept using van Willegen’s hydrology year B (1 Jun $y-1$ to 31 May y ; see F.2). The response is the cluster-centroid mean of {March, April, May} water levels in hydrology year y ; the predictors are the cluster-centroid October-to-February maximum, the cumulative October-to-May rainfall, and the cumulative October-to-May PET. The fits read from `03_regional_averages.csv` — the same Method B baseline that Tool B’s UKCP18 projection uses (see S.18 §Method A and Method B aggregation). Output `11_forecast_spring_transfer_functions.csv` (Table 9) plus a per-cluster calibration scatter at `11_forecast_02_spring_calibration.png`. R^2 ranges 0.73 (Lake Edge) to 0.96 (Coastal Forest). A previous-MSL variant was tested at v1.1.0 and dropped at v1.1.1 on empirical grounds (previous-MSL R^2 0.18–0.44; coefficient non-significant at four of five clusters). The full Tool A treatment — coefficient table, manager workflow, rejected-variant discussion — is in S.18b §S.18b.2.

Site-specific choices (Script 11)

- **Block columns under $k=5$ are one per cluster, not three.** `BLOCK_COLUMNS` is [Lake_Edge, Eastern_Block, Western_Block, Forest, Coastal_Forest]. The historical “Eastern / Western / Forest” three-way split survives only as labelling; the regressions are fitted at cluster-centroid level.
- **Hydrological year starts 1 October.** Standard UK water-year convention.
- **MIN_VALID_MONTHS = 3 per season.** Empirical — large enough to suppress dropout-driven outliers, small enough not to discard years where a month’s reading was missed.
- **EXCLUDED_CLUSTERS is empty under $k=5$.** Legacy from the $k=6$ partition where two singleton clusters were excluded; both were cleaned out at the partition step (F.4). The hook is kept rather than the logic deleted.
- **$h_{\text{target}} = 0$ everywhere in Section 3.** Section 3 solves for the rainfall that brings the water table to the ground surface (slack-floor flooding), not the Curreli SD15b/SD16 winter flooding levels. Solving for those is a one-line change to `model_utils.pflood_lambda`, but the report quotes the slack-floor figure as the management-relevant restoration threshold.

Outputs (Script 11)

File	Contents
<code>11_forecasting_thresholds/11_forecast_01_results.txt</code>	Full transcript of the five-section run
<code>11_forecasting_thresholds/11_forecast_winter_transfer_functions.csv</code>	Table 12 — winter peak prediction equations per cluster

File	Contents
11_forecasting_thresholds/ 11_forecast_summer_transfer_functions.csv	Table 13 — summer drought prediction equations per cluster
11_forecasting_thresholds/ 11_forecast_pflood_threshold_equations.csv	Full iterated P_flood derivation (one row per cluster, 20+ columns). Detail behind report Tables 14/15; not itself a numbered report table
11_forecasting_thresholds/ 11_forecast_pflood_summary.csv	Reviewer-friendly summary: A, B, λ , horizon, P_flood per cluster
11_forecasting_thresholds/ 11_forecast_spring_transfer_functions.csv	Spring MSL transfer functions per cluster (Section 5; Tool A in S.18b). In the report this appears as an inline equation in §3.6, not a numbered table
11_forecasting_thresholds/ 11_forecast_02_spring_calibration.png	Per-cluster calibration scatter for Section 5; in-supplement figure for S.18b

Limitations (Script 11)

- **Transfer-function forecasts inherit the regression residuals.** Sections 2, 4, and 5 are empirical linear regressions on ~20 hydrological years per block. The R^2 per block reported in the CSVs should be quoted alongside any forecast use.
- **The iterated P_flood uses cluster-centroid β only.** Within-cluster heterogeneity in $\beta_1, \beta_2, \beta_3$ is not resolved at this scale; the 11b per-well map is the refinement where it matters. The closed form also assumes uniform rainfall scaling and climatological PET — real winters depart from both, but the SSM's linearity means a more elaborate stochastic forecast would not change the threshold magnitudes much.

Sub-script 11b — Per-well spatial threshold maps

Motivation

Script 11b applies the threshold framework per-well. The core map products are a summer-minima depth map (binned against the Curreli SD15b/SD16 thresholds and the BACI recovery limits), a winter-maxima depth map (against the Curreli winter flooding thresholds), and the P_flood spatial map — the per-well iterated rainfall threshold from §3.6.3, the chapter's principal spatial figure. A flood-

frequency map, a Table 10 spreadsheet export, and an interactive forecaster HTML are written alongside.

Methodology

Shared infrastructure. `load_well_data()` returns a single per-well DataFrame combining reference-network maOD series (with cluster assignments from `03_master_data.csv` and per-well SSM coefficients) and extended-network wells (with cluster assignments from `06_pear_membership_audit_sitewide.csv` and no per-well β). For each well, `_summer_mins()` returns August–September minima per year and the across-years mean is the well’s mean summer minimum; depth below ground is `DEM_Ground_Elev` minus that mean. `_winter_maxima()` computes the October–March maximum per hydrological year with a three-month-minimum-data threshold. Two wells (CEH18, CEH21) scraped in October 2023 are handled by reducing the DEM elevation by the scrape depth (0.50 m and 0.70 m respectively, listed in the script’s `SCRAPED` dict). The full observed record is used for the maOD water-table averaging: maOD readings are invariant to ground-surface scraping (the water table sits at the same absolute elevation before and after material is removed), so pre-2023 maOD observations remain physically valid alongside post-2023 ones. Only the DEM (subtracted to compute depth-below-ground) is era-specific.

Summer minima depth map (`plot_summer_minima_map`). Per-well summer-minimum depths are binned against five zones derived from the Curreli thresholds plus the BACI recovery margins: wet-slack viable (< 0.61 m), SD15b-recoverable (0.61–0.75 m), SD16 dry slack (0.75–0.98 m), SD16-recoverable (0.98–1.20 m), and beyond single-scraping recovery (> 1.20 m). The recovery limits encode the BACI scraping benefit of +0.143 m at CEH36 (S.6) — a shallow scrape recovers ~0.14 m, a deeper scrape ~0.22 m — so a well within those increments of the relevant Curreli threshold is flagged as recoverable. An interpolated background surface is rendered by `map_utils.add_idw_surface()` on the standard 50 m site grid, with the project-standard ridge mask suppressing cells where the DEM exceeds the IDW-interpolated well DEM by more than 1 m (inter-dune ridges that lie between wells and would otherwise be coloured by an unrepresentative interpolation).

Winter maxima depth map (`plot_winter_maxima_map`). Same approach at the winter peak. The five-zone summer scheme collapses to four winter zones because the Curreli winter framework is shorter: flooding (peak above the surface, depth < 0 m), SD15b winter met (0–0.10 m), between SD15b and SD16 winter (0.10–0.25 m), and below SD16 winter (> 0.25 m). The map identifies slacks

where the winter peak no longer reaches the level required by the Curreli wet/dry typology.

P_flood spatial map (plot_pflood_map). The chapter's principal figure. For each well, the iterated closed-form is solved by `model_utils.pflood_lambda()` with the well's mean summer minimum as h_0 ($h_0 = -\text{depth_bg}$ under the SSM sign convention), the cluster-specific October-to-peak-month horizon, and the RAF Valley monthly climatology averaged from `03_regional_averages.csv`. The β coefficients are selected by a two-tier hybrid: per-well $\beta_1, \beta_2, \beta_3$ from `03_master_data.csv` for reference-network wells, falling back to cluster-centroid β from `03_03_cluster_mechanistic_coefficients.csv` for extended-network wells that have no per-well fit. Wells with non-finite P_flood or $\lambda < 0$ are flagged unreachable, excluded from the surface, and written to the per-well CSV with diagnostic columns (`alpha`, `S_P_mm`, `S_E_mm`, `lambda`, `unreachable`). The interpretive frame on the map is the climatological mean winter rainfall of 521 mm (`MEAN_WINTER_RAINFALL_MM`): wells whose P_flood is below 521 mm receive enough rainfall in a mean winter to flood the slack; wells above 521 mm need a wetter-than-average winter, with `P_flood / 521` indicating how much wetter.

Flood-frequency map and P_flood spreadsheet / forecaster.

`plot_flood_frequency_map` computes, for wells with at least five hydrological years of winter-maxima data, the fraction of years where the winter maximum reached the well's DEM (ground surface); rendered as a per-well percentage on the same 50 m grid. The map is the observational companion to P_flood: P_flood is what rainfall *would* flood under climatology, the flood-frequency map is how often it *has* under the observed record. `export_table10_spreadsheet()` re-exports Script 11's full P_flood derivation as the report's compact Table 10 (cluster, horizon, $A \cdot d + B$, P_clim total, spreadsheet formula).

`build_forecaster_html()` writes a single-page interactive tool that presents the report's three forecast equations as a calculator at cluster level: Forecast 1 uses Table 6 (winter peak from summer minimum), Forecast 2 uses Table 7 (summer minimum from winter peak), Forecast 3 uses Table 10 (P_flood from depth). Each forecast renders in three lines — general equation, substituted values, result — so the user sees exactly which published equation is being evaluated and with what inputs. The May 2026 simplification (`CHANGELOG_forecaster_simplification.md`) removed an earlier per-well SSM-iteration mode that the report does not validate; the forecaster is now cluster-level only by design.

The data bundle that backs the forecaster is built by

`_build_forecaster_data_bundle()` and carries: the cluster coefficients ($\beta_1, \beta_2, \beta_3$,

peak_month, slope_A, intercept_B, P_clim_total) merged from Script 03's mechanistic table and Script 11's Table 8; the block transfer functions from Tables 6 and 7; the monthly RAF Valley P climatology; the well metadata (location, DEM, cluster assignment, nearest_cluster_only flag); and per-well twelve-month depth climatologies computed from 01_wells_clean.csv (wells with fewer than 24 monthly observations or missing any calendar month are skipped and the well-details panel falls back to its cluster's climatology). The per-well climatology lets the panel show each well's own typical depths — this month, summer minimum at the well's trough month, winter peak at the well's peak month — above the matching cluster row, so the user can see how a specific dipwell sits relative to its cluster typical. The trough and peak months are computed per well and can differ from the cluster's (CEH9, C2, troughs in September against the C2 cluster trough in August, for instance). _raf_valley_winter_mean fetches the live Met Office RAF Valley monthly record at build time and falls back to 521 mm if the fetch fails.

Site-specific choices (Script 11b)

- **August–September summer-minimum window** rather than the Apr–Sep window used in Script 11 Section 2. The narrower window captures the ecologically critical late-summer minimum more tightly; the broader window is appropriate for the hydrological-year regression structure but would dilute the late-summer signal on a per-well status map.
- **DEM corrections for CEH18 (–0.50 m) and CEH21 (–0.70 m)**, both scraped October 2023. Corrected ground elevation is used for depth-below-ground. The full observed maOD record is used for the water-table averaging (maOD is invariant to surface change); only the DEM is era-specific.
- **Recovery limits derived from the BACI scraping benefit.** SD15b_REC = 0.75 m and SD16_REC = 1.20 m encode the +0.143 m demonstrated benefit at CEH36 (S.6) as a per-well recovery target.
- **Hybrid β architecture for the P_flood PNG map.** Per-well β for reference-network wells (full per-well dynamics), cluster-centroid β fallback for extended-network wells (cluster dynamics, well-specific h_0). The split is data-driven: extended wells have shorter records and were not refitted individually in Script 03.
- **Forecaster is cluster-level only.** The interactive forecaster presents the report's three forecast equations — Tables 6, 7, 10 — as a calculator at cluster level. An earlier per-well SSM-iteration mode was retired in the May 2026 simplification (CHANGELOG_forecaster_simplification.md) because

the report validates predictions only at cluster level. The per-well structure that does survive in the forecaster is the long-term depth climatology displayed in the well-details panel: each dipwell's own typical depths shown above its cluster's, so users can read where a specific well sits relative to its cluster typical.

- **50 m IDW grid** for all surface interpolations — the project-standard grid shared with Scripts 07, 19, 20, 25 (F.5).
- **NEAREST_CLUSTER_ONLY_WELLS = {ceh3, ceh4, ceh7, ceh8, ceh37}** — five wells flagged in the forecaster as “nearest-cluster pattern match” because their hydrograph correlates with a canonical cluster centroid but they sit outside the SSM operational domain (tidal boundary, upstream exclusions). The label is informational; P_flood is still computed.
- **Site-boundary KML preferred for the interpolation mask** (data/site_boundary.kml, the dissolved SAGA stream-cell boundary), with a rectangular sea-boundary fallback.

Outputs (Script 11b)

File	Contents
11b_spatial_thresholds/ 11b_01_summer_minima_depth.png	Per-well summer-minimum depth map, five Curreli + recovery zones
11b_spatial_thresholds/ 11b_02_winter_maxima_depth.png	Per-well winter-maximum depth map, four Curreli winter zones
11b_spatial_thresholds/ 11b_03_pflood.png	Per-well iterated P_flood spatial map (the §3.6.3 figure)
11b_spatial_thresholds/ 11b_03_pflood_per_well.csv	Per-well P_flood, λ , α , S_P, S_E, h_0 , coeff_source, unreachable flag
11b_spatial_thresholds/ 11b_04_flood_frequency.png	Per-well winter-flooding-frequency map (% of years reaching surface)
11b_spatial_thresholds/ 11b_05_table10_pflood_spreadsheet.csv	Report Table 15 — cluster-specific linear P_flood forms (cluster A, B, horizon, P_clim total); legacy table10 filename token
11b_spatial_thresholds/ forecaster.html	Interactive single-page forecaster tool

Limitations (Script 11b)

- **Per-well DEM elevations carry sampling uncertainty.** The DEM is a LiDAR raster; per-well point elevations are sampled from it. The error is

small relative to the Curreli threshold spacing but is not zero, particularly on inter-dune slopes where the raster gradient is steepest.

- **The recovery limits assume scraping at the per-well location is operationally feasible.** Many wells flagged as “recoverable” sit inside the SSSI core, on protected vegetation, or in locations where mechanical scraping is impractical. The map is a hydrological recovery surface, not a management feasibility surface.
- **The BACI scraping benefit is a single-site mean.** The +0.143 m at CEH36 from Script 09a is the empirical evidence base for the recovery margins. Transferability to other locations within the warren is plausible but not directly demonstrated.
- **The 0.22 m SD16 recovery increment is an operational extrapolation, not a direct empirical observation.** The 0.14 m SD15b recovery margin is anchored on the +0.143 m mean benefit at CEH36 directly. The 0.22 m SD16 recovery margin instead represents the assumption that a *deeper* scraping than the CEH36 intervention would achieve a proportionately larger benefit — closer to the maximum observed within-record gain rather than the mean. Both anchors should be read as planning targets rather than as point predictions: shallow excavation is taken to achieve ~0.14 m of recovery on the SD15b side, and a deeper excavation to achieve ~0.22 m on the SD16 side, with neither value carrying a quantified uncertainty band.
- **The IDW surface is a piecewise-linear triangulation, not a true inverse-distance weighting.** `map_utils.add_idw_surface` is named “IDW” but calls `scipy.interpolate.griddata` with `method='linear'` — Delaunay triangulation with linear barycentric interpolation. For zone-status reading, the per-well markers are authoritative; the surface is a visual aid.
- **Per-well β for the P_flood map can be noisy at network edges.** Wells with short records produce per-well β values with wide CIs in `03_master_data.csv`; the P_flood map does not propagate this uncertainty.

Site-specific choices and limitations (suite-level)

- **One threshold framework, two scales.** The Curreli thresholds (F.4), the BACI recovery margins, the iterated P_flood derivation, and the `model_utils.pflood_lambda` implementation are shared between the two scripts. Script 11 produces cluster-level temporal forecasts; Script 11b produces per-well spatial status. Anything keyed by cluster ID — peak months, mechanistic β — flows from Script 03 to both.
- **Iterated closed-form P_flood.** Both scripts use `model_utils.pflood_lambda`. The single-step formulation is retained only

in Script 11's output CSV column `P_flood_old_single_step_mm` and is not used downstream.

- **The `P_flood` threshold is a closed-form linearisation of the SSM.** Exact under the linear recurrence with uniform rainfall scaling and climatological PET; departures from linearity (the saturation behaviour as the water table approaches the ground surface, unmeasured ridge-recharge subsidies at C4/CEH14) are not captured. The threshold should be read as the rainfall that would flood the slack *under the SSM's assumed dynamics*, not as a hydrodynamic forecast.
- **`MEAN_WINTER_RAINFALL_MM = 521`** is the 2005–2026 monitoring-period Oct–Mar mean. Climatologies from the full Met Office record back to 1930 differ slightly. The forecaster fetches the live record at build time and recomputes; the static script value is the report-cited figure.
- **One peak-month per cluster.** Within-cluster heterogeneity in peak timing is not resolved by the cluster-level peak-month CSV. Wells in the spatial map inherit their cluster's peak month as the horizon end.

Outputs (consolidated)

File	Source	Contents
11_forecasting_thresholds/ 11_forecast_01_results.txt	11	Transcript
11_forecasting_thresholds/ 11_forecast_winter_transfer_functions.csv	11	Table 6, winter transfer functions
11_forecasting_thresholds/ 11_forecast_summer_transfer_functions.csv	11	Table 7, summer transfer functions
11_forecasting_thresholds/ 11_forecast_pflood_threshold_equations.csv	11	Table 8, full <code>P_flood</code> derivation per cluster
11_forecasting_thresholds/ 11_forecast_pflood_summary.csv	11	Reviewer-friendly <code>P_flood</code> summary
11b_spatial_thresholds/ 11b_01_summer_minima_depth.png	11b	Summer minima zone map

File	Source	Contents
11b_spatial_thresholds/ 11b_02_winter_maxima_depth.png	11b	Winter maxima zone map
11b_spatial_thresholds/ 11b_03_pflood.png	11b	Per-well P_flood spatial map
11b_spatial_thresholds/ 11b_03_pflood_per_well.csv	11b	Per-well P_flood with diagnostics
11b_spatial_thresholds/ 11b_04_flood_frequency.png	11b	Winter flooding frequency map
11b_spatial_thresholds/ 11b_05_table10_pflood_spreadsheet.csv	11b	Table 10, spreadsheet-ready
11b_spatial_thresholds/ forecaster.html	11b	Interactive forecaster tool

Where the result appears in the report

- §3.6 *Forecasting* — Script 11 methodology and Tables 6, 7, 8, 9 (Table 9 = Section 5 spring MSL transfer function, Tool A in S.18b).
- §3.6.3 *P_flood management tool* — the iterated closed-form derivation; report Tables 14 and 15 (per-cluster P_flood summary and cluster collapsed linear equations); and the per-well P_flood map from 11b.
- §4 — Curreli-zone spatial maps from 11b (summer minima and winter maxima depth maps).
- §4 — winter flooding frequency map from 11b.

Cross-references

- **F.3** — SSM equation form, drainage datum, sign conventions; the basis for the iterated P_flood derivation.
- **F.4** — Curreli ecohydrological thresholds (SD15b, SD16, SD15b_WINTER, SD16_WINTER, SD15b_REC, SD16_REC), MEAN_WINTER_RAINFALL_MM, cluster colours and labels. The new MSL-aggregation constants subsection covers MSL_SPRING_MONTHS, MSL_HYDRO_YEAR_START_MONTH, and the MSL_MIN_* strictness rules consumed by Section 5's hydrology-year-B partitioning.

- **F.5** — `model_utils.pflood_lambda()` (shared by both scripts), `map_utils.add_idw_surface()` for the 50 m grid, `paths.py` for all output paths.
- **S.3** — Script 03 produces `03_03_cluster_mechanistic_coefficients.csv` and `03_cluster_peak_months.csv` consumed by Script 11, and `03_master_data.csv` consumed by 11b for per-well β . The cluster-centroid `03_regional_averages.csv` is the data source for Section 5’s spring MSL fits (the Method B baseline in S.18 §Method A and Method B aggregation).
- **S.6** — Script 09a’s +0.143 m BACI scraping benefit at CEH36 is the basis for 11b’s recovery-limit values ($SD15b_REC = SD15b + 0.14\text{ m}$, $SD16_REC = SD16 + 0.22\text{ m}$).
- **S.8** — Script 14’s seasonal-extremes scatter is the per-well summer-min vs winter-max companion to 11b’s static threshold maps.
- **S.18** — Script 26’s observational MSL5 metric is the 5-year-mean target that Section 5’s annual MSL forecast feeds into; managers add a Section 5 prediction to the rolling four-year history of observed MSLs to update the MSL5 statistic without waiting for end-May.
- **S.18b** — Full Tool A treatment: §S.18b.2 carries the per-cluster coefficient table, the manager workflow, and the rejected previous-MSL variant discussion. Tool B (Script 26b) consumes the same cluster β coefficients as the SSM-coefficient summary in Section 1 of this chapter to project $\Delta MSL5$ under UKCP18 RCP8.5 scenarios.

S.9.3 Script 11c — P_flood achievability map (post-review addition, Phase 3 step 13)

Script 11c was added on 2026-05-29 following the post-review pass on the main report. Conclusion 4 of the report names a priority criterion for scrape-target identification — “*priority targets are the C1/C2/C3 transitional wells where the aquifer base is stable and P_flood thresholds remain achievable (rainfall multiplier $\lambda < 1.5$)*” — but the report does not surface a per-well lookup against this criterion. Script 11c operationalises the criterion by reading the per-well P_flood multipliers already produced by Script 11b (`11b_03_pflood_per_well.csv`) and emitting a per-well categorical map on the canonical site DEM + KML overlay. The script consumes existing Script 11b output and adds no new model fitting; it is a presentation-layer step.

Procedure. Each of the 88 wells in the classified network is binned on its λ value into three bands:

- **Achievable** ($\lambda < 1.5$) — reachable in normal-to-mildly-wet winters.
- **Marginal** ($1.5 \leq \lambda < 2.5$) — reachable only in wet winters.
- **Unreachable** ($\lambda \geq 2.5$) — effectively unreachable under current climate.

The bin edges follow Conclusion 4’s explicit $\lambda < 1.5$ boundary for the achievable band; the marginal-vs-unreachable boundary at $\lambda = 2.5$ is selected to match the abstract’s reference to a 1.5–2.5× rainfall multiplier band as the conservatively wet-winter zone. Wells are rendered as colour-coded markers (green / amber / red) on the canonical DEM hillshade with KML feature overlays (forest boundary, broadleaf restocking block, clearfell footprint, site features), using the same `load_dem_hillshade()` and `add_kml_features()` helpers as Script 11b. Reference and Extended wells are marker-shape-distinguished (circle / diamond).

Headline result (2026-05-29). The categorisation produces a clean operational separation between the open-dune and forest zones. Of the 65 wells in C1 + C2 + C3 (the open-dune clusters), 57 are achievable, 8 are marginal, and none are unreachable. Of the 23 wells in C4 + C5 (the forest clusters), only 3 are achievable, 16 are marginal, and 4 are unreachable. The four unreachable wells are concentrated in C5 Coastal Forest (three wells) and C4 Main Forest (one well). The cluster pattern reflects the underlying mechanism: open-dune clusters carry higher β_1 recharge efficiency (2.5–4.6) and lower summer-minimum baselines; forest clusters carry canopy interception losses and lower β_1 (1.32–2.55), and C5 additionally carries the Section 4.8 coastal-retreat gradient pushing its summer-minimum baseline progressively further below the Curreli thresholds.

Inputs.

- `outputs/11b_spatial_thresholds/11b_03_pflood_per_well.csv` (Script 11b output).
- DEM hillshade and KML features from `data/`, via `map_utils.load_dem_hillshade()` and `map_utils.add_kml_features()`.

Outputs. Sharing `paths.DIR_11B` with Script 11b — outputs sit alongside 11b’s per-well CSV with `11c_` prefix:

- `11c_pflood_achievability.png` — the operational categorical map for §5.9 and Conclusion 4 of the main report.
- `11c_pflood_achievability_per_well.csv` — per-well lookup table with the category column (Achievable / Marginal / Unreachable).
- `11c_pflood_achievability_results.md` — memo with summary tables (counts per cluster × category), report drop-in text, and caveats.

Path constants (new). `paths.OUT_11C_ACHIEVABILITY_MAP`,
`paths.OUT_11C_PER_WELL`, `paths.OUT_11C_RESULTS_MEMO`.

Limitations. The λ values come from Script 11b's per-well calculation; they inherit Script 11b's assumptions about cluster β coefficients and the climatological winter rainfall baseline. The categorical bin edges (1.5 and 2.5) are operational choices, not derived from natural breaks in the data; Conclusion 4's text explicitly identifies the $\lambda < 1.5$ boundary, and the marginal-vs-unreachable boundary at $\lambda = 2.5$ is selected to match the abstract's wet-winter framing. The achievability category describes only whether the cluster summer minimum can be raised above the Curreli threshold by winter recharge alone — scrape-as-drainage geometry effects (S.6) and forest-management interventions (S.14) are separate degrees of freedom in the scenario framework. Wells flagged as scraped in the existing per-well CSV (CEH36, CEH18, CEH21) retain their categorical assignment based on present-day λ ; the category reflects post-intervention behaviour where applicable.

Cross-references.

- §5.9 of the main report — the figure `11c_pflood_achievability.png` and the per-well lookup table land here as the operational figure for Conclusion 4.
- §7 Conclusion 4 of the main report — the cross-reference is added to point the reader at the new figure and CSV.

S.10 Script 15 — Depth-dependent PET sensitivity

Step 17 / 27. Phase 5 in `run_analysis.py`; second chapter under Phase 4 — Climate and Spatial Context in the supplement.

Motivation

The canonical SSM (F.3, Script 03) fits a single β_2 per cluster: PET draws the water table down at a rate proportional to monthly PET, with no dependence on how deep the water table currently sits. This is a strong assumption. Physically, evapotranspiration reaches the saturated zone through capillary rise from the root zone, and capillary connectivity weakens as the column between root zone and water table lengthens. A deep water table should be drawn down by PET less efficiently than a shallow one.

Script 15 tests this by replacing the fixed β_2 with a depth-decaying term, $\beta_2 \cdot \exp(-\lambda \cdot d)$, where d is the depth below ground surface and λ is a free

parameter (m^{-1}) fitted per cluster. The script is a sensitivity analysis, not a replacement model: the canonical fixed- β_2 SSM remains the published headline. What Script 15 establishes is the *bound* on how much PET draw the fixed- β_2 formulation may be over- or under-estimating across the network — a physically interpretable bracket on the headline model. The §5.3 discussion of Thornthwaite-PET limitations in the main report draws on this work; Scripts 19 and 21 consume β_2 in their spatial and forestry-scenario calculations and inherit the same caveat.

Inputs

Filename	Origin
01_wells_clean.csv	Script 01 — cleaned, upstand-corrected monthly dipwell series
00_climate.csv	Script 00 — monthly P and Thornthwaite PET (RAF Valley, 53.25 °N)
02_cluster_stats.csv	Script 02 — k=5 cluster membership
01_locations.csv	Script 01 — well coordinates
01_well_elevations.csv	Script 01 — per-well upstand heights

Methodology

Modified SSM. The depth-coupled equation is

$$\Delta h(t) = \beta_1 \cdot P(t) - \beta_2 \cdot \exp(-\lambda \cdot d(t-1)) \cdot \text{PET}(t) - \beta_3 \cdot (D + h(t-1))$$

The β_1 rainfall and β_3 drainage terms are unchanged from the canonical SSM (F.3). The β_2 atmospheric-draw term is modified by an exponential decay factor in $d(t-1)$, the depth below ground at the start of the month. The functional form has two important properties: at $\lambda = 0$ the model reduces exactly to the canonical SSM ($\exp(0) = 1$), and as λ grows the PET draw is progressively attenuated when the water table is deep. The decay factor is a multiplier on β_2 , not a replacement: a cluster's PET sensitivity at the surface is still β_2 , but the effective β_2 at depth d is $\beta_2 \cdot \exp(-\lambda \cdot d)$.

Two uses of depth, one term. The script's docstring is emphatic that d (depth below ground, used in $\exp(-\lambda \cdot d)$) and the displacement $D + h$ (used in the β_3 predictor column) are not the same quantity. The displacement is referenced to the drainage datum 3.7 m below ground (F.3) and is the hydraulic head available to drive Darcy drainage; d is referenced to the ground surface and is the physical

distance the capillary fringe must reach. The two coincide only in the trivial case where the datum equals the ground surface. Conflating them — for example, by passing displacement into the exponential — would change the physical meaning of λ from “decay length of capillary connectivity from the soil surface” to “decay length above the drainage datum”, which has no obvious interpretation.

Cluster-centroid construction. Per-well fitting was rejected in favour of cluster centroids: the depth-decay signal is weak relative to the headline SSM at any single well, and aggregating to cluster level both stabilises the fit and matches the granularity of the downstream consumers (β_2 enters Scripts 19 and 21 at the cluster level). `build_cluster_centroids()` reads the $k=5$ membership from `02_cluster_stats.csv`, upstand-corrects each member series to ground reference (subtracting the well-specific upstand from the pipe-relative depth), and takes the mean across members for each monthly index. The cluster-mean upstand — the average of member upstand heights — is carried alongside the centroid so the depth-coupling term can convert head values to depths-below-ground consistently across the cluster. Cluster-mean upstands are small (0.06–0.14 m) and dominated by per-well manufacturing variation rather than meaningful spatial signal.

Grid search. `grid_search_lambda()` scans λ over $[0, 6] \text{ m}^{-1}$ in steps of 0.05 — 121 grid points per cluster. At each λ , the term $-\exp(-\lambda \cdot d_{\{t-1\}}) \cdot \text{PET}(t)$ is a known column once $d_{\{t-1\}}$ is fixed, so β_1 , β_2 , β_3 are recovered by no-intercept OLS on a three-column design matrix (`beta_1_recharge`, `beta_2_atmospheric_draw`, `beta_3_drainage`). Physical-sign filtering follows: a grid point is accepted only if all three fitted β values are positive. Grid points with $\beta_3 \leq 0$ are dropped — these are statistical artefacts where the depth-coupling has shifted the OLS solution onto an unphysical drainage term. The accepted (β, λ) pair is then forward-simulated iteratively (`iterative_simulate()`, identical recurrence to Script 08 and `model_utils.simulate_ssm()` apart from the decay factor in the β_2 step), and the iterative Nash–Sutcliffe efficiency (NSE) is computed against the observed centroid. Best λ per cluster is the grid point that maximises iterative NSE.

Why iterative NSE rather than OLS R^2 . One-step OLS R^2 is a diagnostic of how well the model fits each month’s Δh conditional on the previous month’s *observed* state. It is dominated by the climate-driven variance — both the canonical and depth-coupled forms share the same P and PET inputs, so the one-step R^2 gap between them is small by construction (S.5 documents this for the canonical-vs-TLM comparison). Iterative NSE simulates the trajectory forward from a single initial condition, propagating the depth-coupling feedback at every step: a deeper

simulated water table this month produces less PET draw next month, which feeds back into the depth term, and so on. This is the metric where the depth-coupling either earns its parameter or does not. The script inherits both the metric and the implementation idiom from Script 08 (S.5).

Fit window. The script uses the most-recent 100 months (`DATA_LIMIT = 100`), matching the LCSC window of Script 03 (S.3). This privileges recent data — under steady-state climate the choice would be immaterial, but Newborough’s water-table levels have trended downward over the record (S.3 documents the LCSC empirical estimate at 100 months in this context), so using the same window keeps Script 15’s β_2 estimates comparable to the canonical β_2 from `03_master_data.csv`.

Site-specific choices and rationale

- **λ grid range $[0, 6] \text{ m}^{-1}$.** The lower bound is the canonical-SSM limit: $\lambda = 0$ returns the fixed- β_2 model exactly, so the null hypothesis (no depth coupling) is a special case in the search. The upper bound is generous: at $\lambda = 6 \text{ m}^{-1}$ and a water-table depth of 1 m, $\exp(-6) \approx 0.0025$, so PET draw is effectively suppressed. No cluster’s best λ approaches this bound, so the choice of upper limit does not bind any result.
- **Grid step 0.05 m^{-1} .** This resolution is finer than the cross-cluster spread of best λ values (0.15–1.90), so quantising at 0.05 does not blur any cluster’s position. A finer step would change reported λ values at the third significant figure at most.
- **Cluster centroids rather than per-well fits.** Per-well fitting was considered but not implemented. The depth-coupling signal is small in absolute terms ($\Delta \text{NSE} = 0.028$ to 0.182 across clusters), and per-well noise would swamp it at most wells. Aggregating to cluster level is consistent with how β_2 is consumed downstream — Scripts 19 and 21 use one β_2 per cluster — and matches the canonical Script 03 outputs that the chapter benchmarks against.
- **Physical-sign filtering during the grid search.** The $\beta_3 \leq 0$ filter is the principle of failing loudly: a coefficient that flips sign as λ varies is a statistical artefact, and feeding such a fit into the iterative simulator would produce numerically valid but physically uninterpretable trajectories. In practice the filter removes the C4 band $\lambda \in [0.20, 0.95]$, where β_3 goes mildly negative — C4’s best λ at 0.15 sits at the boundary just below this band. C1, C2, C3, and C5 produce physically valid fits at every grid point.

- **Iterative NSE as the headline metric.** Quoted from S.5: “the iterative NSE is where the SSM’s structural advantage is unmasked”. The same logic applies here: the depth-coupling parameter exists to change the multi-step behaviour of the model, so the multi-step metric is the appropriate test.
- **Depth lag.** The decay term uses $d(t-1)$, the depth at the *start* of the month, not the contemporaneous $d(t)$. This is the same simultaneity argument that motivates `h_disp_prev` in the canonical β_3 term (F.3): using the result of the month’s drainage as a predictor of that drainage biases the fit. The depth-coupling formulation inherits the same lag convention without further argument.

Outputs

Filename	Contents
15_00_lambda_profiles_raw.csv	Per-cluster λ -grid sweep with β_1 , β_2 , β_3 , one-step R^2 , iterative NSE at each grid point
15_01_lambda_profile.png	One panel per cluster: iterative NSE vs λ , with dashed horizontal line at the canonical-SSM baseline
15_02_fit_comparison.png	One panel per cluster: observed centroid vs canonical SSM vs best depth-coupled fit (time series)
15_03_benchmark_table.csv	Per-cluster summary: SSM NSE, depth-coupled NSE, Δ NSE, best λ , fitted β values
15_04_best_params.csv	Per-cluster best-fit parameters including cluster mean upstand, alongside canonical β baselines for comparison

The principal downstream consumer is the §5.3 *Discussion / PET formulation* section of the main report. Script 15 outputs are not consumed by any other pipeline script — the depth-coupled model is sensitivity context, not a replacement that propagates forward.

Results to describe at the methodological level

The depth-coupled model improves iterative NSE over the canonical SSM at all five clusters. Differences are graded:

Cluster	Canonical SSM NSE	Depth- coupled NSE	Δ NSE	Best λ (m ⁻¹)	Mean upstand (m)
C1 Lake Edge	0.71	0.90	+0.18	1.90	0.06
C2 Dune	0.81	0.88	+0.07	0.80	0.07
C3 Western Residual	0.86	0.89	+0.03	0.40	0.10
C4 Main Forest	0.65	0.76	+0.10	0.15	0.14
C5 Coastal Forest	0.81	0.85	+0.04	0.50	0.10

Three observations on the pattern. First, the cluster where depth coupling improves the fit most is C1 Lake Edge — the cluster whose water table sits closest to the ground surface for the largest fraction of the record. A high λ at C1 (1.90 m⁻¹, the steepest decay in the network) means that the PET draw is heavily attenuated even at shallow depths: at $d = 0.5$ m, $\exp(-1.90 \cdot 0.5) \approx 0.39$, so the effective β_2 is reduced by more than half. The canonical fixed- β_2 at C1 has to compromise across the seasonal swing between shallow winter levels and deeper summer levels, and the depth-coupling formulation resolves that compromise by letting β_2 vary with the season as the water table drops. Second, the forest clusters (C4, C5) and the western dune (C2, C3) take smaller λ values — between 0.15 and 0.80. These clusters sit deeper on average, and a smaller λ reflects an effective β_2 that decays only modestly with depth across the working range. Third, C4 Main Forest's best λ at 0.15 m⁻¹ sits at a boundary in the OLS solution: $\lambda \in [0.20, 0.95]$ produces a negative β_3 that the physical-sign filter rejects, so the C4 result picks up the largest contiguous band of physically valid fits at the low- λ end. The headline number is robust, but C4 is the cluster where the depth-coupled and canonical formulations are closest to being statistically indistinguishable on coefficient signs alone.

The cross-cluster λ pattern is consistent with the capillary-connectivity hypothesis: clusters where the water table is shallowest on average show steeper decay (because the working range of d is small, a larger λ is needed to produce a meaningful difference between summer and winter PET draw), while clusters where the water table is deepest show shallower decay (the canonical fixed- β_2

already roughly approximates the effective PET draw across the cluster's working depth range). The pattern is suggestive rather than definitive — five points are not enough to fit a quantitative relationship between mean cluster depth and λ — and the depth-coupling formulation is one of several physically motivated alternatives. It is the simplest one-parameter form consistent with monotonic decay of connectivity with depth.

Limitations and known caveats

- **Cluster-centroid fitting only.** Per-well heterogeneity within a cluster — particularly the range of working depths within C2 Dune and C3 Western Residual — is not addressed. A well at the dry end of C2 may behave more like C3, and the cluster-centroid λ averages over this heterogeneity. A per-well sweep would clarify the partition between within-cluster and between-cluster variation in λ but has not been implemented.
- **Exponential decay is one of several plausible functional forms.** Power-law (d^{-p}), sigmoid ($1/(1 + \exp(k(d - d_0)))$), and piecewise-linear (constant above a threshold depth, decaying below) all have physical-mechanism arguments behind them. None has been tested against the exponential here. The exponential was chosen as the simplest one-parameter form consistent with monotonic decay; ranking alternative forms would require a model-comparison framework (AIC, cross-validation) that the script does not implement.
- **The PET data is Thornthwaite-derived.** The sensitivity analysis is to the PET *signal* as Thornthwaite represents it (mean monthly temperature, day length, latitude 53.25 °N), not to a fundamentally improved PET formulation. Penman–Monteith PET would change the absolute magnitudes of both β_2 and the inferred λ . F.4 covers the Thornthwaite choice; §5.3 of the main report discusses the trade-off explicitly.
- **The improvement in iterative NSE is graded, not transformative.** Three clusters (C2, C3, C5) gain Δ NSE between 0.03 and 0.07; two (C1, C4) gain more substantially (0.10–0.18). The canonical fixed- β_2 formulation is therefore an acceptable approximation for most of the network at most depths, and the depth-coupled model is a refinement rather than a fundamentally different description. This is the central reason Script 15 remains a sensitivity analysis: the gain does not justify retiring the simpler model.

- **C4 sits at a coefficient-sign boundary.** The best-fit λ at 0.15 is the last physically valid grid point before β_3 flips negative for a band of intermediate λ values. This is a feature of the OLS geometry at that particular cluster centroid and may not survive perturbations (alternative climate input, longer fit window, different cluster membership). The C4 result is reported but should not be over-interpreted as a quantitative estimate of the cluster's depth-coupling parameter.

Where the result appears in the report

- **§5.3 Discussion / PET formulation** — the Thornthwaite limitations discussion cites Script 15's depth-coupled λ values as evidence that the fixed- β_2 formulation is defensible across the network. The graded improvements (largest at C1, smallest at C3 and C5) appear in the limitations paragraph.
- **Not part of §4 headline results.** The canonical SSM (Script 03, fixed- β_2) remains the published model; Script 15 is sensitivity context for the §5 discussion.

Cross-references

- **F.3** — canonical SSM, displacement formulation, drainage datum, $P(t)$ under $HEADLINE_LAG = 0$.
 - **F.4** — cluster partition, $FOREST_CIDS = (4, 5)$, $DRAINAGE_DATUM = 3.7$ m.
 - **S.3** — canonical cluster-level β values that Script 15 benchmarks against.
 - **S.5** — Script 08's iterative NSE benchmarking idiom, which Script 15 inherits.
 - **S.13** — Scripts 19 and 20 spatial groundwater calculations consume β_2 at cluster level; the depth-coupling is sensitivity context for those.
 - **S.14** — Script 21 forestry scenarios consume β_2 ; the same caveat applies.
-

S.11 Script 16 — Water balance decomposition

Step 19 / 27. Phase 7 — Water Balance Decomposition in `run_analysis.py`; third chapter under Phase 4 — Climate and Spatial Context in the supplement.

Motivation

The state-space model fitted in S.3 gives each cluster three regression coefficients — β_1 , β_2 , β_3 — and a closure statistic (LCSC) that quantifies how much of the

monthly Δh variance the displacement formulation explains. Those coefficients carry the physics, but a reader looking at the numbers in isolation has no straightforward way to ask the question conservation managers actually want answered: *where does the water at this cluster go?* Script 16 answers that question. It reads the cluster-mean β coefficients, computes long-term mean P and PET from the regional-average series, and rewrites the SSM equation at its steady state — $\Delta \bar{h} \approx 0$ — to expose each component as a physical-units flux. Two outputs result: a head-space decomposition reported in m head per month, and a volumetric partition reported in mm per year. Both feed §4.2.3 *Water balance* of the main report and the two-panel Figure 8.

Inputs

File	Origin
03_regional_averages.csv	Script 03 (S.3); regional-mean cluster water-level series with merged P and PET monthly totals
03_03_cluster_mechanistic_coefficients.csv	Script 03 (S.3); per-cluster β_1 , β_2 , β_3 , LCSC, and the drainage datum that was used to fit them

Methodology

The head-space decomposition

The SSM displacement form (see F.3) is

$$\Delta h(t) = \beta_1 \cdot P(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot (D + h(t-1))$$

where $D = \text{DRAINAGE_DATUM} = 3.7 \text{ m}$ and $h_{\text{disp}}(t-1) = D + h(t-1)$ is the displacement of the water table above the drainage base at the start of month t . Taken at long-term means, $\Delta \bar{h} \rightarrow 0$: the water table is not drifting in either direction over the 21-year record. The closure condition becomes

$$\beta_1 \cdot \bar{P} \approx \beta_2 \cdot \bar{PET} + \beta_3 \cdot \bar{h}_{\text{disp}}$$

with the three terms in head-equivalent units (m head per month). Each term has a transparent physical reading. $\beta_1 \cdot \bar{P}$ is the **recharge** — the head added per month by mean rainfall, filtered through whatever fraction of P actually reaches the water table at this cluster. $\beta_2 \cdot \bar{PET}$ is the **evapotranspirative draw** — the head removed per month by mean PET demand. $\beta_3 \cdot \bar{h}_{\text{disp}}$ is the **drainage flux** — the head lost per month to lateral discharge, proportional to the cluster's mean displacement above the datum.

The function `compute_headspace()` does exactly this: it joins the per-cluster water-level series to P and PET on the date index, takes column means after `dropna()`, and computes $recharge = \beta_1 \cdot \bar{P}$, $et_draw = \beta_2 \cdot \bar{PET}$, $drainage = \beta_3 \cdot \bar{h_disp}$ for each cluster. The residual $recharge - (et_draw + drainage)$ is the closure check; in the current pipeline it sits between 0.03 % and 2.28 % of total losses, well inside the 2.5 % the script's docstring quotes as the design tolerance.

Each cluster receives the same forcing — $\bar{P} = 74.4$ mm/month, $\bar{PET} = 55.1$ mm/month from the RAF Valley series — but maps it into different head fluxes through its own β triplet. C1 Lake Edge converts that forcing into roughly 0.34 m/month of recharge, with drainage dominating losses (85 %, β_3 at the high end of the range and $\bar{h_disp}$ deepest under the lake-buffered cluster). C4 Main Forest, at the other extreme, converts the same $\bar{P}$ into only 0.19 m/month of recharge — the β_1 ratio of 2.5 to C1's 4.6 reflects the canopy interception and the deeper unsaturated zone — and the ET draw share rises to 76 % of losses.

From head-space to volume

The head-space components are millimetres of head per month. To compare them with the climate baseline they have to be expressed as millimetres of water per year. The current Script 16 does this directly through the partition fractions, without an explicit specific-yield (S_y) conversion. The reasoning is simple: at the closure condition $\beta_1 \cdot \bar{P} \approx \beta_2 \cdot \bar{PET} + \beta_3 \cdot \bar{h_disp}$, the head-space recharge is itself the head-equivalent of the net water flux into the cluster per month. Dividing the SSM losses into their two head-space components gives a partition fraction — the ratio of $\beta_2 \cdot \bar{PET}$ to total head-space loss is the ET share, and $\beta_3 \cdot \bar{h_disp} / \text{total}$ is the drainage share — and that partition fraction is a dimensionless ratio that transfers cleanly from the head-space to the volumetric description.

`save_volumetric_table()` implements the partition. Annual rainfall ($P_{\text{annual}} = 892.5$ mm/yr, the same for all clusters by construction) is reduced by interception at forest clusters ($I = 0.24 \times P$ per F.4 for C4 and C5; zero elsewhere), giving $P_{\text{net}} = P - I$. P_{net} is then split into ET and drainage according to the cluster's drainage fraction. The result is a per-cluster mm/yr breakdown that can be read alongside the 892.5 mm/yr rainfall input and the 661 mm/yr Thornthwaite PET ceiling.

Recession as an independent check

The head-space partition is one way to split the SSM losses. It rests on the displacement formulation and inherits whatever uncertainty attaches to β_2 and

β_3 separately. To bracket that uncertainty, `compute_recession_partition()` derives an independent partition from the observed seasonal behaviour of Δh itself.

The reasoning is that ET is strongly seasonal — high in summer, low in winter — while drainage is roughly proportional to head and varies more slowly. A winter month with low PET and a falling water table is approximately a drainage-only signal; a summer month with high PET and a falling water table carries drainage plus ET. Taking the average month-on-month decline ($\Delta h < 0$) in winter (Nov–Feb) and in summer (Jun–Sep), the ratio winter / summer estimates the drainage share of summer losses. That ratio is `drain_frac` in `recession[cid]`, with `et_frac = 1 - drain_frac`.

Two independent constraints on the same partition then exist: the SSM head-space ratio and the recession ratio. The script does not enforce convergence — it reports both and uses their midpoint as the central estimate, the range between them as a partition uncertainty band shown as the hatched area on the ET/drainage boundary of Figure 11b. Numerical agreement is cluster-specific: C3 Western Residual produces identical fractions under both methods (0.6/0.6, SSM/recession) and C5 Coastal Forest agrees closely (0.6/0.7), C1 Lake Edge moderately (0.8/0.6, a 20-percentage-point spread), and C2 Dune (0.7/0.4) and C4 Main Forest (0.2/0.5) the most discrepant, both at a 30-percentage-point spread. The figure shows that spread honestly rather than smoothing it; the bracket is the message.

Interception and the energy budget

Forest interception (F.4: `FOREST_INTERCEPTION = 0.24`, `FOREST_CIDS = (4, 5)`, per Freeman (2008) for Newborough’s Corsican pine canopy) appears on both bars of Figure 11b’s volumetric panel. On the rainfall-input bar it sits above the net-rainfall section, reducing the rainfall that actually reaches the soil ($P_{\text{net}} = 0.76 \cdot P$ at forest clusters). On the loss bar it sits above the ET and drainage losses, accounting for the canopy water that returns to the atmosphere without ever reaching the water table. The two appearances are identical and cancel in the net surplus — by design.

The cancellation matters because interception is a partition of the rainfall energy budget at the canopy, not an additive loss on top of soil ET. The Penman-monteith demand at the canopy is taken by intercepted water first; the soil-surface ET draw measured by β_2 implicitly operates on the $0.76 \cdot P$ that actually reaches the soil. Showing interception as an additive loss term would double-count it, so the

script subtracts it from both sides and lets the cancellation be visible. This is the project's standing convention (see F.4) and applies to C4 and C5 only.

Site-specific choices and rationale

- **Long-term means rather than monthly time series.** The water balance is a steady-state description: it asks what the cluster receives and disposes of on average over the record, not how it responds month-by-month. Year-to-year variability is the SSM's domain (S.3); the residual seasonality not captured by the SSM is taken up in S.16 (Script 24). The 21-year mean window the script uses is whatever the upstream `03_regional_averages.csv` carries — March 2005 to February 2026 for the wells with the fullest record, less for clusters whose member wells came online later. The mean is computed after `dropna()` so each cluster's window is its own complete-rows-only window.
- **No reference cutoff filtering inside Script 16.** The script trusts whatever the upstream regional-average file contains. Script 03 applies `REFERENCE_CUTOFF_DATE = 2026-02-01` (F.4), so the means inherited here are consistent with the SSM fit window. The β coefficients in `03_03_cluster_mechanistic_coefficients.csv` and the $\bar{P}$, $PE\bar{T}$, $\bar{h}$ used in this script are from the same window — a guarantee that the head-space balance closes against the same data the SSM was fitted on.
- **Datum imported from the upstream coefficients file, not from `config.py`.** Although `DRAINAGE_DATUM = 3.7 m` is the canonical value in `config.py`, `load_data()` reads the `drainage_datum_m` column from the coefficients CSV and uses that value, with a warning if it disagrees with `config.DRAINAGE_DATUM` by more than 1 cm. The point is to insulate the water-balance computation against any downstream change to the canonical datum: if a future sensitivity sweep re-fits the SSM at a different datum and writes a different value into the coefficients table, Script 16 will follow that value and produce a consistent decomposition, rather than silently mixing β values fitted at one datum with an `h_disp` computed at another.
- **No S_y conversion in the canonical implementation.** The current script's volumetric translation goes directly from the dimensionless SSM partition fraction to mm/yr via the rainfall input. Earlier draft versions of this script produced S_y -weighted volumetric outputs; they were superseded by the partition-fraction approach because the head-space ratio is itself the S_y -invariant quantity. Specific yield enters elsewhere in the pipeline — the

WTF analysis in S.12 — where its uncertainty is given its own treatment. Importing that uncertainty here would double-count it.

- **Forest interception 0.24, hardcoded centrally rather than per-script.** FOREST_INTERCEPTION and FOREST_CIDS are imported from config.py (F.4). The 0.24 value is Freeman (2008) at this site for Corsican pine; it is not a global canopy interception fraction. The two forest clusters (C4 Main Forest, C5 Coastal Forest) share the same canopy type and the same value applies to both.
- **Cluster-mean only, not per-well.** The water balance is at the cluster centroid. Per-well water balances are not produced because β_1 , β_2 , β_3 are noisier at the well level (S.3) and the residual closure tolerance the script enforces would not be met reliably for individual wells. The cluster-mean balance is the right scale for the methodological statement the report makes in §4.2.3.

Outputs

File (in outputs/16_water_balan- ce/)	Contents	Downstream consumer
16_water_bal_table.csv	Table 3a: head-space decomposition per cluster — β triplet, $\bar{P}$, PET , $\bar{h}_{disp}$, recharge, ET draw, drainage, residual, drainage and ET percentages	Report §4.2.3, Table 3a
16_water_bal_vol_table.csv	Table 3b: volumetric partition per cluster — P , PET , I , P_{net} , both partition methods (SSM, recession), midpoint and bracketing range in mm/yr	Report §4.2.3, Table 3b
16_water_bal_bar_ms.png	Water-balance manuscript version (300 dpi, white background, two-panel)	Report Figure 9

File (in outputs/16_water_balan- ce/)	Contents	Downstream consumer
16_water_bal_bar_lay.png	Figure 8 lay version (coloured background, 150 dpi)	Report companion / web version

Limitations and known caveats

- **The closure is steady-state.** Year-to-year variation, drought-recovery dynamics, and the response to single wet or dry years are deliberately absent from this analysis. They are the SSM's domain (S.3) and the residual-seasonality diagnostic's domain (S.16, Script 24). The water balance describes the average behaviour, not the operational range.
- **Thornthwaite PET inherits its own caveat.** The PET used in the head-space balance is the Thornthwaite estimate (F.4), known to underestimate atmospheric demand under forest canopies and at southern aspects on hot dry days. The depth-dependent PET correction in S.10 (Script 15) quantifies how much of the β_2 inter-cluster spread is attributable to substrate properties rather than canopy demand; the water balance presented here uses the canonical fixed- β_2 SSM and therefore inherits whatever direction the Thornthwaite bias pushes the cluster-mean ET share. The volumetric partition's central tendency is robust to the bias direction, but the *magnitude* of the ET draw at the forest clusters in particular is more uncertain than the numbers in Table 3b might suggest in isolation.
- **The SSM/recession bracketing is the partition's honest uncertainty band, and at some clusters it is wide.** C1 Lake Edge differs by 20 percentage points between the two methods (SSM 80 % drainage, recession 60 %); C2 Dune by 30 (SSM 70 %, recession 40 %); C4 Main Forest by 30 (SSM 20 %, recession 50 %). The midpoint reported as the central estimate masks that spread. A reader looking at the volumetric breakdown should treat the hatched range on Figure 11b as the substantive uncertainty, not a decorative band.
- **Specific yield uncertainty is not propagated here, but it does exist.** The choice to use the head-space partition fraction rather than a Sy-weighted volumetric conversion keeps this script's output Sy-invariant, but the upstream β coefficients themselves were fitted on water-level data — Sy enters implicitly through the relationship between water-level change and

recharge in the unsaturated zone. The decomposition presented here treats the fitted β values as fixed inputs; the full uncertainty around them is given proper accounting in the WTF chapters (S.12).

Where the result appears in the report

- **§4.2.3 Water balance** — the section is built around the two tables and Figure 8 from this script.
- **Table 3a** — head-space decomposition (`16_water_bal_table.csv`).
- **Table 3b** — volumetric partition with SSM/recession bracket (`16_water_bal_vol_table.csv`).
- **Figure 8** — two-panel water balance figure (`16_water_bal_bar_ms.png`).

Cross-references

- **F.3** — SSM displacement formulation; the equation Script 16 rewrites at steady state.
- **F.4** — DRAINAGE_DATUM, FOREST_INTERCEPTION, FOREST_CIDS, the centralised constants this script imports.
- **F.5** — `model_utils.py`; not called directly here but underpins the β coefficients consumed.
- **S.3** — produces `03_03_cluster_mechanistic_coefficients.csv` and `03_regional_averages.csv`, the script's only inputs.
- **S.10** — Script 15 depth-dependent PET sensitivity; relevant context for the Thornthwaite caveat above.
- **S.12** — WTF specific-yield analysis; the chapter that does carry the S_y uncertainty propagation Script 16 deliberately defers.
- **S.16** — Scripts 22, 23, 24 residual diagnostics; Script 24 specifically addresses the residual seasonality the steady-state water balance does not capture.

S.12 Scripts 17, 18 — WTF Specific Yield (cluster and spatial)

Steps 18 and 20 / 27. Phase 6 (Script 17) and Phase 8 (Script 18) in `run_analysis.py`; fourth chapter under Phase 4 — Climate and Spatial Context in the supplement.

The Water Table Fluctuation (WTF) method (Healy and Cook, 2002) translates an observed water-table rise into a recharge-equivalent volume, with specific yield (S_y) as the proportionality constant: $S_y = R / \Delta h$. The SSM in chapter S.3 characterises the *dynamic* response of the aquifer (the β coefficients), but its coefficients are in head units; converting any head-equivalent quantity into a

volumetric flux requires S_y . The WTF S_y is the only network-scale, spatially distributed S_y available for Newborough Warren. Betson and Bristow (2002) report a field-measured hydraulic conductivity of $K = 6 \text{ m day}^{-1}$ from a pumping test at the site — the source used throughout the pipeline for K — but do not report a S_y value from that test, and no slug or pumping test has been run at the dipwell locations themselves. A representative S_y cannot be read off the literature for a coastal dune system with this particular substrate gradient: published coastal-dune S_y values span 0.10–0.45 depending on grain size, saturation history, and depth (Baird et al., 2020; Berendsen et al., 2007), a range wide enough to carry material uncertainty into any volumetric calculation. The WTF method provides the only means of producing a well-by-well, cluster-stratified S_y calibrated to the site’s own observed head fluctuations. Scripts 17 and 18 are paired because they use the same WTF event-detection logic on the same monthly time series, but answer different questions: Script 17 produces a single S_y per cluster (the value consumed by Table 3c and by Script 19’s spatial calculations and Script 21’s forestry scenarios), while Script 18 produces a S_y per well and combines it with the well-level β_3 from the SSM to generate the storage–drainage index map $\tau = S_y / \beta_3$ that informs the main report’s discussion of aquifer architecture.

Sub-script 17 — `17_wtf_specific_yield.py` (cluster-level S_y)

Motivation

The cluster-level S_y is the value most consumers downstream actually need: Script 21’s forestry scenarios apply a multiplier to a cluster-mean head response and need to convert that to a volumetric flux at the cluster scale, not the individual-well scale. Per-well S_y values are noisier (Script 18 quantifies this) and not the right granularity for the cluster-mean SSM coefficients in `03_master_data.csv`. Script 17 produces a single S_y per cluster, with two independent methods (Approach A and Approach B), so the reader can see how sensitive the headline number is to the estimator choice.

Methodology

Script 17 reads the regional cluster-mean hydrographs (`03_regional_averages.csv`) and the climate series (`01_climate.csv`), forms the monthly change in cluster-mean head Δh for each cluster, and applies two estimators.

Approach A — drainage-corrected winter OLS. The observed Δh during a winter month reflects both recharge and continuing drainage:

$$\Delta h = R / S_y - \beta_3 \cdot |h_{\text{prev}}|$$

Rearranging to estimate S_y gives $R = S_y \cdot (\Delta h + \beta_3 \cdot |h_{\text{prev}}|)$, so the OLS regression of net recharge R against the *drainage-corrected* rise $(\Delta h + \beta_3 \cdot |h_{\text{prev}}|)$ through the origin recovers **S_y as the slope itself** (R is the dependent variable, $\Delta h + \beta_3 \cdot |h_{\text{prev}}|$ is the independent variable). Cluster-median β_3 values are loaded from `03_master_data.csv` to perform the correction. The regression is restricted to winter months (November–March) with PET below 25 mm/month so that net recharge ($P - \text{PET}$) is a defensible proxy for actual recharge. Standard errors on S_y are the OLS-through-origin slope SE directly (no delta method is required because S_y is the slope, not its reciprocal). The fit metric quoted is uncentred $R^2 = 1 - \text{SS}_{\text{res}} / \Sigma y^2$, appropriate for through-origin regression where the centred form can be spuriously negative even when the line passes through the data well.

Approach B — event-median. For every rising-limb month with $\Delta h > 5$ mm and net recharge > 10 mm, compute $S_{y_i} = R_i / \Delta h_i$ directly. Reject physically implausible values ($S_y < 0.01$ or $S_y > 0.50$). Report the median, Q25, and Q75 across qualifying events. The event method is noisier than Approach A but distribution-free and provides empirical uncertainty bounds.

Interception correction for forest clusters. For Corsican pine (C4 Main Forest, C5 Coastal Forest), R is replaced with the effective recharge $R_{\text{eff}} = (1 - 0.24) \cdot P - \text{PET}$ (Freeman, 2008). The 0.24 fraction is the throughfall-gauge interception loss measured at the C5 forest at Newborough Warren. PET is not also reduced — Thornthwaite PET is an energy-based atmospheric demand independent of land cover, and reducing it would double-count the canopy effect.

Headline values. Approach A and Approach B produce the following cluster S_y estimates (post Script 17 v1.2.0 fix; uncorrected variants):

Cluster	Approach A S_y	SE	R^2 (uncentred)	n	Approach B S_y	IQR	n
C1 (Lake Edge)	0.334	0.043	0.616	38	0.210	[0.130 , 0.265]	60
C2 (Dune)	0.328	0.034	0.666	49	0.268	[0.196 , 0.372]	64
C3 (West)	0.343	0.021	0.841	53	0.325	[0.285 , 0.372]	58

Cluster	Approach A Sy	SE	R ² (uncentered)	n	Approach B Sy	IQR	n
ern Residual)						0.415]	
C4 (Main Forest)	0.297	0.017	0.869	46	0.315	[0.256, 0.395]	51
C5 (Coastal Forest)	0.410	0.022	0.877	49	0.356	[0.319, 0.432]	34

For the two forest clusters, interception-corrected Approach B values (Freeman 2008) are reported alongside: C4 corrected Sy = 0.248 (IQR [0.182, 0.317], n = 62), C5 corrected Sy = 0.321 (IQR [0.243, 0.401], n = 49). Corrected Approach A variants at forest clusters are not fitted in v1.2.0 (deliberately left as NaN in the CSV — see *Limitations*); Table 3c of the main report uses Approach B with interception correction as the canonical cluster-level estimator for the forest clusters.

Site-specific choices

- **Winter restriction for Approach A.** Approach A is the more defensible of the two estimators because winter PET is small enough that any error in the PET estimate is negligible compared to P. The PET_MAX_WINTER = 25 mm/month threshold excludes warm winter months where PET is no longer negligible.
- **Drainage correction in Approach A.** Without the β_3 correction, the regressor (raw Δh) is biased low because the rise is being shaved by continuing drainage; the slope is biased high, and Sy biased low. The correction is essential, not cosmetic — see *Limitations* for what happens when β_3 is itself badly identified.
- **Plausibility filter in Approach B.** Sy values below 0.01 or above 0.50 are dropped as physically implausible for the substrate. The filter interacts with the interception correction: reducing P brings previously-excluded high-Sy events (which had unrealistically large Sy_i because Δh was small

relative to raw P) back into the admissible range. This is why the corrected event pool can be larger than the uncorrected pool.

- **No bootstrap for Approach B.** The uncertainty is reported via the IQR rather than a parametric SE, because the per-event Sy distribution is heavy-tailed and a Gaussian SE understates it.

Outputs

File	Contents	Principal consumer
17_wtf_01_sy_estimate s.csv	Per-cluster Approach A and B estimates with SE/IQR and n; corrected variants for forest clusters	Script 19, Script 21, Table 3c
17_wtf_02_regression. png	OLS regression plots for Approach A, one panel per cluster	Supplementary figure
17_wtf_03_event_boxplot.png	Sy distribution boxplot for Approach B, including the corrected forest variants	Supplementary figure
17_wtf_04_summary.txt	Plain-text summary for report cross-reference	Author reference

Sub-script 18 — 18_wtf_spatial.py (per-well Sy and the τ map)

Motivation

Script 18 brings the WTF estimator down to the individual well, then interpolates the result spatially. Three downstream uses motivate the per-well treatment. First, the supplementary table S1 in the main report needs a per-well Sy so readers can cross-check the cluster-level headline against the spatial pattern. Second, the spatial contour map shows where the Sy assumption is most exposed (low-Sy pockets that the cluster-level mean smooths over). Third, combining per-well Sy with per-well β_3 from 03_master_data.csv gives the storage–drainage index $\tau = \text{Sy} / \beta_3$ (months), a per-well aquifer-architecture diagnostic that is not available from the SSM alone. τ is a deliberately storage-weighted composite — it fuses two independently estimated quantities (Sy from the WTF method, β_3 from the SSM) and therefore carries information beyond β_3 alone. It is **not** a residence time or drainage timescale in the hydraulic sense: the head-space recession e-

folding time is $t_R = 1/\beta_3$ (months), which is the genuinely time-like quantity (the interval over which a recharge perturbation decays). Because $S_y < 1$, $\tau = S_y/\beta_3$ is always shorter than $1/\beta_3$ — for example, C2 has $1/\beta_3 \approx 14$ months but $\tau \approx 4$ months. The two should not be conflated; $t_R = 1/\beta_3$ is reported as an interpretation layer on the β_3 atlas (see *Site-specific choices* below).

What e-folding time means in practice. If a recharge event raises the water table by 200 mm above its equilibrium position, and nothing further happens, the linear-reservoir model predicts that t_R months later the head has decayed to $200/e \approx 74$ mm above equilibrium — it has lost 63 % of the perturbation. After $2 \cdot t_R$ it has lost 86 %, after $3 \cdot t_R$ it has lost 95 %. For C2 Dune ($\beta_3 \approx 0.07 \text{ month}^{-1}$, $t_R \approx 14$ months) a 200 mm winter recharge event has largely dissipated by the following spring. For C4 Main Forest ($\beta_3 \approx 0.02 \text{ month}^{-1}$, $t_R \approx 49$ months) the same event still carries ≈ 148 mm of residual head into the following year and ≈ 110 mm two years later — the forest interior integrates recharge over multi-year windows, which is why the MSL5 metric (Script 26) captures its water-availability state better than a single-year summer minimum. The contrast is also why C4 appears less responsive to the clearfell signal in any single post-fell year: the long memory means the pre-fell history is still present in the water table alongside the post-fell recharge gain.

What $\tau = S_y/\beta_3$ adds beyond t_R . The e-folding time $t_R = 1/\beta_3$ is a purely dynamic property of the drainage rate. The storage–drainage index $\tau = S_y/\beta_3$ brings in the storage capacity: a cluster with a high drainage rate and a large pore volume (high S_y) will drain slowly in volume terms even if it drains quickly in head terms, because each metre of head drawdown releases $S_y \times 1 \text{ m}$ of water. τ captures this — it is, roughly, the time to drain a head column of S_y metres through a drain of rate β_3 — and is therefore sensitive to the contrast between the thin-aquifer, low- S_y forest (C4: $S_y \approx 0.25$, $\tau \approx 14$ months) and the deeper, well-drained dune (C2: $S_y \approx 0.33$, $\tau \approx 4$ months despite a longer t_R). The aquifer-diagnostic synthesis figure (Script 18, Figure 48) exploits this contrast to distinguish substrate-driven slow drainage from storage-driven slow drainage.

Methodology

Script 18 reads the cleaned reference-network hydrographs (01_wells_clean.csv), climate (01_climate.csv), well locations (01_locations.csv), and cluster assignments (02_cluster_stats.csv). For each reference well it applies the same event-based WTF procedure as Script 17's Approach B: $\Delta h > 5 \text{ mm}$, net recharge $> 10 \text{ mm}$, plausibility filter $0.01 < S_{y,i} < 0.50$, median across qualifying events. Forest-cluster wells (C4 and C5, identified

through `FOREST_CIDS` in `config.py`) receive the Freeman (2008) interception correction. A well needs at least five qualifying events to be included; wells with fewer than 15 are flagged as low-confidence in the output table.

Extended-network handling. The reference network of 66 wells is supplemented by 22 additional dipwells whose records are shorter or whose cluster identity is less certain. Cluster assignments for the extended wells come from `06_pear_membership_audit_sitewide.csv` (chapter S.4), specifically the `Best_Match_Cluster` column. Extended wells are analysed in `wtf_extended_wells()` with the same event rules and forest-correction logic, and rendered as open symbols on the extended contour map (`18_wtf_04`) so the reader can see at a glance which contour areas rely on lower-confidence wells.

Spatial interpolation. The per-well S_y values are interpolated onto the 50 m grid used elsewhere in the spatial pipeline via a local true inverse-distance-weighting routine (`power = 2`; the function is defined inline as `idw()` within the plotting functions). Note this is genuine IDW, not the piecewise-linear barycentric interpolation that `map_utils.add_idw_surface` provides elsewhere — the local routine here does what its name implies. The site-boundary mask (`make_site_mask`) excludes the sea, lake, and forestry-bounded interior, matching the convention used in Scripts 07 and 19.

Storage–drainage index $\tau = S_y / \beta_3$. The function `compute_drainage_timescale()` joins the per-well S_y table with `03_master_data.csv` on a normalized well name and computes τ for each well where $\beta_3 > 0$. Wells with negative β_3 (CEH14) or near-zero β_3 that would produce a $> 10\times$ outlier (CEH13) are flagged as excluded; the ridge / slack-floor wells (CEH12 sand–bedrock contact, CEH15 forest slack floor) are also excluded as not representative of the dune-aquifer S_y that the WTF estimates. The remaining wells produce τ values in months that are then interpolated to a contour map (`18_wtf_05_drainage_timescale_map.png`) and reported in `18_wtf_05_drainage_timescale.csv`.

Head-space recession time $t_R = 1/\beta_3$ (interpretation layer). The β_3 atlas produced in chapter S.5 (Script 07) is already a drainage-rate map; $t_R = 1/\beta_3$ is a deterministic monotone transform of β_3 that re-expresses each well's drainage rate as an e-folding timescale in months — the interval over which a recharge perturbation decays to $1/e$ of its initial magnitude in the absence of further forcing. Because $1/\beta_3$ is a one-to-one rescaling, the spatial pattern of t_R is identical to that of β_3 and no new figure is required; Script 18 does not compute t_R as a separate column. The interpretation anchor is: $\beta_3 \approx 0.10 \text{ month}^{-1} \rightarrow t_R \approx$

10 months; $\beta_3 \approx 0.07 \rightarrow t_R \approx 14$ months; $\beta_3 \approx 0.05 \rightarrow t_R \approx 20$ months; $\beta_3 \approx 0.02 \rightarrow t_R \approx 50$ months. Cluster-mean values from the canonical fits: C1 Lake Edge ≈ 10 months, C2 Dune ≈ 14 months, C3 Western Residual ≈ 20 months, C4 Main Forest ≈ 57 months (≈ 4.8 years), C5 Coastal Forest ≈ 22 months. CEH14 has negative β_3 and has no valid t_R ; CEH13's $\beta_3 \approx 0.002$ gives $t_R \approx 526$ months and is treated as off-scale in any tabulation. t_R adds no spatial information beyond the β_3 atlas; its value is the interpretable “years of drainage memory” framing that the raw coefficient does not immediately communicate. The genuinely independent fusion of storage and drainage information remains $\tau = Sy/\beta_3$ (Workstream A).

Aquifer-diagnostic synthesis figure. `plot_aquifer_diagnostic_synthesis()` combines three independently derived per-well metrics into a single scatter: τ (the storage–drainage index Sy/β_3 , from Sy and β_3 here), the iterative LCSC NSE improvement ΔNSE (from chapter S.5's model-benchmarking output `08_lcsc_model_stats.csv`), and Sy itself as the marker size. Points are coloured by cluster; cluster means are anchored by larger star markers. The synthesis is the figure that turns three diagnostics into a single read of aquifer architecture. Note that τ here is the Sy -weighted composite, not the head-space recession time $t_R = 1/\beta_3$; both are discussed in *Methodology* above.

Site-specific choices

- **The storage–drainage index τ is the per-well diagnostic, not the cluster mean.** Cluster-mean τ averages over substantial within-cluster spread (the C2 Dune τ range is 2.0–8.7 months across 24 wells, mean ≈ 4.0 months). The map preserves the per-well variation. Note that $\tau = Sy/\beta_3$ is not a residence time — see the $t_R = 1/\beta_3$ passage above for the head-space recession time and its cluster-mean anchor values.
- **Ridge-zone exclusions for τ .** CEH12 sits on the sand–bedrock contact where Sy is not representative of the dune aquifer; CEH15 sits on a forest slack floor where the Sy estimate is atypical of the surrounding cluster. CEH13's β_3 is near zero (0.0019, which would give $\tau \approx 124$ months — physically defensible as “very slow” but compresses the contour colour bar and dominates the map). CEH14's β_3 is negative, making τ undefined. All four are documented in the exclusion column of the τ CSV. The same exclusions apply to $t_R = 1/\beta_3$: CEH14 has no valid t_R ; CEH13 is off-scale ($t_R \approx 526$ months).
- **Per-well Sy is treated as a diagnostic, not a parameter.** Downstream calculations (Script 19's spatial groundwater work, Script 21's forestry scenarios) all use the *cluster-level* Sy from Script 17, not the per-well values.

The per-well map is for spatial pattern recognition, not for substitution into other calculations.

Outputs

File	Contents	Principal consumer
18_wtf_01_well_sy_estimates.csv	Per-well WTF Sy with Q25/Q75, n_events, confidence flag	Table S1, supplementary
18_wtf_02_spatial_sy_map.png	Per-well Sy point map (paper figure)	Main report
18_wtf_03_sy_contour.png	Sy contour surface, reference wells only	Supplementary figure
18_wtf_04_sy_contour_extended.png	Sy contour surface, reference + extended	Supplementary figure
18_wtf_05_drainage_timestep_map.png	$\tau = \text{Sy}/\beta_3$ storage–drainage index contour map	Main report
18_wtf_05_drainage_timestep.csv	Per-well τ values with Sy, β_3 , cluster, exclusion flags; τ is the storage–drainage index (not a residence time — see $t_R = 1/\beta_3$ in Site-specific choices)	Supplementary
18_wtf_06_aquifer_diagnostic_synthesis.png	τ vs ΔNSE scatter, sized by Sy, by cluster	Main report, §5 discussion
17_wtf_well_sy.csv	Intermediate copy at outputs/ root	Internal pipeline

Site-specific choices and rationale (suite-level)

- **WTF as a complement to the SSM, not a replacement.** The SSM gives the dynamic response (β_1 , β_2 , β_3); the WTF gives the conversion factor that turns a head response into a volumetric flux. Both are fitted to the same monthly time series but answer different questions, and reporting both lets the reader cross-check (for example) the cluster ordering by Sy against the cluster ordering by β_2 — C4 Main Forest is among the lower-Sy clusters, which is part of the explanation for why C4's β_2 is high in head units while

the volumetric water-balance partition does not place forest evapotranspiration in an extreme position (chapter S.11).

- **Forest interception correction is shared between Scripts 17 and 18.** The 0.24 interception fraction (Freeman 2008) is hard-coded in Script 17 and imported from `config.py` (`FOREST_INTERCEPTION`) in Script 18 — the two scripts produce consistent treatment of the forest clusters because they both apply the same correction to the recharge term. `FOREST_CIDS = (4, 5)` in `config.py` (see F.4) defines which clusters receive the correction.
- **Live network counts.** Script 17 works at the cluster level on the five reference clusters under the $k=5$ partition (C1 Lake Edge, C2 Dune, C3 Western Residual, C4 Main Forest, C5 Coastal Forest), each represented by a cluster-mean hydrograph in `03_regional_averages.csv`. Script 18 works at the well level on the 66 wells of the reference network plus the 22-well extended network. The 66/22 split is established in chapter S.4.
- **Event detection rules are identical in Scripts 17 and 18.** Minimum rise 5 mm, minimum net recharge 10 mm, S_y plausibility filter 0.01–0.50, minimum 5 events per well for inclusion. These rules are conservative — the live event count per well averages around 50 — and the script favours throwing out marginal events over including them.

Limitations and known caveats

- **S_y estimates from the WTF method are indicative, not definitive.** Slug or pumping tests at representative wells per cluster would be the gold standard. The WTF approach assumes that monthly net recharge during rising-limb winter months equals actual aquifer recharge — a defensible approximation for this site but not a measurement.
- **Approach A and Approach B answer slightly different questions and are expected to differ within the data's natural spread.** Approach A's drainage correction means it is reporting S_y *given* the SSM's β_3 , while Approach B is an empirical median across rising-limb months. Under the live partition the two estimators agree within ~10–15 % on C3, C4, and C5; they diverge on C1 (Lake Edge) and C2 (Dune). The chapter on Script 16's water balance (S.11) takes the view that any volumetric calculation built on a single S_y value is exposing itself to this spread, which is why Script 16 was designed to be S_y -free.
- **C1 Lake Edge carries materially more uncertainty than the cluster-level summary suggests.** At C1 the two estimators diverge by ~60 % (Approach A $S_y = 0.334$, Approach B $S_y = 0.210$). This reflects the well-documented difficulty of WTF estimation in lake-buffered clusters, where

the water table is decoupled from rainfall on monthly timescales and within-cluster heterogeneity is high — the Pearson correlation between R and the drainage-corrected rise at C1 is effectively zero ($r \approx 0.00$ across the qualifying winter months), against $r = 0.72$ at C3, 0.70 at C4, and 0.76 at C5. At this cluster the per-event median (Approach B) is treated as the more reliable estimator, and the cluster’s headline S_y in Table 3c reflects that choice.

- **Forest-cluster S_y carries additional uncertainty from the interception value.** The 0.24 throughfall-loss fraction is a single Freeman (2008) value, measured under one canopy density in one stand. Its own uncertainty band is not propagated through the WTF estimate. Section S.13 (Scripts 19, 20) and S.14 (Script 21) both consume the corrected forest S_y values without re-propagating that uncertainty; the size of the cluster’s S_y range itself remains the dominant uncertainty term.
- **Per-well S_y in Script 18 is noisy at short-record wells.** Extended-network wells with few rise events produce wide confidence intervals; the spatial contour map smooths over this but the per-well CSV (Table S1) should be read with care. The 15-event threshold for the high-confidence flag is conservative but does not eliminate the issue.
- **The contour interpolation does not extrapolate beyond the convex hull of well locations.** S_y values at the site margins reflect the nearest interior wells, weighted by inverse-square distance, rather than any genuine measurement at those margins. The site-boundary mask hides this where it would mislead (offshore, in the lake) but the user should be aware that the dune fringes are interpolated, not measured.
- **Downstream consumer note — broadleaf β_2 multiplier in Script 19.** The S_y values produced here feed Script 19’s spatial calculations and Script 21’s forestry scenarios. Script 19 imports the broadleaf β_2 seasonal split — $sB2_w = 0.8817$ (Nov–Apr, leaves off), $sB2_s = 1.0750$ (May–Oct, full leaf) — from `utils.config` (canonical values derived from Script 21’s monthly profile, see chapter S.14). §4.9.6 of the main report carries the broadleaf-scenario values.

Where the result appears in the report

- **§4.2.4 (or the equivalent — verify against the report’s section numbering)** — WTF S_y methodology and the cluster-level result.
- **Table 3c** — cluster-level S_y estimates from Script 17’s interception-corrected output.
- **Table 3d / Figure 11b** — volumetric water-balance using the WTF S_y .

- **Table S1** — per-well S_y from Script 18, supplementary.
- **Spatial figures** — Script 18's contour and τ maps; the τ vs ΔNSE synthesis figure appears in the discussion of aquifer architecture.

Cross-references

- **F.3** — SSM displacement formulation; the β coefficients consumed alongside S_y in volumetric calculations.
- **F.4** — `FOREST_INTERCEPTION = 0.24`, `FOREST_CIDS = (4, 5)`.
- **F.5** — shared utility modules and the IDW interpolation convention used by `map_utils.add_idw_surface` (which is *not* used here; the IDW in Script 18 is implemented locally).
- **S.3** — produces the β coefficients used for the $\tau = S_y / \beta_3$ storage–drainage index calculation and for the $t_R = 1/\beta_3$ recession-time interpretation.
- **S.4** — the Pearson audit that provides `Best_Match_Cluster` for extended wells used by Script 18.
- **S.5** — model benchmarking; supplies the ΔNSE used in the τ vs ΔNSE synthesis figure.
- **S.11** — Script 16's water balance deliberately defers S_y uncertainty by choosing a S_y -free partition-fraction approach. The reciprocal of the relationship: Script 16 doesn't consume S_y because Scripts 17 and 18 carry the uncertainty.
- **S.13** — Scripts 19 and 20 spatial groundwater calculations consume the cluster-level S_y values.
- **S.14** — Script 21 forestry scenarios consume the cluster-level S_y values via `clearfell_common.load_clearfell_b2_multiplier()` and elsewhere.

S.13 Scripts 19, 20 — Spatial groundwater analysis

Steps 21 and 22 / 27. Phase 9 — Spatial Groundwater Analysis in `run_analysis.py`; fifth chapter under Phase 4 — Climate and Spatial Context in the supplement.

Scripts 19 and 20 together construct §4.9 *Spatial groundwater* of the main report. They share inputs (the per-well β coefficients from `03_master_data.csv`, monthly maOD heads from `01_wells_clean_maod.csv`, the 50 m project grid, the F.3 displacement formulation, the cluster-mean S_y from S.12) and methodological choices (Freeman 2008 forest interception, Betson 2002 hydraulic conductivity, the F.2 bucketing convention). They differ in deliverable: Script 19 generates `scenario_viewer.html`, a self-contained interactive HTML calculator; Script 20 generates the publication-quality static figures for §4.9. Script 19 at ~1800 lines is

among the longest in the pipeline; Script 20 at ~1200 lines produces the publication spatial figures (ten outputs). The interactive viewer is part of the methods rather than a results artefact: it is the calculator that lets a reader exercise the per-cluster equation engine for any scenario combination, with sliders and parameter values exposed.

Sub-script 19 — Scenario viewer (interactive HTML)

Motivation. The mechanistic content of the report’s forestry and climate scenarios sits in three places: the per-cluster β coefficients in `03_master_data.csv`, the seasonal climate baselines from `01_climate.csv`, and the seasonal cluster-mean heads. A reader who wants to interrogate any single scenario — say, *what does a 30 % winter rainfall increase under UKCP18 2080s do to the C4 Main Forest summer minima?* — needs all three sources combined through the equilibrium-perturbation engine. Script 19 builds that engine into a self-contained HTML file with every parameter exposed as a slider. Unlike the Flood Forecaster (S.9), which fits a temporal model to a single well, and unlike the Newborough Water-Level tool, which displays the monitoring record, the scenario viewer is purely diagnostic: it lets the reviewer see what the calibrated SSM implies for any combination of climate and forest-management perturbations.

Methodology. The engine is the same monthly equilibrium perturbation that Script 21 (S.14) uses for the published scenario figures. For each well with cluster ID `c1`, β coefficients (b_1, b_2, b_3) from the master CSV and seasonal head h (winter, summer or annual mean), `_well_dh()` computes the equilibrium Δh as the difference between the perturbed and baseline net water-balance fluxes:

$$\Delta h = [\beta_1 \cdot P_{\text{eff}}(\text{scenario}) - \beta_2(\text{scenario}) \cdot \text{PET}(\text{scenario}) - \beta_3 \cdot |h|] - [\beta_1 \cdot P_{\text{eff}}(\text{baseline}) - \beta_2 \cdot \text{PET}(\text{baseline}) - \beta_3 \cdot |h|]$$

The $\beta_3 \cdot |h|$ terms cancel exactly, leaving a monthly perturbation in the recharge and atmospheric-draw fluxes:

$$\Delta h = \beta_1 \cdot [P_{\text{eff}}(\text{scenario}) - P_{\text{eff}}(\text{baseline})] - [\beta_2(\text{scenario}) \cdot \text{PET}(\text{scenario}) - \beta_2 \cdot \text{PET}(\text{baseline})]$$

For non-forest clusters P_{eff} equals raw P and β_2 is unscaled. For forest clusters (C4 Main Forest and C5 Coastal Forest, identified through `is_forest = (c1 == 4 or c1 == 5)`), three transformations apply: (1) baseline P is reduced by the Freeman (2008) canopy interception ($1 - 0.24$); (2) under a scenario the canopy interception fraction may be replaced (clearfell $\rightarrow 0$, broadleaf $\rightarrow 0.15$, thinning $\rightarrow$

0.12); (3) β_2 may be scaled by a season-specific multiplier $sB2_w$ (winter) or $sB2_s$ (summer). The forest scaling is identical for C4 and C5 — the BACI analysis (S.7) does not separate them. The annual response is the unweighted mean of the winter and summer responses, with winter defined as November–April and summer as May–October — the canopy-state phenological windows ($\beta_2 < 1$ in winter months, $\beta_2 \geq \sim 1$ in summer months) that the broadleaf β_2 derivation uses.

The Python function `_well_dh()` and its JavaScript twin `dhOne()` implement this identically. Python-side values (well coordinates, β coefficients, seasonal climate means, cluster-mean heads, KML polygons, the DEM hillshade as a base64-encoded PNG, a coarse DEM grid for ridge masking) are interpolated into a static HTML template at generation time. The duplication is deliberate: JavaScript drives the interactive sliders, and the Python mirror writes `19_scenario_summary.csv` so the same numerical result is available as a citeable file.

The scenarios. Six preset scenarios are wired to buttons in the sidebar, with the slider state for each defined in the `SCENARIO_PARAMS` dictionary at the top of the script:

Preset	P scaling	PET scaling	Forest interception	Forest β_2 scaling
Baseline (2005–2026)	1.00, 1.00	1.00, 1.00	0.24	1.00, 1.00
UKCP18 2050s RCP8.5	1.10, 0.85	1.05, 1.20	0.24	1.00, 1.00
UKCP18 2080s RCP8.5	1.20, 0.70	1.10, 1.35	0.24	1.00, 1.00
Clearfell	1.00, 1.00	1.00, 1.00	0.00	dyn., dyn.
Broadleaf conversion	1.00, 1.00	1.00, 1.00	0.15	0.8817, 1.0750
Forest thinning (50 %)	1.00, 1.00	1.00, 1.00	0.12	dyn., dyn.

The clearfell and thinning forest- β_2 multipliers (shown as “dyn.” above) are not literals — they are loaded at script startup through `clearfell_common.load_clearfell_b2_multiplier()`, which reads the BACI-corrected Edge-tier ratio derived in S.7’s Script 10e. Version 2.5.0 of Script 19

(April 2026) replaced an earlier hardcoded 1.20 with this dynamic load. The thinning multiplier is the same load with the BACI shift halved. This is the closest the viewer comes to the project's no-hardcoded-values discipline: the values flow from the upstream pipeline through `pipeline_params.py` to both the Python summary CSV and the embedded JavaScript at HTML-generation time.

Sliders. Beneath the preset buttons, six climate sliders (Winter P, Summer P, Winter PET, Summer PET, and the C4 and C5 canopy interception controls) and two β_2 scaling sliders (Winter $\beta_2 \times C4+C5$, Summer $\beta_2 \times C4+C5$) expose every input to the engine. Each takes a 0.5–1.5 $\times$ multiplicative range except the interception sliders, which span 0.0–0.4 in absolute units (so a “0.00” position means “no canopy”, as in the clearfell preset, rather than “no perturbation”). The 0.5–1.5 $\times$ range brackets the UKCP18 end-century probabilistic envelope for Wales under RCP8.5 while staying inside the linear steady-state domain of the fitted SSM.

The seasonal β_2 split for broadleaf. Version 2.6.0 (May 2026) merged earlier per-cluster `sB2_c4`, `sB2_c5` sliders into seasonal `sB2_w`, `sB2_s` sliders shared across C4 and C5. Version 2.7.0 (May 2026) consolidated the values to a single source of truth in `utils.config`: the deciduous-conversion preset now uses `sB2_w = 0.8817` (leaves off, reduced atmospheric draw) and `sB2_s = 1.0750` (full canopy, elevated atmospheric draw), derived from Script 21's monthly β_2 profile averaged over the canopy-state phenological windows (Nov–Apr for winter, May–Oct for summer — see chapter S.14 for the 12-month profile and the rationale for the window choice). Clearfell and thinning multipliers are non-seasonal — canopy removal itself is non-seasonal — so the same value is written to both seasons.

Well exclusions and ridge mask. Two wells are excluded: `ceh12` (on the bedrock ridge, depth signal the SSM does not represent — see Script 23 in S.16) and `ceh15` (forest-slack-edge well whose β coefficients fit poorly and would bias the C4 cluster mean). Both exclusions match the pattern in Script 07 (S.5) for the static β maps. The viewer's spatial interpolation is then ridge-masked at runtime: each pixel is checked against a coarse 120×110 DEM grid (`build_dem_grid()`) and suppressed where the DEM exceeds the interpolated head surface by more than `RIDGE_MASK_THRESHOLD = 1.0` m. This is the same threshold used by `map_utils.add_idw_surface()`.

Outputs. A single self-contained HTML file (`scenario_viewer.html`, typical size 250–300 KB including the embedded base64 hillshade) and a tidy summary CSV (`19_scenario_summary.csv`, one row per scenario $\times$ season $\times$ spatial unit, where spatial unit is each of C1–C5 plus a site-wide mean). The CSV mirrors the JavaScript calculation through the shared `_well_dh()` engine. To anchor with one

scenario: for UKCP18 2080s annual the site-wide mean Δh is -33 mm, with the largest impacts in C4 Main Forest (-46 mm) and C3 Western Residual (-34 mm) and the smallest in C1 Lake Edge (-17 mm). Forest clusters show the steepest summer drawdown (C4 -118 mm, C5 -71 mm) and the most muted winter response (C4 $+25$ mm, C5 $+26$ mm), reflecting the high β_2 that dominates the forest water balance. Clearfell produces the inverse — C4 $+38$ mm annual, C5 $+38$ mm annual — with winter recovery exceeding summer recovery in both forest clusters (C4 winter $+47$ mm vs summer $+29$ mm; C5 winter $+45$ mm vs summer $+31$ mm). These values appear directly in §4.9.6 of the main report.

Δ MSL5 row — v2.8.0 (2026-05-27). A new top row was added to the viewer's per-cluster scenario summary table, reporting the van Willegen et al. (2025) 5-year mean spring water-level shift defined as the mean of monthly Δh over March, April, and May. The row uses the same pure-climate perturbation formula $\Delta h(m) = \beta_1 \cdot P(m) \cdot (sP(m)-1) - \beta_2 \cdot PET(m) \cdot (sPET(m)-1)$ that Script 26b uses, evaluated at *per-well* β coefficients arithmetically averaged within each cluster — the existing viewer convention. Canopy interception is *not* applied to the Δ MSL5 row (the existing Δh /storage rows above do apply it for C4/C5); the asymmetry is documented in a footnote under the viewer table. MSL5 is in any case most ecologically relevant for the non-forest clusters C1–C3, where canopy interception does not enter the calculation. Under the pure-climate framing the four land-use presets (baseline, clearfell, broadleaf, thinning) produce Δ MSL5 = 0 identically; only the two UKCP18 climate presets carry non-zero Δ MSL5 values. The scenario summary CSV gains a corresponding season="msl5" block (6 rows per scenario: 5 clusters plus a well-count-weighted SITE row; 12 non-zero rows for the two UKCP18 scenarios, 24 zero-valued rows for the four non-climate presets, consistent with the pure-climate framing). The CSV's season="summer" block is the data source for Script 26c's §4.10.1 Δ summer-minimum bars.

Why per-well aggregation, and how it relates to Script 26b. The viewer's per-well-averaged β does not algebraically reduce to Script 26b's cluster-centroid OLS, even on the same wells: the two aggregations differ by 0.5–3.7 mm per cluster per UKCP18 scenario, with the largest gap in C1 (lake-edge, $n = 7$, the most heterogeneous reference-network cluster). To anchor the viewer row against a matching canonical reference, Script 26b v1.1.0 (2026-05-27) added a parallel per-well aggregation pathway that writes `26b_msl5_ukcp18_projection_summary_perwell.csv` alongside the canonical centroid summary. The viewer's CSV-side computation includes a cross-script validation block that loads the 26b per-well CSV if present, iterates the 10 (cluster, UKCP18 scenario) pairs, and prints the maximum absolute difference; current

state is 0.042 mm worst case (well within the 0.5 mm acceptance tolerance, a rounding artefact of the viewer CSV's 4-dp round()). The canonical report numbers for §3.7.5 / §4.8.4 / §4.10.1 remain anchored to the centroid-fitted 26b_msl5_ukcp18_projection_summary.csv consumed by Script 26c (S.18c); the per-well CSV is a secondary artefact whose role is the viewer-row validation target.

Sub-script 20 — Publication spatial figures

Motivation. Script 20 generates ten spatial outputs that the viewer alone cannot replace: a head-surface map with the stream network and flow vectors (20_head_surface_streams.png); a map of the SSM water-balance residual (20_residual_ssm.png); an independent topographic check from a smoothed-DEM ridge-slope map (a supporting diagnostic, not carried as a numbered report figure); a flow-weighted forest-drawdown propagation map (20_drawdown_propagation_nohead.png, filled no-head form since v1.4.0); a three-figure coastal-process family added in May 2026 (coastal-erosion drawdown, sea-level-rise head response, and the cell-by-cell SLR-erosion net coastal change — the coastal-figures family, cited in §3.4.5 and §4.9.4 of the main report); a dune-scape topographic-drain drawdown field at the CEH36 site (20_scrape_drawdown_nohead.png, the §5.4.2 “scrape as drain” reading); a composite clearfell pre-fell baseline drawdown that overlays the scrape on the Storm Brendan coastal retreat (20_clearfell_baseline_drawdown.png, the §5.4.2/§5.4.3 confounder framing); and a public-summary three-driver comparison panel (20_public_drivers_panel.png, a lay-summary figure not numbered in the main report; forest canopy / dune scrape / coastal erosion on the shared sequential drawdown scale). Script 20 generates all ten at publication DPI.

Methodology. These figures share a common stack: load the DEM hillshade through `map_utils.load_dem_hillshade()` (the canonical basemap, F.5); compute or load surfaces on the 50 m grid (`np.arange(240200, 243800, 50) × np.arange(362200, 365000, 50)`); overlay KML site features through `draw_kml_features()`; and place well markers at (E, N) with cluster-coloured fills from `config.CLUSTER_COLOURS`. These figures differ in what occupies the central panel:

Head surface with stream network and flow vectors

(20_head_surface_streams.png). The mean annual head is interpolated to the 50 m grid using the `idw_surface()` helper, which despite its name calls `scipy.interpolate.griddata` with `method='linear'` — Delaunay-triangulation piecewise-linear interpolation, as in `map_utils.add_idw_surface()` (S.5). The

interpolation is augmented with zero-head anchor points every 200 m along the southern, eastern and western sea boundaries, constraining the interpolation at site edges to the saltwater datum. The SAGA-derived stream network is loaded from `data/streams.kml` as polygon cells and drawn as filled polygon outlines. Flow vectors are computed from the head-surface gradient, with magnitudes in the top 5 % suppressed to mask ridge interpolation artefacts, then **normalized to unit length** for display. Contours at 1 m intervals are labelled in m AOD.

Per-well mean head: all-record method and cross-check. The mean head at each well is computed as the arithmetic mean of the full observed maOD record — every valid monthly row from `01_wells_clean_maod.csv` from the well's first measurement to `REFERENCE_CUTOFF_DATE`. The all-record method is deliberately preferred over a windowed approach for three reasons: (i) the head surface is a spatial-geometry product, not a climate-trend product — it describes where the water table sits, not how it is changing; (ii) a short window (e.g. 2010–2025) would exclude 4–6 early-record years from several wells, including 4 reference C3 wells (CEH39–42) which fall just below a 12-of-16-year completeness threshold at a knife-edge exclusion that would be harder to defend than accepting the full record; (iii) the spatial pattern of mean head — the SW-to-NE head gradient, the ridge-margin feature, the C5 coastal low — is robust to the window choice.

A windowed cross-check (water years 2010–2025, ≥ 12 of 16 years with ≥ 10 months/year) was conducted against the all-record means (64 of 88 wells qualifying). The per-well differences are small: median shift -9 mm, maximum shift ~ 112 mm. The ~ 112 mm outlier is **NW9**, the central well of the C5 Coastal Forest cluster. The shift is not a measurement artefact — it reflects a coherent C5-wide downward shift (all five C5 wells shift negative under the windowed method), because the 2006–2009 sub-period carries comparatively high water levels in the coastal-forest zone that the windowing excludes. The effect on the C5 cluster-mean head is ≈ 55 mm cluster-wide (of which NW9 contributes ≈ 14 mm), negligible against the metre-scale inter-cluster head contrasts. The interior aquifer geometry — the flow vectors, the gradient directions, the residual-field pattern — is unchanged between methods. The all-record method is therefore retained as the primary, with this windowed comparison as the empirical defence against the question of record-length bias.

SSM water-balance residual (`20_residual_ssm.png`). For each well, the steady-state residual $\alpha = \beta_2 \cdot \bar{P}_{\text{ET}} + \beta_3 \cdot h_{\text{disp}} - \beta_1 \cdot \bar{P}_{\text{eff}}$ is computed at the long-term seasonal mean climate and the well's mean displacement $h_{\text{disp}} = \text{DRAINAGE_DATUM} + h_{\text{depth}}$ (F.3). A positive residual means atmospheric draw

plus drainage exceeds rainfall recharge — physically, the well requires an external (lateral) input to balance. The residual is interpolated to the 50 m grid by the same Delaunay-linear scheme, anchored at zero along the sea boundaries. The colour scale is divergent (RdBu_r) with TwoSlopeNorm anchored at zero and the 95th-percentile of $|\text{residual}|$ at the limits. Flow vectors on this panel are independent of the residual: they come from the head-gradient calculation of the head-surface map (20_head_surface_streams.png), included so a reviewer can read the residual pattern against the flow field without flipping between figures.

Ridge hillslope gradient — supporting topographic check (not a numbered report figure). Slope (degrees) from a 50 m smoothed LiDAR DEM (scipy.ndimage.uniform_filter), clipped to the study extent with slopes $< 1^\circ$ masked to leave the flat dune plain transparent. The figure is DEM-only — no β coefficients, no interpolation of well measurements — and serves as a falsifiable cross-check on the residual interpretation: if ridge-derived lateral recharge is the explanation for the positive residual pattern, it should coincide with regions of significant ridge slope.

Forest drawdown propagation (20_drawdown_propagation_nohead.png). A flow-weighted cost-distance is computed via Dijkstra's algorithm on the LiDAR DEM (downsampled to 10 m), seeded from forest-boundary cells; cell-to-cell costs are weighted by alignment with the DEM gradient (downhill cheap, uphill expensive). The drawdown signal decays with characteristic length $\lambda = \sqrt{(K \cdot b / (S_y \cdot \beta_3))}$, where $D = K \cdot b / S_y$ is the hydraulic diffusivity. The decay length depends on aquifer properties of the *propagation medium*, not of the forest itself. Since the drawdown signal propagates outward from the forest edge into the surrounding open dune, S_y and β_3 are sourced from C3 (Western Residual): $S_y \approx 0.31$ from the C3 cluster median of the per-well WTF estimates (Script 17, S.12) and β_3 from the C3 centroid SSM coefficient (Script 03, S.3) — currently $\approx 0.058 \text{ month}^{-1}$ under the limit=1 interpolation policy. With $K = 6 \text{ m/day}$ from Betson (2002) and saturated thickness $b = 5 \text{ m}$, this gives $\lambda \approx 223 \text{ m}$. The $\Delta h = H_0 \cdot \exp(-d/\lambda)$ field with $H_0 = 150 \text{ mm}$ (forest interception deficit) is overlaid on the mean head surface, with λ annotated as a horizontal bar. Both S_y and β_3 are read from the live pipeline CSVs at generation time (Script 20 v1.0.1), so the figure tracks the current pipeline state; pre-2026-05-17 the script used hardcoded snapshot values ($S_y = 0.25$ mid-range, $\beta_3 = 0.060 \text{ month}^{-1}$) that gave $\lambda = 245 \text{ m}$.

Coastal-process figures (erosion, sea-level-rise, and net coastal change)

The forest drawdown figure (20_drawdown_propagation_nohead.png) was joined in May 2026 by three companion fields that illustrate western-margin coastal

influences on the dune water table. All three are single-mechanism illustrative constructions on the eroding Caernarfon Bay shore only, sharing the DEM hillshade base, the F.5 spatial-figure machinery, and a site-boundary clip applied through `data/site_boundary.kml` so the fields terminate at the coast and the warren edges rather than extrapolating offshore or onto the bedrock ridge. The construction of each field is documented in §3.4.5 of the main report; this subsection carries the full parameter exposition that §3.4.5 deliberately defers to the supplement.

Erosion drawdown. Applies the Script 25 forest-free linear-capped coastal-retreat fit $\delta(d) = \delta_0 \cdot (1 - d/L)$ to a single discrete shoreline-retreat event of magnitude `COAST_RETREAT_M` = 6 m (the Storm Brendan early-2020 exemplar). The retreat magnitude is converted to an edge water-table drawdown via $h_0 = \text{COAST_RETREAT_M} \cdot (\delta_0 / \text{COAST_RETREAT_RATE})$, where `COAST_RETREAT_RATE` = 8.3 m yr⁻¹ is the storm-inclusive long-term retreat rate (Forgrave 2020 / North Wales Live, 22 January 2020 — Walker-Springett’s then-ongoing Bangor coastal-erosion measurements, approximating the ≈ 50 m of retreat between 2014 and 2020). The field decays inland from a dune-edge front derived by offsetting the DEM mean-waterline contour (`COAST_SHORE_LEVEL_M` = 0.5 m AOD) landward by `COAST_DUNE_OFFSET_M` = 100 m to the dune toe. δ_0 and L are read live from `25_01_panel_fit_parameters.csv` (Script 25, S.15) at generation time — currently $\delta_0 \approx 0.13$ m and $L \approx 1.5$ km — so the field tracks the live coastal-gradient fit. The retreat rate is *storm-inclusive*: a single Brendan-class event normalized by a storm-inclusive rate is the correct construction for a single-event illustration. It is **not** a chronic background retreat rate; the chronic rate at the southern frontage (≈ 1.5 – 1.8 m yr⁻¹ per Pye & Blott 2024) would be the wrong normaliser here.

Sea-level-rise head response. Models the gradual rise as a finite-window boundary perturbation diffusing inland according to $\Delta h(d) = \Delta \text{SLR} \cdot \text{erfc}(d / (2 \cdot \sqrt{D \cdot t}))$, where $D = K \cdot b / S_y$ is the hydraulic diffusivity. The boundary is referenced to mean sea level (`SLR_SHORE_LEVEL_M` = 0 m AOD), as appropriate for a water-table head response (a tidal-inundation construction would instead reference MHWS $\approx +2.9$ m AOD from the Caernarfon tide tables, but is not used here). $\Delta \text{SLR} = \text{SLR_RISE_M} = 0.02$ m over `SLR_WINDOW_YEARS` = 5 yr matches the UKCP18 north-Wales near-term central estimate (≈ 4 mm yr⁻¹). Aquifer parameters are `SLR_K` = 6 m day⁻¹ (Betson 2002) and `SLR_B` = 5 m saturated thickness; S_y is read live as the C3 (Western Residual) cluster median of per-well WTF estimates from `17_wtf_well_sy.csv` (currently ≈ 0.311), matching the propagation-medium

convention used for the forest drawdown figure. The figure annotates $\sqrt{(D \cdot t)}$ on the inland axis.

Net coastal change. The cell-by-cell difference (SLR head gain minus erosion drawdown) over the matched window, on a diverging colour scale. No additional parameters: the field is purely arithmetic on the two preceding fields. It is *not* a closed water budget — discussion caveats appropriate to that point appear in COASTAL_NET_VS_EASTING_MEMO.md and the §5 weave-in.

Parameter table. All controlling parameters for the three coastal-process figures, with their sources and the figures in which each appears. Live-pipeline values are read on each pipeline run and revise if their upstream outputs change; the snapshot values stated here apply at time of writing under the live pipeline state.

Parameter	Symbol	Value	Source	Field
Forest interception deficit at edge	H_0	150 mm	Freeman (2008); estimate at site	Forest drawdown
Hydraulic conductivity	K	6 m day ⁻¹	Betson et al. (2002); literature	Forest drawdown, SLR, Net
Saturated thickness	b	5 m	Estimate (aquifer architecture; literature)	Forest drawdown, SLR, Net
Specific yield (C3 propagation medium)	S_y	≈ 0.311 (live; C3 WTF median from 17_wtf_well_sy.csv)	Live: Script 17, S.12	Forest drawdown, SLR, Net
SSM drainage coefficient (C3)	β_3	≈ 0.058 month ⁻¹ (live; C3 centroid from 03_03_cluster_mechanistic_coefficients.csv)	Live: Script 03, S.3	Forest drawdown
Storm-	COAST_RETRE	6 m	Storm	Erosion, Net

Parameter	Symbol	Value	Source	Field
magnitude shoreline retreat (exemplar)	AT_M		Brendan early-2020 exemplar	
Long-term retreat rate (storm-inclusive)	COAST_RETRE AT_RATE	8.3 m yr ⁻¹	Forgrave (2020); storm-inclusive normaliser	Erosion, Net
Dune-toe offset (inland of waterline)	COAST_DUNE _OFFSET_M	100 m	Estimate (DEM-derived dune-toe position)	Erosion, Net
DEM waterline contour	COAST_SHOR E_LEVEL_M	0.5 m AOD	DEM-derived	Erosion, Net
Coast-edge anomaly	δ_0	live (Script 25 fit)	Live: Script 25, S.15	Erosion, Net
Erosion-decay characteristic length	L	live (Script 25 fit)	Live: Script 25, S.15	Erosion, Net
Mean-sea-level rise (window total)	SLR_RISE_M	0.02 m	UKCP18 north-Wales central, near-term	SLR, Net
Response window	SLR_WINDO W_YEARS	5 yr	Near-term horizon	SLR, Net
SLR boundary datum	SLR_SHORE_L EVEL_M	0 m AOD	Physical convention (water-table head, not inundation)	SLR, Net

Storm-inclusive versus chronic rate. The 8.3 m yr⁻¹ retreat rate is the *storm-inclusive* normaliser for a single-event Brendan-class illustration — the only physically correct choice when the construction is a single discrete retreat event of storm magnitude. Pye & Blott (2024) report chronic background retreat rates at

the southern frontage of the warren of approximately $1.5\text{--}1.8\text{ m yr}^{-1}$; that chronic rate is the right number for long-term contextual statements about cumulative shoreline change but is the wrong normaliser for this construction. The two rates are both correct, for different purposes, and the chapter is careful not to conflate them. The chronic rate appears in §5 of the main report (the Pye & Blott discussion weave-in); the storm-inclusive rate appears here and in §3.4.5.

Caveats specific to the coastal-process figures. Each field is a single-mechanism illustration on a single coastal frontage. The erosion field combines an episodic event with a chronic-rate normaliser; the SLR field combines a gradual climate process with a finite five-year window. Neither field is a deterministic forecast. The net field is a cell-by-cell difference of the two and inherits the caveats of both; it is not a closed water budget, and “net coastal change” should be read as relative spatial reach of the two opposing processes rather than as a balance. The construction depends on assumed retreat-rate, diffusivity, datum and window parameters, all of which are flagged in the table above; sensitivity to any of these can be assessed by re-running with revised constants. See COASTAL_NET_VS_EASTING_MEMO.md (project store) for the easting-correction comparison analysis and §5 of the main report for the wider interpretive caveats.

Equation forms. The model forms are stated in §3.4.5 of the main report in LibreOffice StarMath syntax ($\Delta h = H_0 \cdot \exp(-d/\lambda)$, $\delta(d) = \delta_0 \cdot (1 - d/L)$, $\Delta h(d) = \Delta \text{SLR} \cdot \text{erfc}(d/(2\sqrt{D \cdot t}))$). The supplement does not duplicate these forms; cross-reference §3.4.5 of the main report for equations.

Dune-scrape topographic-drain drawdown (20_scrape_drawdown_nohead.png)

Script 20 v1.23.0–v1.24.0 (2026-06-30): drawdown model updated from single-cut envelope to multi-cut additive superposition with method-of-images coastal boundary correction. See CHANGELOG_delta_2026-06-30_scrape_drawdown_physics.md.

The scrape-drain drawdown field models all eight GPS-traced dune-scrape cuts (Feb 2013: CEH40, CEH41, CEH42; Apr 2015: CEH36, Scrape A, Scrape B; Oct 2023: CEH18, CEH21) as permanent co-active features contributing simultaneously to the long-term equilibrium water table. The field is computed as a leaky-aquifer superposition with method-of-images coastal boundary correction:

$$dd(x,y) = \sum_i H_{0i} \cdot [\exp(-d_i/\lambda) - \exp(-d'_i/\lambda)]$$

where d_i is the Euclidean distance from grid point (x, y) to the footprint of cut i , d_i' is the distance to the mirror image of cut i reflected across the High Water Mark coastline, $\lambda = \sqrt{(Kb/Sy\beta_3)}$ is the leaky-aquifer decay length (live C3 pipeline values; $K = 6 \text{ m d}^{-1}$, Betson & Bristow 2002; $b = 5 \text{ m}$; $\lambda \approx 225 \text{ m}$), and H_{0i} is the edge drawdown magnitude for cut i .

The image term enforces the physical condition that the coast is a fixed-head boundary: the sea maintains hydraulic head and drawdown must decay to zero at the High Water Mark. The approximation is exact for a straight boundary and locally accurate for the gently curved Caernarfon Bay coast. The coastline geometry is loaded from `data/geo/coastline_hwm.geojson` (OpenStreetMap `natural=coastline`, Caernarfon Bay MHW and Menai Strait, EPSG:27700, ODbL, extracted 2026-06-30; Malltraeth estuary excluded). If `data/geo/coastline_hwm.geojson` is absent, `_scrape_field()` falls back to unbounded superposition (no image term) with a printed warning.

The edge drawdown H_{0i} is set to the matched-control BACI response (Script 09a, S.7) for three monitored cuts: CEH36 +130 mm (Pure_Scraping vs CEH4), CEH21 +74 mm (After_Scraping vs CEH22), CEH18 +9 mm (After_Scraping vs CEH4). For the five unmonitored cuts (CEH40, CEH41, CEH42, Scrape A, Scrape B — all pre-dating the dipwell network or lacking an in-cut well) H_0 is assumed equal to the CEH36 value and flagged as such in the figure annotation. The CEH36 BACI response also yields an inferred excavation depth of $H_0/Sy \approx 0.42 \text{ m}$ via the live C3 specific yield (Script 17, S.12), consistent with the 0.40 m design depth.

The scrape interior (footprint plus a 10 m collar) is masked: each cut slack rises at the excavation, not falls, so the field displayed is the surrounding drawdown — the dipole counterpart of the slack rise observed at the monitored cuts (the §5.7 scraping BACI step). The modelled drawdown reach is not independently resolved in the 88-well network (Script 09b distance-decay, $p = 0.54$).

The spatial reach of the drawdown field is modelled (leaky-aquifer superposition); the edge magnitudes at three of the eight cuts are empirically anchored to measured BACI responses. The same field is used by `plot_scrape_coastal_net()`, `plot_net_state_map()`, and `plot_public_panel()` via the shared `_scrape_field(gx, gy)` helper. The figure is the methodological evidence underpinning §5.4.2's "scrape as topographic drain" reading of the wet-slack BACI response, and contributes the scrape-zone half of the clearfell pre-fell baseline composite below.

Clearfell pre-fell baseline drawdown composite
(20_clearfell_baseline_drawdown.png)

Script 20 v1.25.0 (2026-06-30): Epoch filter applied — Oct 2023 cuts (CEH18, CEH21) excluded from this figure. See CHANGELOG_delta_2026-06-30_scrape_drawdown_physics.md.

The clearfell BACI documented in §4.6 and §5.7 controls for a number of confounders through the easting $\times$ time covariate, the CWB main effect, and the three-tier control hierarchy. The clearfell-baseline composite figure makes the confounder-set visible as a single drawdown field: scrape + Storm Brendan retreat, both head losses, both rendered on the shared sequential drawdown scale. The composite displays *what the clearfell BACI is correcting against*.

The scrape component of this figure includes only the Feb 2013 (CEH40, CEH41, CEH42) and Apr 2015 (CEH36, Scrape A, Scrape B) cuts — five cuts in total. The Oct 2023 cuts (CEH18, CEH21) are explicitly excluded: they postdate the clearfell event (December 2017) by six years and played no part in the hydrological conditions at the time of felling. The same leaky-aquifer superposition with method-of-images coastal boundary correction is applied as for the equilibrium drawdown figure above, but restricted to the two pre-clearfell epochs via the epochs parameter of `_scrape_field()`. The Storm Brendan retreat component is the coastal-erosion drawdown evaluated at $\text{COAST_RETREAT_M} = 6 \text{ m}$ (a single Brendan-class event), with δ_0 and L from the live Script 25 fit. The composite is $\text{net} = \text{scrape} + \text{erosion}$ (both positive drawdowns; not a diverging-scale subtraction).

This figure is *not* a closed water budget. It is a paired-confounder construction: the two head losses that overlap in the western clearfell zone and that the §5.4.2/§5.4.3 framing names as the BACI's principal confounders. It replaces an earlier v1.8.0 attempt at a three-driver $\text{net} = \text{SLR} - \text{erosion} - \text{scrape}$ diverging-scale composite, which conflated head gain (SLR) with two head losses on a diverging colormap and required a $\pm 50 \text{ mm}$ cap that obscured the scrape's own $\sim 129 \text{ mm}$ magnitude (v1.10.0 framing rationale). The clearfell-baseline composite drops SLR — SLR head gain remains available on its own figure (the SLR head response, 20_slr_response.png, and the SLR – erosion net, 20_coastal_net_effect.png). The composite is rendered on the shared sequential drawdown scale so the scrape and the storm-retreat drawdown read colour-to-colour against each other and against the forest drawdown field (20_drawdown_propagation_nohead.png) — a like-for-like magnitude comparison across the three driver mechanisms.

Public summary three-driver comparison panel (20_public_drivers_panel.png)

A display-only companion figure produced at Script 20 v1.11.0, intended for the lay-summary report and presentations. The panel is a 1-over-2 portrait layout: coastal erosion spans the top of the page, with the forest canopy drawdown and the dune scrape drawdown side-by-side below. Each subpanel uses the shared discrete six-colour banded drawdown scale; the single horizontal colorbar at the bottom of the figure is sourced from the forest panel so the full 2–150 mm range is shown (coastal erosion alone reaches only ~21 mm; scrape ~129 mm; forest spans to ~150 mm). Faint context dipwells are drawn behind each subpanel for orientation; labelled band-boundary contour lines (the `DRAWDOWN_LINE_LEVELS` constants 5, 10, 25, 50, 100 mm with white-haloed labels) overlay each subpanel so the low bands read clearly regardless of fill tint. The plain-language title and caption reinforce the central scientific message: same colour = same amount, no single driver is the whole story. The figure is reproducible directly from `plot_public_panel()` and reuses the underlying field helpers (`_forest_field`, `_scrape_field`, `_erosion_field`) so the public-facing figure is byte-for-byte consistent with the technical figures elsewhere in the chapter.

Net water-table state map (20_net_state_map.png)

Script 20 v1.26.0 (2026-06-30): Scrape rise zones now contribute +H₀ gain to the net state field (previously treated as no change). See `CHANGELOG_delta_2026-06-30_scrape_drawdown_physics.md`.

The net water-table state map combines all five driver mechanisms into a single cell-by-cell balance: SLR head gain, clearfell canopy-removal gain, and scrape-rise gain on the positive side; forest drawdown loss and coastal-erosion drawdown loss on the negative side. The scrape component enters the net field as a dipole: each cut's footprint (plus a 10 m collar) contributes a gain of +H_{0i} mm (the measured BACI response for monitored cuts, assumed = CEH36 value for unmonitored cuts), while the surrounding area contributes a drawdown loss via the superposition-with-images field described above. The net formula is:

$$\text{net}(x,y) = \text{SLR_gain} + \text{clearfell_gain} + \text{scrape_rise} - \text{forest_loss} - \text{erosion_loss} - \text{scrape_drawdown}$$

where `scrape_rise` is +H₀ at rise-zone pixels and zero elsewhere, and `scrape_drawdown` is the leaky-aquifer superposition field (zero at rise-zone pixels). This correctly represents the dipole nature of each cut — the surrounding water table is drawn down, but the slack itself rises — and ensures the rise-zone patches in the figure reflect the actual net field values rather than a purely visual

overlay. All eight cuts are included at their equilibrium epoch (no epoch filter here — the figure represents the full current-state scrape network, not the pre-clearfell baseline).

Shared drawdown scale across the four drivers

Script 20 v1.4.0 introduced a single shared drawdown colour scale across the forest, scrape, coastal-erosion, and clearfell-baseline composite figures, with refinements through v1.11.1 to the discrete band ramp. The scale is `DRAWDOWN_FILL_LEVELS = [2, 5, 10, 25, 50, 100, 150]` mm, banded at the low end (each band is an equal-width colour segment on the colorbar, denser at low magnitudes where the small-magnitude maps reside) and capped at 150 mm with an over-range `set_over` colour. The colours are a discrete six-step warm ramp (`DRAWDOWN_BAND_COLOURS`: cream `#fff3b0` → yellow → gold → amber → orange → brown, then `#5c1d02` for >150 mm) chosen so adjacent bands are visually distinguishable at $\alpha = 0.62$, replacing a continuous `YlOrBr` ramp whose 2–5 / 5–10 / 10–25 mm bands rendered as near-identical pale creams (v1.11.1 fix). The honest-magnitude consequence is that the forest panel reads dark (it spans to ~150 mm) while the erosion panel reads pale (it reaches only ~21 mm) — that is the intended scientific comparison, not a deficiency in the colour scale. The SLR map (`20_slr_response.png`) keeps its own linear `GnBu` colormap because head gain is a different quantity from head loss; no shared-scale comparison applies.

Updated parameter table — additions for the v1.1.0–v1.11.3 figures. The coastal-process parameter table above documents the erosion, SLR, and net fields. The additional parameters introduced by the scrape and composite figures are:

Parameter	Symbol	Value	Source	Field
Scrape centre easting	<code>SCRAPE_CENT_RE_E</code>	241 161 m OSGB36	CEH36, the documented 2015 scrape site	Scrape, Clearfell-baseline, Public
Scrape centre northing	<code>SCRAPE_CENT_RE_N</code>	363 306 m OSGB36	CEH36, the documented 2015 scrape site	Scrape, Clearfell-baseline, Public
Scrape long-axis length	<code>SCRAPE_LON_G_M</code>	60 m	Estimate (Newborough Warren scrape)	Scrape, Clearfell-baseline, Public

Parameter	Symbol	Value	Source	Field
			footprint)	
Scrape short-axis length	SCRAPE_SHORT_M	30 m	Estimate (Newborough Warren scrape footprint)	Scrape, Clearfell-baseline, Public
Scrape long-axis bearing	SCRAPE_BEARING_DEG	45° from N (NE)	DEM-derived; long axis points into the warren	Scrape, Clearfell-baseline, Public
Scrape edge drawdown magnitude	H _{0i} (scrape)	CEH36 +130 mm, CEH21 +74 mm, CEH18 +9 mm (live BACI shifts); unmonitored cuts assumed = CEH36 value	Live: Script 09a, S.7	Scrape, Clearfell-baseline, Public
Inferred scrape excavation depth	D	≈ 0.42 m (back-calculated; H ₀ / S _y)	Inferred output (not an input)	Scrape (informational only)
Up-gradient propagation flag	SCRAPE_FAVOR_UPGRADIENT	True	Coastal-drain physical reasoning	Scrape, Clearfell-baseline, Public
Seaward truncation flag	SCRAPE_TRUNCATE_SEAWARD	False (CEH36 is inland)	Site-specific (active only for coastal-margin scrape scenarios)	Scrape, Clearfell-baseline, Public
Seaward minimum elevation	SCRAPE_SEAWARD_MIN_ELEV_M	1.5 m AOD	Foreshore-clip threshold (active only when	Scrape (flag-dependent)

Parameter	Symbol	Value	Source	Field
			truncation is on)	
Shared drawdown fill levels	DRAWDOWN_ FILL_LEVELS	[2, 5, 10, 25, 50, 100, 150] mm	Banded for low-magnitude resolution + high-end ceiling	Forest, Scrape, Erosion, Clearfell-baseline, Public
Shared drawdown fill alpha	DRAWDOWN_ ALPHA	0.62	Adjacent-band contrast at the low end	Forest, Scrape, Erosion, Clearfell-baseline, Public

Site-specific choices.

- **Three zero-head sea boundaries.** Anchor points are placed every 200 m along the southern Menai Strait edge and the eastern and western Caernarfon Bay edges. Without them the linear interpolation would extrapolate the on-site head pattern off the warren. The anchors are at saltwater elevation (0 m AOD) and reflect the tide-controlled boundary on three sides.
- **FOREST_INTERCEPTION from config.py.** Script 20 imports the 24 % value and applies it to $\bar{P}$ only for clusters 4 and 5 in the residual computation. Other clusters use raw $\bar{P}$. The choice matches the SSM design matrix construction in Script 03 (S.3) and the interception treatment documented in F.4.
- **CEH14 annotation.** The residual map carries a fixed annotation for CEH14 (a northern-edge well that consistently shows the largest positive residual). The annotation prints the well's actual α at runtime rather than hardcoding the value. The CEH14 signal — large positive water-balance residual co-located with the bedrock ridge — is the qualitative anchor for the ridge-recharge interpretation taken up in Script 23 (S.16).

Site-specific choices and rationale (suite-level)

- **50 m grid as the spatial resolution.** Scripts 07, 11b, 19 and 20 all use the same grid. Cell size is large enough to suppress single-well noise in the IDW

and small enough that contours read smoothly at the warren's 3.5×2.6 km extent. It is not a project-wide convention — `10b_spatial_step_maps.py` uses 40 m (flagged in S.5) — but it is the spatial-figure convention.

- **`is_forest = (cl == 4 or cl == 5)`**. The forest-cluster test is the same in both scripts, matching `FOREST_CIDS = (4, 5)` in `config.py` (F.4). Script 19 uses an inline literal; Script 20 imports `FOREST_INTERCEPTION` from `config.py` but uses an inline membership check.
- **The viewer's equation engine is shared with Script 21.** Both use the same monthly equilibrium perturbation, and ship the same clearfell and thinning multipliers (loaded via `clearfell_common`). The broadleaf β_2 values were briefly divergent between the two source files (Script 19 v2.6.0 used local literals; v2.7.0 consolidated to `utils.config`) — see *Limitations*.
- **Forest-drawdown propagation constants from C3.** The forest-drawdown map's diffusivity inputs are sourced from the live pipeline: $S_y \approx 0.31$ from the C3 cluster median of `17_wtf_well_sy.csv` (Script 17, S.12), and β_3 from the C3 row of `03_03_cluster_mechanistic_coefficients.csv` (Script 03, S.3) — $\approx 0.058 \text{ month}^{-1}$ under the `limit=1` policy — giving $\lambda \approx 223 \text{ m}$. $K = 6 \text{ m/day}$ (Betson 2002) and saturated thickness $b = 5 \text{ m}$ remain literature/estimate literals. The S_y and β_3 loads are dynamic, so the figure tracks pipeline state.

Limitations and known caveats

- **Displayed flow vectors carry direction only.** Both the head-surface map (`20_head_surface_streams.png`) and the residual map (`20_residual_ssm.png`) normalize (U, V) to unit length before plotting. The docstring's "normalized Darcy quiver" is accurate as to direction; the Darcy $K = 6 \text{ m/day}$ does not enter the displayed vectors. A reader expecting quiver length to scale with flow magnitude will not see that. $K = 6 \text{ m/day}$ is used in the forest-drawdown propagation λ length scale, not in the static-figure quivers.
- **The `idw_surface()` name is misleading.** It calls `scipy.interpolate.griddata(method='linear')` — Delaunay piecewise-linear interpolation, not Shepard inverse-distance weighting. The misnomer is shared with `map_utils.add_idw_surface()` and the S.5 chapter; consistency with the older scripts argues against renaming.
- **The broadleaf β_2 values are consolidated to a single source of truth.** `utils/config.py` carries `BROADLEAF_B2_WINTER = 0.8817` and `BROADLEAF_B2_SUMMER = 1.0750` — derived from Script 21's 12-month broadleaf β_2 profile (chapter S.14) averaged over the canopy-state

phenological windows (Nov–Apr for winter, May–Oct for summer). Script 19 imports these names directly from config. §4.9.6 of the main report carries these values.

- **Remaining hardcoded literals in the forest-drawdown map.** $K = 6.0$ m/day (Betson 2002), $b = 5.0$ m saturated thickness, $H_0 = 150$ mm, $FLOW_WEIGHT = 0.4$, and $UPHILL_PENALTY = 2.0$ are literals in `plot_drawdown_propagation()`. S_y and β_3 are now read from the live pipeline (Script 20 v1.0.1; see *Site-specific choices*). K has an authoritative literature source (Betson 2002); the saturated thickness b and initial drawdown H_0 are site-mean estimates with no published per-cluster value at this site and remain reasonable defensive literals — but should be read explicitly in any sensitivity discussion. The flow-weighting parameters are tuning constants for the Dijkstra cost surface, not physical quantities.
- **Broadleaf preset uses annual-mean interception, with phenology folded into β_2 .** The broadleaf preset sets $sI_c4 = sI_c5 = 0.15$ for both winter and summer, but deciduous canopy intercepts $\approx 0\%$ in winter and $\approx 25\%$ in summer. The annual mean 0.15 is correct; the seasonal split is absorbed into $sB2_w / sB2_s$ rather than into interception itself. A reviewer inspecting the slider values may find this counter-intuitive; the engine still computes the correct equilibrium because the β_2 split absorbs the phenology.
- **Broadleaf interception is a literature average, not a site measurement.** Komatsu et al. (2011) is a deciduous-forest annual mean across multiple sites and species. Unlike the Freeman (2008) 24 % value, which is a Newborough Warren measurement on Corsican pine, the broadleaf value is generic. S.14 inherits the same constraint.
- **Linear-domain assumption.** All scenarios remain inside the steady-state equilibrium framework. Within-year dynamical trajectories — the timing of summer minima and winter peaks — are not resolved. The 0.5–1.5× slider range stays within the linear domain of the fitted β coefficients.
- **The 50 m grid does not extrapolate.** All interpolations are bounded by the well-network convex hull (plus the sea-boundary anchors). Map areas near the network margins are interpolated rather than measured, particularly along the northern bedrock ridge where well density drops to zero. The static figures and the viewer’s ridge mask flag these zones; map values at the network margins should not be read as observations.

Outputs

File	Contents	Downstream consumer
19_spatial_groundwater/ r/ scenario_viewer.html	Self-contained interactive HTML calculator (250–300 KB)	GitHub Pages site (online supplementary tool); §4.9 reference
19_spatial_groundwater/ r/ 19_scenario_summary.csv	Per-scenario × season × cluster Δh table (6 scenarios × 3 seasons × 6 spatial units = 108 rows)	§4.9.6 <i>Forestry scenarios</i>
20_spatial_figures/ 20_head_surface_streams.png	Mean annual head surface with stream network and flow vectors	§4.9 <i>Spatial groundwater</i> (Figure)
20_spatial_figures/ 20_residual_ssm.png	SSM water-balance residual with independent flow vectors and CEH14 annotation	§4.9 <i>Spatial groundwater</i> (Figure)
20_spatial_figures/ 20_slope_gradient.png	50 m smoothed DEM ridge slope, independent topographic check	§4.9 supporting diagnostic; not a numbered report figure
20_spatial_figures/ 20_drawdown_propagation_nohead.png	Flow-weighted forest drawdown propagation with λ length-scale annotation; filled-contour no-head form on the shared discrete drawdown scale (primary form emitted by <code>main()</code> since v1.4.0; the with-head form is callable but not in the default run)	§4.9.6 <i>Forestry scenarios</i> (Figure)
20_spatial_figures/ 20_coastal_erosion.png	Coastal-erosion drawdown field for a single Brendan-class retreat event; dune-edge	§3.4.5 <i>Drawdown-Field Visualisation</i> / §4.9.4 <i>Coastal-Margin Processes</i> (Figure)

File	Contents	Downstream consumer
	front; δ_0 and L from live Script 25 fit	
20_spatial_figures/ 20_slr_response.png	Sea-level-rise head response over a finite five-year window; erfc transient, mean-sea-level boundary; $D = K \cdot b / S_y$ with S_y live from C3 WTF	§3.4.5 / §4.9.4 (Figure)
20_spatial_figures/ 20_coastal_net_effect.png	Cell-by-cell net coastal change (SLR head gain minus erosion drawdown); diverging scale	§3.4.5 / §4.9.4 (Figure)
20_spatial_figures/ 20_scrape_drawdown_no_head.png	Dune-scrape drawdown field, all eight cuts, leaky-aquifer superposition with method-of-images coastal boundary correction; H_0 empirically anchored at three monitored cuts (CEH36, CEH21, CEH18) from live Script 09a BACI shifts; shared discrete drawdown scale	§3.4.5 / §5.4.2 (Figure 52)
20_spatial_figures/ 20_clearfell_baseline_drawdown.png	Composite drawdown on the clearfell pre-fell baseline = Feb 2013 + Apr 2015 scrape cuts only (five cuts; Oct 2023 cuts excluded as post-clearfell) + Storm Brendan coastal retreat; both head losses on the shared sequential drawdown scale; paired-	§3.4.5 / §5.4.2 / §5.4.3 (Figure)

File	Contents	Downstream consumer
	confounder construction underpinning the §5.4.2/§5.4.3 BACI framing	
20_spatial_figures/20_public_drivers_panel.png	Public-summary three-driver comparison panel (forest canopy / dune scrape / coastal erosion); portrait 1-over-2 layout; shared drawdown scale; band-boundary contour lines	Lay-summary report (Figure)

Where the result appears in the report

- §3.4.5 *Drawdown-Field Visualisation* — the methodological home of the drawdown-field figures (forest drawdown, coastal erosion, sea-level-rise head response, SLR-erosion net coastal change, dune-scrape topographic-drain drawdown, and the clearfell pre-fell baseline composite). The methods-register equation forms appear here; the supplement carries the full parameter exposition above.
- §4.9 *Spatial groundwater* — the head-surface, residual, and slope maps from Script 20 (20_head_surface_streams.png, 20_residual_ssm.png, 20_slope_gradient.png).
- §4.9.4 *Coastal-Margin Processes and the Easting Gradient* — the three coastal-process figures (20_coastal_erosion.png, 20_slr_response.png, 20_coastal_net_effect.png) and their interpretive context.
- §4.9.6 *Forestry scenarios* — values from 19_scenario_summary.csv for the clearfell, broadleaf and thinning rows; the forest drawdown propagation map (20_drawdown_propagation_nohead.png) from Script 20.
- §4.9.7 *UKCP18 climate scenarios* — values from 19_scenario_summary.csv for the 2050s and 2080s rows.
- §5.4.2 *The scrape-as-drain reading of the wet-slack BACI* — the dune-scrape drawdown figure (20_scrape_drawdown_nohead.png), all eight cuts, with H_0 anchored empirically at three monitored cuts from live Script 09a BACI shifts, supplies the propagation framing for the §5.4.2 interpretation.
- §5.4.3 *Clearfell BACI confounder framing* — the clearfell pre-fell baseline composite (20_clearfell_baseline_drawdown.png) renders the scrape +

Storm Brendan retreat composite that the BACI's easting $\times$ time correction implicitly controls for.

- Lay-summary report — the public-summary three-driver comparison panel (20_public_drivers_panel.png) is the primary lay-facing figure communicating that no single driver is the whole story.

Cross-references

- **F.3** — SSM displacement formulation; the β coefficients the engine consumes.
- **F.4** — DRAINAGE_DATUM, FOREST_INTERCEPTION, FOREST_CIDS, BROADLEAF_INTERCEPTION, BROADLEAF_B2_WINTER/SUMMER, the canonical cluster constants.
- **F.5** — `map_utils.load_dem_hillshade`, `map_utils.add_idw_surface`; the shared spatial-figure machinery.
- **S.3** — per-cluster β coefficients (03_03_cluster_mechanistic_coefficients.csv).
- **S.5** — established the 50 m grid convention and the `idw_surface` naming.
- **S.7** — Script 10e's BACI-corrected clearfell β_2 multiplier, consumed by Script 19 via `clearfell_common.load_clearfell_b2_multiplier()`. Also Script 09a's BACI shifts (09_baci_shifts.csv), consumed live by Script 20 to set the empirically anchored edge drawdown magnitudes H_{0i} for three of the eight scrape cuts (CEH36 Pure_Scraping vs CEH4, CEH21 After_Scraping vs CEH22, CEH18 After_Scraping vs CEH4); this empirical anchoring is what underwrites the scrape-as-drain reading at §5.4.2 and the clearfell pre-fell baseline composite at §5.4.3.
- **S.12** — WTF cluster S_y estimates consumed for the volumetric translations in §4.9.
- **S.14** — Script 21 forestry scenarios; uses the same monthly perturbation engine. The broadleaf β_2 value reconciliation between Scripts 19 and 21 is owed to that chapter.
- **S.16** — Script 23 ridge-recharge lag hypothesis test; the qualitative anchor for the CEH14 residual annotation on the residual map (20_residual_ssm.png).

S.14 Script 21 — Forestry scenarios

Step 23 / 27. Phase 10 — Forestry Scenario Analysis in `run_analysis.py`; sixth chapter under Phase 4 — Climate and Spatial Context in the supplement.

Motivation

Section 4.9.6 of the main report quantifies how the C4 Main Forest groundwater regime would respond to three management alternatives — full clearfell, 50 % thinning, and conversion to broadleaf — under the calibrated SSM. Script 21 is the analysis behind that section. It produces a synthetic mean-year hydrograph that shows each scenario as a monthly perturbation from the observed C4 baseline, alongside an empirical benchmark derived from the BACI clearfell evidence accumulated in Script 10 (chapter S.7), and a series of observed summer-minimum depth distributions that document what actually happened across the site through the 2018 clearfell intervention.

The script is paired with Script 19 (chapter S.13) but answers a complementary question. Script 19’s `scenario_viewer.html` is the interactive engine — the reader-facing calculator that exposes every scenario knob through sliders. Script 21 is the publication artefact: a fixed, citeable figure that pins specific scenario shifts to specific months of the calendar year, and overlays the BACI-observed clearfell displacement as the empirical reference. The gap between the modelled and observed clearfell responses is itself an output of the analysis, not a defect: it represents the multi-year drainage adjustment that the single-step monthly perturbation deliberately does not capture, and it informs the report’s framing of what the SSM-based scenarios should be read as.

Script 21 is also the canonical source for the broadleaf monthly β_2 profile that Script 19 and the wider pipeline consume in seasonal-mean form. The 12 monthly values are defined inside `build_scenarios()` and propagate downstream through `config.BROADLEAF_B2_SUMMER` and `config.BROADLEAF_B2_WINTER`, and through `pipeline_params.csv`. Where this chapter and S.13 disagree on summary values, S.14 is the canonical reference.

Inputs

File	Origin
<code>03_master_data.csv</code>	Script 03 — per-well β coefficients (recharge, atmospheric draw, drainage) and cluster ID
<code>01_climate.csv</code>	Script 01 — monthly P and PET (RAF Valley)
<code>03_regional_averages_maod.csv</code>	Script 03 — cluster-mean head time series in m AOD
<code>Well_locations_height.csv</code>	Raw data input — DEM ground

File	Origin
	elevation and pipe-top elevation per well
01_wells_clean.csv (+ extended)	Script 01 — quality-controlled monthly depth series; used by the distribution and BACI-violin figures
pipeline_scenario_params.csv	Script 01's consolidated parameter file — cluster β , S_y , h_{disp} , peak month, BACI-corrected β_2 multipliers, UKCP18 climate scaling
10a_report_numbers.csv	Script 10a — ANCOVA BACI clearfell step for Forest Impact (annual displacement)
10e_01_coefficient_shifts.csv	Script 10e — per-well Before / After β coefficients across the five BACI tiers (fallback source for the β_2 multiplier)
09c_01_summer_minima.csv	Script 09c — per-well annual summer minima (used by the scraping-era and BACI-violin functions for consistency with the scraping analysis)

The script does not refit the SSM. All coefficients are loaded from upstream pipeline outputs; the master CSV is the fallback path when `pipeline_scenario_params.csv` is unavailable.

Methodology

The Option 3 monthly perturbation. The script's central operation is a single-step monthly perturbation:

$$\Delta h(m) = \beta_1 \cdot [P_{\text{eff}}(\text{scen}, m) - P_{\text{eff}}(\text{base}, m)] - [\beta_2(\text{scen}, m) - \beta_2(\text{base})] \cdot \text{PET}(m)$$

This is implemented in `model_utils.monthly_perturbation()` and applied for each of the 12 calendar months. P_{eff} is the effective rainfall after canopy interception; the baseline corresponds to the Corsican pine canopy with `FOREST_INTERCEPTION = 0.24` (Freeman, 2008). $\beta_2(\text{base})$ is the cluster-mean atmospheric-draw coefficient from `03_master_data.csv`; $\beta_2(\text{scen}, m)$ is the scenario value, which may vary by month for broadleaf conversion and is a

uniform multiple of $\beta_2(\text{base})$ for clearfell and thinning. Each scenario therefore yields a length-12 vector of monthly Δh values relative to the pine baseline.

The β_3 drainage term does not appear. This is not an oversight. The formulation prices the first-month response of the water table to a forcing change: the level at the start of the month has not yet moved in response to the new canopy condition, so the drainage flux $\beta_3 \cdot (D + h(t-1))$ is identical under baseline and scenario and cancels in the difference. The cancellation is documented in the function docstring and is the same logical step that Script 19's `_well_dh()` makes. What this means in practice is that the modelled Δh is the immediate forcing response, not the steady-state new equilibrium. Across many months the drainage term will adjust as h itself drifts; the perturbation deliberately does not follow that adjustment forward, which keeps it free of the intercept drift that troubles a literal forward SSM simulation.

The synthetic mean-year hydrograph. For Figure 1 (OUT_21_HYDROGRAPH), the scenario Δh vectors are added to the observed C4 mean monthly depth-below-ground cycle, which is computed from the `03_regional_averages_maod.csv` cluster-mean head series referenced to the C4-mean DEM elevation. The result is four 12-month hydrographs — Corsican pine baseline (observed), full clearfell (observed + Δh_{cf}), 50 % thinning (observed + Δh_{thin}), and broadleaf conversion (observed + Δh_{bl}). They are plotted on a single axis with the wet-slack winter threshold SD15b and the dry-slack summer threshold SD16 (Curreli et al., 2013) drawn for reference, along with the C1 and C2 observed cycles as spatial context. From Script 21 v1.2.0 the figure is accompanied by a companion data file (`21_forestry_01_hydrograph.csv`, `OUT_21_HYDROGRAPH_CSV`) so the scenario separations can be cited from a pipeline output rather than read off the figure. It carries the plotted monthly depths for every line in tidy long format, and a trough/separation summary that reports each series' deepest month, its separation from the observed C4 baseline at the baseline trough month, and the largest Jun–Sep summer separation with the month it occurs. The two-anchor design is deliberate: the scenario shifts move the deepest month — the observed baseline troughs one month later than the shifted scenarios — so a single unqualified “trough separation” would be ambiguous.

The BACI-observed clearfell displacement is overlaid as a benchmark band: a horizontal shift applied to the C4 baseline that brackets the empirical post-2018 displacement at the Forest Impact well WMC3 (and at the wider Edge tier; chapter S.7 has the full BACI account). On the live pipeline data the ANCOVA Forest-Impact clearfell step is +0.120 m annual (95 % CI [+0.050, +0.189], $p < 0.001$,

N = 162 months). The summer BACI benchmark band is the clearfell step from a Jun–Sep refit of the Script 10a ANCOVA specification, fit on N = 52 months under the same model terms as the annual headline: intercept, climate-water-balance covariate, distance-weighted scraping, clearfell dummy, CWB × clearfell interaction, and easting × time. The directly-fitted summer step is +0.046 m (95 % CI [−0.069, +0.162], $p = 0.44$, $R^2 = 0.337$) — not significant at conventional thresholds against the Jun–Sep subsample. A CWB-dropped sensitivity variant (+0.123 m, $p = 0.058$) is emitted to `10a_report_numbers.csv` as `ANCOVA_Forest_Impact_clearfell_step_summer_noCWB`; the full specification is preferred on Δ AIC grounds. The annual headline is the durable BACI-observed clearfell signal; the summer step corroborates the direction but is not retained as a separate statistical claim. This whole construct replaces an earlier arithmetic construct ($\text{BACI_ANNUAL} \times 1.5034$) preserved transiently as a fallback in Script 21 v1.0.2 and removed in v1.0.3; the legacy construct is documented as Defect 14 in the project flags log. The chapter writes the headline as displacement towards ground (water table closer to surface = positive in the script’s convention).

The broadleaf monthly β_2 profile (canonical). Broadleaf conversion is the one scenario where β_2 varies through the year. Deciduous phenology — bare in winter, full canopy in summer — gives lower transpirative draw than evergreen pine when leaves are off and higher draw at peak LAI. The canonical 12-month profile is defined directly in `build_scenarios()`:

Month	β_2 multiplier	Phenological state
Jan	0.85	Leaves off
Feb	0.85	Leaves off
Mar	0.88	Bud burst beginning
Apr	0.92	Partial leaf
May	0.98	Approaching full leaf
Jun	1.08	Full leaf, high ET
Jul	1.12	Peak ET
Aug	1.15	Peak ET draw
Sep	1.10	Late season, leaves turning
Oct	1.02	Early leaf fall
Nov	0.92	Mostly bare
Dec	0.87	Dormant

The seasonal means consumed by downstream consumers are derived from this profile. With the canonical seasonal windows used by `config.py` — winter = Nov–Apr, summer = May–Oct — the means are 0.8817 (winter) and 1.0750 (summer). These are the values exported as `BROADLEAF_B2_WINTER` and `BROADLEAF_B2_SUMMER` in `config.py` and consumed by `scraping_common.compute_scenario_bars()` and by the figure-5 scenario-comparison panel within Script 21 itself. They round to 0.88 and 1.08 in any column where two decimal places are reported. The seasonal-window choice is phenologically aligned: by May the broadleaf canopy is essentially functional ($\beta_2 \geq \sim 1$ for May through October), and through October the canopy is still operative; April and November are leaf-off shoulder months and are grouped with winter accordingly. The choice is broadleaf-specific — other seasonal-window definitions in the pipeline (Script 17’s PET-negligible Nov–Mar, Script 11b’s summer-minimum Jun–Sep) reflect different physical questions and are unchanged.

Broadleaf interception is treated separately from pine and constant through the year: `BROADLEAF_INTERCEPTION = 0.15` (Komatsu et al., 2011, deciduous annual mean). This is a single annual-mean value rather than a seasonal split because the year-round interception loss for broadleaf — roughly 20 % growing-season, zero in winter — averages to about 15 % when the canopy is present for half the year.

The BACI-corrected β_2 multiplier load. The clearfell β_2 multiplier is not hardcoded. The script reads it through `_load_baci_params()`, which first attempts `pipeline_params.load_params()` (the consolidated `pipeline_scenario_params.csv` written by Script 01), and on failure falls back to direct read of `10e_01_coefficient_shifts.csv`. The mathematics of the BACI correction itself live in `clearfell_common.load_clearfell_b2_multiplier()`, which computes:

$$\text{clearfell_mult} = \text{Edge_ratio} - \text{Climate_Ctrl_ratio} + 1.0$$

where each ratio is the tier-mean of `b2_after / b2_before` across the wells of that BACI tier (Edge: CEH31, CEH20, CEH30, CEH16; Climate Ctrl: CEH9, NW7, NW6, NW5, WMC2). The Edge tier is used in preference to the Impact well WMC3 because Impact-side post-felling β_2 is suppressed by the canopy that has been removed — the Edge wells retain pine canopy and receive lateral moisture from the cleared compartment, which gives the cleaner detection of the β_2 shift attributable to felling rather than to the canopy removal itself. The Climate Ctrl correction subtracts the post-2017 background drift. The chapter on Script 10 (S.7) develops the BACI tier structure in full.

On the live pipeline data the loaded multiplier is 1.0315 (Edge ratio 0.9830 minus Climate Ctrl ratio 0.9515, plus 1.0), with the 50 % thinning multiplier defined as half the clearfell perturbation: $\text{thinning_mult} = 1.0 + (\text{clearfell_mult} - 1.0) / 2 = 1.0157$. Each tier ratio is computed as the ratio of the tier-mean `b2_after` to the tier-mean `b2_before` (i.e. the pooled mean of the after-period values divided by the pooled mean of the before-period values — ratio of means, not mean of ratios). These values flow into the `b2` arrays in `build_scenarios()` and onwards into the monthly perturbation. The fallback value of 1.20 hard-coded in `_FALLBACK_B2` reflects an older empirical estimate from a pre-BACI-correction era and is only used if `10e_01_coefficient_shifts.csv` is missing.

The summer-minimum distribution figure. Figure 2 (`OUT_21_DISTRIBUTIONS`) is observational, not modelled. For each cluster (C1, C2, C3) and for C4 split into pre-felling (2005–2017) and post-felling (2018–2025) phases, the script computes annual summer-minimum depths from the cluster-mean monthly depth series (the well-mean within each cluster, then the annual maximum depth across June–September). The display is a strip/violin plot with the Curreli thresholds drawn as horizontal reference lines. The accompanying CSV (`21_forestry_02_distributions_means.csv`) carries phase-level mean, median, and the percentage of summers in which the cluster mean fell below SD16, which is the simplest summary the figure aims to convey.

The scraping-era, BACI-violin, and scenario-comparison panels. Three further figures sit alongside the headline two. The scraping-era figure (`OUT_21_SCRAPING`) shows annual summer-minimum depths for the three scraped wells (CEH36, CEH18, CEH21) and an unscraped control (CEH4) across four eras (Pre-2015, 2015–17, Post-fell 2018–23, Post-rescrape 2024+), drawing summer minima per well from `09c_01_summer_minima.csv` so the numbers reconcile with chapter S.6. The BACI-violin figure (`OUT_21_BACI_VIOLIN`) shows the same summer-minima distributions but aggregated by BACI tier (Impact, Edge, Forest Ctrl, Coastal Ctrl, Climate Ctrl) across the Pre-2015, 2015–17, and Post-fell phases; tier definitions are loaded from `clearfell_common.TIERS` and match the suite-shared 17-well network. From Script 21 v1.1.0 the per-zone summer minima are computed through the shared `clearfell_common.annual_summer_minimum()` estimator — each well’s annual Jun–Sep minimum taken individually with the provenance / `min_measured = 2` filter, then averaged across the wells of the zone (per-well-then-aggregate), on a window floored at 2011 — so the violin matches the canonical Script 10d clearfell-BACI summer-minimum method by construction; the single-well Impact zone (WMC3) reduces to that well’s own series. The scenario-comparison panel (`OUT_21_SCENARIO_COMPARE`) is the cross-pipeline

summary grouped bar chart: per-cluster volumetric Δ in mm w.e./month for clearfell, thinning, broadleaf, and the two UKCP18 climate scenarios (Dry and Wet, 2050s RCP8.5, summer means). The computation runs through `scraping_common.compute_scenario_bars_from_params()`, which uses the same Option 3 formulation but with the full $\beta_1 \cdot \Delta P - \Delta \beta_2 \cdot \text{PET} - \beta_3 \cdot \Delta h_{\text{disp}}$ form (no β_3 cancellation, because the comparison includes climate scenarios that perturb both P and PET and yield non-trivial h_{disp} shifts when written in equilibrium terms). The forestry rows in this panel use `BROADLEAF_B2_SUMMER = 1.0750` from `config.py` — the same canonical value derived from the monthly profile above.

Volumetric companion (21_forestry_06). Alongside the volumetric panel, `plot_scenario_comparison()` also emits `21_forestry_06_summer_scenario.csv`: the per-cluster equilibrium volumetric metric `Delta_vol_summer_mm_per_month` for the three forest-management scenarios (clearfell, thinning, broadleaf), taken directly from the monthly scenario values and byte-identical with the forestry rows of `09b_05` by construction. The former summer-minimum amplification conversion (`scraping_common flux` $\rightarrow$ `head` $\div$ `Sy` $\rightarrow$ `summer minimum` $\times$ `amplification factor`) was removed on 2026-07-02. This gives the \$4.9.6 forestry narrative a forestry-module source for the volumetric ecological metric rather than having to reach into the scraping suite for it; the climate-scenario comparison remains with `09b_05`.

Site-specific choices and rationale

- **Monthly perturbation, not forward SSM simulation.** The SSM contains an implicit intercept term that closes the water balance at the well's long-term mean. A literal forward simulation from an arbitrary initial condition does not include that intercept and accumulates drift that has no physical meaning. The Option 3 perturbation circumvents this by expressing the scenario response as a difference against the observed baseline — the intercept cancels, the drainage term cancels, and what remains is the first-order forcing-change response. This is the first-year-adjustment magnitude, not a steady-state prediction.
- **The BACI benchmark band as an empirical reference, not a target.** The figure overlays the BACI clearfell observation alongside the modelled clearfell trajectory. The two are not expected to coincide. The modelled value is a first-year forcing-change response; the BACI observation is an integrated displacement over multiple years of post-felling adjustment, during which the drainage flux responds to the new water-table position and the canopy regrowth, soil moisture, and groundwater–subsurface coupling evolve. The gap between modelled and BACI is the multi-year

drainage feedback that the perturbation framework deliberately does not capture. Both are useful: the modelled value bounds the immediate response, the BACI value documents what happened.

- **The β_2 multiplier loaded dynamically through pipeline_params.** The clearfell β_2 multiplier flows from Script 10e to pipeline_scenario_params.csv (Script 01) to Script 21 at runtime via _load_baci_params(). There is no hardcoded scenario constant in the script; the BACI evidence drives the scenario directly. The value 1.0315 in the current pipeline is the BACI-corrected Edge-tier ratio (0.9830) with the Climate Ctrl drift (0.9515) subtracted, giving a multiplier of 1.0315.
- **Forest interception treatment.** The FOREST_INTERCEPTION = 0.24 (Freeman 2008) value applies to the Corsican pine baseline and to 50 % thinning (taken as 0.12, i.e. half). Broadleaf uses BROADLEAF_INTERCEPTION = 0.15. Clearfell uses 0 (no canopy). Interception is a partition of incoming rainfall, not an additive term added to PET — it reduces effective rainfall reaching the water table. The treatment is consistent with the SSM design matrix construction in Script 03 (chapter S.3) and is documented in F.4.
- **C4 only for the synthetic hydrograph.** The hydrograph figure shows C4 (Main Forest) only. C5 (Coastal Forest) is forested and shares the same β_2 multipliers in the scenario-comparison panel, but it has no BACI clearfell observation — the 2018 felling was the C4-side compartment, not C5. The hydrograph's value lies in the BACI overlay, so plotting C5 alongside C4 without an empirical reference would mislead. C5 does appear in the scenario-comparison panel (Figure 5) where the modelled response is presented at face value without a BACI anchor.
- **Cluster-mean β coefficients, not per-well.** All scenario calculations use the cluster mean of beta_1_recharge, beta_2_atmospheric_draw, and beta_3_drainage from 03_master_data.csv. The scenario response is a cluster-level quantity; per-well projections would over-represent the spread within each cluster, which is captured separately by Script 19's spatial viewer (chapter S.13).

Outputs

Filename (relative to outputs/21_forestry_s cenarios/)	Contents	Downstream consumer
21_forestry_01_hydrog raph.png	Synthetic C4 mean-year hydrograph: pine baseline, full clearfell, 50	Main report §4.9.6

Filename (relative to outputs/21_forestry_s cenarios/)	Contents	Downstream consumer
	% thinning, broadleaf conversion; BACI benchmark band; SD15b / SD16 reference lines; C1, C2 observed cycles as context	
21_forestry_01_hydrog raph.csv	Companion data file for the hydrograph figure (Script 21 v1.2.0+). Block 1 — the 12-month depth- below-ground series for every plotted line (four management scenarios, observed C4 baseline, both BACI benchmarks, observed C1/C2 context) in tidy long format. Block 2 — a trough/separation summary: each series’ deepest month and depth, its separation from the observed C4 baseline anchored at the baseline trough month, and the largest Jun–Sep summer separation with the month it occurs	Main report §4.9.6 / §5.7.4
21_forestry_02_distri butions.png	Summer-minimum depth distributions per cluster, with C4 split pre/post-felling	Main report §4.9.6
21_forestry_02_distri butions_means.csv	Phase-level mean, median, SD, min, max summer-minimum depths and percentage of	Reference table

Filename (relative to outputs/21_forestry_s cenarios/)	Contents	Downstream consumer
	summers below SD16	
21_forestry_03_scrapi ng_eras.png	Annual summer minima per scraped well (CEH36, CEH18, CEH21) and unscraped control (CEH4), four eras	Main report §4.5 / §4.9.6 cross-link
21_forestry_03_scrapi ng_era_means.csv	Era-level summaries for the same wells	Reference table
21_forestry_04_baci_z one_violin.png	Summer-minimum violins by BACI tier (Impact, Edge, Forest Ctrl, Coastal Ctrl, Climate Ctrl) across three phases	Main report §4.6 cross- link
21_forestry_04_baci_z one_means.csv	Tier-level summaries	Reference table
21_forestry_05_scenar io_comparison.jpg	Per-cluster volumetric Δ (mm w.e./month) for clearfell, thinning, broadleaf, UKCP18 Dry, UKCP18 Wet	Main report §4.9.6
21_forestry_05_scenar io_comparison.csv	Same data in tidy form	Reference table
21_forestry_06_summer _scenario.csv	Per-cluster equilibrium volumetric metric Delta_vol_summer_mm_p er_month for clearfell, thinning, broadleaf — byte-identical with 09b_05 forestry rows; former summer- minimum amplification conversion removed	Main report §4.9.6

2026-07-02

Limitations and known caveats

- **First-year adjustment, not steady state.** Every Δh produced by `monthly_perturbation()` is the immediate response to a forcing change with h held at its baseline trajectory. The water table will continue to adjust in subsequent months as the drainage flux responds to the new equilibrium head. The synthetic hydrograph is therefore a useful magnitude reference rather than a prediction of the post-management cluster mean depth in year 5 or year 10.
- **The BACI gap is a feature.** The empirically-observed clearfell displacement at WMC3 and Edge tier (currently +0.120 m annual on the live data, with the directly-fitted summer step at +0.046 m, $p = 0.44$, $N = 52$) exceeds the single-month modelled Δh because the BACI step integrates multi-year drainage feedback that the perturbation does not. The figure presents both values; the gap is part of the result. The summer step is not significant at $\alpha = 0.05$ against the Jun–Sep subsample, and is not retained as a separate statistical claim — see the headline-values paragraph above and the Defect 14 entry in the flags log.
- **Broadleaf β_2 — canonical here, consolidated in config.** The 12 monthly multipliers above are the canonical source. The seasonal means consumed by downstream code are 0.8817 (Nov–Apr) and 1.0750 (May–Oct), and these are what `config.BROADLEAF_B2_WINTER` and `config.BROADLEAF_B2_SUMMER` export. Script 19 imports the same names directly from config. §4.9.6 of the main report carries these values.
- **Thornthwaite PET caveat.** The script uses the Thornthwaite-estimated monthly PET from RAF Valley (chapter S.10) without revision. The Thornthwaite formulation is temperature-only and is known to underestimate summer PET at this site relative to Penman–Monteith; the scenario differentials are therefore conservative in the summer scaling. Chapter S.10 carries the depth-dependent sensitivity check on this; the implications carry through to the C4 summer-minimum projections in §4.9.6.
- **C5 absent from the synthetic hydrograph.** No BACI observation exists for C5; the hydrograph plots only what can be anchored to an empirical reference. C5 appears in the scenario-comparison panel where its modelled response is presented on its own terms, and chapter S.13 carries the Script 19 viewer where C5 can be interrogated interactively.
- **Sy enters at the scenario-comparison panel only.** The hydrograph and the distributions are in head units (m) throughout. The volumetric

scenario-comparison panel converts through `cluster.Sy` from `17_wtf_01_sy_estimates.csv` (chapter S.12); the interception correction on forest-cluster `Sy` is the one Script 17 carries, with the uncertainty discussed there. The scenario-comparison panel inherits the same `Sy` uncertainty.

- **Fallback parameters in `_load_baci_params()`.** There are no arithmetic BACI fallbacks: missing `10a_report_numbers.csv`, missing `ANCOVA_Forest_Impact_clearfell_step`, or missing `ANCOVA_Forest_Impact_clearfell_step_summer` each raise an explicit error with a remediation message rather than silently substituting an arithmetic construct. The β_2 multiplier fallback (1.20) is independent and remains in place — it activates only if `10e_01_coefficient_shifts.csv` is missing, and the script prints a `WARNING:` message when it does; chapter consumers should check the log for this message before quoting a β_2 multiplier.

Where the result appears in the report

- **§4.9.6 *Forestry scenarios*** — Figures 21-01 (synthetic hydrograph) and 21-02 (distributions); the headline narrative on clearfell, thinning, and broadleaf at C4; the BACI-modelled gap.
- **§4.9.6** — Figure 21-05 (scenario comparison panel) for the cross-cluster volumetric summary.
- **§4.5 and §4.6 cross-links** — Figures 21-03 (scraping-era) and 21-04 (BACI tier violins) where the forestry chapter draws on the scraping and BACI evidence for context.

Cross-references

- **F.3** — SSM displacement formulation; β_3 cancellation in the perturbation derivation.
- **F.4** — `FOREST_INTERCEPTION`, `BROADLEAF_INTERCEPTION`, `BROADLEAF_B2_WINTER`, `BROADLEAF_B2_SUMMER` constants.
- **F.5** — `monthly_perturbation()`, `clearfell_common.load_clearfell_b2_multiplier()`, `scraping_common.compute_scenario_bars_from_params()` shared utilities.
- **S.3** — Origin of the per-well β coefficients in `03_master_data.csv`.
- **S.6** — Scraping intervention; complement to forestry as a direct-surface management lever; `09c_01_summer_minima.csv` is the shared minima source for Figure 21-03.

- **S.7** — Clearfell BACI; origin of the β_2 multiplier and of the +0.120 m annual displacement that anchors Figure 21-01’s benchmark band, along with the directly-fitted +0.046 m summer band (Script 10a v1.3.0).
- **S.10** — Depth-dependent PET sensitivity; carries the Thornthwaite caveat that propagates here.
- **S.12** — WTF Sy values that the scenario-comparison panel consumes for the volumetric conversion.
- **S.13** — Script 19’s `scenario_viewer.html` is the interactive companion; the Option 3 engine is shared, and the broadleaf β_2 seasonal-mean divergence (defect 11) is documented here as the canonical anchor.

Phase 5 — Post-pipeline supplementary analyses

The eleven Phase 4 chapters (S.8–S.14) cover the climate-and-spatial scripts that feed §4 of the report. Phase 5 collects six further chapters whose methodological purpose is supplementary: each consumes pipeline intermediates, none feeds downstream pipeline scripts, and the conclusions they produce inform the report’s discussion rather than its core results chain. S.15 covers Script 25 (coastal-retreat gradient, step 26/43, Phase 11 in `run_analysis.py`). S.16 covers Scripts 22, 23, and 24 (residual diagnostics, steps 27–29/43, Phase 12 in `run_analysis.py`). S.18 covers Script 26 (van Willegen 5-year MSL aggregation, step 30/43, Phase 13). S.18b covers Script 11 Section 5 (Tool A — spring MSL transfer function, embedded in Script 11 at step 22/43, Phase 1) together with Script 26b (Tool B — UKCP18 RCP8.5 MSL5 projections, step 31/43, Phase 13). S.18c covers Script 26c (MSL5 report-format figures, step 32/43, Phase 13). S.19 covers Scripts 28, 29 and 30 (cluster framework diagnostics, steps 33–35/43, Phase 14). S.20 covers Scripts 32, 33 and 35 (observed differential change and climate-response envelope, steps 36–38/43, Phase 15). S.21 covers Scripts 24b, 31, 31b and 34 (supplementary standalone diagnostics, steps 39–42/43, Phase 16). S.17 is appendices. All scripts in S.15, S.16, S.18, S.18b, S.18c, S.19, S.20, and S.21 are part of the pipeline orchestrated by `run_analysis.py`; the “post-pipeline supplementary” framing reflects their role in the report (discussion-feeding rather than results-feeding), not their orchestration status. Scripts 26c (MSL5 report-format figures, Phase 13) and 27 (greyscale figures, Phase 17) are display/utility steps excluded from the 41-step analytical count; Script 26c is documented in S.18c and Script 27 in Appendix A.

S.15 Script 25 — Coastal-retreat gradient

Step 24 / 30. Phase 11 — Coastal-Retreat Gradient Analysis in `run_analysis.py`; first chapter under Phase 5 — Post-pipeline supplementary analyses in the supplement.

Motivation

The BACI ANCOVA in S.7 (Script 10a) returned a small but highly significant easting $\times$ time interaction. Eastings west of the clearfell block were drifting downward against time at a rate the BACI's other covariates — cumulative water balance, scraping, the clearfell step itself — could not absorb. That term is what allows the BACI to recover an unbiased clearfell-step coefficient; but it raised a separate question. What was easting $\times$ time actually correcting for? The candidate answer was the retreat of the Caernarfon Bay shoreline immediately west of the site, episodic but acute (Forgrave 2020; ongoing monitoring by Walker-Springett, Bangor University; ~50 m between 2014–2020 with most of the loss concentrated in Storm Brendan, January 2020). If the eroding shoreline is exporting freshwater storage westward as the dune front loses width, wells closer to the coast should show steeper summer-minimum declines than inland wells, with the steepness decaying inland on a length scale set by the aquifer's diffusive response.

Script 25 tests that hypothesis. It fits a network-scale, physics-based non-linear regression of per-well summer-minimum trends against perpendicular distance to the eroding Caernarfon Bay shoreline, recovers a forest-independent distance-decay signal (coast-edge anomaly $\approx -29 \text{ mm yr}^{-1}$ on a far-field climate background of $\approx -6 \text{ mm yr}^{-1}$, inland reach $\approx 900 \text{ m}$), and uses the fit to corroborate the BACI ANCOVA's easting $\times$ time term.

Inputs

File	Origin
<code>data/well_distance_to_coast.csv</code>	Versioned data input, computed once out-of-pipeline from OS Open Map Local TidalBoundary
<code>outputs/01_wells_clean.csv</code>	Script 01 — reference-network depth time series
<code>outputs/01_wells_extended.csv</code>	Script 01 — extended-network depth time series
<code>outputs/01_locations.csv</code>	Script 01 — well coordinates
<code>outputs/01_climate.csv</code>	Script 01 — RAF Valley monthly P, PET

File	Origin
outputs/03_master_data.csv	Script 03 — cluster assignments
outputs/14_climate_projections/ 14_summer_trend_stats.csv	Script 14 — cluster-centroid summer- minimum slopes
outputs/10_clearfell_baci/ 10a_02_ancova_full_coefficients.c sv	Script 10a — BACI ANCOVA coefficients for the corroboration check

Methodology

Functional forms. Two candidate decay forms are fitted, each with three parameters: a coast-edge anomaly δ_0 (mm yr⁻¹), an inland reach L (m), and a far-field climate background c (mm yr⁻¹).

The linear-with-cutoff form follows a Dupuit–Forchheimer strip-aquifer interpretation, in which the water-table response to a steadily retreating boundary decays linearly with distance from that boundary and asymptotes to the climate background beyond the reach L :

$$\delta(d) = \max(\delta_0 \cdot (1 - d/L), 0) + c$$

(For $\delta_0 < 0$ — the physically expected sign for an eroding boundary — the max becomes a min so that the gradient does not change sign at $d > L$.)

The exponential decay form follows a diffusive / transient interpretation, in which the response falls off smoothly with no hard cutoff:

$$\delta(d) = \delta_0 \cdot \exp(-d/L) + c$$

Both forms are tried at each specification, and AIC is used to compare them.

Three specifications. The fits are run at three increasingly stringent specifications to eliminate forest cover as a confound:

1. **Full network.** All clusters, with the 17 clearfell-zone wells (Impact + Edge + Forest controls + Coastal controls, imported from `clearfell_common.py`) dropped to prevent the felling-induced rise from contaminating the gradient signal. FE1–4 (broadleaf-restock) and CEH36 (scraping) are also excluded as direct intervention effects.
2. **Forest-free network.** C1, C2, and C3 only; C4 and C5 excluded entirely. If the distance-decay signal survives the removal of both forest clusters, forest cover cannot be the principal driver of the apparent gradient. This is

the *headline* specification — used downstream for cluster attribution and BACI corroboration.

3. **C3 only.** The single non-forested cluster with wells at intermediate distances. The closest C3 well sits at $d \approx 400$ m, so a 3-parameter fit on this restricted distance range is under-identified — c is held to the forest-free value. The C3-only fit is the sensitivity check: if the gradient is real, it should reproduce on a single cluster.

Per-well summer-minimum slopes. Script 25 fits its own per-well annual summer-minimum slopes; it does *not* load per-well slopes from Script 14 (which produces cluster-centroid slopes only). For each well with at least 8 hydrological years (`PANEL_OBS_MIN_YEARS = 8`) of April–September observations, an OLS regression of summer-minimum depth on hydrological year returns a slope (m yr^{-1}) and standard error. These appear in the scatter visualization (panel a of the diagnostic figure) and as a per-well mean reported alongside the Script-14 centroid in the cluster partition.

The panel fit method. Non-linear fits run on the monthly long-form panel, not on the per-well slopes directly. Each panel observation has three components: well- and-month fixed effects absorbed by within-well demeaning (Frisch–Waugh–Lovell), a centred-cumulative-water-balance covariate ($P - PET$ anomaly cumsum, mm), and a constrained decay term $\delta(d_w) \cdot t$ whose coefficient is fixed at 1 — non-linear search runs over the three parameters of $\delta(d_w)$ alone. Fitting is by `scipy.optimize.least_squares` with bounds on each parameter; the Jacobian gives a covariance matrix and Wald 95% CIs.

Cluster attribution. The headline (forest-free linear-capped) fit is applied to each cluster’s mean distance-to-coast to compute a *gradient-only* component ($\delta_0 \cdot (1 - \bar{d}/L)$), clipped at zero beyond the cutoff). This is added to the climate background c and subtracted from the observed Script-14 centroid summer-minimum slope; the remainder is the *residual* — the part of the cluster’s deepening that neither gradient nor far-field climate explains.

BACI corroboration. The BACI ANCOVA fits `delta_easting × months_since_intervention` as a covariate to absorb monotonic spatial drift. Its coefficient (units: m per m-easting per month) implies an absorbed differential deepening rate at the impact zone relative to each control tier, which can be compared directly against the gradient model’s predicted differential $\delta(d_{\text{impact}}) - \delta(d_{\text{control}})$. For each (control tier × impact zone) pair, Script 25 computes both, runs a z-test of one against the other, and labels the pair “consistent” if $|z|$

< 2. The Forest-Impact comparison is the principal check: if the BACI's easting $\times$ time term is doing nothing but coastal-retreat correction, BACI absorption and model prediction should agree.

Site-specific choices and rationale

- **Caernarfon Bay west-facing coast only.** Only the ~15 km Caernarfon Bay west-facing shoreline is included. The Menai Strait north-east coast is a tidal channel, not subject to the SW-prevailing-wind erosion regime, and per project knowledge is not retreating. Llanddwyn Island is a bedrock islet, hydrogeologically separate. The Malltraeth Sands estuary is sheltered estuarine. The included polyline wraps around Abermenai Point at the SE end of the bay, which matters for the geometry of several eastern wells that would otherwise be misallocated.
- **Perpendicular distance via shapely.** `shapely.geometry.Point(easting, northing).distance(coastline)` in EPSG:27700 — the minimum perpendicular distance to any segment, geometrically correct for the irregular coastline. Computing once out-of-pipeline and versioning the result avoids putting geopandas and shapely into the main pipeline's dependency set for a single analytic question.
- **Clearfell-zone exclusion.** All 17 BACI wells (WMC3 Impact + 4 Edge + 5 Forest controls + 2 Coastal controls + 5 Climate controls) are dropped from the full and forest-free fits. The December 2017 clearfell drives a positive water-table rise at Impact and Edge wells that would appear as a non-monotonic distance perturbation if left in. The list is imported from `clearfell_common.py` so any BACI design update propagates automatically.
- **C3-only fit holds c fixed.** With no C3 well at $d \rightarrow 0$, the three-parameter fit becomes under-identified: δ_0 and c trade off. Holding c to the forest-free value (-6.40 mm yr^{-1} for linear-capped; -5.24 mm yr^{-1} for exponential) reduces the C3-only problem to two parameters and tests whether the same coast-edge anomaly is recoverable from C3 wells alone.
- **Storm Brendan is not separated.** The coastal-retreat rate is treated as a long-run mean. Newborough's retreat is in fact episodic — order-of-magnitude losses concentrated in single storms, with Storm Brendan (January 2020) the principal event of the study period. The mean rate is the methodologically defensible quantity given the data; the implications are taken up in *Limitations*.

Results to describe at the methodological level

All six fits (3 specifications $\times$ 2 functional forms) are written to 25_01_panel_fit_parameters.csv. Under the forest-free specification the linear-capped and exponential forms are near-indistinguishable on AIC ($\Delta\text{AIC} = -0.5$, $\text{exp} - \text{lin-cap}$, i.e. a marginal exponential preference); the linear-capped form remains the headline on the strength of its Dupuit–Forchheimer strip-aquifer interpretation rather than a decisive AIC margin. The three linear-capped fits return δ_0 values in the range -26.1 to -29.2 mm yr $^{-1}$, L values in the range 894–972 m, and c values around -6.5 mm yr $^{-1}$. The closeness of the C3-only δ_0 (-26.1 mm yr $^{-1}$, SE 2.55) to the forest-free network δ_0 (-29.0 mm yr $^{-1}$, SE 1.92) is the chapter’s primary identification claim: the distance-decay signal is recoverable from a single non-forested cluster, ruling out forest cover as the principal driver.

The per-cluster attribution under the headline fit is in 25_03_cluster_partition.csv. C5 (Coastal Forest, mean $\bar{d} \approx 420$ m) carries by far the steepest observed slope (-37.7 mm yr $^{-1}$ at the Script-14 centroid); the gradient model attributes roughly 41 % of that to coastal retreat. C3 (Western Residual, $\bar{d} \approx 800$ m) has a modest observed slope (-6.8 mm yr $^{-1}$), of which the gradient accounts for about 32 %. C1, C2, and C4 sit beyond or near the inland reach L ; the gradient component is zero or near-zero, and their observed deepening (~ -10 to -12 mm yr $^{-1}$) is left as climate plus an unexplained residual.

The BACI corroboration in 25_04_baci_corroboration.csv is the chapter’s substantive payoff. For Forest-Impact — the principal BACI clearfell test — the easting $\times$ time coefficient absorbs an implied differential deepening of -16.8 mm yr $^{-1}$ (SE 5.0); the gradient model predicts -11.0 mm yr $^{-1}$. The two agree at $z = -1.18$, “consistent”. Climate-Impact is the one comparison the live data flags as *not* consistent ($z = 2.17$): the BACI absorbs $+10.0$ mm yr $^{-1}$ where the gradient model — with impact and climate-control wells at almost identical distances to coast — predicts essentially zero (-0.9 mm yr $^{-1}$). The Forest-Edge comparison is consistent ($z = -0.29$) and the Climate-Edge comparison sits just inside the threshold ($z = 1.96$). The Impact-zone Forest comparison is what matters for the BACI clearfell step’s interpretation, and it is consistent; the Climate-Impact discrepancy reflects the easting $\times$ time covariate at the Climate-control tier absorbing spatial drift in components the coastal-retreat model does not represent. Headline numbers are repeated in 25_report_numbers.csv for downstream cross-referencing.

Limitations and known caveats

- **The coastal-retreat rate itself is not fitted.** The model fits a per-well *response* to distance from a *known eroding boundary*; it does not derive the retreat rate from the water-level data. The literature value (~50 m over 2014–2020, with acute losses in Storm Brendan, January 2020) is the assumed driver.
- **Episodic retreat is averaged to a mean rate.** Storm-event losses are absorbed into a long-run mean. Defensible for the question asked (does retreat *signal* in the data?) but understates year-to-year variability and any qualitative non-linearity in the dune-front aquifer's response.
- **The C3-only sample is modest.** 19 wells after the clearfell-zone exclusion (the live 21-well C3 minus the WMC3 BACI Impact well and the CEH36 scrape site). Three wells sit at $d < 400$ m — CEH4 (221 m), CEH21 (232 m) and CEH18 (302 m) — anchoring the inward end of the fit; the remaining 16 are at $d \geq 458$ m. The δ_0 recovered is a partial extrapolation from intermediate-to-inner distances. Closeness to the forest-free δ_0 is the reassurance; under-identification (hence c held fixed) is the caveat.
- **The cluster partition's per-well subsample is small in the forested clusters.** After the clearfell-zone exclusion, the per-well slope panel contains only two C4 wells (CEH13, CEH14) and one C5 well (NW9). The cluster attribution accordingly uses the Script-14 centroid slope (computed from the full cluster hydrograph) for the *observed* column, with the per-well mean reported alongside. C5's `gradient_pct_of_observed` (40 %) should be read with that in mind.

Outputs

File	Contents	Downstream consumer
25_coastal_gradient/ 25_01_panel_fit_parameters.csv	All six fits (3 specs $\times$ 2 forms): δ_0 , L , c with SE and 95% CI, AIC, n	Report §5.4.3, §5.7.2
25_coastal_gradient/ 25_02_per_well_summer_min_slopes.csv	Per-well annual summer-minimum OLS slopes (m yr^{-1}), SE, p , R^2 , n_{years} , easting, northing, cluster, distance to coast	Diagnostic figure panel (a)
25_coastal_gradient/	Cluster attribution:	Report §5.8, §5.9

File	Contents	Downstream consumer
25_03_cluster_partition.csv	observed centroid slope, observed per-well mean, predicted gradient + climate + residual, gradient pct	
25_coastal_gradient/ 25_04_baci_corroboration.csv	BACI easting × time absorption vs model prediction for each (control × impact zone) pair, with z-test and consistency verdict	Report §5.4.3, §5.7.2
25_coastal_gradient/ 25_05_fit_diagnostic.jpg	Two-panel figure: (a) per-well slopes vs distance with three fits, (b) cluster stacked-bar decomposition	Report figure
25_coastal_gradient/ 25_06_baci_corroboration_chart.jpg	Forest plot of BACI absorption vs model prediction per pair	Report figure
25_coastal_gradient/ 25_07_cluster_decomposition.png	Horizontal stacked-bar figure: each cluster's observed centroid slope decomposed into climate + coastal + residual. New at v1.1.0 (2026-05-29, folded in from retired Script 30)	Report §4.8.2 figure
25_coastal_gradient/ 25_report_numbers.csv	Headline numbers in project-standard Parameter, Well, Era, Value, Unit, Note format	Report cross-referencing

All paths resolve through `utils/paths.py` (`OUT_25_FIT_PARAMETERS`, `OUT_25_PER_WELL_SLOPES`, `OUT_25_CLUSTER_PARTITION`, `OUT_25_BACI_CORROBORATION`, `OUT_25_FIT_DIAGNOSTIC`, `OUT_25_BACI_CHART`, `OUT_25_CLUSTER_DECOMP_FIG`, `OUT_25_REPORT_NUMBERS`, `DIR_25`).

Update — 2026-05-29 (Script 25 v1.0.1 → v1.1.0). Two changes were folded into Script 25 in the post-review cascade. First, `25_03_cluster_partition.csv` gains two percentage-share columns parallel to the existing `gradient_pct_of_observed`: `climate_pct_of_observed` and `residual_pct_of_observed`. All existing columns and numerical values remain byte-identical. Second, a new figure `25_07_cluster_decomposition.png` renders the per-cluster decomposition as a horizontal stacked-bar visualization, with each bar showing the cluster’s observed centroid summer-minimum slope (diamond marker) decomposed into climate-uniform background (blue), coastal-retreat gradient (red), and unattributed residual (grey), and the cluster’s mean perpendicular distance to coast and (n-in-fit / n-total) annotated to the right. This figure lands in §4.8.2 of the main report as the visual companion to the per-cluster decomposition table inserted in that section. The fold-in retired a short-lived standalone script (`30_cluster_slope_decomposition.py`, 2026-05-27 to 2026-05-29) that produced the same figure from already-existing Script 25 outputs; the decomposition logic is now resident in `cluster_partition()` and the rendering in `plot_cluster_decomposition()`. See also §S.19 for the post-review cluster framework diagnostics that motivated the surfacing of this decomposition as a report-level table and figure.

Where the result appears in the report

- §5.4.3 — coastal-retreat gradient context for the BACI clearfell step.
- §5.7.2 — corroboration of the BACI’s easting $\times$ time covariate as a coastal-retreat correction at the Impact zone.
- §5.8 — cluster-level summer-trend partition (C5 Coastal Forest attribution, C3 Western Residual attribution).
- §5.9 — implications for the whole-site interpretation of summer-minimum decline.
- Conclusions — coastal retreat as a network-wide confound on long-term trend interpretation.

(The acknowledgement edits to §5.4.3, §5.7.2, §5.8, §5.9 and Conclusions regarding coastal retreat’s network-wide implications are in progress at the time of writing.)

Cross-references

- **F.3** — SSM displacement formulation; the per-well slopes in Script 25 are independent of the SSM but the cluster attribution lands against an SSM-fitted picture.

- **S.3** — Script 03’s cluster assignments are the basis for the per-cluster partition.
- **S.7** — Script 10 suite (BACI); Script 25’s BACI corroboration directly references Script 10a’s $\text{easting} \times \text{time}$ coefficient.
- **S.13** — Script 19’s residual and per-well slope figures show the same coast-to-inland gradient from an independent visualization. Script 20’s coastal-process figures (`20_coastal_erosion.png`, `20_slr_response.png`, `20_coastal_net_effect.png`) consume δ_0 and L from this chapter’s `25_01_panel_fit_parameters.csv` live at generation time; see `COASTAL_NET_VS_EASTING_MEMO.md` (project store) for the SLR-extended comparison against the BACI $\text{easting} \times \text{time}$ coefficient.
- **S.16** — Scripts 22/23/24 residual diagnostics, which benefit from the coastal-gradient framing established here.

S.16 Scripts 22, 23, 24 — Residual diagnostics

Steps 27, 28, 29 of 35 in the orchestrator (Phase 12 — Residual Diagnostics in `run_analysis.py`); inside the supplement’s 41-step analytical count; second chapter under Phase 5 — Post-pipeline supplementary analyses in the supplement.

Motivation

Section §4.4.2 and §5.3 of the main report frame the spatial pattern of the SSM water-balance residual as ridge-derived recharge: wells along the northern bedrock ridge flank, with CEH14 as the canonical example, carry persistent positive residuals that the three-term SSM does not represent. The interpretation is a physically plausible reading of the spatial map (Figure 55, see S.13) — wells nearest the ridge carry the largest unmodelled inputs, consistent with lateral subsidy from the rock outcrop. But spatial correlation with the ridge is not, on its own, mechanistic evidence that water is travelling from the ridge to the wells. A model that misspecifies the rainfall response, for example, could produce the same spatial pattern if the misspecification happens to be largest near the ridge.

Scripts 22, 23 and 24 are the analytical apparatus that subjects the ridge-recharge interpretation to mechanistic testing. They share a single source — the SSM residual series $e(t)$ at each well — and ask three complementary questions about that series. Script 22 characterises the residual: how time-structured is it well-by-well, and do the wells with the largest mean residuals (the intercept α from Model B) also have the most autocorrelated residuals? Script 23 puts the residual to a physical test: if it carries genuine ridge-to-well travel time, the cross-correlation

peak lag against rainfall should increase with distance from the ridge. Script 24 asks the alternative-hypothesis question: does the residual carry a seasonal signature consistent with the Thornthwaite PET estimate misrepresenting summer demand, or is the seasonal structure flat (consistent with steady ridge baseflow), or something else again? Together the three diagnostics constrain what the residual can and cannot be, and in doing so constrain what the ridge-recharge interpretation can safely claim.

Inputs

File	Origin	Used by
01_wells_clean.csv	Script 01 — cleaned wide well-level matrix	22, 23, 24
01_climate.csv	Script 01 — monthly P_m and PET	22, 23, 24
01_locations.csv	Script 01 — well coordinates (E, N, OSGB36)	22, 23, 24
02_clusters.csv	Script 02 — cluster ID per well (k=5 partition)	22, 23, 24
RAF_Valley_Climate.csv	raw data input — monthly sunshine hours	24 only

Sunshine hours are loaded from the raw RAF Valley climate CSV directly (DATA_CLIMATE_RAW) rather than from a pipeline intermediate. The pipeline’s 01_climate.csv carries P and PET only — sunshine hours are not used anywhere else in the pipeline and so were not propagated into the cleaned climate file. Script 24 parses the original MMM YY date format (e.g. “Jan 25” = January 2025) directly from the raw file.

Methodology — Script 22 (residual lag analysis)

Script 22 refits Model B — the SSM with a constant intercept — on every well with at least 140 months of usable data, on the full record rather than the 100-month recent window used by Script 07. The model is the canonical displacement-formulation SSM with an additive constant:

$$\Delta h(t) = \alpha + \beta_1 \cdot P(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot (D + h(t-1))$$

The fit is performed via the shared `fit_ssm_intercept()` in `model_utils.py` (see F.5), with `HEADLINE_LAG = 0` from `config.py`. The intercept α absorbs whatever

constant part the unmodelled hydrological inputs carry; the residual series $e(t) = \Delta h_{\text{observed}} - \Delta h_{\text{predicted}}$ then represents only the *time-varying* component of the unmodelled contribution. This decomposition is methodologically important. If the residual at a ridge-adjacent well averages, say, +0.030 m/month, that constant offset is by construction absorbed into α — it cannot also show up in $e(t)$. Any signal that $e(t)$ carries must therefore vary in time, which is the only kind of signal that can carry a lag structure against rainfall.

Choosing the full record over Script 07's 100-month window is deliberate: lag analysis at the monthly timestep needs as many degrees of freedom as the record will allow, and rolling-window stability of the lag signal (a stage 3 analysis not implemented in Script 22 itself) requires that windows of different lengths be available. The trade-off is that Model B's β fits in this script are full-record averages, whereas Table 2 / Figure 45b in the main report use the recent 100-month window. The two should be similar but will not be identical — Script 22 is an analytical companion, not a replacement for the headline fits.

For each well's residual series, Script 22 then computes an AR(1) diagnostic: it regresses $e(t)$ on $e(t-1)$ via OLS and reports the AR(1) coefficient ϕ , its p-value, and the residual standard deviation. The decision criterion `AR1_WHITE_THRESHOLD = 0.3` flags wells whose residuals are sufficiently autocorrelated that pre-whitening would be required before any cross-correlation analysis. A second diagnostic — the α -vs- ϕ scatter — tests whether the wells with the largest persistent subsidies (high α) are also the wells with the most time-structured residuals (high $|\phi|$). If both conditions held, the unmodelled input would be both high in magnitude and variable in time, the physical signature expected from a sustained lateral flux that itself responds to seasonal or interannual climate.

The Script 22 outputs are five files: a wide residuals CSV (rows = months, columns = wells, values = $e(t)$), a per-well fits table with α , β_1 , β_2 , β_3 , R^2 , AR(1) statistics, mean and standard deviation of the residual, and four figures — an AR(1) histogram by cluster, a spatial map of AR(1) coefficient, the α -vs- ϕ scatter, and example residual time series for one well per cluster (chosen as the longest-record well with a valid AR(1) fit).

Methodology — Script 23 (ridge recharge lag test)

Script 23 is the principal mechanistic test of the ridge-recharge interpretation. The physical hypothesis is straightforward: if the water-balance residual reflects genuine lateral recharge from the northern rock ridge, then water arriving at any given well had to travel from the ridge to that well. That travel has a time, and

that time must increase with distance from the ridge. The cross-correlation peak lag between the residual series and rainfall — at what lag does the residual best correlate with past rainfall? — should therefore increase systematically with distance from a fixed ridge reference point.

The methodological subtlety is what Script 22's diagnostic implies for the test design. A single-period SSM leaves any rainfall response that takes more than one month to propagate through the vadose zone in the residual. At the monthly timestep this generic vadose-zone lag is non-negligible: at every well across the network, $e(t)$ will correlate to some extent with $P(t-1)$. This generic signal would dominate any ridge-specific structure if the test were run on the Model B residuals directly. Script 23 deals with this by fitting an *extended* model that includes both contemporaneous and lag-1 rainfall:

$$\Delta h(t) = \alpha + \beta_{10} \cdot P(t) + \beta_{11} \cdot P(t-1) - \beta_2 \cdot PET(t) - \beta_3 \cdot (D + h(t-1))$$

The $\beta_{11} \cdot P(t-1)$ term absorbs the generic monthly-timescale vadose-zone response. The residual from this extended model is, by construction, free of the generic lag-1 confound, and any remaining lag structure against rainfall is the candidate ridge signal. (A current design note in the docstring records that, with HEADLINE_LAG now configurable, the two-month spanning window of $P(t)+P(t-1)$ is retained pending a scientific review of whether the same absorb-the-generic-lag purpose could be achieved with an alternative shift.) The β_2 and β_3 from this fit are not propagated back to any pipeline output — Script 23 is a diagnostic refit, and the canonical β values in `03_master_data.csv` remain authoritative.

The ridge reference point is fixed at E = 241750, N = 364500 (OSGB36) — the northern rock-ridge centroid as estimated from the site DEM and the ridge KML overlay. Distance from the ridge to each well is computed as Euclidean distance in OSGB36 metres. A coordinate-validity filter (`MAX_RIDGE_DISTANCE_M = 3000` m) discards any well whose ridge distance exceeds 3 km, which removes records carrying garbage coordinates (the location CSV contains some non-well rows with placeholder coordinates 300+ km from the site) without affecting any real Newborough well — the most distant real well sits at roughly 2 km from the reference point.

For each well, Script 23 then computes the cross-correlation between the extended-model residual and rainfall at lags 0 to 12 months. Both series are pre-whitened before the CCF is computed: rainfall is pre-whitened once globally using its own AR(1) coefficient, and any residual series whose AR(1) coefficient exceeds $|0.2|$ is additionally pre-whitened to remove its own autocorrelation. The peak-

correlation lag N^* is the lag at which $|r|$ is largest. Bartlett's 95 % confidence threshold is computed per-well as $1.96/\sqrt{n_{\text{eff}}}$ where n_{eff} is the smallest sample size across the computed lags; wells whose peak $|r|$ falls below this threshold are classified as not showing a significant peak.

The hypothesis test itself is a Spearman rank correlation of peak lag against ridge distance, restricted to wells with a significant peak. The choice of Spearman rather than Pearson reflects the nature of the prediction: the physical theory predicts monotone increase, not linear proportionality, and the lag axis is discrete (integer months) rather than continuous. The verdict at the bottom of the summary is recorded in three branches: H1 supported (positive ρ , $p < 0.05$), an unexpected-direction outcome (negative ρ , $p < 0.05$), and H0 not rejected. The third branch carries the methodological caveat that a null result does not by itself rule out steady ridge baseflow — a constant subsidy that is effectively smoothed in time by the time it reaches the dune field would be observationally indistinguishable from the constant α that Model B already absorbs.

A structural caveat on the test design. Investigation of Script 23 in May 2026 established that the test as constructed cannot resolve the mechanism it is designed to detect, for three structural reasons: (a) monthly time resolution against a ~ 2 km transect collapses the expected lag range onto a 2–3 month discrete grid, leaving only a handful of distinguishable lag values across the network's distance range; (b) peak-lag is a poor cross-correlation summary, discarding amplitude and shape information that would carry the lateral-flux signature if it were present; and (c) the residual signal the mechanism would explain is approximately 2.5 % of annual flux, at or below the per-well intercept α uncertainty in the headline SSM. A reframed α -geography test was also explored — correlating per-well α against ridge distance and against elevation — and was inconclusive due to physical collinearity between elevation and ridge distance: the ridge is the high ground, so any elevation-related residual structure is partly indistinguishable from a ridge-distance-related one. The script is retained in the pipeline as documentation of the analysis attempted; its null result should not be interpreted as evidence against ridge recharge, only as evidence that the present monitoring design cannot resolve the question. Sub-monthly water-level monitoring at the ridge-adjacent wells, paired with rainfall at the same resolution, would be the design needed to test the mechanism cleanly; that data does not exist at this site for the analysis period.

Methodology — Script 24 (residual seasonality)

Script 24 is the third diagnostic. It refits Model B via `fit_ssm_intercept()` for every well (so the residuals are directly comparable with Script 22's), and asks whether those residuals carry a systematic seasonal signature. Two physically distinct hypotheses produce contrasting predictions. If Thornthwaite PET — computed from temperature only — underestimates true summer atmospheric demand by some proportion, the residual should be systematically negative in summer (JJA) and approximately zero in winter (DJF), with a sinusoidal climatology peaking near the temperature low. If the unmodelled input is instead a flat year-round ridge baseflow, the climatology should be approximately flat across all months at a constant positive offset — and that offset is, again, already absorbed into α , leaving the climatology of $e(t)$ close to zero everywhere.

The diagnostic computes a monthly climatology per well — the mean of $e(t)$ across all instances of each calendar month — and fits a single-harmonic sinusoid to the twelve monthly means:

$$\bar{e}(m) = a_0 + a_1 \cdot \cos(2\pi \cdot m/12) + a_2 \cdot \sin(2\pi \cdot m/12)$$

The fitted offset (a_0), amplitude ($\sqrt{a_1^2 + a_2^2}$), and phase (the calendar month at which the sinusoid peaks) summarize each well's seasonal residual structure. The winter-minus-summer contrast (mean of DJF months minus mean of JJA months) is reported alongside as a model-free check. The convention matches Paper 1 and the main report's residual-field discussion, and matches the cluster-stratified diagnostic introduced below.

The chapter-defining methodological detail is the sunshine-hours correlation. A naïve correlation of $e(t)$ against PET would be zero by construction — OLS makes residuals orthogonal to every regressor, so any test that uses PET as the seasonality proxy is testing nothing. To test whether real ET departs from the Thornthwaite estimate in a way that the fitted β_2 has not absorbed, Script 24 correlates $e(t)$ against monthly *sunshine hours* — an independent radiation-based ET proxy that is not in the regression. A systematic negative correlation across the network would indicate that high-insolation months carry extra ET losses that the temperature-only Thornthwaite estimate has not captured and that β_2 has therefore not been able to fit. A network-wide null result (correlations scattered within Bartlett bands) is direct evidence that Thornthwaite's seasonal bias, whatever it is, has been absorbed by β_2 and is not what the residual is reflecting.

A C3 split diagnostic — partitioning the Western Residual cluster by ridge distance at 1000 m — is retained as a within-cluster check that the seasonal

amplitude is not systematically different in the forest-adjacent subset versus the warren interior. The 1000 m threshold is a legacy parameter from the $k=4$ partition era, where the C3 cluster contained both forest-adjacent and interior wells; under the $k=5$ partition the forest-adjacent subset of the old C3 has been split out as C5 (Coastal Forest), and the diagnostic is now an *intra*-C3 robustness check rather than an across-cluster comparison.

The phase analysis uses a circular mean rather than an arithmetic mean to handle the December/January wrap correctly, and is plotted on a hydrological-year axis (September → August) to make the winter–early-spring versus summer signal easier to read off.

Diagnostic — cluster-stratified residual climatology (Script 24b, non-pipeline)

Script 24b (`24b_residual_climatology.py`) is step 39/43 in the orchestrated pipeline, wired into Phase 16 alongside Scripts 31, 31b, and 34 (§S.21). It runs after the canonical residual-diagnostics suite (Scripts 22–24) and reads their committed outputs. Its purpose is to ask whether the seasonal residual signature uncovered by Script 24 carries any cluster-level structure that would discriminate between candidate mechanisms — in particular, whether the winter-spring residual is concentrated in the forested clusters (as a canopy-interception over-estimation would predict) or in the open-dune cluster (as a recharge-nonlinearity would predict). Full documentation is in §S.21.1.

The method reads Script 22’s per-well residuals $e(t)$ directly — there is no SSM re-fit, so the diagnostic inherits whatever the canonical Model B has fitted at each well. For every well meeting Script 22’s 140-month minimum, it computes a per-well *winter-minus-summer contrast* defined as the mean of $e(t)$ across all calendar DJF months (December, January, February) minus the mean across all calendar JJA months (June, July, August). The five clusters are then aggregated by a well-level bootstrap with 1000 resamples within each cluster, producing a cluster-mean contrast, a 95 % confidence interval on the cluster mean, and a two-sided one-sample *t*-test of the cluster mean against zero. Per-well distance-to-ridge and a signed distance-to-forest-edge are attached as covariates — the forest-edge geometry comes from the “Forest” polygon in `Features.kml`, reprojected to OSGB36 / EPSG:27700, with positive values denoting wells inside the forest and negative values denoting wells outside. The ridge reference point is defined locally in the diagnostic, mirroring Script 24’s coordinates, so the script can run standalone against a clean main with no shared-file edits.

The cluster-stratified result ($k = 5$ partition, current `22_residuals_wide.csv`):

Cluster	Winter – Summer (mm) <i>p</i>		Interpretation
C3 Western Residual	+7.4	< 0.001	Substantive positive contrast
C4 Main Forest	+6.2	0.086	Marginal positive contrast
C1 Lake Edge	+7.3	0.198	Magnitude similar to C3 but small <i>n</i> and high variance
C2 Dune	+3.4	0.152	Small positive contrast, not significant
C5 Coastal Forest	–2.8	0.138	Small negative contrast, not significant

Within the 14 forest wells ($C4 \cup C5$), the contrast strengthens toward the ridge: regressing per-well contrast against ridge distance gives Pearson $r = -0.63$, $n = 14$ — wells closer to the ridge carry larger winter-minus-summer residuals.

The result is conservative against a canopy-interception over-estimation. If the $F = 0.24$ Freeman interception fraction were too large, the over-correction would concentrate the winter residual in the forested clusters ($C4$ and $C5$), because that is where the correction is applied. Instead, the largest contrast sits in $C3$ — an unforested cluster — and the $C3 > C4 > C5$ ordering places the smallest contrast in $C5$, which is the most ridge-distant of the two forest clusters. Canopy-interception miscalibration cannot account for that ordering; the diagnostic provides no data-driven case to revise $F = 0.24$. The within-forest gradient toward the ridge is consistent with a ridge-derived component but, on its own, cannot be separated from a site-wide recharge nonlinearity — the $C3$ result demonstrates that the same winter-spring excess exists where there is no canopy and no immediate ridge proximity.

Taken with Script 24's network-wide null on the sunshine-hours correlation, this places the seasonal residual signature as consistent with a winter-phased recharge mechanism rather than with either a Thornthwaite PET misspecification or a canopy-interception parameterization error. The mechanistic attribution beyond that — separating ridge-derived from site-wide

winter recharge — is left to future work, since the present monitoring resolution at monthly time-steps and ~2 km transect cannot discriminate cleanly between candidates that respond at similar bandwidths. Paper 1 §5.4 and SI S9.3 carry the same finding under the same terminology; this diagnostic is the methodological back-reference for those claims.

The diagnostic emits three CSVs (per-well contrasts with covariates; cluster bootstrap summary; forest-only gradient regression), a single 3×2 PNG panel figure (per-cluster contrast distributions, the spatial map of per-well contrasts, the forest-only contrast-vs-ridge-distance scatter with regression line, the per-cluster bootstrap CI bars, the per-cluster boxplots, and a forest-edge-signed-distance panel), and a plain-text interpretation file. Outputs are listed below in the chapter’s *Outputs* table under the “Script 24b — cluster-stratified residual climatology” group; full documentation is in §S.21.1.

Site-specific choices

- **Full record vs Script 07’s 100-month window.** Scripts 22 and 24 use the full per-well record to maximize statistical power for the lag and seasonality analyses, which need as many monthly observations as possible. The trade-off is that the Model B fits here are not identical to Table 2 / Figure 45b in the main report. The supplement’s headline coefficients remain the 100-month-window fits; Scripts 22, 23, 24 are diagnostic companions, not revisions.
- **140-month minimum record (MIN_MONTHS = 140).** Roughly 11.7 years of continuous data. Wells with fewer observations are dropped from all three scripts. This excludes shorter-record monitoring points without compromising the Bartlett confidence threshold calculation, which becomes unreliable below 100 monthly observations.
- **Common exclusion set (EXCLUDED_WELLS_NORM = {ceh3, ceh4, ceh7, ceh8, ceh37}).** CEH7, CEH8 and CEH37 carry over the upstream exclusions from Script 07. CEH3 sits at the tidal boundary and is outside the SSM’s operational domain (report §S4.4.2). CEH4 carries both a coastal-erosion drift and a post-2017 clearfell drawdown pulse (S.7); both confound any lag or seasonality signal. The same set is shared across 22, 23 and 24 so that the three diagnostics describe a coherent well population.
- **Ridge reference point (E = 241750, N = 364500).** Estimated from the site DEM and the ridge KML feature. The point sits on the bedrock ridge crest at the northern site boundary, approximately equidistant from the eastern and western extents of the ridge feature. The diagnostic is not sensitive to small displacements of this point along the ridge axis — what matters is the

gradient of distance across the well network, which is dominated by the perpendicular distance from the ridge axis.

- **MAX_RIDGE_DISTANCE_M = 3000 m.** A coordinate-validity filter, not a methodological exclusion. Real Newborough wells all sit within ~2 km of the ridge reference; anything beyond 3 km is a placeholder coordinate that the location CSV carries for non-well records.
- **AR(1) pre-whitening with threshold $|\phi| \geq 0.2$ in Script 23.** A higher-order or non-parametric pre-whitening (e.g. ARMA-based or wavelet) could be defended; the AR(1) choice is what the standard CCF literature uses and is conservative against over-fitting at small effective sample size. The residual AR(1) coefficients across the network sit in a narrow band around -0.12 (see Script 22), so the additional pre-whitening rarely activates in practice.

Results to describe at the methodological level

Script 22's per-well AR(1) diagnostics show that the network's Model B residuals are close to white. Across the 64 wells with valid fits, the AR(1) coefficient sits at a small negative value at most wells (network mean $\phi \approx -0.13$, median ≈ -0.13), and only three wells exceed $|\phi| = 0.3$. The α -vs- ϕ scatter shows essentially no relationship (Pearson $r \approx -0.08$): wells with large persistent subsidies (high α) are not preferentially the wells with the most time-structured residuals. The constant part of the unmodelled input — whatever it is — sits in α , and the small remaining time-varying part is not strongly clustered in any particular well or cluster.

Script 23's hypothesis test returns a null result. With 50 of 63 wells showing a significant peak in the cross-correlation against rainfall, the Spearman rank correlation of peak lag against ridge distance is $\rho = -0.006$, $p = 0.97$. The mean peak lag (2.8 months across all significant wells) does not vary systematically with distance from the ridge. The cluster ordering of mean peak lag (C1, C2 ≈ 2.0 months; C3 ≈ 3.0 ; C4 ≈ 4.4) does not parallel the ridge-distance ordering: C4 and C5 sit at the southern forest margin, not adjacent to the northern ridge. The b_{10} / b_{11} pattern by cluster is informative in its own right: forest clusters carry a larger lagged-rainfall coefficient (mean $b_{11} \approx 0.7$ in C4, 0.5 in C5, versus near zero in C1–C3), consistent with greater vadose-zone storage and longer monthly-scale rainfall propagation under tree canopies, but unrelated to ridge transport.

The Spearman test on its own does not corroborate the §4.4.2 / §5.3 ridge-recharge interpretation. As the structural caveat on Script 23's test design records, however, the test as constructed cannot resolve the mechanism either

way: the null result is consistent with both (a) the residual being largely model error and (b) ridge recharge being delivered as a near-steady baseflow that is observationally indistinguishable from the constant α that Model B already absorbs — but it is also consistent with (c) the test design lacking the temporal resolution to detect the lag pattern even if it were present. The chapter for the main report's §5.3 reflects this: ridge recharge is retained as a candidate mechanism in the discussion, but the present monitoring design cannot quantitatively confirm or rule it out. Script 23's null result documents the analytical attempt and the limit of what the present data can support; it should not be read as positive evidence against the mechanism.

Script 24's seasonal diagnostic supports neither the Thornthwaite-PET-misspecification hypothesis nor the dominant-unmodelled-summer-ET interpretation. Across all 63 wells, no well's correlation between $e(t)$ and sunshine hours exceeds the Bartlett threshold ($|r| < 0.15$ at every well); the network mean is -0.03 . The winter-minus-summer contrast at the cluster mean is small in magnitude across all clusters (between -0.003 and $+0.007$ m). The seasonal amplitude per well is small (cluster mean amplitudes between 0.009 and 0.015 m). What the diagnostic *does* find is a strong phase preference: 47 of 63 wells have their residual climatology peaking in the November–March window (winter to early spring), and zero wells peak in the May–August window. This is not the signature of unmodelled summer ET — it is consistent with threshold or nonlinear recharge behaviour not captured by the linear $\beta_1 \cdot P$ term: in mid-winter with saturated soils, rainfall reaches the water table with higher efficiency than the cluster-mean β_1 represents. The Thornthwaite-misspecification hypothesis can be set aside; what remains is a small but coherent winter-spring residual that is not ridge-related.

Taken together, the three diagnostics narrow the space of admissible interpretations of the residual but do not by themselves settle it. The spatial concentration of large residuals along the northern ridge flank, of which CEH14 is the canonical example, remains real and is the empirical observation that motivates the ridge-recharge framing. Script 24 establishes that the residual does not carry a Thornthwaite-bias signature, ruling out the unmodelled-summer-ET interpretation. Script 23 establishes that no month-scale distance-dependent lag pattern is detectable — but, as the structural caveat above records, the test design at monthly resolution cannot resolve a sub-monthly lag if one were present, and the residual amplitude itself sits at the noise floor of per-well α uncertainty. What survives across all three diagnostics is a residual that is (a) constant in time at the level the model can resolve, (b) absorbed into α at any well where it dominates,

and (c) carries a small winter-spring recharge nonlinearity at the wider network. Whether the residual is genuinely lateral inflow from the ridge, a different unmodelled mechanism, or model error, cannot be resolved from these data alone — sub-monthly water-level monitoring at ridge-adjacent wells would be the design needed to test the mechanism cleanly.

Limitations

- **Single-station climate.** The diagnostic relies on RAF Valley as the sole source for rainfall, PET, and sunshine hours. Any seasonal-bias diagnostic at the residual level is conditional on RAF Valley being representative; the chapter's null result on the Thornthwaite-misspecification hypothesis is therefore an absence of evidence given the available data, not a positive demonstration that Thornthwaite is well-specified at the dune-field scale.
- **140-month threshold and cluster representation.** With the 140-month minimum and the five-well exclusion set, Scripts 23 and 24 carry 63 wells. C5 Coastal Forest is represented by five wells in Script 24 and only one in the Script 23 significant-peak population — too few to make a within-C5 statement. The cluster-mean lag and amplitude figures should be read with this n in mind.
- **Spatial sparsity for the distance-lag test.** The 50 significant-peak wells in Script 23 span a ridge-distance range of roughly 270 to 2000 m, distributed unevenly across the gradient. The test has reasonable power to detect a strong monotone trend; it would be less reliable against a weak trend, and a non-rejection here is not the same as a precise estimate that the slope is zero.
- **AR(1) pre-whitening.** Higher-order or non-parametric whitening could in principle reveal lag structure that AR(1) does not remove. The residual AR(1) magnitudes are small enough (most wells $|\phi| < 0.2$) that this is not a likely confound, but the choice is principled rather than verified.
- **Lag-extended SSM in Script 23 vs canonical Model B.** Script 23's extended model differs from the canonical SSM in including $P(t-1)$ as a second rainfall regressor. The β_2 and β_3 from this fit are not propagated downstream, and the canonical β values in `03_master_data.csv` remain the report's source of truth.

Outputs

File	Contents	Downstream consumer
22_residuals_wide.csv	Wide Model B residuals, rows = months, columns	Stage 2/3 lag analyses (planned); cross-script

File	Contents	Downstream consumer
	= wells	reuse
22_model_b_fits.csv	Per-well Model B fits: α , β_1 , β_2 , β_3 , R^2 , AR(1) ϕ , p , σ	Script 22 plots; cross-script reuse
22_residual_lag_analysis/ 22_01_ar1_histogram.png	AR(1) coefficient histogram by cluster	§S6 supplementary note
22_residual_lag_analysis/ 22_02_ar1_spatial_map.png	Spatial map of AR(1) coefficient per well	§S6 supplementary note
22_residual_lag_analysis/ 22_03_alpha_phi_scatter.png	α vs AR(1) ϕ scatter, coloured by cluster	§S6 supplementary note
22_residual_lag_analysis/ 22_04_example_residuals_by_cluster.png	Example $e(t)$ time series, one well per cluster	§S6 supplementary note
23_residuals_extended_wide.csv	Wide extended-model residuals (after $P(t) + P(t-1)$ fit)	Reserved for stage 3
23_ridge_lag_fits.csv	Per-well extended-model fits, ridge distance, peak lag, peak r , significance	Script 23 plots; §5.3
23_ridge_recharge_lag_test/ 23_01_ccf_headline_ridge_wells.png	CCF curves overlaid for four headline ridge-adjacent wells	§5.3 mechanistic test
23_ridge_recharge_lag_test/ 23_02_peak_lag_vs_ridge_distance.png	The key figure: peak lag vs ridge distance, with Spearman ρ annotated	§5.3 mechanistic test
23_ridge_recharge_lag_test/ 23_03_peak_lag_spatial_map.png	Spatial map of peak lag, ridge reference marked	§5.3 mechanistic test
23_ridge_recharge_lag_test/ 23_04_b10_b11_by_cluster.png	β_{10} vs β_{11} by cluster — diagnostic check on the extended model	§5.3 mechanistic test
23_ridge_recharge_lag	Plain-text summary of	§5.3 mechanistic test

File	Contents	Downstream consumer
<code>_test/23_05_hypothesis_test_summary.txt</code>	the hypothesis test result	
<code>24_residual_climatology.csv</code>	Per-well climatology summary: offset, amplitude, phase, summer–winter, sun correlation	Script 24 plots; §5.4
<code>24_residual_seasonality/24_01_climatology_panels_by_cluster.png</code>	Per-cluster mean climatology with per-well overlays	§5.4 PET-bias caveat
<code>24_residual_seasonality/24_02_seasonal_amplitude_map.png</code>	Spatial map of seasonal amplitude per well	§5.4 PET-bias caveat
<code>24_residual_seasonality/24_03_sun_residual_correlation.png</code>	Sunshine-hours vs residual correlation per well, by cluster	§5.4 PET-bias caveat
<code>24_residual_seasonality/24_04_phase_by_cluster.png</code>	Per-well phase month, grouped by cluster, on hydrological-year axis	§5.4 PET-bias caveat
<code>24_residual_seasonality/24_05_diagnostic_summary.txt</code>	Plain-text interpretive summary	§5.4 PET-bias caveat

Where the result appears in the report

- §4.4.2 *Water-balance residual* — the spatial pattern that motivates the ridge-recharge framing. Scripts 23 and 24 supply the mechanistic test of that framing.
- §5.3 *Ridge-derived recharge interpretation* — Script 23 is the principal test; its null result on the distance-lag relationship is the substantive finding that constrains what §5.3 can claim.
- §5.4 *PET-bias caveat* — Script 24 supplies the diagnostic that the Thornthwaite-misspecification interpretation can be set aside, and that a winter-spring recharge nonlinearity is the residual signal that survives.
- §5 conclusions — the integrated finding from the three diagnostics: the residual is observationally consistent with a steady or near-steady subsidy

at ridge-adjacent wells (captured by α) plus a small winter-spring recharge nonlinearity at the wider network, but not with month-scale ridge-to-well lateral transport or with Thornthwaite PET misspecification.

Cross-references

- **F.3** — SSM formulation; the displacement-formulation equation that Scripts 22 and 24 fit via `fit_ssm_intercept()` and that Script 23 extends with a lag-1 rainfall term.
 - **F.4** — `HEADLINE_LAG = 0`, `DRAINAGE_DATUM = 3.7 m`, cluster labels and colours.
 - **F.5** — shared utility modules. `model_utils.fit_ssm_intercept()` is the authoritative Model B fit used by Scripts 22 and 24; `model_utils.build_ssm_frame()` is the underlying data-alignment routine.
 - **S.3** — origin of the per-well α values that Script 22's α -vs- ϕ scatter cross-correlates against; the canonical β coefficients that Scripts 22, 23, 24 do not revise.
 - **S.5** — Script 07's 100-month windowed fit, contrasted with Script 22's full-record fit.
 - **S.13** — Figure 48 (SSM water-balance residual map) is the visual the diagnostic suite is interrogating; the CEH14 annotation on that figure is the headline well around which the §5.3 interpretation is anchored.
 - **S.15** — coastal-retreat gradient as an alternative explanation for some southern-network residual structure; Scripts 23 and 24 are the northern-ridge analogues to that southern-boundary diagnostic.
-

End of chapter S.16.

S.18 Script 26 — Van Willegen MSL aggregation

Step 30 / 35. Phase 13 — Van Willegen MSL Analyses in `run_analysis.py`; paired with S.18b which covers the forecasting tools that operate on this chapter's outputs and with S.18c which covers the report-format figures that render this chapter's trajectory output for §4.8.4.

Motivation

Curreli et al. (2013) is the foundational reference for ecohydrological thresholds in lowland dune slacks. That paper calibrated absolute groundwater-depth thresholds (SD15b at -0.61 m and SD16 at -0.98 m below ground) against vegetation community composition at fifteen Welsh dune sites, including

Newborough, using a 4-year mean of winter water levels as the hydrological metric. The thresholds remain the operational standard for conservation management across the SAC network — they are the numbers cited by Natural Resources Wales when reporting site condition against vegetation criteria.

Van Willegen et al. (2025) revisited the framework with a larger dataset and a wider range of candidate metrics. The paper tested eighty hydrology metrics against community-mean Ellenberg EbF moisture response across 17 dune-slack vegetation quadrats at Newborough, drawing on vegetation surveys from 2010 to 2019 (453 relevées) and contemporaneous groundwater monitoring from 2007 to 2019. The best-performing predictor was the unweighted 5-year mean spring water level — MSL5, the metric this chapter documents. The refinement over Curreli is modest in shape but practically significant: the window is 5 years rather than 4 (sensitivity-tested by van Willegen et al. (2025), who found cross-correlation stabilises at 5), and the seasonal sub-window shifts from winter to spring (March through May) to capture conditions during the active vegetative growth period. The paper does not derive new absolute thresholds; the Curreli SD15b and SD16 values remain the calibration anchor against which sites are reported.

The Newborough connection runs deeper than co-location alone. The author of the present study is a co-author on van Willegen et al. (2025), and the paper's piezometer dataset is the same long-term Newborough monitoring network this pipeline operates on. The methodological framework is therefore directly applicable, and the present chapter implements the metric on the full project network — the 66 reference and ~18 extended-network wells, of which 17 carry co-located vegetation quadrats (the van Willegen quadrat subset). At those 17 wells the calibrated link between MSL5 and vegetation response applies directly; at the other ~67 wells, MSL5 is a hydrological monitoring metric rather than a calibrated vegetation predictor, useful for tracking spatial pattern but not directly tied to a quantified ecological response.

The chapter sits as a complement to, not a replacement for, the report's existing summer-minimum predictive framework. The Scripts 11 / 11b transfer functions, the iterated P_flood derivation in §3.6.3 of the main report, the BACI clearfell summer-minimum step in Scripts 10a / 10d, the CEH36 scraping benefit in Script 09c, the climate projections in Script 14 — all of these run in summer-minimum space, against the Curreli SD15b / SD16 thresholds. MSL5 serves a monitoring role rather than a predictive one: managers who collect spring readings can monitor MSL5 directly each year, and the 5-year integrated window damps single-year

extremes. The two frameworks measure the same multi-year hydrological state through different seasonal windows, and the Empirical relationship to summer minima section below quantifies their cross-correlation at the network scale (Pearson $r = 0.95$ between MSL5 and the parallel 5-year summer-minimum mean) and the constant offset between them (≈ 0.54 m, the seasonal amplitude of the typical Newborough water table).

Inputs

File	Origin
outputs/01_wells_clean.csv	Script 01 — reference-network monthly depths
outputs/01_wells_extended.csv	Script 01 — extended-network monthly depths
outputs/01_well_elevations.csv	Script 01 — Upstand_m for pipe-top to ground-level conversion
outputs/01_locations.csv	Script 01 — easting / northing for the MSL5 spatial map
outputs/01_wells_provenance.csv	Script 01 — interpolated-cell flags (S.1 limit=1 policy)
outputs/ 02_07_cluster_membership_k5.csv	Script 02 — reference cluster identifiers
outputs/ 06_pear_membership_audit_sitewide.csv	Script 06 — extended cluster identifiers
outputs/03_regional_averages.csv	Script 03 — cluster-centroid monthly series (used by the Method B output; see the Method A and Method B aggregation section below)

Methodology

Spring window and hydrology year. The spring window is calendar months 3, 4, and 5 — March, April, May — following van Willegen 2025 Table 2. The hydrology year y (van Willegen’s “hydrology year B”) runs from 1 June ($y-1$) to 31 May (y), so spring of hydrology year y is the three months immediately preceding the year’s close. A reading dated 2010-04 therefore belongs to hydrology year 2010 (spring of 2010); a reading dated 2009-07 belongs to hydrology year 2010 (summer of the preceding calendar year). The annual MSL_y for a single piezometer is the unweighted mean of the three spring months of hydrology year

y. The 5-year mean MSL5(end-year y) is the unweighted mean of the five consecutive annual MSLs {MSL_{y-4}, MSL_{y-3}, MSL_{y-2}, MSL_{y-1}, MSL_y}. The convention follows Curreli 2013 (who used a 4-year winter-mean version) and is sensitivity-tested at 5 years in van Willegen et al. (2025).

Strictness rules. Two completeness rules are applied. The minimum-months-per-spring rule requires all three of {March, April, May} to be present in the cleaned monthly series for an annual MSL to be valid (3 of 3, strict). The minimum-years-in-window rule requires all five constituent annual MSLs to be valid for a 5-year mean to be valid (5 of 5, strict). Single-month interpolation under the S.1 limit=1 cleaning policy is allowed to count toward the 3/3 requirement; the per-well CSVs flag interpolation in an `n_interpolated_spring` column so analysts can audit reliance on filled cells. These rules are stricter than van Willegen 2025 itself, which permitted lower completeness in their published analysis. The project's preference for cleanliness over coverage is consistent with the BACI summer-minima panel in Scripts 09c and 10d, which use a similar two-of-four months policy. Annual MSL rows that fail the 3/3 rule are excluded; 5-year windows that contain any excluded annual MSL are excluded; the per-well trajectory line is broken with a true gap rather than bridged with a straight diagonal (see the Site-specific choices section below).

Sign conventions. The pipeline's native depth frame is below the pipe top — `01_wells_clean.csv` reports `level_pipe`, with the water level negative when below the top of the standpipe. The paper convention is below ground — `level_bg = level_pipe + Upstand_m`, where `Upstand_m` is the per-well height of the pipe top above the ground at that well (from `01_well_elevations.csv`, typically 0.1 to 0.4 m). Every output CSV from Script 26 carries both frames in adjacent columns (`MSL_m_pipe` and `MSL_m_bg`, and the same pairing for `MAX`, `MSL5`, `MAX5`). All trajectory and spatial figures use `_m_bg` to match the paper.

Annual maxima as a secondary metric. Van Willegen 2025 reports `MAX` (annual maximum water level over the hydrology year) as the second-best-performing metric in their Table 2. Each Script 26 output CSV carries `MAX5` (5-year mean annual maximum) alongside `MSL5` for cross-reference. The chapter does not develop `MAX5` separately — the paper notes that topographic truncation of peaks at individual slacks (water can pond at the surface in wet years, capping the measurable maximum) makes `MAX` inferior to `MSL` for cross-site comparison — but the column is retained for any analyst who wants to reproduce the secondary panel of the van Willegen et al. (2025) analysis.

Cluster aggregation. Per-cluster MSL5 trajectories are computed as the simple unweighted mean of constituent wells' per-well MSL5 by (cluster, end-year), pooling the reference and extended-network wells. Cluster identifiers follow the k=5 partition from Script 02 (reference wells) and the Script 06 Pearson membership audit (extended wells assigned to the same k=5 clusters). C1 is the Lake Edge cluster; C2 Dune; C3 Western Residual; C4 Main Forest; C5 Coastal Forest. The Method A and Method B aggregation section below documents a second aggregation method introduced at v1.1.2 — the cluster-centroid trajectory from `03_regional_averages.csv`, which is the SSM-consistent companion used by the forecasting tools documented in S.18b. The per-well method described here is the headline monitoring metric.

Network coverage. The script processes 1,433 (well, hydrology year) annual rows, of which 1,304 (91 %) pass the strict 3/3 rule. From these are derived 884 (well, end-year) 5-year window rows, covering 84 wells (66 reference + 18 extended). 16 of the 17 van Willegen quadrat wells are covered (T41 fails the rule because of an insufficient recent record). The cluster trajectory restricts to window-ends 2014 onwards (see the Site-specific choices section below) and contains 79 (cluster, end-year) rows across the five clusters and sixteen valid window-end years.

Site-specific choices

Three editorial decisions warrant explicit defence in this chapter, since they depart from the most literal van Willegen 2025 protocol.

Trajectory restriction to window-ends $\geq$ 2014. The reference and extended monitoring network expanded materially between 2007 and 2010 as CEH instrumentation came online. Valid annual MSL counts by hydrology year are 7 (2005), 10 (2006), 24 (2007), 31 (2008), 34 (2009), and then 60 from 2010 onwards. Pre-2010 5-year windows draw from a network of fewer than 35 wells, materially differently composed from the post-2010 network of ~60 wells, and any cluster-mean trajectory line that mixes the two creates the visual appearance of a hydrological shift around 2010 that is in fact a network-composition shift. The first 5-year window drawn entirely from the post-2010 network closes at end-year 2014 (covering hydrology years 2010 through 2014). The per-well CSVs retain the full record from window-end 2009 onwards for any analyst who needs early data; only the cluster trajectory plot and the per-quadrat-well plot are clipped to end-year 2014 onwards. This restriction is consistent with van Willegen 2025 themselves, whose published analysis period is 2010–2019 for the same network-coverage reason.

Pooling reference and extended networks at cluster level. The cluster aggregations include extended-network wells that the SSM-fitting analyses (Script 03 onward) do not use. The rationale is that spatial coverage matters more than network purity for a monitoring lens: the extended-network wells provide more piezometers in the coastal-edge sub-cluster of C5 and in the lake margin of C1, two areas where the reference network alone is thin. Per-well CSV rows are flagged with a network column $\in \{\text{Reference, Extended}\}$ so analysts can split the network if needed for a more conservative analysis. The Method B output introduced at v1.1.2 (see the Method A and Method B aggregation section below) uses only the reference network and serves as the SSM-consistent companion.

Curreli SD15b / SD16 reference lines on the MSL5 trajectory. The two horizontal reference lines drawn on every Script 26 trajectory plot (SD15b at -0.61 m, SD16 at -0.98 m, both below ground) are the Curreli ecohydrological thresholds, calibrated against summer minima, not against MSL5. Strictly, they are not threshold lines for the MSL5 metric. They are retained on the MSL5 plot because they are the most-recognized ecological reference points in the literature and a manager reading the figure expects to see them. The figure caption notes the offset explicitly — a cluster-mean MSL5 line touching SD15b corresponds to a 5-year summer-minimum mean roughly at SD16, given the ≈ 0.54 m offset documented in the Empirical relationship to summer minima section below.

Analysis-specific exclusion of CEH13 and CEH14 from MSL5 (v1.2.0, 2026-06-25). Two wells — CEH13 and CEH14 — are excluded from every derived MSL5 product: the Method A cluster-mean trajectory, the latest-per-well table, the IDW change map, and the quadrat figure. Both wells have a degenerate SSM drainage coefficient (CEH13 near-zero β_3 , a τ outlier; CEH14 negative β_3 , SSM failure, $\text{NSE} = -3.21$) that gives them anomalously long drainage memory. Within a five-year window, this autocorrelation makes their spring readings correlated across years, so their MSL5 value — and especially their MSL5 change between windows — is unreliable as a monitoring metric. The exclusion criterion is the MSL5-side application of the same drainage-memory rule that already excluded CEH13 and CEH14 from the $\tau = S_y/\beta_3$ storage–drainage index map (§S.12): both are excluded from the τ map on $\beta_3 \leq 0$ or near-zero grounds; the same rationale applies here to the five-year windowed spring metric. The excluded wells are held in `config.MSL5_EXCLUDED_WELLS` (keys lowercase to match the normalized well column in the per-well CSV). The rows are **retained** in `26_msl_5yr_per_well.csv` with two new columns, `msl5_excluded` (boolean) and `msl5_excluded_reason`, for transparency; a filtered subset with excluded rows removed feeds all downstream products. This is an analysis-specific exclusion that does not affect

either well's membership of the clustering, SSM, BACI, or any other pipeline analysis.

Empirical relationship to summer minima

A direct cross-check between MSL5 and the parallel 5-year mean of summer minima (5-yr SM5) was run as part of script verification. At the individual-year scale (annual MSL vs annual summer minimum, per well), the Pearson correlation across all valid (well, year) rows is 0.77. At the 5-year window scale (MSL5 vs the corresponding 5-year mean summer minimum, per well across window-ends 2014 onwards), the correlation rises to 0.945 ($n = 829$ (well, end-year) rows where both metrics are valid). The constant offset between the two metrics, computed as the mean of (5-yr SM5 – MSL5) across the same rows, is 0.54 m with a standard deviation of 0.15 m. The 5-yr SM5 is roughly 0.54 m deeper than MSL5 — the seasonal amplitude of a typical Newborough water table between spring and the late-summer minimum.

The offset is approximately constant across clusters, varying only over the range 0.51 to 0.67 m at window-end 2025:

Cluster	n wells	MSL5 (m below ground)	5-yr SM5 (m below ground)	Gap (5-yr SM5 – MSL5)
C1 Lake Edge	8	–0.24	–0.82	–0.58
C2 Dune	30	–0.27	–0.89	–0.62
C3 Western Residual	12	–0.78	–1.50	–0.72
C4 Main Forest	8	–0.83	–1.35	–0.52
C5 Coastal Forest	26	–0.62	–1.18	–0.56

The two metrics measure essentially the same multi-year hydrological state through different seasonal windows. MSL5 is shallower because spring water tables, refreshed by winter recharge, sit roughly half a metre above the late-summer minimum at each cluster. The implication for the report's analytical structure is that MSL5 does not add new *predictive* capacity beyond the existing summer-minimum forecasting framework — the two are essentially co-linear at the 5-year window scale. What MSL5 does add is the *calibrated link to the van Willegen vegetation framework* at the 17 quadrat wells, plus a metric that can be

computed from spring readings alone for managers who prefer that monitoring cadence.

This complementarity is not a contradiction. The two frameworks describe the same hydrological state through different seasonal windows, and a reader who encounters substantial summer-minimum threshold-crossing predictions in §4.9.6 of the report alongside more modest MSL5 climate projections in §S.18b should not interpret the contrast as one framework overturning the other. Section S.18b.3.7 expands on this point in the context of the UKCP18 climate scenarios: the spring window's structural cancellation between increased winter rainfall and increased summer PET produces a smaller projected climate shift than the summer-minimum window, which has no compensating winter-rainfall effect. Both readings are correct; they sample the climate signal at different points of its seasonal structure.

Climate context for the post-2024 trajectory lift

The cluster MSL5 trajectory shows a marked upward lift at window-ends 2024 and 2025. The lift is real, but it is largely climate-driven rather than evidence of management recovery. Hydrology year 2024 is the wettest in the 2007–2026 record at Newborough — 1,143 mm annual rainfall against a long-term mean of 855 mm (+34 %), and 234 mm spring rainfall against a long-term mean of 161 mm (+45 %). The 2024 spring is the wettest spring in the record. By contrast, hydrology year 2025 returned to dry conditions: 746 mm annual (–13 %) and 121 mm spring rainfall (–25 %).

Because MSL5 at window-end 2024 averages hydrology years 2020 through 2024 — and window-end 2025 averages 2021 through 2025 — the entry of the record-wet 2024 year into both windows lifts the cluster means substantially. The expected lift from a single wet year carrying ~7 % of the window's weight, multiplied by the 73 mm spring anomaly and the local β_1 recharge coefficient, is consistent in scale with the observed end-2024 lift across the network. The post-2024 trajectory is not evidence of management recovery alone; the climate component is doing most of the work at the right edge of the figure. As hydrology year 2024 ages out of the rolling window in subsequent years, MSL5 should fall back closer to its pre-2024 level unless one of the intervening years matches 2024's anomaly.

This is exactly the kind of single-year extreme influence that the 5-year window is designed to dampen, and the framework does dampen it — the lift is real but smaller than the 2024 single-year anomaly would produce on an annual basis.

The window smooths; it does not eliminate. The chapter’s limitations section makes this caveat explicit, and the §5 discussion in the main report should mirror the same caution: trajectory readings from the most-recent window-ends should be interpreted with the underlying climate sequence in mind.

Intervention markers and the management-response horizon

The cluster trajectory and per-quadrat-well plots draw three intervention events as paired vertical lines, encoding the temporal logic of the 5-year window:

Event	Date	First post-intervention HY	Solid line at end-year	Dashed line at end-year
Scrape (CEH36)	April 2015	2015	2015	2019
Clearfell	December 2017	2018	2018	2022
Re-scrape (CEH18/21)	October 2023	2024	2024	(out of range)

The solid vertical line marks the first window-end that contains any post-intervention spring data — the earliest window in which an intervention can register at all under a 5-year aggregation. The dashed vertical line marks the first window-end where all five constituent years are fully post-intervention — the earliest window in which the intervention is the only post-intervention signal in the average.

For the December 2017 clearfell, the first fully-post-clearfell window closes at end-year 2022, covering hydrology years 2018 through 2022. This is the management-relevant horizon for expecting a vegetation response under van Willegen’s 5-year framework. Vegetation monitoring rounds before the end-2022 window are observing partial-recovery hydrology (the 5-year mean still includes pre-clearfell years); rounds at or after the end-2022 window are observing the integrated post-clearfell hydrological state. A practical implication for site managers is that monitoring at year-end 2022 onwards is the first time the framework would expect the clearfell signal to be visible in MSL5 at any cluster; expecting earlier response is asking the metric to do something the 5-year smoothing was specifically designed to avoid.

Intervention dates are imported from `utils.scraping_common` as the canonical pipeline constants (`SCRAPING_DATE = 2015-04-01`, `INTERVENTION_DATE = 2017-`

12-01, SCRAPING_DATE_2 = 2023-10-01); intervention line colours come from `utils.config` (`INTERVENTION_COLOUR_SCRAPE` = "#7b3294" purple for scraping, `INTERVENTION_COLOUR_CLEARFELL` = "#e66101" orange for clearfell). Both are sourced from configuration rather than hardcoded locally.

Method A and Method B aggregation

Script 26 produces two per-cluster MSL5 aggregation methods alongside the per-well aggregation described above. The reason two methods exist is an internal consistency requirement: the SSM β coefficients in Script 03 are fitted against the cluster-centroid monthly series in `03_regional_averages.csv`, which uses the LCSC reference network only (the 66 reference wells, not the ~84 reference + extended pool). The MSL5 forecasting tools introduced in S.18b — Section 5 of Script 11 (Tool A) and Script 26b (Tool B) — would otherwise sit in a different baseline than the per-well-aggregated trajectory documented above. To make the forecasting tools internally consistent with the SSM coefficients they read from, Script 26 produces a second per-cluster CSV alongside the per-well-aggregated one.

The terminology used in the codebase and across the supplement is:

- **Method A:** per-well annual MSL $\rightarrow$ arithmetic cluster mean across all wells assigned to the cluster (reference + extended). Written to `26_msl_5yr_per_cluster.csv`. This is the headline monitoring metric documented in the preceding sections of this chapter, used in the §4.8.4 trajectory figure and spatial map.
- **Method B:** cluster-centroid monthly series from `03_regional_averages.csv` $\rightarrow$ annual MSL $\rightarrow$ 5-year MSL5 on the centroid. Reference network only. Written to `26_msl_5yr_per_cluster_centroid.csv`. This is the SSM-consistent companion used by the forecasting tools in S.18b.

The two methods can differ by tens of centimetres at any single (cluster, year) pair because they describe different network compositions, not different aggregation algebra. The empirical comparison across all 77 common (cluster, window-end) rows shows mean $|\text{Method B} - \text{Method A}|$ of 0.30 m and maximum 0.78 m (at C4 Main Forest in window-end 2011); the Pearson correlation between the two trajectories is 0.69. The sign of the difference varies by cluster: Method B is deeper than Method A at C1, C2, C4, and C5 (because the extended network adds shallower coastal-edge and lake-margin wells that pull the per-well mean

upward), but shallower at C3 (because the C3 reference network omits some particularly deep wells that the extended network includes).

Both methods are mathematically correct. They answer different questions and describe different cluster populations. Method A is “what is the typical well in this cluster experiencing?” — the natural aggregation for monitoring, closest to van Willegen’s per-piezometer calibration framework. Method B is “what is the cluster’s reference-network signal?” — the natural aggregation for any analysis that sits on top of the SSM coefficients. The chapter retains both and the report’s editorial convention is that §4.8.4 figures cite Method A and any numerical claim from Tool A or Tool B carries the implicit Method B provenance. Each cluster MSL5 number quoted in the report should be qualified with the method that produced it; the supplement’s chapter S.18b restates this point in its own context.

Limitations

A small set of limitations should be noted alongside the chapter’s results, all of which are operational rather than methodological challenges to the metric itself.

MSL5 is an aggregation, not a model. The script does not attribute the cluster-mean trajectory to particular drivers (clearfell, climate, coastal retreat). Attribution work sits in Scripts 10a and 10d (BACI clearfell), Script 25 (coastal-retreat gradient), and Script 14 (climate projections), all of which operate in summer-minimum space against the Curreli thresholds. The MSL5 trajectory should be read as a monitoring record, not as a causal decomposition.

Direct ecological calibration applies only at the 17 van Willegen quadrat wells where co-located permanent vegetation quadrats exist. At the other ~67 wells in the network MSL5 is a hydrological monitoring metric only, useful for tracking spatial pattern but not directly tied to a quantified vegetation response. Sixteen of the seventeen quadrat wells pass the strict 3/3 + 5/5 rules; T41 is excluded because of an insufficient recent record.

The 5-year window lags the underlying hydrological state by approximately half the window length. A change in conditions in year y will not be fully reflected in MSL5 until the window closes at end-year $y + 4$. This is by design — the smoothing is the point — but it means that the metric is not a real-time indicator. Tool A in S.18b addresses this by providing an empirical transfer function that lets managers predict the next year’s annual MSL from monthly readings collected through end-February; that prediction can be added to a rolling four-year history of observed annual MSLs to update the MSL5 statistic without waiting until end-May for the actual reading.

The post-2024 trajectory lift documented in the Climate context section is climate-influenced rather than purely management-driven. Conclusions about clearfell or scraping recovery drawn from the most-recent window-end alone should be made cautiously; the broader trajectory shape is the more defensible reading. As hydrology year 2024 ages out of the rolling window in subsequent years, the lift should attenuate unless one of the intervening years matches its anomaly.

The Curreli SD15b and SD16 reference lines on the trajectory plot are calibrated against summer minima, not MSL5. They are retained on the MSL5 plot for visual familiarity, with the ≈ 0.54 m offset between the two metrics flagged in the figure caption and quantified in the Empirical relationship to summer minima section above. A site whose MSL5 trajectory crosses SD15b is operationally in the SD16-equivalent state when read in summer-minimum space, and the management implications follow correspondingly.

No UKCP18 forward projection of MSL5 is produced by this chapter. The predictive forecasting framework runs in summer-minimum space (Scripts 11, 11b, 14). A separate perturbation overlay applying UKCP18 RCP8.5 multipliers to the MSL5 trajectory is documented in chapter S.18b as Tool B; that capability sits in the Method B baseline (cluster centroid from 03_regional_averages.csv) and produces small projected shifts of 1 to 4 cm at the 2050s and 2080s. The smallness of the projected MSL5 shifts is structural — a feature of the spring window straddling the seasonal partition of the UKCP18 multipliers — and is not in tension with the larger projected summer-minimum shifts that drive the report’s climate-vulnerability framing.

Outputs

File	Description
outputs/26_van_willegen_msl/ 26_msl_annual_per_well.csv	Per (well, hydro_year) annual MSL and MAX in both pipe-top and below-ground frames, with completeness flags
outputs/26_van_willegen_msl/ 26_msl_5yr_per_well.csv	Per (well, end_year) 5-year MSL5 and MAX5 with cluster identifiers, network flag, and msl5_excluded / msl5_excluded_reason columns (CEH13 and CEH14 flagged; all rows retained — downstream products filter on msl5_excluded = False)
outputs/26_van_willegen_msl/	Method A cluster-mean trajectory:

File	Description
26_msl_5yr_per_cluster.csv	mean, median, std per (cluster, end_year). Headline monitoring metric.
outputs/26_van_willegen_msl/26_msl_5yr_per_cluster_centroid.csv	Method B cluster-centroid trajectory from 03_regional_averages.csv. SSM-consistent companion to Tools A and B.
outputs/26_van_willegen_msl/26_msl_5yr_latest_per_well.csv	Most-recent valid MSL5 per well — input to the spatial map
outputs/26_van_willegen_msl/26_msl_5yr_trajectory.png	Cluster-mean MSL5 trajectory with SD15b/SD16 reference lines and intervention markers (Method A; main-report figure)
outputs/26_van_willegen_msl/26_msl_5yr_map.png	IDW-interpolated MSL5 surface with DEM hillshade, KML overlays, ridge mask; van Willegen quadrat wells flagged (main-report figure)
outputs/26_van_willegen_msl/26_msl_5yr_quadrat_wells.png	Per-well MSL5 trajectories at the 17 van Willegen quadrat wells (supplement figure)
outputs/26_van_willegen_msl/26_msl_results.txt	Run transcript with cluster-level summary and per-quadrat-well numerical values

Where the result appears in the report

Report section	Content
§4.8.4 <i>Five-Year Mean Spring Water Level (MSL5)</i>	Table 16 (cluster-mean MSL5 at window-end 2025 with SD15b/SD16 threshold-crossing counts, source 26_msl_5yr_per_cluster.csv), the cluster trajectory plot (Figure 39) and the spatial MSL5 map (Figure 40); the results narrative
§5 <i>Discussion</i>	Curreli ↔ van Willegen framing paragraph; the climate-context caveat for the post-2024 lift; the management-

Report section	Content
	response horizon framing under the 5-year window; one-sentence cross-reference to the climate-projection results in S.18b

Cross-references

- **F.4** — Curreli SD15b / SD16 constants; the MSL_* constants; VW_QUADRAT_WELLS; INTERVENTION_COLOUR_*
- **F.5** — paths.OUT_26_*, paths.DIR_26
- **S.1** — limit=1 cleaning policy underlying the strict 3/3 spring rule
- **S.6** (Script 09c) — summer minima at CEH36 scraping; the metric MSL5 is offset from
- **S.7** (Scripts 10a, 10d) — BACI clearfell summer-minimum step; the predictive complement
- **S.9** (Scripts 11, 11b) — spatial threshold maps in summer-minimum space; cross-correlation with MSL5 at the 5-year scale ($r = 0.945$, see Empirical relationship section above)
- **S.15** (Script 25) — coastal-retreat gradient contributing to the western-margin pattern visible in the MSL5 map
- **S.18b** (Scripts 11 Section 5, 26b) — forecasting tools that operate on this chapter's Method B output: empirical spring MSL transfer function (Tool A) and UKCP18 RCP8.5 climate projection (Tool B)

End of chapter S.18.

S.18b — Spring MSL forecasting tools (Script 11 Section 5, Script 26b)

Step 31 / 35 for Script 26b (Phase 13 — Van Willegen MSL Analyses in run_analysis.py); Script 11 Section 5 lives in Phase 1 at step 22/43 inside 11_forecasting_thresholds.py. Companion to S.18; followed in the supplement by S.18c which documents Script 26c's display-only report-format figures.

S.18b.1 Purpose and editorial weighting

This chapter documents two complementary forecasting capabilities that pair with the observational 5-year mean spring water level (MSL5) metric introduced

in S.18. The first is a per-cluster empirical transfer function predicting next-year MSL from monthly readings collected through end-February (Tool A, implemented as Section 5 of Script 11_forecasting_thresholds.py). The second is a perturbation-overlay capability producing per-cluster Δ MSL5 estimates under UKCP18 RCP8.5 50th-percentile climate scenarios for the 2050s and 2080s (Tool B, implemented as Script 26b_van_willegen_msl_projections.py).

The two tools sit at different levels of editorial weight in the report. Tool A produces fits with R^2 between 0.73 and 0.96 across the network and a clean operational use case for site managers; the equations are reported as Table 9 in §3.6 and the calibration scatter as a §3.6 figure. Tool B is a documented robustness / climate-sensitivity capability whose projected magnitudes (1–4 cm at 2050s and 2–4 cm at 2080s) sit within observed interannual variability and well below the forest-management BACI signal at this site. Tool B is therefore presented in this supplement chapter with full method detail but is summarized in the main report in a single discussion sentence rather than as a headline figure. This editorial weighting reflects the empirical scale of the result rather than a methodological reservation about the technique.

S.18b.2 Tool A: empirical spring MSL transfer function

S.18b.2.1 Specification

For each of the five clusters (C1 Lake Edge, C2 Dune, C3 Western Residual, C4 Main Forest, C5 Coastal Forest) the transfer function takes the form

$$\text{MSL}_y = \alpha \cdot h_{\text{max_winter}} + \beta \cdot P_{\text{win_to_spr}} + \gamma \cdot \text{PET}_{\text{win_to_spr}} + \text{intercept}$$

where MSL_y is the mean of March, April, and May water levels in hydrology year y (the van Willegen 2025 spring window), $h_{\text{max_winter}}$ is the maximum cluster-centroid head over October $y-1$ to February y , and $P_{\text{win_to_spr}}$ and $\text{PET}_{\text{win_to_spr}}$ are cumulative totals over October $y-1$ to May y . The hydrology year follows van Willegen’s “hydrology year B” (1 June $y-1$ to 31 May y), so the response and all three predictors fall within the same hydrology year by construction. Coefficients are fitted independently per cluster by ordinary least squares with an intercept term.

S.18b.2.2 Data source

The fit uses Script 03’s cluster-centroid monthly series in 03_regional_averages.csv — five columns (Lake_Edge, Eastern_Block, Western_Block, Forest, Coastal_Forest) corresponding to the five clusters under

Script 03's BLOCK_MAP. This is the same baseline that the report's state-space coefficients β_1 , β_2 , β_3 are fitted against (S.3 of this supplement) and the same baseline as Tool B's perturbation overlay. The choice ensures internal consistency between the empirical transfer function, the SSM mechanism, and the projection companion. Section S.18b.4 below addresses the relationship between this cluster-centroid baseline (referred to in the codebase as "Method B") and the per-well-aggregated baseline used in S.18's headline trajectory figure ("Method A").

The annual-aggregation strictness rules match those of S.18: three of three spring months must be present for an annual MSL to be valid, and the rolling 5-year MSL5 statistic (not used by Tool A directly, but inherited from the framework's data source) requires all five constituent annual MSLs to be valid.

S.18b.2.3 Results

The five-cluster table of fitted coefficients is reproduced below (full numerical values are written to `outputs/11_forecasting_thresholds/11_forecast_spring_transfer_function s.csv`).

Cluster	α (h_max_ winter)	β (P_win_to _spr)	γ (PET_win _to_spr)	Intercept	R ²	n
C1 Lake Edge	+0.116	+0.00113	−0.00141	−0.689	0.727	19
C2 Dune	+0.384	+0.00128	−0.00171	−0.708	0.843	20
C3 Western Residual	+0.619	+0.00099	−0.00100	−0.764	0.887	20
C4 Main Forest	+0.830	+0.00089	+0.00042	−1.000	0.955	19
C5 Coastal Forest	+0.760	+0.00039	−0.00028	−0.609	0.964	19

Coefficients on h_max_winter, P_win_to_spr are positive everywhere and statistically significant ($p < 0.01$) at all clusters except Lake Edge, where the winter-peak coefficient is small (+0.12) and non-significant ($p = 0.54$).

PET_win_to_spr coefficients are negative at all clusters except Main Forest, where the value (+0.0004) is essentially zero and statistically non-significant ($p = 0.75$).

Physical interpretation: rainfall over the October-to-May window raises the next spring's MSL; potential evapotranspiration lowers it; and the previous winter peak — when statistically distinguishable from zero — carries information about the antecedent groundwater state.

Five-cluster R^2 values range from 0.727 (Lake Edge) to 0.964 (Coastal Forest), with the two forest clusters (C4 and C5) fitted essentially deterministically. The Lake Edge cluster has the lowest R^2 of the five but the fit is still strongly predictive; its lower R^2 is consistent with the lake's hydrological buffering, where the lake stage moderates year-to-year sensitivity of the dune water table to the winter peak.

S.18b.2.4 A rejected variant

An earlier two-variant implementation (Script 11 v1.1.0) also fitted a “previous-MSL input” form of the transfer function using the previous year's MSL as the antecedent state in place of the winter peak. That variant was empirically weaker: R^2 values ranged from 0.18 (Forest) to 0.44 (Coastal Forest), and the previous-MSL coefficient was statistically non-significant ($p > 0.5$) at four of five clusters. Only Coastal Forest, with its high persistence and long memory, gave the previous-MSL form real predictive weight. The previous-MSL variant was dropped at v1.1.1 in favour of the single winter-peak form retained here. The decision was made on empirical grounds: a manager-accessible form with weak predictive power is not a useful tool, and the winter-peak form requires no additional measurement effort beyond the routine monthly dipwell programme that managers already operate.

S.18b.2.5 Manager workflow

The intended use case is illustrated by the following workflow. A site manager monitoring spring readings each year accumulates a rolling history of observed MSL values $\{\text{MSL}_y-4, \text{MSL}_y-3, \text{MSL}_y-2, \text{MSL}_y-1\}$. To project the next 5-year MSL5 forward in early spring of year y — before taking the actual May reading — the manager takes the cluster's winter peak from October-to-February monthly readings, the cumulative October-to-May rainfall, and the corresponding PET, and plugs the values into the appropriate cluster's transfer function from the table above. The predicted MSL_y is then added to the rolling history to produce an updated 5-year MSL5 estimate. The estimate becomes verifiable when end-May readings arrive, at which point it can be replaced by the observation.

The workflow is most operationally useful for early-warning purposes: a predicted MSL_y that crosses into a depth-of-concern band before May allows managers to consider intervention or attention scheduling in advance. Coupling

the transfer function to van Willegen’s Ellenberg EbF response curves (for clusters where per-piezometer calibrations are available) extends the early warning into a vegetation-risk indicator.

S.18b.3 Tool B: UKCP18 RCP8.5 MSL5 climate projection

S.18b.3.1 Specification

Script 26b_van_willegen_msl_projections.py produces per-cluster projected MSL5 trajectories under UKCP18 RCP8.5 50th-percentile Wales scenarios for the 2050s and 2080s. The script does not run the state-space model forward in time. Instead, it computes a monthly Δh perturbation from each cluster’s fitted SSM coefficients and the UKCP18 multipliers, and applies the perturbation as a constant vertical shift to the observational Method B trajectory produced by Script 26.

S.18b.3.2 Method

For each cluster and each scenario the monthly Δh perturbation is

$$\Delta h(m) = \beta_1 \cdot (P_scen(m) - P_base(m)) - \beta_2 \cdot (PET_scen(m) - PET_base(m))$$

where β₁ and β₂ are the cluster’s SSM coefficients from Script 03 (S.3 of this supplement), P_base and PET_base are the long-term monthly climatology over the monitoring period (2005-04 to 2026-02), and P_scen and PET_scen apply seasonal UKCP18 multipliers to P_base and PET_base. The pattern matches the single-step monthly perturbation function documented in Script 21 (utils.model_utils.monthly_perturbation), with the simplification that for a pure climate scenario the scenario β₂ equals the baseline β₂ — no land-use change — so the PET term reduces to the form shown.

The UKCP18 multipliers used by Tool B are the central-estimate Wales 50th-percentile values for RCP8.5 at the 2050s and 2080s time horizons, identical to those exposed by Script 19’s SCENARIO_PARAMS dictionary in the scenario viewer. Specifically:

Scenario	sP_winter	sP_summer	sPET_winter	sPET_summer
2050s (2040–2069)	1.10	0.85	1.05	1.20
2080s (2070–2099)	1.20	0.70	1.10	1.35

Winter is taken as November through March; summer as May through September. April and October are shoulder months, and Tool B assigns each shoulder month the mean of the winter and summer multipliers — a documented choice for transitional months that straddle the canonical seasonal windows. April matters because it is one of the three spring-window months; assigning it the mean rather than either pure winter or pure summer is the least biased choice when the spring window straddles the seasonal partition.

Because the perturbation is linear in P and PET and the multipliers are constant year-on-year (climatology shift, not interannual sequence), the resulting MSL5 shift is a single constant per cluster per scenario equal to the mean of the spring Δh values:

$$\Delta MSL5 = \text{mean}(\Delta h_{\text{Mar}}, \Delta h_{\text{Apr}}, \Delta h_{\text{May}})$$

The projected MSL5 trajectory is therefore the observational Method B trajectory shifted vertically by this constant. The observed line and the projected lines differ in absolute position but share the year-to-year shape — same wet years, same dry years, same intervention markers in the same places. The horizontal offset between observed and projected trajectories is the scenario sensitivity at each cluster.

S.18b.3.3 What the projection is and is not

Tool B is a perturbation overlay. It answers the question “what would the observed MSL5 trajectory over the monitoring period have looked like if the UKCP18 2050s (or 2080s) climate had been in force throughout?” The horizontal offset is the scenario sensitivity; the year-to-year shape is the observed climatology.

Tool B is *not* a forward-in-time forecast of what MSL5 will be observed in 2050. UKCP18 projects shifts in climatology, not shifts in interannual variability; the actual 2050 record could include individual years wetter or drier than any in the 2014–2025 observed window. The projection is a climatological response under stationary observed variability, not a single-realisation forecast.

Tool B’s single-step perturbation approach also avoids the SSM forward-integration drift problem documented in Script 21. Running the SSM forward from an arbitrary initial condition accumulates drift because the water-balance intercept α that closes the SSM at fitting time is not part of the single-step propagation rule. The single-step perturbation pattern works as a forcing-shift

overlay on the observed record rather than as an integrated time projection, and is drift-free by construction.

A standard UKCP18 caveat applies: the multipliers used are 50th-percentile central estimates. The 5th-to-95th percentile ranges span considerably wider intervals at end-century, and the projection magnitudes shown below would be correspondingly wider if computed under those ranges. Tool B is run on the central estimates as a single best-guess perturbation; the report does not extend it to the full UKCP18 uncertainty range.

S.18b.3.4 Results

Per-cluster projected ΔMSL5 values are reproduced below (full numerical values are written to `outputs/26b_van_willegen_msl_projections/26b_msl5_ukcp18_projection_summary.csv`).

Cluster	β_1	β_2	Spring Δh mean 2050s (m/month)	ΔMSL5 2050s (m)	ΔMSL5 2080s (m)
C1 Lake Edge	4.58	0.96	−0.012	−0.012	−0.021
C2 Dune	3.98	1.77	−0.017	−0.017	−0.031
C3 Western Residual	3.57	1.85	−0.017	−0.017	−0.031
C4 Main Forest	2.52	2.55	−0.021	−0.021	−0.039
C5 Coastal Forest	2.41	1.32	−0.012	−0.012	−0.022

The projected MSL5 shifts are modest at all five clusters: between 1.2 and 2.1 cm under the 2050s scenario and between 2.1 and 3.9 cm under the 2080s scenario. Main Forest (C4) has the largest shift because its high β_2 makes it most responsive to PET increases; Lake Edge and Coastal Forest (lower β_2) have the smallest shifts. The signs are negative everywhere — net drying — because the PET-driven term dominates the rainfall-driven term across the spring window even when winter rainfall is projected to increase.

S.18b.3.5 The spring-window structural cancellation

A non-trivial feature of the projection is the partial cancellation between winter and summer climate signals within the spring window. The spring months span both the winter (March) and the summer (May) UKCP18 partition windows, with April as the shoulder month. Across these three months the rainfall multiplier shifts from +5% in March through +10% in April (shoulder mean) to +10% in May (summer); the PET multiplier shifts from +5% in March through +13% in April (shoulder mean) to +20% in May. The two trends largely offset for spring totals, and the net Δh emerges as the residual.

This structural cancellation is specific to the spring window. The same UKCP18 multipliers applied to summer minima would produce much larger shifts, because the summer-minimum window (June through September) sits entirely within the summer partition and the +20–35% PET increase has no compensating winter-rainfall offset. The existing Script 11 Section 4 summer transfer function captures the summer signal explicitly. The implication for managers using MSL5 as a monitoring metric is that the metric is relatively climate-resilient compared to the summer minimum, which is a feature for monitoring stability but is *not* a finding that the site is insensitive to climate change overall.

S.18b.3.6 Climate signal in context

The projected Δ MSL5 magnitudes (1–4 cm) are smaller than three other quantities that bear on the report's interpretation:

The forest-management BACI clearfell signal at the Forest Impact tier is +0.120 m (raising spring water tables; Chapter S.7 of this supplement, BACI ANCOVA results). The clearfell signal is three to seven times larger than the central-estimate UKCP18 spring climate signal. Within the spring monitoring window and at the decadal timescale considered, forest management decisions matter more than central-estimate climate change for cluster-mean MSL5 at this site. Over longer (multi-decadal) horizons the climate signal accumulates and could become comparable, but within the framing of van Willegen's 5-year monitoring statistic the management signal dominates.

The observed interannual variability of MSL5 across the 2014–2025 monitoring window is on the order of tens of centimetres at each cluster — substantially larger than the projected climate shifts. A single dry or wet year shifts MSL5 by more than the 2080s projection moves the climatology mean. Interpreting the projection therefore requires care: the climatology shifts but a manager

observing a particular year will continue to see the dominant interannual signal on top of it.

The aggregation-method difference between Method A and Method B (S.18b.4 below) is itself on the order of tens of centimetres at some clusters — mean $| \text{Method B} - \text{Method A} |$ of about 30 cm across the network, maximum of 78 cm at Main Forest in 2011. The methodological choice between aggregations therefore moves cluster MSL5 estimates by more than the projected climate signal, which is one of several reasons the chapter retains both methods rather than declaring one canonical.

These three quantities — management signal, interannual variability, aggregation choice — all exceed the central-estimate climate signal by an order of magnitude or more. This sets the appropriate weight for Tool B in the report: a documented capability whose results are reported as evidence of climate-signal scale at this site rather than as a foregrounded prediction.

S.18b.3.7 Relationship to the Curreli summer-minimum predictions

The report's principal climate-vulnerability framing (§4.9.6 and the discussion in §5) uses summer minima and the Curreli SD15b and SD16 ecological thresholds. Under UKCP18 RCP8.5 50th-percentile central estimates, summer minima at most clusters are projected to cross those thresholds during the 2040s and beyond. Tool B, in contrast, finds that spring MSL5 shifts only by 1–4 cm under the same UKCP18 scenarios. The two findings can appear contradictory: how can summer minima be projected to fall through ecological thresholds while spring MSL5 barely moves?

The two findings are not contradictory. They are the same UKCP18 climate signal projected onto two different observation windows, and the difference arises from the seasonal asymmetry of the climate signal itself. UKCP18 RCP8.5 projects winter (Nov–Mar) wetter ($P \times 1.10$ to 1.20) and slightly elevated PET ($\times 1.05$ to 1.10), while it projects summer (May–Sep) drier ($P \times 0.85$ to 0.70) and substantially elevated PET ($\times 1.20$ to 1.35). The spring window (Mar–May) straddles the boundary: March and parts of April sample winter conditions, while May samples summer conditions. Within the spring window the wet-winter and dry-summer signals partially offset (S.18b.3.5). The summer-minimum window (Jun–Sep), by contrast, sits entirely inside the dry-summer signal with no offsetting winter-rainfall contribution.

For direct comparison: Script 19's spatial-groundwater scenarios — which apply the same UKCP18 multipliers to seasonal cluster water balances — project

summer-season Δh values of approximately 6 to 7 cm at the 2050s and 11 to 13 cm at the 2080s across the five clusters, compared with Tool B's spring-window ΔMSL5 of 1 to 2 cm at the 2050s and 2 to 4 cm at the 2080s. The summer-season projections are three to five times larger than the spring projections, with the precise ratio depending on the cluster's β_2 and the relative weight of the winter offset in its spring window. This three-to-five-fold ratio between summer and spring climate sensitivity is the structural feature of the UKCP18 signal at this site, not a discrepancy between two competing methodologies.

The implication for monitoring practice is that the two metrics measure different aspects of ecological stress. Summer minima measure the annual depth-of-drying peak — the seasonal stress maximum that pushes vegetation toward the dry end of its tolerance band — and are the appropriate metric for the Curreli SD15b and SD16 thresholds, which were calibrated against summer-stress responses. The 5-year mean spring water level (MSL5) measures the integrated water availability during the spring growing window — the conditions for active vegetative growth — and is the appropriate metric for the van Willegen 2025 Ellenberg-EbF framework, which was calibrated against per-piezometer spring conditions across multiple sites. Both metrics are ecologically valid; they characterize distinct stresses; and they respond differently to a seasonally-structured climate signal. The MSL5 metric is structurally climate-resilient because the spring window straddles the seasonal partition; the summer-minimum metric is structurally climate-sensitive because it samples only the side of the year that is drying. Neither finding overturns the other.

For the report's main-text discussion this means three things. First, the Curreli summer-minimum threshold-crossing projections stand: they are the correct climate-vulnerability framing for the SD15b and SD16 thresholds, and the projected 2040s+ crossings are real. Second, the modest MSL5 shifts in Tool B are also correct: they are the spring-window response to the same climate signal, and they characterize the climate resilience of the spring monitoring metric specifically. Third, the right reading is that **MSL5 is a stable monitoring backbone while the summer-minimum framework captures the climate-vulnerability dimension**, and both belong in the report's monitoring recommendations. Managers using MSL5 for routine annual tracking should also continue to track summer minima as the climate-stress indicator; the two together give a more complete picture of dune-slack water-table dynamics under change than either gives alone.

S.18b.3.8 A parallel per-well aggregation pathway

Tool B as documented above fits the SSM β coefficients on the cluster-centroid hydrograph: one OLS per cluster, with β_1 and β_2 taken from the cluster-centroid `03_regional_averages.csv` series. This is the canonical pathway. Its outputs anchor §3.7.5 of the main report (the MSL5 methods framework), §4.8.4 (Script 26c’s cluster trajectory figure), and §4.10.1 (Script 26c’s Δ MSL5 vs Δ summer-minimum contrast); the Method B aggregation rationale (S.18b.4) gives the reasoning.

A parallel per-well aggregation pathway was added at Script 26b v1.1.0 (2026-05-27) producing the secondary CSV

`26b_msl5_ukcp18_projection_summary_perwell.csv`. The per-well pathway fits the SSM β coefficients per well on `03_master_data.csv` and then arithmetically averages β_1 and β_2 within each cluster, applying the same pure-climate perturbation formula $\Delta h(m) = \beta_1 \cdot P(m) \cdot (sP(m) - 1) - \beta_2 \cdot PET(m) \cdot (sPET(m) - 1)$ and the same Mar–May spring window. The output is a separate CSV with 12 rows per scenario (5 cluster rows plus a well-count-weighted SITE row), structurally parallel to the centroid summary.

The per-well aggregation does not reduce algebraically to the centroid OLS, even on the same wells: the two aggregations differ by 0.5 to 3.7 mm per cluster per UKCP18 scenario, with the largest gap in C1 (lake-edge, $n = 7$, the most heterogeneous reference-network cluster). Both are defensible summaries of the same SSM. The motivation for the parallel pathway is the Script 19 v2.8.0 scenario viewer (S.13), whose new Δ MSL5 row uses per-well-averaged β throughout — consistent with the rest of the viewer’s per-cluster table — and which therefore cannot validate against the centroid CSV on principle. The per-well CSV serves as the matching canonical reference for the viewer row, validating to ≤ 0.5 mm per (cluster, scenario) pair (current state: 0.042 mm worst case, a rounding artefact of the viewer CSV’s 4-dp formatting). The canonical centroid CSV remains unchanged byte-for-byte at the v1.1.0 commit; Script 26c continues to read it, and the §3.7.5 / §4.8.4 / §4.10.1 report numbers are not affected by the addition of the per-well pathway.

The two CSVs therefore sit side-by-side as parallel summaries: the centroid CSV is the report-anchored canonical, and the per-well CSV is a secondary artefact whose primary role is the viewer-row validation target. A future report-aggregation decision can compare them directly.

S.18b.4 Method A and Method B aggregation

A consequence of paired Tools A and B both operating on Script 03's cluster-centroid monthly series is that the report carries two MSL5 aggregations side by side. Script 26 produces both as parallel CSV outputs:

`26_msl_5yr_per_cluster.csv` is the **Method A** trajectory, computed by taking the per-well annual MSL5 for every well assigned to the cluster (under the $k=5$ partition extended through Script 06's Pearson membership audit) and then taking the mean across wells per cluster per window-end-year. Method A aggregates the extended cluster network — on the order of 25 wells in C5, for example. It is the natural aggregation for monitoring because it uses the maximum spatial coverage available and is closest to the per-piezometer framework that van Willegen uses for vegetation calibration. The trajectory figure in §4.8.4 of the report and the spatial MSL5 map in §4.8.4 are both Method A.

`26_msl_5yr_per_cluster_centroid.csv` is the **Method B** trajectory, computed by taking the cluster-centroid monthly series from `03_regional_averages.csv` (which is the LCSC reference network only, on the order of 5 wells in C5) and applying the same $3/3 + 5/5$ strictness rules used by Method A. Method B is the aggregation that the SSM β coefficients are fitted against and is therefore the internally consistent baseline for Tool A's empirical OLS fits and Tool B's perturbation overlay. The Method B trajectory is referenced by Tools A and B; it is not a headline observational figure in the report.

The two methods are both valid; they answer different questions and describe different network compositions, not different aggregation algebra. The Method A cluster mean is “the typical well in this cluster”, aggregating across all monitored points; the Method B cluster centroid is “the cluster's reference-network signal”, aggregating across the subset that the SSM coefficients describe. When network composition is stable the two coincide; at Newborough the extended cluster network includes shallower coastal-edge wells (in C5 particularly) that the reference network does not, so the mean depths differ. Mean $|\text{Method B} - \text{Method A}|$ across all (cluster, window-end) pairs is approximately 0.30 m; maximum 0.78 m at Main Forest in 2011. C3 Western Residual is the one cluster where Method B is shallower than Method A (the C3 reference network omits some particularly deep wells that the extended network includes); the other four clusters have Method B deeper than Method A.

The chapter S.18 of this supplement covers the Method A side as the headline monitoring metric. This chapter (S.18b) uses Method B because it is the SSM-consistent baseline for the predictive tools, and Tools A and B both read from `03_regional_averages.csv` (Method B's underlying monthly series) for that reason. Each cluster MSL5 number quoted in the report should be qualified with the method that produced it; the main-report convention is that §4.8.4 figures cite Method A and any number from Tool A or Tool B carries the implicit “Method B” provenance.

S.18b.5 Pipeline integration

Tool A and Tool B are integrated into the analytical pipeline managed by `run_analysis.py` as follows:

Tool A is Section 5 of Script `11_forecasting_thresholds.py` (step 22 of the pipeline, Phase 1). It runs whenever Script 11 runs and adds two outputs (one CSV, one calibration figure) to the existing five-output Script 11 result set. Section 5 is invoked from Script 11's `run_models()` orchestrator function after the four pre-existing sections (state-space coefficient summary, winter transfer functions, summer transfer functions, P_flood iteration); existing outputs are unchanged.

Tool B is Script `26b_van_willegen_msl_projections.py` (step 29 of the pipeline, Phase 13). It runs alongside Script 26 (step 28) and produces four CSV outputs and one figure to its own output directory `outputs/26b_van_willegen_msl_projections/`. The script reads its observational baseline from Script 26's Method B CSV (`26_msl_5yr_per_cluster_centroid.csv`), so Script 26 must run first; this dependency is enforced by run order within Phase 13. The four CSVs are the centroid-fitted projection summary (the canonical one consumed by Script 26c and used in §3.7.5 / §4.8.4 / §4.10.1), the parallel per-well-aggregated summary (added at Script 26b v1.1.0 as a validation target for Script 19's scenario-viewer Δ MSL5 row — see §S.18b.3.8 below), the full 12-month Δh -per-cluster matrix, and the run transcript.

Script `26c_msl5_report_figures.py` (step 30 of the pipeline, Phase 13) follows Script 26b and renders the two report-format MSL5 figures cited in §4.8.4 and §4.10.1 of the main report; it is documented in S.18c and reads only canonical outputs from Scripts 26, 26b, and 19, so all three must have run.

The greyscale figure-conversion utility (`27_greyscale_figures.py`) is step 43 of the pipeline and sits in Phase 17 as a single-script post-processing phase.

S.18b.6 Data sources and reproducibility

UKCP18 RCP8.5 multipliers used by Tool B are central-estimate (50th-percentile) Wales values for the 2050s and 2080s time horizons. The 2050s constants also exist in `utils.config` as `UKCP18_DRY_*` / `UKCP18_WET_*` pairs (carried for cross-reference with Script 19's scenario viewer); the 2080s constants are hardcoded in Script 26b's `UKCP18_SCENARIOS` dictionary at the top of the file. Centralising the 2080s constants in `utils.config` alongside the 2050s pairs, so Scripts 19 and 26b share a single source, is noted as a follow-up.

Source: van Willegen et al. (2025), *Ecological Indicators* 170, 113016 — methodological anchor for the MSL5 statistic and its hydrology-year convention. Met Office (2018), UKCP18 Land Projections, Regional 12 km ensemble — source of the climate multipliers. Curreli et al. (2013) — source of the SD15b and SD16 reference thresholds shown on the trajectory figures (note that these are calibrated against summer minima, not MSL5; the ~0.54 m offset at the 5-year window scale is documented in S.18 of this supplement).

End of chapter S.18b.

S.18c Script 26c — Report-format MSL5 figures

Step 32 / 35. Phase 13 — Van Willegen MSL Analyses in `run_analysis.py`; display-only companion to S.18 and S.18b.

Script 26c is a display-only companion to Scripts 26 and 26b. It produces two figures cited in §4.8.4 and §4.10.1 of the main report and reads only the canonical outputs of Scripts 26, 26b, and 19; no analysis is recomputed. The methodological framework underlying both figures is the MSL5 framework documented at §3.7.5 of the main report and at §S.18 and §S.18b of this supplement. This chapter records only the display step.

The first output is `fig_msl5_trajectory_report.png`, the report-format cluster-mean MSL5 trajectory over 2014–2025. The figure plots Script 26's Method A trajectory against the Curreli (2013) SD15b (–0.61 m) and SD16 (–0.98 m) reference values, with the SD16 dry-slack zone shaded for visual emphasis of cluster-time combinations sitting inside that ecohydrological regime. The figure is a report-format companion to the methods-context trajectory `26_msl_5yr_trajectory.png` produced by Script 26: both show the same data; the difference is editorial. The methods-context figure retains the intervention

markers (2015 scrape, 2017 clearfell, 2023 re-scrape) for methodological clarity; the report-format figure omits them and adds the SD16 shading and 2025 value labels for direct ecological readability. Both figures are valid in their respective contexts.

The second output is `fig_msl5_vs_summer_min_projection.png`, a two-panel horizontal-bar contrast of ΔMSL5 against $\Delta\text{summer-minimum}$ for the five clusters under UKCP18 RCP8.5, with the 2050s in the top panel and the 2080s in the bottom. ΔMSL5 values are read from Script 26b's `26b_msl5_ukcp18_projection_summary.csv`; $\Delta\text{summer-minimum}$ values are read from Script 19's `19_scenario_summary.csv` (rows where `season = summer`, column `dh_mean_m` — Script 19's "summer" Δh is the seasonal mean of monthly Δh over the `SUMMER_MONTHS` window, treated here as the closest SSM correlate to the annual summer minimum). The figure makes one scientific point: the spring baseline metric MSL5 is substantially better buffered against the projected climate trajectory than the summer-minimum metric, by a factor of three to six across all five clusters and both horizons. The cross-reference to §4.10.1 of the main report carries the discussion of this result.

End of chapter S.18c.

S.19 Scripts 28, 29, 30 — Cluster framework diagnostics (Phase 14)

Steps 33, 34 and 35 / 36. Phase 14 — Cluster framework diagnostics.

The two scripts in this chapter were added on 2026-05-29 following the post-review pass on the main report (Hollingham 2026). They test post-Script-25 implications for the cluster framework documented in §5.1 of the report and supply the quantitative validation paragraphs in §5.1.1 (gap D in the post-review priorities list). Both scripts consume already-produced pipeline outputs and write into their own output directories; both run after Phase 13 and before the Phase 15 differential-change scripts.

The methodological motivation for both diagnostics is the same. §5.1 of the main report frames C2 and C3 as eastern and western positions on a substrate-thickness gradient — shallow sand over till in the east, deeper sand in the west as the basal till sheet dips and merges with the Menai Strait estuarine clays. The Ward's clustering on $(1 - \text{Pearson } r)$ distance imposes a discrete cluster boundary on what is, on this substrate-architecture picture, a continuous behavioural gradient. The framing rests on Stratford et al. (2007) and Grootjans et

al. (2004) but is not, in the published version of the report, validated against the project's own data. Two questions follow naturally. First: is the C2/C3 behavioural distinction reproducible by adding a coastal-erosion drift onto C2 hydrographs — i.e., can the distinction be folded back into C2 by a single reductive operation acting on the level signal? Second: given that C3 is internally heterogeneous in spatial coefficients (visible in Script 07's per-well β maps), what is the structure of that heterogeneity, and does it resolve a smooth gradient or warrant a finer cluster partition? Script 28 addresses the first question; Script 29 addresses the second. Together they form a diagnostic dyad: Script 28 rules out one reductive alternative, Script 29 provides a spatial-gradient signature that the substrate-architecture picture predicts.

S.19.1 Script 28 — C3 detrend check

Motivation. Test whether the C3 cluster is mechanistically C2 with a western-margin coastal-erosion drift superimposed (H1) or whether it is a behaviourally coherent cluster on a westward-thickening substrate gradient (H0). Under H1, removing the predicted coastal drift from each C3 well's hydrograph should cause the de-trended series to migrate to a C2 best-match against the un-de-trended cluster centroids; under H0, C3 wells should remain best-matched to C3 regardless of de-trending. The test rules out the simplest reductive explanation (a linear-in-time drift on level, added to otherwise C2-like behaviour); it does not by itself distinguish a discrete-architecture C3 from a continuous-substrate-gradient C3, which Script 29 addresses.

Inputs.

Input	Source script	Role
02_cluster_stats.csv	Script 02	Cluster assignments and reference network membership
01_wells_clean_maod.csv	Script 01	Monthly hydrographs in mAOD
03_regional_averages.csv	Script 03	Cluster centroid time series for re-classification
25_01_panel_fit_parameters.csv	Script 25	Live δ_0 , L, c from the forest-free linear-capped headline fit
25_02_per_well_summer	Script 25	Per-well dist_coast_m

Input	Source script	Role
<code>_min_slopes.csv</code>		for evaluating $\delta(d)$

Methodology. For each well, the predicted coastal-erosion drift rate is $\delta(d) = \delta_0 \cdot \max(0, 1 - d/L)$ using the live Script 25 forest-free linear-capped fit ($\delta_0 = -29.03$ mm/yr, $L = 894$ m, $c = -6.40$ mm/yr at the headline state of 2026-05-29). The well's monthly hydrograph is de-trended by subtracting the linear trend of slope $\delta(d)$ over the observation window: since δ is negative (a decline rate), the correction is positive in time, undoing the drift the model attributes to coastal-erosion proximity. The de-trended series is then re-classified against the un-de-trended cluster centroids using correlation distance on monthly anomalies ($1 - \text{Pearson } r$, the same metric Script 02 uses to construct the clusters). The well is assigned to the cluster whose centroid it correlates with most strongly post-de-trending.

The diagnostic rule of decision is calibrated against the C3 cluster size of $n = 21$ wells (with two intervention wells excluded — CEH36, the 2015 dune-scape site, and WMC3, the 2017 clearfell Impact well — leaving 19 testable C3 wells). If ≥ 17 of 19 ($\approx 90\%$) C3 wells migrate to a C2 best-match after de-trending, H1 is strongly supported. If 11–16 wells migrate, the result is partial. If ≤ 10 wells migrate, H1 is rejected — the reductive “C2 + coastal drift” explanation does not account for the C2/C3 distinction.

Five sensitivity variants are run to guard against under-specification of the drift model: the headline forest-free monthly-uniform variant; a summer-only Jun–Sep variant that applies the drift correction only to summer-window months; a full-network δ_0 variant that uses the full-fit gradient (which includes forest wells) instead of the forest-free fit; an $L = 500$ m variant (shorter inland reach); and an $L = 1500$ m variant (longer inland reach). A C2 sanity check accompanies each variant: C2 wells should not migrate to other clusters under the same procedure, since the gradient model attributes negligible drift to them (C2 wells are mostly beyond L). Low C2 retention would indicate that the de-trending procedure is contaminating hydrographs rather than testing a hypothesis.

Outputs.

File	Contents
<code>outputs/28_c3_detrend/28_c3_detrend.csv</code>	Per-well: original best-match, de-trended best-match (headline + 4 sensitivities), correlation deltas, <code>dist_coast_m</code> , applied $\delta(d)$

File	Contents
outputs/28_c3_detrend/ 28_c3_detrend_results.md	Memo with headline result, decision rule, full sensitivity table, C2 sanity-check retention, excluded-wells list
outputs/28_c3_detrend/ 28_c3_detrend_panel.png	Four-panel figure: (a) representative C3 hydrograph before and after de-trending; (b) classification scatter showing correlation deltas; (c) sensitivity bar chart across the five variants; (d) C2 sanity-check retention bars

Path constants (new). `paths.DIR_28`, `paths.OUT_28_DETREND_TABLE`, `paths.OUT_28_DETREND_MEMO`, `paths.OUT_28_DETREND_PANEL`.

Headline result (2026-05-29). H1 rejected. Of 19 testable C3 wells, **0 genuinely migrate** from a C3 best-match to a C2 best-match as a result of de-trending. One nominal migrant (NW13 under the headline variant) was already C2-best-matched before any de-trending and is therefore not a genuine migrant. The result is robust across all five sensitivity variants — the highest migration count in any variant is 2 of 19, well below the H1 threshold of 17. C2 sanity-check retention is 24 of 24 wells (100 %), confirming the de-trending procedure does not contaminate the underlying hydrographs. The reductive “C2 + linear coastal drift” explanation is rejected; the result is compatible with both a discrete-architecture C3 and a continuous-substrate-gradient C3, and Script 29 (below) provides additional evidence favouring the gradient interpretation.

Implication for the report. The C2/C3 behavioural distinction cannot be reproduced by removing a linear coastal-erosion drift from C2 hydrographs — it is not the residue of an east-to-west drift superimposed on identical underlying behaviour. This is consistent with the §5.1 aquifer-architecture framing — shallow sand over till in the east, deeper sand in the west — without by itself resolving whether the substrate change is a discrete architectural boundary or a continuous westward-thickening gradient as the basal till sheet dips and merges with the Menai Strait estuarine clays. Script 29’s spatial-structure findings within C3 (below) favour the gradient interpretation. The result feeds the first §5.1.1 paragraph in the main report.

Limitations. The de-trending is linear in time, applying the steady-state $\delta(d)$ over the full observation window. Any transient coastal-erosion signal (episodic

retreat events such as Storm Brendan 2020) is not separately resolved. The drift correction operates only on the western-margin coastal-erosion mechanism on the level signal; if other site-wide mechanisms contribute to the C2/C3 distinction — most notably an east-to-west substrate-thickness gradient as the basal till dips and merges with the Menai Strait estuarine clays, producing a continuous shift in the SSM coefficient field rather than a drift in level — they are not represented in the test. The procedure tests a specific H1 ($C3 = C2 + \text{coastal drift on the level signal}$); it does not falsify a continuous-substrate-gradient explanation, which Script 29 is better positioned to address through the within-C3 spatial-structure analysis.

S.19.2 Script 29 — Within-C3 variance attribution

Motivation. Given that Script 28 rejects the reductive “ $C3 = C2 + \text{coastal drift}$ ” hypothesis, what is the structure of variation *between* C3 wells, and does it favour a discrete-architecture or continuous-substrate-gradient interpretation? The per-well β maps from Script 07 visibly show within-cluster heterogeneity in β_1 , β_2 , β_3 , and τ across the C3 wells. The question is whether this within-cluster variance follows a spatial or hydrogeological axis that the cluster framework treats as noise but that warrants explicit treatment in the report’s discussion of the §5.1 cluster boundaries, and — if it does — whether the axis location is geometrically consistent with the dipping-till / Menai-estuarine-clays substrate-gradient picture.

Inputs.

Input	Source script	Role
02_cluster_stats.csv	Script 02	C3 cluster membership
01_wells_clean_maod.csv	Script 01	Monthly hydrographs for slope and amplitude metrics
01_locations.csv	Script 01	Easting, northing, ground elevation per well
07_spatial_coefficients/ 07_coeff_maps_data.csv	Script 07	Per-well β_1 , β_2 , β_3 , τ (storage–drainage index S_y/β_3 ; not a residence time)
25_01_panel_fit_parameters.csv	Script 25	δ_0 , L for the exponential and linear-capped coastal predictors

Input	Source script	Role
25_02_per_well_summer_min_slopes.csv	Script 25	Per-well dist_coast_m and slope_m_yr
data/Features.kml	versioned data input	Forest polygon for the dist_forest predictor

Methodology. Two panels are constructed for each well in C3:

- **Behavioural metrics (n = 9):** annual summer-minimum slope slope_m_yr, per-well coefficients β_1 , β_2 , β_3 , storage–drainage index $\tau = S_y / \beta_3$ (not a residence time — see §S.12 for the $t_R = 1/\beta_3$ recession-time distinction), long-term mean head mean_head_maod, summer-minimum depth summer_min_depth_m, winter-maximum depth winter_max_depth_m, and seasonal amplitude as the p90 – p10 spread of the monthly mean hydrograph (matching Script 02’s amplitude descriptor).
- **Predictors (n = 5):** the Script 25 exponential coastal predictor $\delta_0 \cdot \exp(-d/L)$ (using the forest-free exponential parameters $\delta_0 = -40.24$ mm/yr, $L = 407$ m, $c = -5.24$ mm/yr); distance to CEH36 (the 2015 dune-scraper site at 241,161 E / 363,306 N — chosen as a topographic and hydrogeological anchor in the south-western interior of the warren); distance to the forest polygon edge from the Features.kml feature set (the C4/C5 boundary that bounds forest-management influence); ground elevation from 01_locations.csv (a topographic axis correlated with northing and elevation); and depth-to-water as ground_elev – long_term_mean_head (a hydrogeological covariate separating depth-to-water response from raw ground elevation).

The predictor space has substantial known collinearity: northing and ground elevation correlate at $r \approx 0.98$ in the C3 well set, and elevation correlates with the coastal-exponential predictor at $r \approx 0.79$. The analysis therefore reports both univariate R^2 per (metric, predictor) pair and drop-one unique contributions in the full five-predictor model — the drop-one Δ identifies the predictor that carries unique signal not absorbed by any other predictor.

Outputs.

File	Contents
outputs/29_within_c3_variance/29_within_c3_variance.csv	Per-well: nine behavioural metrics + five predictors + ancillary variables
outputs/29_within_c3_variance/	Single-predictor R^2 matrix (metric $\times$

File	Contents
29_univariate_R2.csv	predictor)
outputs/29_within_c3_variance/ 29_drop_one.csv	Drop-one unique contribution matrix (metric $\times$ predictor)
outputs/29_within_c3_variance/ 29_within_c3_variance_results.md	Memo: full R^2 and drop-one matrices with headline interpretation and caveats
outputs/29_within_c3_variance/ 29_within_c3_variance_panel.png	Six-panel figure: per-metric scatter against the strongest unique predictor; drop-one heatmap; collinearity matrix

Path constants (new). `paths.DIR_29`, `paths.OUT_29_PANEL_CSV`,
`paths.OUT_29_UNIVARIATE_R2`, `paths.OUT_29_DROP_ONE`, `paths.OUT_29_MEMO`,
`paths.OUT_29_PANEL_FIG`.

Headline result (2026-05-29). Within-C3 variance is well-explained by spatial position, with the strongest fits at the SSM coefficients:

Metric	R^2 (full)	adj- R^2	Strongest unique predictor (drop-one Δ)
β_1 recharge	0.813	0.741	dist_CEH36 ($\Delta = +0.166$)
β_3 drainage	0.736	0.634	dist_CEH36 ($\Delta = +0.156$)
τ storage– drainage index	0.700	0.585	dist_CEH36 ($\Delta = +0.317$)
β_2 atmospheric draw	0.672	0.546	dist_CEH36 ($\Delta = +0.114$)
slope_m_yr	0.696	0.579	delta_coast_exp ($\Delta = +0.060$)
seasonal_amplitu de	0.724	0.617	dist_CEH36 ($\Delta = +0.112$)

The summer/winter depth and mean-head metrics return near-perfect R^2 (≥ 0.97) because of their definitional dependence on `depth_to_water` — these are sanity-check results rather than findings.

Distance to CEH36 emerges as the strongest unique predictor across all four SSM coefficients and across the storage–drainage index and seasonal amplitude. The `dist_CEH36` axis represents a hydrogeological gradient within C3 anchored near the south-western interior of the warren, distinct from the network’s coastal-erosion axis (which is the strongest unique predictor for the summer-minimum slope itself but does not dominate the SSM coefficients).

A methodological aside emerges from the regression: the Script 25 exponential coastal predictor lands at a regression coefficient of +1.046 against per-well slopes, an independent face-value validation of the exponential functional form (S.15 reports both linear-capped and exponential fits; the exponential is favoured at marginal AIC).

Implication for the report. Within-C3 heterogeneity is structured — not noise — and the structure aligns along an axis anchored near the south-western interior of the warren rather than along the coastal margin. This anchor location is **predicted by**, not merely compatible with, the westward-thickening substrate-gradient hypothesis: a basal till sheet dipping westward and merging with the Menai Strait estuarine clays would produce the thickest overlying sand (and therefore the most distinctive SSM coefficient values) in exactly the south-western interior the regression identifies. The $k = 6$ sensitivity check performed in the same session (not in Script 29’s outputs, but documented in the session log) confirms that this structure is internally coherent: at $k = 6$, Ward’s clustering does not split C3 (all 21 wells remain together) but instead splits C2 (24 wells → 17 + 7 by seasonal amplitude). C3 is “structured but not strongly enough to split” on the $(1 - \text{Pearson } r)$ distance metric — exactly the behaviour expected from a smooth substrate gradient that varies the SSM coefficient field continuously without producing a stepped discontinuity. Read together with Script 28’s rejection of the reductive “C2 + coastal drift” hypothesis, the two diagnostics together support the picture of a continuous substrate-gradient architecture with the C2/C3 cluster boundary as a behavioural cut on the continuum rather than a discrete geological boundary. The result feeds the second §5.1.1 paragraph in the main report and is cross-referenced from §5.7.2 (the discussion of within-cluster spatial structure).

Limitations. The regression uses $n = 19$ wells (the C3 cluster minus CEH36 and WMC3, which carry direct intervention signals), so degrees of freedom are limited and the unique contributions of correlated predictors are not always cleanly separable. The `dist_CEH36` axis is operationally defined by the 2015 scrape-site location, which is a hydrogeological anchor of opportunity rather than a parameter fitted to the data — its physical interpretation (proximity to a

topographic and hydrogeological transition within C3) is a candidate explanation rather than a tested one. The analysis is C3-specific by motivation; analogous within-cluster diagnostics for C1, C2, C4 and C5 are not produced, on the grounds that the original motivating question (the C3 cluster’s distinctness and internal structure) is the §5.1-relevant one.

Cross-references

- **F.4** — Cluster partition constants, $k=5$ anchor wells, partition history.
 - **S.2** — Ward’s hierarchical clustering on $(1 - \text{Pearson } r)$ distance; the metric used for the re-classification step in Script 28.
 - **S.3** — Cluster centroid time series (`03_regional_averages.csv`) used by Script 28 for re-classification.
 - **S.7** — Per-well β maps (Script 10 spatial coefficients) showing within-C3 heterogeneity; the visual motivation for Script 29.
 - **S.15** — Script 25 coastal-retreat gradient parameters (δ_0 , L , c) feeding both scripts’ coastal predictors.
 - §5.1.1 of the main report — primary destination for both scripts’ headline findings (gap D in the 2026-05-29 post-review cascade).
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S.19.3 Script 30 — C4 constrained- β_3 triangulation sensitivity and water-balance partition

Step 35/43, Phase 14. Added 2026-06-23 (v1.0.0); water-balance partition block added 2026-06-25 (v1.1.0). (The filename `30_c4_constrained_fit.py` reuses a number freed when a short-lived cluster-slope-decomposition script of the same number was retired into Script 25 on 2026-05-29 — see the §S.15 update note; the two scripts are unrelated.)

Motivation. The unconstrained per-well SSM identifies the drainage coefficient β_3 cleanly across most of the network, but at C4 Main Forest it is weakly identified: the cluster centroid returns $\beta_3 = 0.020 \text{ month}^{-1}$ (the lowest in the network), one well (CEH14) returns a physically inadmissible negative value, and the atmospheric-draw coefficient β_2 is correspondingly inflated. Script 30 quantifies the effect of this degeneracy on the forest-interior coefficients, drainage timescale, and water-balance partition, and reports the result as a labelled sensitivity bound — not a correction to the canonical record.

Inputs.

Input	Source	Role
03_regional_averages.csv	Script 03	C4 centroid monthly hydrograph
03_03_cluster_mechanistic_coefficients.csv	Script 03	Unconstrained centroid β coefficients
16_water_balance/ 16_01_water_balance_summary.csv	Script 16	Cluster-mean P, PET, h_disp for partition
03_master_data.csv	Script 03	Per-well β_3 values for the identified-wells summary

Methodology. The C4 centroid hydrograph is re-fitted through `model_utils.fit_ssm(fixed_beta_3=...)` with β_3 held at a triangulation anchor: $C4 \beta_3 \approx C2 \beta_3 (0.073) + \Delta\beta_{3_tree}$, where $\Delta\beta_{3_tree} = -0.015$ is inferred from the inland-C5 minus C3 contrast (coastal-corrected), giving an anchor of 0.058 month^{-1} . The script then partitions the C4 loss budget at three β_3 values — the unconstrained centroid, the residual-minimizing β_3 , and the anchor — reading cluster means (P, PET, h_disp) from Script 16’s table so the unconstrained partition is identical to Table 4a of the main report (verified: the centroid re-fit at the unconstrained β_3 reproduces Script 16’s $\beta_2 = 2.548$). The partition convention follows Script 16 exactly: drainage = $\beta_3 \cdot h_disp$, atmospheric draw = $\beta_2 \cdot PET$, residual = $\beta_1 \cdot P - (\text{drainage} + \text{draw})$. No values are hardcoded — the means come from Script 16 and the anchor is recomputed at run time.

Headline result. Per-well fitting reveals a β_2 – β_3 trade-off. Five of the nine C4 wells constrain β_3 (NW10 0.043, CEH20 0.030, CEH30 0.024 at $p \leq 0.05$; CEH2 0.023, CEH33 0.020 at $p \approx 0.10$); three do not (CEH13, CEH32, CEH34); and CEH14 fails the SSM outright (iterative NSE = -3.21). The identified wells centre on 0.025–0.030, below the anchor (0.058). Holding the centroid at the anchor moves β_2 from 2.55 to 1.86 — close to the open-dune values (C2, 1.66; C3, 1.77) — and shortens $t_R = 1/\beta_3$ from 49 to 17 months. The water-balance partition remains atmospheric-draw-led across the identified range: drainage takes 24 % of the loss budget at the unconstrained centroid, approximately 30 % at the residual-minimizing β_3 (0.0255), and crosses into a drainage-led 56 % only at the anchor. The closure residual is smallest at $\beta_3 \approx 0.026$ (essentially zero) and grows in both directions — +0.8 % of the loss budget at the unconstrained value, –4 % at the anchor (the largest closure residual in the network).

Implication. The forest-interior drainage coefficient is bounded, not pinned. The identified wells, the β_2 – β_3 trade-off, and the water-balance closure converge on $\beta_3 \approx 0.026$ ($t_R \approx 39$ months), so C4 drains more slowly than the open dune (C2 $t_R \approx 14$ months) — a real canopy-and-substrate effect — but remains atmospheric-draw-dominated at the identified range, neither as sluggish as the unconstrained centroid nor as drainage-led as the anchor. The report retains the unconstrained fit as its primary record; this script’s numbers appear in main-report §5.2.3 only.

Outputs.

File	Contents	Downstream consumer
30_c4_constrained_fit / 30_c4_constrained_perwell.csv	Per-well β_3 , β_2 , p-value, identification status, exclusion flag	§5.2.3 per-well table
30_c4_constrained_fit / 30_c4_constrained_fit.png	β_2 – β_3 trade-off and water-balance partition sensitivity figure	§5.2.3 figure
30_c4_constrained_fit / 30_c4_constrained_report_numbers.csv	25 named keys — single source for all §5.2.3 citations	§5.2.3

Trace map (§5.2.3 citation → CSV key). β_2 2.55 → c4_beta2_centroid_uncon; β_2 1.86 → c4_beta2_centroid_anchor; β_3 0.020 → c4_beta3_centroid_uncon; β_3 0.058 → c4_anchor_beta3; β_3 0.026 → c4_beta3_bestclosure; t_R 49/17/39/14 months → c4_tr_uncon/c4_tr_anchor/c4_tr_bestclosure/c2_tr; drainage partition 24 %/30 %/56 % → c4_drainage_pct_uncon/_bestclosure/_anchor; closure +0.8 %/–4 % → c4_wb_resid_uncon/c4_wb_resid_anchor; per-well β_3 → 30_c4_constrained_perwell.csv.

Limitations. The triangulation anchor is a cross-cluster construct; its C5–C3 tree-effect proxy demonstrably under-estimates the suppression seen at the wells that do constrain β_3 , so it functions as an upper bound on the forest drainage rate rather than a best estimate. Per-well fits are individually noisy (≈ 100 -month windows). CEH14 is excluded throughout (ridge-flank SSM failure, NSE –3.21). The sensitivity is diagnostic only — it is not propagated into the canonical coefficient masters, the water balance, or the spatial maps, which all retain the unconstrained fit.

Cross-references.

- §S.3 — canonical unconstrained SSM fits; 03_master_data.csv and 03_03_cluster_mechanistic_coefficients.csv.
 - §S.11 — Script 16 water-balance decomposition; partition convention and Table 4a.
 - §S.12 — $t_R = 1/\beta_3$ head-space recession time; $\tau = S_y/\beta_3$ storage–drainage index.
 - §5.2.3 of the main report — the only location in the report that cites Script 30’s numbers.
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End of chapter S.19.

S.20 Scripts 32, 33, 35 — Observed differential change and the climate-response envelope

Phase 15 (steps 36–38/43). Three figures that characterize observed network change directly, independently of the single-mechanism driver model of §S.13–S.14: where the spring water table is moving relative to the site (Script 32, Figure 59), how far it swings between climate extremes (Script 33, Figure 60a), how deep it sits in dry years against the ecological thresholds (Script 33, Figure 60b), and a per-well, frame-independent coefficient extending the amplification metric to short-record wells (Script 35). All three read committed pipeline outputs and re-fit nothing in the SSM; their value layers are observational.

S.20.1 Script 32 — Secular differential water-table movement (Figure 59)

Motivation. The two-window MSL5 comparison (Figure 58) cannot separate secular change from window choice (Script 34, §S.21.4). The differential-movement map isolates the secular component by fitting, for each well, the trend of its spring level relative to the site mean. Subtracting the site mean removes the common climate signal, leaving the relative drift — the slow inland mound holding its position while the fast-draining lake and coastal margins decline. The map is explicitly a differential-recession field, not an absolute-drying map and not a management-signature map.

Inputs. outputs/01_wells_clean.csv (per-well monthly levels, depth below ground); outputs/01_locations.csv (well Easting/Northing); outputs/03_master_data.csv (per-well cluster id). The SSM is not re-fitted.

Methodology. A spring value is formed per well-year as the mean of the available March–May readings (config.MSL_SPRING_MONTHS). The well’s anomaly

is its spring level minus the site-mean spring level in the same year. The site-mean reference panel is held to the wells present in at least a fixed majority fraction of the period's springs (PANEL_MIN_FRACTION), so the reference does not drift with coverage. The metric is the ordinary-least-squares slope of the anomaly against year, in mm yr^{-1} , requiring a minimum number of spring-years per well. Significance is a lag-1 autoregression-corrected t-test on the effective sample size, cross-checked against a moving-block bootstrap confidence interval; significant wells are drawn solid, non-significant wells hollow. Two periods are mapped: 2011–2025 (primary; 66-well site-mean panel) and 2005–2025 (robustness; 21-well panel). Only the Llyn Rhos-Ddu lake gauge is excluded — CEH13 and CEH14, withheld from the SSM for state-space reasons, are retained because the differential anomaly trend is observational. The per-well slopes are interpolated to a 50 m grid by inverse-distance weighting (power 2), masked beyond 450 m of the nearest well.

Headline result. Over 2011–2025, 8 of 74 mapped wells move significantly, almost all sinkers on the south-western coastal and western margin (CEH22 -26.5 , CEH21 -15.1 , CEH11 -13.3 mm yr^{-1}). Over 2005–2025 the pattern firms to 13 of 77, the coastal and western sinkers (CEH22, NW8, CEH3, NW9, CEH17) strengthening. The forest interior holds its position, with the forest control wells (CEH32 $+20.5$, CEH34 $+17.2$, CEH2 $+14.9 \text{ mm yr}^{-1}$) among the few drifting upward.

Outputs. outputs/32_differential_movement/:

32_differential_movement_per_well.csv (slope, significance, both periods);

32_differential_movement_2011_2025.png (Figure 59 primary);

32_differential_movement_2005_2025.png (robustness); 32_results.txt.

Limitations. The differential framing answers “where relative to the site”, not “how much in absolute terms” — the absolute site-mean trend is separately and non-significantly -7.0 mm yr^{-1} (Script 32 site-mean panel, AR-corrected, $p = 0.52$). The per-well metric is an OLS slope; the AR correction enters the significance test only. Sparse-coverage wells are flagged via the panel rule rather than dropped.

Report location. Figure 59; main-report §4.9.8 (description) and §5.7.5 (interpretation).

S.20.2 Script 33 — Climate-swing amplification and dry-year spring floor (Figure 60)

Motivation. A window-independent companion to the window-sensitivity caution (Script 34) and to Script 32's secular drift: rather than differencing two marginal windows, it compares genuine dry and wet end-members. From the

envelope it derives two products — a relative amplification field (how much each area magnifies or damps the shared climate swing) and an absolute drought-floor surface (how deep the dry-year spring table sits against the ecological thresholds).

Inputs. `outputs/01_wells_clean.csv`; `outputs/01_locations.csv`; `outputs/03_master_data.csv`. The coefficient is observational and does not use the SSM.

Methodology. Spring values are formed as for Script 32. Dry and wet extreme states are the per-well mean spring level over fixed extreme-year sets selected by site-mean spring extremity and antecedent-rainfall consistency — dry {2011, 2012, 2019} and wet {2014, 2021, 2024}. The year 2006 is excluded as a wet extreme because its 2004–2005 antecedent was the driest in the record, so the slow- τ forest wells had not refilled by that spring (it is not an antecedent-matched wet state). Each well's swing is its wet state minus its dry state; its amplification coefficient (Figure 60a) is that swing divided by the network-mean swing, with the common-mode swing removed, so values above one magnify and below one damp the shared forcing. The drought-floor surface (Figure 60b) is the dry-state depth to water, kept in absolute depth below ground and contoured against the Curreli et al. (2013) SD15b (0.61 m) and SD16 (0.98 m) thresholds, which are absolute distances to surface. Only the lake gauge is excluded; CEH13 and CEH14 are included (`config ENVELOPE_METRIC_EXCLUDE = lake only`) because the coefficient is observational. The amplification field is interpolated to a 50 m grid by linear interpolation (`scipy griddata`), masked beyond 450 m; the drought-floor surface is likewise linearly interpolated with a ridge mask, and the single raised inter-slack well CEH10 is shown as a distinct slack-edge marker but excluded from the interpolated slack-floor surface, which would otherwise smear its raised-ground depth across the neighbouring slacks.

Headline result. The site-mean dry-to-wet spring swing is ≈ 752 mm (0.75 m). The slow-draining C4 forest interior amplifies it to $\approx 1.72\times$ ($\approx 1.7\times$), the lake edge damps it to $\approx 0.61\times$ ($\approx 0.6\times$), and the open dune is close to unity. The amplification field correlates with the per-well β_2 coefficient, a marker of the deep, slow-recession store. In the dry years the forest interior and western residual block already sit below the SD16 dry-slack threshold.

Outputs. `outputs/33_envelope_amplification/`: `33_envelope_per_well.csv` (dry/wet state, swing, amplification, cluster); `33_amplification_field.png` (Figure 60a); `33_dry_spring_depth.png` (Figure 60b); `33_results.txt`.

Limitations. The amplification is a relative measure and the extreme-year selection is fixed and antecedent-screened rather than data-adaptive. The drought-floor contour is a strict, conservative lower bound on the area below ecological viability: because spring is the seasonal high, a spring level already below a summer-minimum threshold guarantees the summer minimum is deeper still, so the true late-summer sub-threshold footprint is larger. Head swing is not specific-yield-normalized — the figures report observed head movement directly, for ecological and observational directness.

Report location. Figure 60a/b; main-report §4.9.8 and §5.7.5.

S.20.3 Script 35 — Per-well climate-sensitivity coefficient

Motivation. A discrete, frame-independent per-well companion to Script 33's interpolated amplification field, extended to wells with short or inconsistent records that the matched surface and the SSM cannot reach. It produces no surface — a coefficient table, an SSM-calibration figure, and a discrete marker map — so it does not duplicate Script 33.

Inputs. outputs/01_wells_clean.csv; outputs/01_locations.csv;
outputs/03_master_data.csv ($\beta_1/\beta_2/\beta_3$ + cluster, for calibration);
outputs/06_pear_membership_audit_sitewide.csv (cluster fallback for unclustered wells).

Methodology. Each well's dry-to-wet spring swing is normalized co-temporally: the reference core is the wells with full dry-extreme coverage and adequate wet coverage, and the coefficient is the well's swing divided by the core's mean swing recomputed over that well's own extreme years. This cancels the common climate signal window by window, so wells measured on different subsets of extreme springs remain comparable; it reproduces the matched-window amplification (validated $r \approx 0.98$) while removing coverage artefacts. Each coefficient carries a confidence tier (A: ≥ 2 dry and ≥ 2 wet years; B: ≥ 1 of each; C: a single year on one side) and a delete-one-extreme-year jackknife 90% interval. The coefficient is calibrated against the independently fitted SSM response — amplification versus β_2 ($\approx +0.74$) and versus β_3 (≈ -0.45) — with the SSM-unreliable wells shown as hollow markers (“shown, not fitted”) and dropped from the calibration regression.

Honesty note. The coefficient is validated only where β_2 exists (long-record wells); short-record wells are both the use case and the place it cannot be directly verified. The tiers, jackknife CIs, and β_2/β_3 calibration are how that extrapolation

is kept honest. Language throughout is “consistent with the fitted drainage/draw response”, never “confirms”.

Outputs. outputs/35_amplification_metric/: 35_per_well_amplification.csv (coefficient, CI, confidence tier); the SSM-calibration figure; the discrete per-well marker map.

Report location. Main-report §5.7.5 (and Paper 1).

End of chapter S.20.

S.21 Scripts 24b, 31, 31b, 34 — Supplementary standalone diagnostics

Phase 16 (steps 39–42/43). Four standalone diagnostics wired into the orchestrator so they regenerate whenever upstream data change, each addressing a specific robustness question raised elsewhere in the analysis: the mechanism behind the seasonal SSM residual (Script 24b), whether the $k=5$ partition is corroborated by evidence the clustering never used (Scripts 31 and 31b), and whether the two-window MSL5 method can resolve absolute site-wide change (Script 34). None re-fits the SSM; all read committed pipeline outputs.

S.21.1 Script 24b — Cluster-stratified residual climatology

Motivation. Script 24 reports a seasonal structure in the SSM residual field. Three mechanisms could produce it, each with a distinct spatial signature: (1) winter-phased nonlinear recharge that the linear $\beta_1 \cdot P$ term under-represents in heavy-rainfall months — site-wide, including the open dune; (2) ridge-derived lateral input from the metamorphic bedrock ridge to the north — ridge-proximal, concentrated at C4 and forest-margin wells; (3) over-estimation of the $F = 0.24$ canopy-interception correction in winter — forest-confined, at both C4 and C5. Script 24b discriminates among them; it does not pre-judge.

Inputs. The per-well per-month SSM residuals (Script 22), the cluster assignment and Easting/Northing (Script 03 master data), and the “Forest” KML polygon (for distance-to-forest-edge). The SSM is not re-fitted.

Methodology. The residual climatology is stratified across the $k=5$ partition; the discriminant is the per-cluster winter-minus-summer residual contrast together with a within-forest ridge-distance gradient. A site-wide contrast points to mechanism (1); a contrast concentrated at C4 with a ridge-distance gradient

points to (2); a contrast shared by C4 and C5 without a ridge gradient points to (3). Reported neutrally.

Outputs. outputs/24b_residual_climatology/:

24b_01_cluster_climatology.csv (per-cluster mean residual by month) and the accompanying diagnostic figures.

Report location. Supports the discussion of the SSM residual field and the ridge-recharge hypothesis in the main report; supplementary diagnostic, not a headline result.

S.21.2 Script 31 — Independent k=5 partition validation

Motivation. The canonical clusters are formed in Script 02 by Ward's linkage on (1 – Pearson correlation) distance between well hydrographs. Script 31 asks whether that partition is corroborated by evidence the clustering never used, organized by how independent each line of evidence actually is.

Methodology. Four tiers. Tier 1 (external): data orthogonal to the hydrographs — geography, the forest-canopy polygon, distance-to-coast, elevation. Tier 2 (metric-independent): magnitude descriptors of the hydrographs — mean depth, amplitude, summer minima, dry depth — which are largely orthogonal to the correlation (shape/timing) structure the clustering used. Tier 3 (convergent): the same water-level series via a different estimation method — SSM betas, WTF Sy, LCSC — supporting but not independent. Tier 4 (robustness): whether k=5 survives a different linkage or distance metric (average, complete; Spearman, DTW), measured by the Adjusted Rand Index against the canonical Ward+Pearson partition. All canonical numbers are read from live pipeline CSVs.

Outputs. outputs/31_cluster_validation/: 31_validation_summary.csv (one row per test — tier, statistic, p, independence); 31_method_robustness_ari.csv (ARI of each alternative clustering vs canonical); 31_forest_confusion.csv (cluster × forest-polygon crosstab + Cohen's kappa); 31_forest_borderline.csv (wells within the edge band, signed distance); 31_cluster_validation_panel.png (4-panel figure).

Report location. Supports the k=5 cluster-framework justification in the main report (§4.2 / §5.1); supplementary diagnostic.

S.21.3 Script 31b — Cluster separation versus recoverability

Motivation. A standalone companion to Script 31 making one point with one figure: the clusters differ on the independent variables (so they are real), but

those variables do not by themselves reconstruct the clusters (because the hydrograph timing carries information no static attribute holds). High separation does not imply high recoverability.

Methodology. For each independent variable X the figure places side by side a separation statistic — η^2 , the variance in X explained by the pre-formed partition (“do the clusters differ on X ?”) — and a recoverability statistic — the Adjusted Rand Index of a fresh Ward $k=5$ clustering on standardized X against the canonical partition (“does X alone rebuild the clusters?”). Distance-to-coast is taken to the Caernarfon Bay mean-high-water shoreline (the Menai Strait excluded), from the consolidated well metadata.

Outputs. outputs/31_cluster_validation/
31b_separation_vs_recoverability.csv;
31b_separation_vs_recoverability.png.

Report location. Companion to Script 31; supplementary diagnostic.

S.21.4 Script 34 — MSL5 two-window sensitivity demonstration

Motivation. A deliberate cautionary demonstration: how strongly an apparent “site-mean water-table change” depends on which two five-year spring windows are differenced. The §4.9.8 headline differences window-end 2017 against window-end 2023 and reports -96.8 mm; Script 34 places that single comparison inside the envelope of every admissible window pair, so §5.7.5 can show from a committed, reproducible figure that the two-window MSL5 method cannot resolve absolute site-wide change.

Inputs. The committed per-well annual spring MSL (outputs/26_van_willegen_msl/26_msl_annual_per_well.csv, column MSL_m_bg), valid rows only; the committed annual climate summary (for the window-axis rainfall-deviation annotations). Exclusions: config.MSL5_EXCLUDED_WELLS (CEH13, CEH14), matching Script 26.

Methodology. Per-well MSL5 for a window ending in year Y is the mean of the well’s annual spring MSL over years $Y-4$ to Y ; a well qualifies only if all five spring-years are present. For each ordered pair ($W_i < W_j$) the site-mean change is the mean over the common panel (wells qualifying in both windows) of the per-well ($W_j - W_i$) change — the panel is held fixed across the pair, so composition change cannot inflate it. A pair is admissible when its common panel reaches config.MSL5_WINDOW_MIN_PANEL = 40 wells, which excludes only the thin seven-well 2005–2009 baseline, so the spread arises from window choice alone. All

admissible pairs are retained, including those whose current window contains the anomalously wet 2024 spring — admitting them is the point of the demonstration.

Headline result. The site-mean change spans -0.14 to $+0.22$ m (-136 to $+221$ mm) across the 66 admissible pairs and changes sign — 19 falling, 47 rising. The 2017→2023 pair reproduces the -96.8 mm headline (it returns -96.5 mm, $n = 60$; a one-well coverage-rule nuance). The most negative pairing is 2017→2020 (-136 mm) and the most positive 2015→2024 ($+221$ mm).

Outputs. `outputs/34_window_sensitivity/`: `34_window_matrix.csv` (every admissible pair: baseline-end, current-end, change, `n_common`, admissible); `34_results.txt` (anchor check, envelope range, sign split); `34_window_sensitivity.png` (Panel A all-pairs matrix; Panel B site-mean spring trajectory with interannual SD).

Limitations. Panel B’s own-panel OLS trend is descriptive; the canonical secular trend is Script 32’s AR-corrected -7.0 mm yr⁻¹ ($p = 0.52$), and both are non-significant. The demonstration deliberately admits the wet-2024 windows, so the envelope is wider than a “defensible-windows” subset would give — that is the intended message, not a defect.

Report location. Main-report §5.7.5 (window-sensitivity figure).

End of chapter S.21.

S.17 Appendices

Reference and post-pipeline material. Final chapter of the supplement.

This chapter closes the Methods Supplement. It covers two pieces of material that belong with the document but sit outside the main script-by-script chapter sequence (S.1–S.18b). Appendix A documents the one remaining pipeline script not yet given a chapter — the post-pipeline greyscale figure utility. Appendix B is the canonical-sources index: a reference table mapping each recurring concept across the supplement to the place where it is authoritatively defined.

A. Greyscale figure post-processing — Script 27

`27_greyscale_figures.py` is a post-pipeline rendering utility, not an analytical step. It exists to produce a journal-ready black-and-white bundle of the pipeline’s colour figures without re-running any analytical script. Reviewers, journal

proofs, and print compatibility occasionally require this; the conversion runs once and produces a parallel `outputs_bw/` tree alongside the canonical `outputs/` tree, preserving the directory structure so that any figure has the same relative path in both.

The script offers two conversion modes. The default uses ITU-R BT.709 perceptual luminance weights ($0.2126 R + 0.7152 G + 0.0722 B$), preserving the alpha channel where present and compositing onto white first to avoid grey halos around transparent edges. This is the right choice for line plots, bar charts, scatter plots, and most hydrograph figures — the kind of figure whose informational content is in shape and position rather than colour. The `--enhanced` mode applies contrast-limited adaptive histogram equalisation (CLAHE-style auto-contrast with a 0.5 % cutoff, plus light sharpening) on top of the luminance conversion. It is intended for figures where the colour palette compresses into a narrow luminance range, typically diverging colormaps on spatial maps.

Two classes of figure do not survive naive greyscale conversion and are flagged in the output with [REVIEW]. Script 11b's categorized ecological-zone maps (`11b_01_summer_minima_depth`, `11b_02_winter_maxima_depth`, `11b_03_pflood`) use four or five colour bands that collapse to similar mid-greys under any luminance transform — distinguishing them in print needs native B&W rendering with hatched zones rather than post-processing. The Script 19 difference maps (`difference_maps/diff_*`) use diverging RdBu colormaps with colourbar labels reading “blue = wetter / red = drier” baked into the raster; the labels become meaningless in greyscale and cannot be post-edited. Both classes are still converted by default (the output is usable, just suboptimal), and `--exclude-problem` skips them entirely.

The CLI is

```
python src/27_greyscale_figures.py [--enhanced] [--dpi DPI] [--skip-  
maps]                                     [--exclude-problem] [--dry-run]
```

with `--dpi` overriding the source DPI, `--skip-maps` excluding the spatial-map outputs that benefit from manual review, and `--dry-run` listing the files that would be converted without writing anything. Script 27 is orchestrated by `run_analysis.py` as PHASE_17 (step 43/43) but is a post-analysis utility rather than an analytical step — it is excluded from the supplement's 41-step analytical pipeline count and lives here in Appendix A. The script is idempotent: deleting `outputs_bw/` and re-running rebuilds it cleanly.

Script 27's filename prefix (27_) and orchestrator step number (43/43) deliberately do not match — the same convention applied to Script 26 (26_van_willegen_msl.py at step 30/43), Script 26b (26b_van_willegen_msl_projections.py at step 31/43), and Script 26c (26c_msl5_report_figures.py at step 32/43). The filename groups Script 27 alphabetically with the other 2x_ scripts; the orchestrator number reflects its position in the run order after Phase 14. Under the supplement's analytical-step count (41 steps, Phases 1–16 of run_analysis.py), Script 27 has no analytical-step number — it is the post-analysis utility documented in this appendix. Script 26c, the MSL5 report-format figure-rendering companion, is similarly excluded from the analytical count and is documented in §S.18c.

B. Canonical sources of truth — reference table

The supplement is long enough that a reader who has read it once and needs to find where a particular concept, parameter, or function is defined will not always remember which chapter to open. This table is the navigational index. For each recurring concept, it points to the single canonical source — a chapter, a front-matter section, or a specific file in the repository — where the concept is defined and explained. Where a constant is read from config.py, the table names the constant; where a function is the canonical implementation, it names the module.

The convention throughout the supplement is that there is one place to change a value and one place to look it up. The table reflects that. Entries are organized into thematic groups — model formulation, partition and cluster constants, intervention machinery, scenario engine, ecological thresholds, spatial machinery, MSL aggregation, and provenance and orchestration.

Model formulation

Concept / parameter	Canonical source	Notes
SSM equation	F.3	$\Delta h(t) = \beta_1 \cdot P(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot (D + h(t-1))$
SSM displacement formulation	F.3	$h_disp = DRAINAGE_DATUM + h_depth$
$DRAINAGE_DATUM = 3.7\text{ m}$	F.4 / config.py	Sensitivity analysis in Script 03 (03_08_datum_sensitivity.csv)
$HEADLINE_LAG = 0$	F.4 / config.py	Contemporaneous

Concept / parameter	Canonical source	Notes
		rainfall; lag-1 regime is historical
Bucketing convention	F.2	Day $\leq 15 \rightarrow$ previous month; day $> 15 \rightarrow$ same month
Sign conventions	F.3	All three β reported positive; sign baked into design matrix
MIN_OBS = 30	F.5 / model_utils.py	Minimum aligned rows for any per-well SSM fit
LCSC_DATA_LIMIT = 100	F.5 / model_utils.py	Recent-window length for intercept-audit fits
assert_physical_signs()	F.3 / F.5 / model_utils.py	Hard violations ($\beta_1 \leq 0$, $\beta_2 \leq 0$); soft warnings ($\beta_3 \leq 0$)
build_ssm_frame()	F.3 / F.5 / model_utils.py	Canonical data alignment; no script reimplements
fit_ssm() / fit_ssm_intercept()	F.3 / F.5 / model_utils.py	No-intercept (Model A, headline) / with-intercept (Model B, diagnostic)
simulate_ssm()	F.5 / model_utils.py	Forward simulation with displacement recurrence
Two regimes (A / B)	F.3 / S.5	Model A is the headline; Model B used by Scripts 07, 08, 22, 24

Partition and cluster constants

Concept / parameter	Canonical source	Notes
CLUSTER_LABELS (C1–C5)	F.4 / config.py	k=5 Ward's partition
CLUSTER_COLOURS, CLUSTER_COLOURS_BW, CLUSTER_MARKERS	F.4 / config.py	Canonical palette and B&W equivalents
CLUSTER_ID_ANCHORS	F.4 / 02_clustering.py	Pins Ward's raw output

Concept / parameter	Canonical source	Notes
Anchor wells per cluster	F.4 / config.py	to canonical IDs C1: ceh5, ceh11; C2: d10; C3: nw1; C4: ceh2; C5: ceh16, nw9
Reference network (66 wells)	S.2 / 01_data_prep.py	REFERENCE_CUTOFF_DATE = "2026-02-01"
Identity-vs-integer keying principle	F.4	Survives partition changes if applied correctly
Partition history (k=6 → k=5)	F.4	Old C5 ≠ new C5; Lake cluster dropped as singleton
Cluster β coefficients	S.3 / 03_master_data.csv	Per-well $\beta_1, \beta_2, \beta_3$; canonical source for downstream
Per-cluster mean β coefficients	S.3 / 03_03_cluster_mechanistic_coefficients.csv	Used by Scripts 07, 11, 16, 19, 21
BW_MODE rendering	F.4 / config.py	Toggled by NRG_BW_MODE environment variable

Specific yield

Concept / parameter	Canonical source	Notes
Fetter mass-balance Sy	F.4 / S.11 / config.py	Operational Sy for Script 16 water balance
WTF cluster median Sy (Approach B)	S.12 / 17_wtf_01_sy_estimates.csv	Canonical WTF cluster Sy; Approach A also valid in the current pipeline — chapter S.12 quotes both
WTF per-well Sy	S.12 / 18_wtf_per_well_sy.csv	Used by Script 19 volumetric conversion
Storage–drainage index τ = S_y/β_3	S.12 / 18_wtf_05_drainage_timescale.csv	Deliberately storage-weighted composite; not a residence time. Head-space recession time is

Concept / parameter	Canonical source	Notes
		$t_R = 1/\beta_3$ (see S.12 <i>Site-specific choices</i> ; cluster means $C1 \approx 10$, $C2 \approx 14$, $C3 \approx 20$, $C4 \approx 57$, $C5 \approx 22$ months)
Interception correction for forest Sy	S.12	Cluster vs well-level reconciliation

Forest and forest scenarios

Concept / parameter	Canonical source	Notes
FOREST_CIDS = (4, 5)	F.4 / config.py	Both forested clusters under k=5
FOREST_INTERCEPTION = 0.24	F.4 / config.py	Freeman (2008), Newborough Corsican pine
BROADLEAF_INTERCEPTION = 0.15	F.4 / config.py	Komatsu et al. (2011) annual mean
BROADLEAF_B2_SUMMER, BROADLEAF_B2_WINTER	F.4 / config.py / S.14	Live values: 1.0750 / 0.8817 (May–Oct / Nov–Apr)
Interception treatment under SSM	F.4	Partition of PET energy budget, not additive
load_clearfell_b2_multiplier()	F.5 / clearfell_common.py	BACI-corrected Edge-tier ratio; read dynamically from 10e_01_coefficient_shifts.csv
Thinning β_2 multiplier	F.5 / clearfell_common.py	Derived as half-perturbation from clearfell multiplier

Intervention machinery

Concept / parameter	Canonical source	Notes
BACI 5-tier network	S.7 / clearfell_common.py	Impact / Edge / Forest Ctrl / Coastal Ctrl / Climate Ctrl

Concept / parameter	Canonical source	Notes
Clearfell well tier definitions	F.5 / <code>clearfell_common.py</code>	IMPACT_WELLS, EDGE_WELLS, FOREST_CONTROL_WELLS, COASTAL_CONTROL_WELLS, CLIMATE_CONTROL_WELLS
INTERVENTION_DATE = 2017-12-01	F.5 / <code>clearfell_common.py</code>	Clearfell intervention boundary
SCRAPING_DATE, SCRAPING_DATE_2	F.5 / <code>clearfell_common.py</code>	2015-04-01 and 2023-10-01
Scraping suite well groups	F.5 / <code>scraping_common.py</code>	IMPACT_WELLS, PAIRED_CONTROLS_MAP, CLIMATE_CONTROLS, DONOR_CANDIDATES
Scraping era definitions	F.5 / <code>scraping_common.py</code> / S.6	Per-well WELL_ERAS dict
Scraping $\lambda = 300$ m	F.5 / <code>clearfell_common.py</code>	Exponential distance weight for scraping propagation
INTERVENTION_COLOUR_S CRAPE, INTERVENTION_COLOUR_C LEARFELL	F.4 / <code>config.py</code> / S.18	Purple #7b3294 for scraping (2015 CEH36, 2023 CEH18/21), orange #e66101 for the 2017 clearfell; used by Script 26 trajectory plots

Scenario engine

Concept / parameter	Canonical source	Notes
<code>monthly_perturbation()</code>	F.5 / <code>model_utils.py</code> / S.14	Option 3 single-step forcing-change response; replaces equilibrium $\Delta h / \beta_3$
<code>compute_scenarioBars()</code>	F.5 / <code>scraping_common.py</code>	Per-cluster scenario engine consumed by Scripts 09d, 19, 21
<code>summer_amplification_factors()</code> , <code>scenario_cluster_sy()</code> ,	F.5 / <code>scraping_common.py</code>	Flux $\rightarrow$ summer-minimum helpers (retained but unused)

Concept / parameter	Canonical source	Notes
flux_to_summer_min_mm (), scenario_summer_min_b ars() pflood_lambda()	F.5 / model_utils.py / S.9	after 2026-07-02 volumetric re-basis; marked for removal)
UKCP18 2050s scaling factors	F.4 / config.py	RCP8.5 Wales, 50th percentile, 2050s; UKCP18_DRY_* / UKCP18_WET_* constants in config.py
UKCP18 2080s scaling factors	F.4 / S.18b / 26b_van_willegen_msl_ projections.py	Currently hardcoded in Script 26b's UKCP18_SCENARIOS dict; follow-up to centralise as UKCP18_2080s_* in config.py
pipeline_scenario_par ams.csv	F.4 / pipeline_params.py / S.1	Producer-consumer architecture for cross- script parameters
load_params()	F.5 / pipeline_params.py	Read by Scripts 09b, 09d, 19, 21
Tool A — spring MSL transfer function	S.9 (§Sub-script 11 Methodology) / S.18b (§S.18b.2) / 11_forecasting_thresh olds.py Section 5	$MSL_y =$ $\alpha \cdot h_{max_winter} +$ $\beta \cdot P_{win_to_spr} +$ $\gamma \cdot PET_{win_to_spr} +$ intercept; R^2 0.73–0.96; previous-MSL variant dropped at v1.1.1
Tool B — UKCP18 MSL5 perturbation overlay	S.18b (§S.18b.3) / 26b_van_willegen_msl_ projections.py	Single-step $\Delta h(m) =$ $\beta_1 \cdot \Delta P - \beta_2 \cdot \Delta PET$; drift- free by construction; $\Delta MSL5$ 1–4 cm
Spring-window structural cancellation	S.18b §S.18b.3.5	Why $\Delta MSL5$ modest at 1– 4 cm despite +20–35 % summer PET; the spring window straddles the UKCP18 seasonal

Concept / parameter	Canonical source	Notes
partition		
<i>Ecological thresholds</i>		
Concept / parameter	Canonical source	Notes
Curreli summer thresholds	F.4 / config.SD15b, SD16	Wet slack 0.61 m, dry slack 0.98 m
Curreli winter flooding limits	F.4 / config.SD15b_WINTER, SD16_WINTER	Wet slack 0.10 m, dry slack 0.25 m
Recovery / excavation limits	F.4 / config.SD15b_REC, SD16_REC	Wet slack 0.75 m, dry slack 1.20 m
Ecological zone categorisation	S.9 / 11b_spatial_threshold s.py	Native rendering of zone maps
<i>Coastal and spatial</i>		
Concept / parameter	Canonical source	Notes
Coastal-retreat gradient parameters	S.15 / 25_coastal_gradient.py	$\delta_0 = -28.6$ mm/yr, $L = 894$ m, $c = -6.3$ mm/yr
Coastline provenance	S.15	OS Open Map Local TidalBoundary, EPSG:27700
well_distance_to_coast.csv	S.15 / paths.DATA_DIST_COAST	Generated once, out-of-pipeline
Site geography (drainage paths, ridge)	F.2 / site_geography.md	Bedrock ridge as northern boundary; bipartite drainage
add_idw_surface()	F.5 / map_utils.py / S.5	50 m grid IDW with optional ridge mask
load_dem_layer(), load_dem_hillshade()	F.5 / map_utils.py	DEM rendering helpers
add_kml_features()	F.5 / map_utils.py	Features, streams, clearfell overlays
plot_metric_map()	F.5 / map_utils.py	High-level publication map wrapper

MSL aggregation

Concept / parameter	Canonical source	Notes
Spring window (Mar–May)	S.18 / config.MSL_SPRING_MONTHS = (3, 4, 5)	Van Willegen 2025 Table 2
Hydrology year B (1 Jun y–1 to 31 May y)	F.2 / S.18 / config.MSL_HYDRO_YEAR_START_MONTH = 6	Van Willegen 2025 convention; co-exists with the October-start convention used in Script 11 Sections 2 and 4
5-year MSL window	S.18 / config.MSL_DEFAULT_WINDOW_YEARS = 5	Van Willegen et al. (2025) sensitivity-tested this choice
Strictness: 3/3 spring, 5/5 window	S.18 / config.MSL_MIN_MONTHS_PER_SPRING = 3, MSL_MIN_YEARS_IN_WINDOW = 5	Stricter than van Willegen 2025; consistent with 09c, 10d summer-minima min_measured=2 pattern
Trajectory restriction to window-ends $\geq$ 2014	S.18 / config.MSL_TRAJECTORY_START_YEAR = 2014	Network-composition shift around 2010; per-well CSVs retain full record from 2009
Van Willegen quadrat wells (17 piezometers)	S.18 / config.VW_QUADRAT_WELLS	16/17 covered under strict rules; T41 excluded for insufficient recent record
Method A (per-well cluster mean)	S.18 / paths.OUT_26_5YR_PER_CLUSTER	Headline monitoring metric; extended network
Method B (cluster-centroid from 03_regional_averages.csv)	S.18 / S.18b / paths.OUT_26_5YR_PER_CLUSTER_CENTROID	SSM-consistent companion; reference network only; baseline for Tools A and B
MSL5 ↔ summer-minima 5-yr offset (~0.54 m)	S.18 §Empirical relationship to summer minima	Pearson $r = 0.945$ at the 5-year window scale ($n = 829$ well-year rows)
Curreli SD15b/SD16 on	S.18 §Site-specific	Reference lines retained

Concept / parameter	Canonical source	Notes
MSL5 plots	choices	for visual familiarity; calibrated against summer minima, not MSL5 — figure captions flag the offset

Residuals and diagnostics

Concept / parameter	Canonical source	Notes
SSM residual structure	S.16 / Script 22	Network mean residual $AR(1) \approx -0.12$
Ridge-recharge lag null result	S.16 / Script 23	Spearman $\rho = +0.010$ on lag-vs-distance; test design statistically degenerate against monthly resolution (see §S.16 structural caveat) — null result documents the analytical attempt, not positive evidence against the mechanism
Residual seasonality diagnostics	S.16 / Script 24	Phase 12 residual diagnostics; 47/63 wells peak Nov–Mar
LCSC intercept audit	S.5 / Script 08	Model A vs Model B comparison per well

Climate and field data

Concept / parameter	Canonical source	Notes
RAF Valley climate station	F.2 / RAF_VALLEY_LAT_DEG = 53.25	53°14'32"N, ≈ 16 km from site
Thornthwaite PET	F.2 / S.8 / Script 00	Computed at RAF Valley latitude
Dipwell field protocol	F.2	End-of-month readings, day-15 bucketing
<code>clean_well_series()</code>	F.5 / <code>data_utils.py</code>	Drops > 4.0 m depths; linear interp single-

Concept / parameter	Canonical source	Notes
MAX_PHYSICAL_DEPTH = 4.0 m	F.5 / data_utils.py	month gaps (limit=1) Deepest plausible water table
normalize_well_name()	F.5 / data_utils.py	Used wherever well names join across sources

Orchestration and rendering

Concept / parameter	Canonical source	Notes
paths.py OUT_* / INT_* / DIR_*	F.5 / utils/paths.py	All file paths; no script hardcodes a path
make_all_dirs()	F.5 / paths.py	Directory creation helper
run_analysis.py	S.1 (introduction)	Pipeline orchestrator; CLI flags for phases
Two-pass workflow	F.4 / PIPELINE_README.md	First pass uses fallbacks; second uses canonical Sy / β_2 multipliers
Greyscale post-processor	Appendix A (this chapter) / 27_greyscale_figures.py	Phase 17 in run_analysis.py (step 43/43); outside the 41-step analytical count
Script 11c / 14b / 25 / 26 / 26b / 26c / 27 / 28 / 29 naming convention	F.4	Filename prefix, orchestrator step number, and chapter assignment aligned
paths.DIR_26, paths.OUT_26_*	F.5 / utils/paths.py / S.18	All Script 26 outputs (Method A + Method B parallel CSVs; quadrat-wells figure; MSL5 map)
paths.DIR_26B, paths.OUT_26B_*	F.5 / utils/paths.py / S.18b	All Script 26b outputs (UKCP18 projection figure, summary table, monthly Δh table, results transcript)
paths.DIR_26C, paths.OUT_26C_*	F.5 / utils/paths.py / S.18c	All Script 26c outputs (trajectory report figure,

Concept / parameter	Canonical source	Notes
		Δ MSL5 vs Δ summer-min projection figure, results transcript)
paths.OUT_11_TABLE_SP RING, paths.OUT_11_SPRING_C ALIBRATION	F.5 / utils/paths.py / S.9 / S.18b	Tool A outputs (Script 11 §5): Table 9 coefficients CSV and per-cluster calibration scatter figure
paths.OUT_11C_*	F.5 / utils/paths.py / S.9.3	Script 11c P_flood achievability outputs (sharing DIR_11B with Script 11b): map PNG, per-well category CSV, results memo
paths.DIR_14 (shared with Script 14b)	F.5 / utils/paths.py / S.8.5	Script 14b bootstrap year-of-crossing outputs share Script 14's directory; no new path constants added
paths.DIR_28, paths.OUT_28_*	F.5 / utils/paths.py / S.19.1	All Script 28 C3 detrend check outputs (per-well CSV, results memo, four-panel figure)
paths.DIR_29, paths.OUT_29_*	F.5 / utils/paths.py / S.19.2	All Script 29 within-C3 variance outputs (per-well panel CSV, univariate R^2 matrix, drop-one matrix, results memo, six-panel figure)
paths.OUT_25_CLUSTER_ DECOMP_FIG	F.5 / utils/paths.py / S.15	Script 25 v1.1.0 fold-in: per-cluster decomposition stacked-bar figure (§4.8.2 of the main report)

Closing remarks

The Methods Supplement closes here. The chapters S.1–S.21 together document the 41-step Newborough Warren analytical pipeline, the design choices behind each step, the rationale for site-specific parameters, and the verification chain by which pipeline outputs feed the main report. Script 26c (`26c_msl5_report_figures.py`, Phase 13 in `run_analysis.py`) is a display-only figure-rendering companion to Scripts 26 and 26b, outside the 41-step analytical count, covered in §S.18c; Script 27 (`27_greyscale_figures.py`, Phase 17 in `run_analysis.py`) is a post-analysis figure-conversion utility, also outside the 41-step analytical count, covered in Appendix A. Readers needing a specific topic should consult the canonical-sources table in Appendix B; readers needing the canonical implementation of any function or constant should consult the live main branch of https://github.com/newbroman/Newborough_Hydrology, which remains the source of truth. The supplement is a guide to what the repository contains and why each choice was made; the repository itself is the deliverable.

End of the Methods Supplement.